NSP
Network Services Platform

NSP Device Configuration Intent Type Catalog

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1 Introduction

1.1 Introduction

1.1.1 Scope

This document lists and describes the device configuration intent types available for use with NSP. It is a catalog comprising brief descriptions of the NSP product intent type artifacts delivered on the Nokia NSP software download site.

The guide was formerly named the NSP ICM Intent Type Catalog.

1.1.2 Delivery

This document is updated each time new or modified device configuration artifact bundles are delivered, which can happen off-cycle from NSP releases.

1.1.3 About device configuration intent types

NSP product intent types for device configuration are commercially available artifacts developed by Nokia and delivered in software bundles on the Nokia NSP software download site.

Device configuration intent types may be either:

- **Unified**: designed for NEs that can be managed by either NSP classic (SNMP) or model-driven (MD) mediation, and supports configuring a shallow range of attributes that are common to both forms of mediation
- **Global**: designed for NEs that can be managed by either NSP classic (SNMP) or model-driven (MD) mediation, and supports configuring a deeper range of attributes than unified intents, focused on specific areas
- **NE-specific**: designed to support specific NEs and covering meditation-specific attributes. The following categories of NE-specific device configuration intent types are available:
  - MD SR OS: denoted with "msros" or "mixr" in their name
  - Classic SR OS: denoted with "csros", "cixr", or "csas" in their name
  - SRL: denoted with "srl" in their name
  - Wavence: denoted with "wavence" in their name

Development effort is shifting from unified intents to NE-specific and global intents to allow for greater depth of attribute coverage.

1.1.4 Compatibility

Device configuration intent types are typically forward-compatible with the next two major NSP releases. For example, device configuration intent type artifacts developed and delivered for NSP Release 23.x will work with NSP 24.x and NSP 25.x at the original 23.x feature set level.

This document identifies the "Supported from" NSP release for each artifact; that release can be considered the "N" release in the N+2 forward compatibility scenario, regardless of when the artifact is delivered.

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From an SR OS version point of view, device configuration intent type artifacts typically offer N-3 and N+2 compatibility.

Classic and MD SR OS intent types have their initial NE and NSP "N" release support appended to the name of the intent type. For example, lag_msros_23-10-1_24-4 support is introduced for MD SR OS node version 23.10.R1 and NSP release 24.4. Classic intents are certified for the NFM-P release that supports the corresponding NE release. For example, Release 23.10 of the NE is supported in Release 23.11 of the NFM-P.

1.1.5 Document content

Available device configuration intent types are listed alphabetically within each chapter of this guide. For each intent type, the following information is provided:

- Name of the intent type
- Brief description
- Supported NE configuration options
- NSP release in which support is introduced
- Compatible (tested) NE types and releases
- Advisories (limitations or notable information about support)

1.1.6 Further reading

See the "Intent types" chapter of the *NSP Network Automation Guide* for information about obtaining intent types and importing them into NSP.

See the Nokia Developer Portal for API documentation related to device configuration: https://network.developer.nokia.com/api-documentation/

See the router configuration guides for detailed descriptions of the attributes configurable in NSP through the addition of a pluggable artifact.

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1.2 Release Information

1.2.1 Introduction

The following sections list which intents were added or removed in each issue of this catalog for the current year.

1.2.2 Issue 15, November 2025

SRL intent types

Added the following intents:

- interface_srl_24-10-1_25-4
- routing-community-set_srl_24-10-1_25-4
- routing-prefix-set_srl_24-10-1_25-4
- subinterface-routed_srl_24-10-1_25-4

MD mode intent types

Added the following intents:

- router-bgp-group_msros_24-10-1_24-11
- router-bgp-peer_msros_24-10-1_24-11
- router-bgp_msros_24-10-1_24-11
- router-mpls-labels_msros_23-10-1_24-11
- router-mpls_msros_24-10-1_24-11
- subscriber-mgmt-sla-profile_msros_24-10-1_24-11
- subscriber-mgmt-sub-profile_msros_24-10-1_24-11

Classic mode intent types

Added the following intents:

- subscriber-mgmt-sla-profile_csros_23-10-1_24-11
- subscriber-mgmt-sub-profile_csros_23-10-1_24-11
- router-mpls_csros_23-10-1_24-11

Unified intent types

The following intents have been withdrawn:

- icm-qos-sapegress-srqos
- icm-qos-sapingress-srqos
- icm-router-communitypolicy-srrouter
- icm-router-prefixlist-srrouter

1.2.3 Issue 14, September 2025

This issue includes several updates to intents related to Wavence nodes.

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1.2.4 Issue 13, August 2025

SRL intent types

Added the following intents:

- qos-classifiers-dot1p-policy_srl_24-10-1_25-4
- qos-classifiers-dscp-policy_srl_24-10-1_25-4
- qos-policer-template_srl_24-10-1_25-4
- qos-rewrite-rules-dot1p-policy_srl_24-10-1_25-4
- qos-rewrite-rules-dscp-policy_srl_24-10-1_25-4

MD mode intent types

Added the following intents:

- routing-flex-algo-def_msros_24-10-1_24-4
- segment-routing-mpls_msros_24-10-1_24-4
- segment-routing-policies_msros_24-10-1_24-4
- segment-routing-srv6_msros_24-10-1_24-4
- router-mpls-path_msros_23-10-1_23-11
- router-mpls-lsp_msros_24-10-1_25-4
- policy-options-as-path_msros_23-10-1_23-11
- policy-options-as-path-group_msros_23-10-1_23-11

Classic mode intent types

Added the following intents:

- router-mpls-dynamic-lsp_csros_23-10-1_25-4
- router-mpls-path_csros_23-10-1_23-11
- router-mpls-srte-lsp_csros_23-10-1_25-4
- policy-options-as-path-group_csros_23-10-1_23-11
- policy-options-as-path_csros_23-10-1_23-11

1.2.5 Issue 12, June 2025

MD mode intent types:

Added the following intents:

- ipsec-ike-policy_msros_24-10-1_24-4
- ipsec-ike-transform_msros_24-10-1_24-4
- ipsec-transport-mode-profile_msros_24-10-1_24-4
- isa_msros_24-10-1_24-4
- lag_mixr_24-10-1_24-4
- lag_msros_24-10-1_24-4
- port-eth_mixr_24-10-1_24-4

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- port-eth_msros_24-10-1_24-4
- qos-ingress-classification_mixr_24-10-1_24-4
- qos-sap-egress_msros_24-10-1_24-4
- router-interface_mixr_24-10-1_24-4
- router-interface_msros_24-10-1_24-4
- router-isis-interface_msros_24-10-1_24-4
- router-ospf-interface_msros_24-10-1_24-4
- sec-pki_msros_24-10-1_24-4

1.2.6 Issue 11, May 2025

Wavence intent types:

Added the following intents:

- icm-wavence-system-preference
- icm-wavence-radio-port-specifics
- icm-wavence-radio-lag-specifics
- icm-wavence-network-element-config
- icm-wavence-feature-capacity
- icm-wavence-epc-power-saving
- icm-wavence-1588_ptp
- icm-wavence-time-synchronization

1.2.7 Issue 10, March 2025

Classic mode intent types:

Added the following intents:

- sec-aaa-local-profile_csros_23-10-1_23-11
- sec-client-tls-profile_csros_23-10-1_23-11
- sec-maf_csros_23-10-1_23-11
- sec-server-tls-profile_csros_23-10-1_23-11
- sec-ssh_csros_23-10-1_23-11_v1
- sec-tls-cert-profile_csros_23-10-1_23-11
- sec-tls-client-cipher-list_csros_23-10-1_23-11
- sec-tls-server-cipher-list_csros_23-10-1_23-11
- sec-tls-trust-anchor-profile_csros_23-10-1_23-11
- security-user_csros_23-10-1_23-11
- security-zone_csros_23-10-1_23-11
- log-syslog_csros_23-10-1_23-11

MD mode intent types.

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Added the following intents:

- log-syslog_msros_23-10-1_23-11
- sec-aaa-local-profile_msros_23-10-1_23-11
- sec-client-tls-profile_msros_23-10-1_23-11
- sec-maf_msros_23-10-1_23-11
- sec-server-tls-profile_msros_23-10-1_23-11
- sec-ssh_msros_23-10-1_23-11
- sec-tls-cert-profile_msros_23-10-1_23-11
- sec-tls-client-cipher-list_msros_23-10-1_23-11
- sec-tls-server-cipher-list_msros_23-10-1_23-11
- sec-tls-trust-anchor-profile_msros_23-10-1_23-11
- sec-user-params_msros_23-10-1_23-11
- system-login-control_msros_23-10-1_23-11

1.2.8 Issue 9, January 2025

This issue makes several editorial clarifications and fixes.

Classic mode intent types.

Added the following intents:

- qos-vlan-policy_cixr_23-10-1_23-11

1.2.9 Issue 8, January 2025

Classic mode intent types.

Added the following intents:

- port-eth_cixr_23-10-1_24-4
- routing-policy-statement_cixr_23-10-1_24-4
- router-network-interface_cixr_23-10-1_24-4
- qos-shared-policer_cixr_23-10-1_23-11
- qos-ingress-classification_cixr_23-10-1_23-11
- qos-egress-remark-policy_cixr_23-10-1_23-11
- service-bgp-evpn_csros_23-10-1_24-4
- lag_cixr_23-10-1_24-4

MD mode intent types.

Added the following intents:

- lag_mixr_23-10-1_24-4
- port-eth_mixr_23-10-1_24-4
- policy-options-statement_mixr_23-10-1_24-4
- router-interface_mixr_23-10-1_24-4

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- qos-shared-policer_mixr_23-10-1_23-11
- qos-vlan-policy_mixr_23-10-1_23-11
- qos-ingress-classification_mixr_23-10-1_23-11
- qos-egress-remark-policy-mixr_23-10-1_23-11
- service-system-bgp-evpn_msros_23-10-1_23-11

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2 MD mode intent types

2.1 Name of the Intent: bfd-sbfd-reflector_msros_23-10-1_23-11

2.1.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750-SR, 7950-XRS, 7450-ESS and 7250 IXR MD SROS of node version 23.10

2.1.2 Description:

bfd-sbfd-reflector_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Seamless-BFD reflector.

The following configurations are supported per deployment:

```
- Description
- Admin State
- Local State
- Discriminator
```

CLI context: configure > bfd > seamless-bfd > reflector

2.1.3 Advisory:

- Unique discriminator should be assigned for every node.

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2.2 Name of the Intent: filter-ml-ip-prefix-list_msros_23-10-1_23-11

2.2.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS MD SR OS node version 23.10

2.2.2 Description

filter-ml-ip-prefix-list_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of filter match-list IP-prefix list policies.

The following configurations are supported per deployment:

```
- Description
- Creation of IP Prefix List:
  - Apply Path: BGP Peers [Criterion Index (1 to 255), Group (1 to
255/regex), Neighbor (1 to 255/regex), Router Instance (VPRN
Service/Base)]
  - Prefix: IP Prefix
  - Prefix Exclude: IP Prefix
```

CLI context: filter > match-list > ip-prefix-list

2.2.3 Advisory

- BGP Peers: the Group, Neighbor, and Router Instance elements are immutable. Modifying these elements re-creates the BGP Peers with the new values.
- Apply-Path and Prefix-Exclude capabilities are mutually exclusive.
- Apply Path is not supported on 7250 IXR.
- Prefix Exclude is not supported on 7250 IXR-e, IXR-ec, IXR-R6d, IXR-R6dl, IXR-e2, IXR-e2c and IXR-x.

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2.3 Name of the Intent: filter-ml-ipv6-prefix-list_msros_23-10-1_23-11

2.3.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SR OS node version 23.10

2.3.2 Description

filter-ml-ipv6-prefix-list_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of filter match-list IPv6-prefix list policies.

The following configurations are supported per deployment:

```
- Description
- Creation of IPv6 Prefix List:
  - Apply Path: BGP Peers [Criterion Index (1 to 255), Group (1 to
255/regex), Neighbor (1 to 255/regex), Router Instance (VPRN
Service/Base)]
  - Prefix: IPv6 Prefix
  - Prefix Exclude: IPv6 Prefix
```

CLI context: filter > match-list > ipv6-prefix-list

2.3.3 Advisory

- BGP Peers: The Group, Neighbor, and Router Instance elements are immutable. Modifying these elements re-creates the BGP Peers with the modified values applied.
- Apply-Path and Prefix-Exclude capabilities are mutually exclusive.
- Apply Path is not supported on 7250 IXR.
- Prefix Exclude is not supported on IXR-R6d, IXR-R6dl and IXR-x/x3/x1.

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2.4 Name of the Intent: filter-ml-port-list_msros_23-10-1_23-11

2.4.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS MD SR OS node version 23.10

2.4.2 Description

filter-ml-port-list_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Port List.

The following configurations are supported per deployment:

```
- Create Port List with Port List Name:
  - Description
  - Port Value
  - Port Range (Start to End)
```

CLI context: filter > match-list > port-list

2.4.3 Advisory

N/A

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2.5 Name of the Intent: filter-redirect-policy_msros_23-10-1_23-11

2.5.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7450/7950 MD SR OS node version 23.10

2.5.2 Description

filter-redirect-policy_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of filter redirect policy.

The following configurations are supported per deployment:

```
- Create Redirect Policy with Redirect Policy Name
- Admin State, Description, Router Instance, Sticky DEST, Notify DEST
Change
- Destination:
  - Destination Address
  - Admin State
  - Description
  - Priority
  - Unicast Route Reachability Test
  - Ping Test:
    - Source Address
    - Interval
    - Time Out
    - Drop Count
    - Hold Down
```

2.5.3 Advisory

- The user should ensure that each deployment must be configured to use a singular type of destination addresses exclusively, either IPv4 or IPv6, but not both.
- The user should ensure that both the Destination Address and Source Address are of the same type, either IPv4 or IPv6.
- The user should ensure that separate deployments must be performed for disabling "Ping Test" and enabling "Unicast Route Reachability Test" or disabling "Unicast Route Reachability Test" and enabling "Ping Test".
- The user should ensure that "Router Instance" should be provided when we select "Unicast Route Reachability Test" for this Redirect Policy.

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2.6 Name of the Intent: ipsec-cert-profile_msros_24-10-1_24-4

2.6.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7450 ESS MD SROS node version 24.10

2.6.2 Description

ipsec-cert-profile_msros_24-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of Ipsec cert-profile with CERT-PROFILE-NAME as target identifier. The following configurations are supported per deployment:

```
- Admin State
- Entry : ID, Cert, Key, RSA Signature,Compare Chain Include
         Send Chain: CA Profile
```

CLI context: ipsec > cert-profile

Prerequisite:

- Need to configure CA profile (configure system security pki ca-profile) before using in the intent.

2.6.3 Advisory

- Ipsec Cert profile is not supported on IXR and XRS nodes.
- cert and key both values should be different.

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2.7 Name of the Intent: ipsec-ike-policy_msros_24-10-1_24-4

2.7.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SROS, 7450 ESS MD node version 24.10

2.7.2 Description:

The ipsec-ike-policy_msros_24-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of an ike-policy with name as the ike-policy targetIdentifier.

The following configurations are supported per deployment:

```
-IKE Version 1: Auth Method, Own Method, Ike Mode, PH1 Responder Delete
Notify
-IKE Version 2: Auth Method, Own Method, Auto EAP Method, Own Auto EAP
Method, Send IDR After EAP Success, PPK required
-IKEV2 Fragment: MTU, Reassembly Timeout (seconds)
-Limit Init Exchange: Admin State, Reduced Max Exchange Timeout (seconds)

-DPD: Reply Only, Interval (seconds), Max Retries
-Nat Traversal: Force, Keep Alive Interval (seconds), Force Keep Alive
-Lockout: Failed Attempts, Duration (minutes), Block (minutes), Max Port
Per IP
-PFS : DF Group
-Relay Unsolicited CFG Attribute: Internal IPv4 Address, Internal IPv4
Netmask, Internal IPv4 DNS, Internal IPv6 Address, Internal IPv6 DNS
```

CLI context: configure > ipsec > ike-policy

Prerequisites: - IKE Transform value should be preconfigured on the node

2.7.3 Advisory

- The Limit Init Exchange feature is supported only when IKE Version 2 is configured.
- Ikev1 and Ikev2 attributes are mutually exclusive elements cannot be set together

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2.8 Name of the Intent: ipsec-ike-transform_msros_24-10-1_24-4

2.8.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7450 ESS MD SROS node version 24.10

2.8.2 Description

ipsec-ike-transform_msros_24-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of IPSec Transform with IPSecTransform ID as target identifier. The following configurations are supported per deployment:

```
- Configuration of IPSec IKE Transform attributes as below
  DH Group, IKE Auth Algorithm, IKE Encryption Algorithm, IKE PRF
Algorithm, ISAKMP Lifetime
```

CLI context: ipsec > ike-transform

Prerequisite: NA

2.8.3 Advisory

- IKE-Transform is not supported on 7250 IXR, 7950-XRS, and 7750 SR A4/A8.

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2.9 Name of the Intent: ipsec-transport-mode-profile_msros_24-10-1_24-4

2.9.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7450 ESS MD (SROS version 24.10)

2.9.2 Description:

The ipsec-transport-mode-profile_msros_24-10-1_24-4 intent enables creation, modification, discovery, audit, and alignment of ipsec-transport-mode-profile configurations using the transport-mode-profile-name as the identifier.

The following configurations are supported per deployment:

```
- Description, Replay Window.
- Key Exchange (Dynamic):
    - Auto Establish
    - IKE Policy
    - IPsec Transform
    - Pre-Shared Key
    - Cert:
        - Cert Profile
        - Trust Anchor Profile
        - Status Verify:
            - Default Result
            - Primary
            - Secondary
    - Ppk:
        - List
        - ID
    - ID (choice):
        - FQDN
        - IPv4
        - IPv6
- Max History Key Records:
    - IKE
    - ESP
```

CLI context: configure > ipsec > ipsec-transport-mode-profile.

Prerequisite: Below profile/list must be pre-configured on the node.

- cert-profile < *configure ipsec cert-profile* >.
- trust-anchor-profile < *configure ipsec trust-anchor-profile* >.
- ike-policy < *configure ipsec ike-policy* >.
- ipsec-transform < *configure ipsec ipsec-transform* >.
- ipsec ppk-list < *configure ipsec ppk-list* >.

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2.9.3 Advisory:

- IPsec-transport-mode-profile is not supported on 7950 XRS, 7750 SR-a4 and 7750 SR-a8 systems.
- IPsec transport mode does not support IKE version 1 (IKEv1).
- The pre-shared key (as a password-protected attribute) is excluded from audit operation for security reasons.

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2.10 Name of the Intent: isa_msros_24-10-1_24-4

2.10.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, and 7450 ESS MD SROS node version 24.10

2.10.2 Description

isa_msros_24-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of ISA with ISA as target identifier. The following configurations are supported per deployment:

```
- Tunnel Group:ID, Admin State, Description, ISA Scale Mode, IPsec
Responder Only, Strict ESP Sequence Number Ordering, SPI Range Index
    - Reassembly: Max Wait Time (milliseconds)
    - Stats Collection: ISA DP CPU Usage, Redundancy
    - Multi Active: Active ISA Number, Group Members [ESA, ISA]

- Tunnel Member Pool: Name, Description, High Availability, Group
Members[ESA, ISA]

- Map T Group: ID, Admin State, Description, Fragments Per packet, ESA
[ESA ID, VM]

- NAT Group: ID, Admin State, Description, Radius Accounting Policy,
Scaling Profile
    - Redundancy: Active MDA Limit, Redundancy Mode[Inter Chassis, Intra
Chassis]
    - Log: Suppress LSN Sub Blocks Free, Suppress LSN Events
    - Session Limits: Reserved, UPNP Mappings
    - Watermarks: Low(percent), High(percent)
    - MDA: MDA ID
    - ESA: ESA ID, VM

- WLAN-GW Group: ID, Admin State, Description, Redundancy[MDA, IOM],
Active IOM Limit, Active MDA Limit, Port Policy,
    Tunnel Port Policy, Scaling Profile, FWA Checksum Assist, IOM, MDA,
ESA[ESA ID, VM]
    - NAT: LSAN, Radius Accounting Policy
        - Log: Suppress LSN Sub Blocks Free, Suppress LSN Events
        - Session Limits: Reserved, UPNP Mappings
        - Watermarks: Low(percent), High(percent)
    - Distributed Sub Mgmt: ISA AA Group, ISA AA Oversubscription Factor
    - Watermarks: Entity, Low(percent), High(percent)

- LNS Group: ID, Admin State, Description, Port Policy, MDA[MDA
ID,Drain], ESA[ESA ID, VM, Drain]
```

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```
- Application Assurance Group: AA Group ID, Admin State, Description, AA
Sub Scale, Minimum ISA Generation, HTTP Enrich MAX Packet Size
    Overload Cut Through, Failure Mode, ISA Capacity Cost Low Threshold,
ISA Capacity Cost High Threshold
    Divert FC, VM Traffic Distribute By IP, VM Traffic Distribute By
TEID, AA Sub Distribute Traffic By IP
    Overload Sub Quarantine: Admin State, Flow Attribute
    - Shared Resources: GTP Tunnel Database, TCP Advanced Functions, URL
Filter Web Service Cable
    - Primary: ESA[ESA ID, VM], ISA[MDA ID]
    - Backup: ESA[ESA ID, VM], ISA[MDA ID]
    - Transit Prefix Limits: Ipv4 Entries, Ipv6 Entries, Ipv4 Remote
entries, Ipv6 remote entries
    - Statistics: Stats Type [ISA AA Group Stats, Collect Stats,
Accounting policy]
    - QOS:Egress
        - From Subscriber: Port Scheduler Policy, Queue Policy, WA Shared
Low Wmark (percent), WA Shared High Wmark (percent)
            - Buffer Pool: Name, Resv CBS, Slope Policy
        - To Subscriber: Port Scheduler Policy, Queue Policy, WA Shared
Low Wmark (percent), WA Shared High Wmark (percent)
            - Buffer Pool: Name, Resv CBS, Slope Policy

- Video Group: Video Group ID, Admin State, Description, FCC Server,
Local RT Server, Resv Ret (megabps), Stream Selection, MDA [MDA ID], VAPP
[ESA ID, VAPP ID]
    - Watermark
        - Bandwidth: FCC(percent), Ret(percent), Total(percent)
        - Session: FCC(percent), Ret(percent), Total(percent)
```

CLI context: isa >

Prerequisite:

- configure ESA along with VM to bind in the intent.
- configure Application Assurance Group ID
- configure below policies in order to use in the intent: Radius Accounting Policy, Oper Group, Port Policy, Tunnel Port policy, Port Scheduler Policy, Queue Policy

2.10.3 Advisory

- In order to configure host port for ESA the port must be ethernet mode hybrid, encap-type dot1q and tunneling enabled.
- Cannot have both an MDA and an ESA-VM configured in lns-group ID.
- Based on redundancy of WLAN-GW Group either active iom limit or active MDA limit can be configured.
- Inorder to delete the ISA configurations , first need to delete the reference objects binded to the config. e.g: Binded Application assurance group must be deleted first before deleting the ISA config.

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2.11 Name of the Intent: lag_mixr_23-10-1_24-4

2.11.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR node version 23.10 MD SROS.

2.11.2 Description

lag_mixr_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of LAG with LAG name as target identifier. The following configurations are supported per deployment:

```
- Configure the lag name :lag-<number or string>
- Configuration of Description, Admin State, Encap Type
[null/dot1q/qinq], Mode[Access,Network,Hybrid], LACP Xmit Interval, LACP
Xmit Stdby,MAX Ports,MAC Address,Dynamic Cost,Standby Signalling[lacp,
Poweroff],Hold Time Down,LACP Mux Control,Monitor oper Group
- Configure the Scheduler attributes as below
    a.Mode
    b.Low Latency
    c.VLAN QOS Policy
- Configure the Port Threshold,LACP,Access,Selection Criteria,Hash Weight
Threshold
- Configure the BFD Liveness
    a.Soft Reset Extension
    b.IPv4
    c.IPv6
```

CLI context: lag > lag-<name or ID

Prerequisite:

- Configure Oper-Group and vlan-qos-policy
- Port Mode and Encap should configured on the node

2.11.3 Advisory

- LAG name should always start with lag- and can contain numeric or alphanumeric characters. However, if the LAG name is lag-(only numeric) then the numeric value should match the LAG ID.
- standby-signaling power-off requires selection-criteria best-port or monitor-oper-group on lag
- LACP Administrative State cannot be enabled when Standby Signalling is set to Power Off
- Port sub-group and monitor-oper-group is an invalid combination
- Selection Criteria: best-port does not support
- The ports in a LAG must have the same characteristics, speed, duplex, etc.
- Up to 8 ports can be bundled in a LAG, and up to 64 LAGs can be on a chassis.
- All involved ports must work at the same rate.

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- All involved ports must have the same configurations regarding AutoNegotiation and Flow Control.
- All aggregated links must be point-to-point (multicast is not allowed).
- All involved interfaces must work in full-duplex modality.
- Port type and Encap type act as filters to list the available ports in the Add Port configuration.
- Hybrid Port type does not support Null encap type.
- Add Port section in deployment creation window will list all the ports; it is the user's responsibility to add relevant ports as LAG members.
- Both greenfield and brownfield deployments allow the same target identifier deployments in different templates.
- Pre-21.2 SROS nodes support only LAG ID, and post-21.2 SROS nodes support lag-name for MD node.
- Deployment will be created in misaligned state if LACP Admin key is not unique.
- Default value of max-port is 32 for IXR nodes.
- Port sub-group auto-iom/auto-mda is not allowed on IXR
- LAG ID range will change based on the IXR chassis
- BFD IPv6 supports only IXR-R6dl/R6/x3/x
- Scheduler is not supported on 7250 IXR-6/IXR-10
- Low-latency scheduler is not supported on 7250 IXR-s, IXR-6, IXR-10, IXR-R6, IXR-e, IXR-R4, IXR-ec

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2.12 Name of the Intent: lag_mixr_24-10-1_24-4

2.12.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR node version 24.10 MD SROS.

2.12.2 Description

The lag_mixr_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of LAG, with LAG name as the target identifier. The following configurations are supported per deployment:

```
- Configure the lag name :lag-<number or string>
- Configuration of Description, Admin State, Encap Type
[null/dot1q/qinq], Mode[Access,Network,Hybrid], LACP Xmit Interval, LACP
Xmit Stdby,MAX Ports,MAC Address,Dynamic Cost,Standby Signalling[lacp,
Poweroff],Hold Time Down,LACP Mux Control,Monitor oper Group,Port
Threshold,LACP,Access,Selection Criteria,Hash Weight Threshold,BFD
Liveness,LLDP Member Template
- Configure the Scheduler attributes as below
    a.Mode
    b.Low Latency
    c.VLAN QOS Policy
```

Prerequisite:

- Configure Oper-Group and vlan-qos-policy
- Port Mode and Encap should configured on the node

CLI context: lag

2.12.3 Advisory

- LAG name should always start with lag- and can contain numeric or alphanumeric characters. However, if the LAG name is lag-(only numeric) then the numeric value should match the LAG ID.
- Setting standby-signaling power-off requires setting selection-criteria best-port or monitor-oper-group on lag
- LACP Administrative State cannot be enabled when Standby Signalling is set to Power Off
- Port sub-group and monitor-oper-group is an invalid combination
- Selection Criteria: best-port is not supported
- The ports in a LAG must have the same characteristics, speed, duplex, etc.
- Up to 8 ports can be bundled in a LAG, and up to 64 LAGs can be on a chassis.
- All involved ports must work at the same rate.
- All involved ports must have the same configurations regarding AutoNegotiation and Flow Control.
- All aggregated links must be point-to-point (multicast is not allowed).

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- All involved interfaces must work in full-duplex modality.
- Port type and Encap type act as filters to list the available ports in the Add Port configuration.
- Hybrid Port type does not support Null encap type.
- Add Port section in deployment creation window will list all the ports; you must add relevant ports as LAG members.
- Both greenfield and brownfield deployments allow the same target identifier deployments in different templates.
- Pre-21.2 SROS nodes support only LAG ID, and post-21.2 SROS nodes support lag-name for MD node.
- Deployment will be created in misaligned state if LACP Admin key is not unique.
- Default value of max-port is 32 for IXR nodes.
- Port sub-group auto-iom/auto-mda is not allowed on IXR.
- LAG ID range will change based on the IXR chassis.
- BFD IPv6 supports only IXR-R6dl/R6/x3/x.
- Low-latency scheduler is not supported on 7250 IXR-s.
- LACP Fallback timeout cannot be configured if LACP fallback mode is not set to static.
- Receive and tunnel-nearest-bridge both cannot be true together.

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2.13 Name of the Intent: lag_msros_23-10-1_24-4

2.13.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7950/7450/7750 MD SR OS node version 23.10

2.13.2 Description

lag_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of LAG.

The following configurations are supported per deployment:

```
- Configuration of the LAG Name: LAG-<Number or String>
- Configuration of Description, Admin State, Encap Type
[Null/Dot1q/QinQ], Mode [Access, Network, Hybrid], Port Type, LACP Xmit
Interval, LACP Xmit Stdby, MAX Ports, MAC Address, Dynamic Cost, Standby
Signaling [LACP, Power Off], Hold Time Down, LACP Mux Control, Monitor
Oper Group
- Port configuration:
  - Port ID, Priority, Sub-Group*, Hash Weight*
- Link Map Profile configuration:
  - Link Map Profile ID*, Description, Failure Mode [Per-Link-Hash,
Discard]
  - Create Link by Port ID, Port Type [Primary/Secondary]
- Port Threshold configuration:
  - Value, Action [Down, Dynamic-Cost, Static-Cost], Cast
- LACP configuration:
  - Mode, System ID, System Priority, Administrative Key
- Adaptive Load Balancing configuration:
  - Tolerance, Interval, Bandwidth Threshold
- Access configuration:
  - Bandwidth, Booking factor, Per FP ING Queuing, Per FP SAP Instance*,
Per FP Egr Queuing
- Adapt QoS configuration:
  - Mode, Include EGR Hash CFG
- Selection Criteria configuration:
  - Mode, Slave To Partner, Sub Group Hold Time
- Per Link Hash configuration:
  - Subscriber Hash Mode, SAP Hash Mode, Auto Rebalance
- Hash Weight Threshold configuration:
  - Value, Action, Cost
- BFD Liveness configuration:
  - Soft Reset Extension
  - IPv4/IPv6 [Admin State, Local IP Address, Remote IP Address, Receive
Interval, Transmit Interval, Max Admin Down Time (seconds), Max Setup
Time (milliseconds), Multiplier]
- ETH CFM
```

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```
  - MEP ID, MA/MD Admin Name, Admin State, Description, VLAN, One Way
Delay Threshold, MAC Address, Low Priority Defect, CCM, CCM LTM Priority,
CCM Padding Size, CCM TLV Ignore, Facility Fault, Collect LMM Stats
  - CSF Multiplier
  - Alarm Notification FNG Alarm Time, FNG Reset Time
  - AIS Client MEG Level, Interface Support, Interval, Low Priority
Defect, Priority
  - ETH Test Bit Error Threshold
  - Test Pattern, CRC TLV
  - Grace Max RX Defect Window, Priority, RX ETH ED, TX ETH ED, RX ETH
VSM Grace, TX ETH VSM Grace
```

Prerequisites:

- Port Mode and Encap should be configured in the node.
- System ETH-CFM, Domain, and Monitor-Oper-Group should be confiugred in the node if used in LAG.

2.13.3 Advisory

- LAG name should always start with lag- and it can be either numeric or alphanumeric. However, If LAG name is lag-(only numeric) then the numeric value should match the LAG ID.
- LAG Access Mode does not support link-mapping profile
- LAG Access Mode does not support access adapt-qos
- LAG Access Mode does not support per link hash weighted
- standby-signaling power-off requires selection-criteria best-port or monitor-oper-group on lag
- LACP Administrative State cannot be enabled when Standby Signalling is Power Off
- Invalid combination port sub-group with monitor-oper-group
- Cannot enable MEP Grace TX and MCC-ED TX at the same time.
- Adaptive-Load-Balancing and Per-FP-EGR-Queuing are mutually exclusive.
- Adaptive-Load-Balancing and Per-FP-SAP-Instance are mutually exclusive.
- Adaptive-Load-Balancing and Per-Link-Hash are mutually exclusive.
- Adaptive-Load-Balancing and Hash-Weight are mutually exclusive.
- Selection Criteria: Best-Port is not supported.
- VLAN: Facility tunnel MEP is only supported on a QinQ Encap Port.
- CSF is only supported on Primary-VLAN Port and LAG MEPs.
- Sub-Second MEPs do not support CCM-Padding.
- PXC Port should be configured for Ethernet DOT1X Tunneling attribute.
- LAG should be in hybrid mode for PXC Port.
- LAG Sub-Group: Hash-Weight is not supported for PXC Port.
- Cannot add Member Port of the same Sub Port in LAG means that Add Member Port PXC-1.a and PXC-1.b cannot be added in the same LAG.
- The ports in a LAG must have the same characteristics, speed, duplex, etc.

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- In all, eight ports can be bundled in a LAG, and up to 64 LAGs can be on a chassis.
- All involved ports must work at the same rate.
- All involved ports must have the same configurations regarding Auto-Negotiation and Flow Control.
- All aggregated links must be point-to-point (multicast is not allowed).
- All involved interfaces must work in Full-Duplex Modality.
- Port Type and Encap Type act as filters to list the available ports in the Add Port configuration.
- Hybrid Port Type does not support Null Encap Type.
- If System Priority, Sub Group, and Hash Group are not added during deployment creation, ICM will take default values and create the deployment for the classical node.
- LAG name should always start with "LAG-" and it can be either numeric or alphanumeric. However, if LAG name is LAG-(only numeric), then the numeric value should match with the LAG ID.
- Add Port section in the deployment creation window will list all the ports; it is the user's responsibility to add relevant ports as LAG members.
- Both greenfield and brownfield deployments allow for the same target identifier deployments in different templates.
- Releases earlier than 21.2 SR OS nodes support only LAG ID, and releases later than 21.2 SR OS nodes support LAG-Name for the MD node.
- Deployment will be created in the misaligned state if the LACP Admin key is not unique.
- Max-Port restriction is applicable only to the SR MD Mode and LAG ID.
- Max-Port value for LAG ID: LAG-ID 1 to 64 Max-Port value is 64 and LAG-ID 65 to 800 Max-Port value is 32.
- Max-Port restriction is applicable only to the SR Family SR OS MD nodes, not 7250 IXR nodes.
- The default value of Max-Port is 32 for SR OS nodes.

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2.14 Name of the Intent: lag_msros_24-10-1_24-4

2.14.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7950/7450/7750 MD SR OS node version 24.10

2.14.2 Description

The lag_msros_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a LAG.

The following configurations are supported per deployment:

```
- Configuration of the LAG Name: LAG-<Number or String>
- Configuration of Description, Admin State, Encap Type
[Null/Dot1q/QinQ], Mode [Access, Network, Hybrid], Port Type, LACP Xmit
Interval, LACP Xmit Stdby, MAX Ports, MAC Address, Dynamic Cost, Standby
Signaling [LACP, Power Off], Hold Time Down, LACP Mux Control, Monitor
Oper Group
- Port configuration:
  - Port ID, Priority, Sub-Group*, Hash Weight*
- Link Map Profile configuration:
  - Link Map Profile ID*, Description, Failure Mode [Per-Link-Hash,
Discard]
  - Create Link by Port ID, Port Type [Primary/Secondary]
- Port Threshold configuration:
  - Value, Action [Down, Dynamic-Cost, Static-Cost], Cast
- LACP configuration:
  - Mode, System ID, System Priority, Administrative Key
  - Fallback:
    - Mode, Timeout
- Adaptive Load Balancing configuration:
  - Tolerance, Interval, Bandwidth Threshold
- Access configuration:
  - Bandwidth, Booking factor, Per FP ING Queuing, Per FP SAP Instance*,
Per FP Egr Queuing
- Adapt QoS configuration:
  - Mode, Include EGR Hash CFG
- Selection Criteria configuration:
  - Mode, Slave To Partner, Sub Group Hold Time
- Per Link Hash configuration:
  - Subscriber Hash Mode, SAP Hash Mode, Auto Rebalance
- Hash Weight Threshold configuration:
  - Value, Action, Cost
- BFD Liveness configuration:
  - Soft Reset Extension
  - IPv4/IPv6 [Admin State, Local IP Address, Remote IP Address, Receive
Interval, Transmit Interval, Max Admin Down Time (seconds), Max Setup
```

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```
Time (milliseconds), Multiplier]
- ETH CFM
  - MEP ID, MA/MD Admin Name, Admin State, Description, VLAN, One Way
Delay Threshold, MAC Address, Low Priority Defect, CCM, CCM LTM Priority,
CCM Padding Size, CCM TLV Ignore, Facility Fault, Collect LMM Stats
  - CSF Multiplier
  - Alarm Notification FNG Alarm Time, FNG Reset Time
  - AIS Client MEG Level, Interface Support, Interval, Low Priority
Defect, Priority
  - ETH Test Bit Error Threshold
  - Test Pattern, CRC TLV
  - Grace Max RX Defect Window, Priority, RX ETH ED, TX ETH ED, RX ETH
VSM Grace, TX ETH VSM Grace
- LLDP Member Template:
  - Dest MAC:
    - MAC Type, Notification, Port ID Subtype, Receive, Transmit, Tunnel
Nearest Bridge
    - Tx TLVs:
      - Port Desc, Sys Name, Sys Desc, Sys Cap
    - Tx Mgmt Address:
      - Mgmt Address System Type, Admin State
- PTP Timestamping:
  - Admin State, IPv4 Address, IPv6 Address
```

CLI context: lag

Prerequisites:

- Port Mode and Encap should be configured in the node.
- System ETH-CFM, Domain, and Monitor-Oper-Group should be configured in the node if used in LAG.

2.14.3 Advisory

- LAG name should always start with lag- and it can be either numeric or alphanumeric. However, If LAG name is lag-(only numeric) then the numeric value should match the LAG ID.
- LAG Access Mode does not support link-mapping profile
- LAG Access Mode does not support access adapt-qos
- LAG Access Mode does not support per link hash weighted
- Setting standby-signaling power-off requires setting selection-criteria best-port or monitor-oper-group on lag
- LACP Administrative State cannot be enabled when Standby Signalling is Power Off
- LACP Fallback timeout cannot be configured if LACP fallback mode is not set to static
- Invalid combination port sub-group with monitor-oper-group
- Cannot enable MEP Grace TX and MCC-ED TX at the same time.
- Adaptive-Load-Balancing and Per-FP-EGR-Queuing are mutually exclusive.
- Adaptive-Load-Balancing and Per-FP-SAP-Instance are mutually exclusive.

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- Adaptive-Load-Balancing and Per-Link-Hash are mutually exclusive.
- Adaptive-Load-Balancing and Hash-Weight are mutually exclusive.
- Selection Criteria: Best-Port is not supported.
- VLAN: Facility tunnel MEP is only supported on a QinQ Encap Port.
- CSF is only supported on Primary-VLAN Port and LAG MEPs.
- Sub-Second MEPs do not support CCM-Padding.
- PXC Port should be configured for Ethernet DOT1X Tunneling attribute.
- LAG should be in hybrid mode for PXC Port.
- LAG Sub-Group: Hash-Weight is not supported for PXC Port.
- Cannot add Member Port of the same Sub Port in LAG means that Add Member Port PXC-1.a and PXC-1.b cannot be added in the same LAG.
- The ports in a LAG must have the same characteristics, speed, duplex, etc.
- In all, eight ports can be bundled in a LAG, and up to 64 LAGs can be on a chassis.
- All involved ports must work at the same rate.
- All involved ports must have the same configurations regarding Auto-Negotiation and Flow Control.
- All aggregated links must be point-to-point (multicast is not allowed).
- All involved interfaces must work in Full-Duplex Modality.
- Port Type and Encap Type act as filters to list the available ports in the Add Port configuration.
- Hybrid Port Type does not support Null Encap Type.
- If System Priority, Sub Group, and Hash Group are not added during deployment creation, ICM will take default values and create the deployment for the classical node.
- LAG name should always start with "LAG-" and it can be either numeric or alphanumeric. However, if LAG name is LAG-(only numeric), then the numeric value should match with the LAG ID.
- Add Port section in the deployment creation window will list all the ports; it is the user's responsibility to add relevant ports as LAG members.
- Both greenfield and brownfield deployments allow for the same target identifier deployments in different templates.
- Releases earlier than 21.2 SR OS nodes support only LAG ID, and releases later than 21.2 SR OS nodes support LAG-Name for the MD node.
- Deployment will be created in the misaligned state if the LACP Admin key is not unique.
- Max-Port restriction is applicable only to the SR MD Mode and LAG ID.
- Max-Port value for LAG ID: LAG-ID 1 to 64 Max-Port value is 64 and LAG-ID 65 to 800 Max-Port value is 32.
- Max-Port restriction is applicable only to the SR Family SR OS MD nodes, not 7250 IXR nodes.
- The default value of Max-Port is 32 for SR OS nodes.

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2.15 Name of the Intent: log-syslog_msros_23-10-1_23-11

2.15.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SROS node version 23.10

2.15.2 Description

log-syslog_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Syslog with SYSLOG-NAME name as target identifier. The following configurations are supported per deployment:

```
- Syslog
  Configuration of Description, Address, Facility, Severity, Log Prefix,
TLS Client Profile, Port

- HostName
  Hostname[Use System Name,Value]
```

CLI context: log > syslog

Prerequisite:

- Configure client-tls-profile [configure system security tls]

2.15.3 Advisory

- Syslog tls-client-profile can not be defined together with port
- Syslog name must be numeric in range 1..10, or alphanumeric not starting with underscore and without trailing spaces

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2.16 Name of the Intent: macsec-ca_msros_23-10-1_23-11

2.16.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7450/7750/7250 node version 23.10 MD SROS

2.16.2 Description

macsec-ca_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of MACSec connectivity_association Policy.

The following configurations are supported per deployment:

- Creation of MACSec with name
- Configuration of Connectivity Association attributes as follows: Admin-state, Description, Replay Window Size, Replay Protection, Macsec Encrypt, Clear Tag Mode[none,single-tag,dual-tag], Encryption Offset, Delay Protection, Cipher Suite[gcm-aes-128,gcm-aes-256,gcm-aes-xpn-128,gcm-aes-xpn-256], AnySec
- Configuration of Static CAK attributes such as Active PSK, Mka Hello Interval, Mka Key Server Priority
- Creation of PreShared Key: PSK Id[1,2], Encryption Type[aes-128-cmac,aes-256-cmac], CAK, CAK Name

Prerequisite: N/A

2.16.3 Advisory

- Creation of MACSec deployment fails if MACSec Name is not unique; that is, MACSec Name for the selected target should not be used already.
- Brownfield discovery will fail if MACSec Name does not match.
- Admin-state enable when Mka Hello Interval value 500ms,Delay Protection attributes required.
- CAK Name should comprise an even number of Hex characters, for example, ABCD or DE1234.
- For deployments on multiple nodes, the user must enter the CAK Name in the suggest field.
- Audit will not be shown as misaligned for new entries added from the node.
- CAK will display in the PSK row table.
- CAK depends on the encryption type:
- If aes-128-cmac, then CAK will be 32 hex character
- If aes-256-cmac, then CAK will be 64 hex character For example: For aes-256-cmac: CAK: ABCDEABCDEABCDEABCDEABCDEABCDE12ABCDEABCDEABCDEABCDEABCDEABCDE12 CAK Name: ABCD For aes-128-cmac: CAK: ABCDEABCDEABCDEABCDEABCDEABCDE12 CAK Name: ABCD
- Anysec is applicable on few SR variants. Anysec cannot be enabled with the admin-state enabled if the following attributes are not set to their default configurations: replay-window-size (Default: 0), replay-protection (Default: false), clear-tag-mode (Default: none), and encryption-offset (Default: 0).

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2.17 Name of the Intent: policy-options-as-path-group_msros_23-10-1_23-11

2.17.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7250 IXR, 7950 XRS, and 7450 ESS MD SROS Node version 23.10

2.17.2 Description:

The policy-options-as-path-group_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of as-path-group, with name as target identifier.

The following configurations are supported per deployment:

```
- Entry:
    Entry, Expression
```

CLI context: policy-options > as-path-group

Prerequisites:

- N/A

2.17.3 Advisory

- N/A

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2.18 Name of the Intent: policy-options-as-path_msros_23-10-1_23-11

2.18.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS and 7250 IXR MD SROS node version 23.10

2.18.2 Description

The policy-options-as-path_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of as-path, with NAME as the target identifier. The following configurations are supported per deployment:

`- Expression [1..255 characters]`

CLI context: router > policy-options > as-path

Prerequisite:

- N/A

2.18.3 Advisory

- N/A

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2.19 Name of the Intent: policy-options-community_msros_23-10-1_23-11

2.19.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS MD SR OS node version 23.10

2.19.2 Description:

policy-options-community_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Policy Options Community.

Below configurations are supported per deployment:

```
- Members
- Expression: Expression, Exact
```

CLI context: configure > policy-options > community

2.19.3 Advisory:

- Members and Expression are mutually exclusive.
- Format information for Members and Expression is provided on GUI.
- Following are some examples for Members and Expression.
- For Members:
- ipv6-redirect:22:22:22:22:22:455:4542:876
- ext:ffaf:fffffffffffe
- target:2.2.2.2:4
- origin:2.2.2.2:6
- target:56:78
- no-advertise
- 4.4.4.4:1234
- 65535:65535
- llgr-stale
- ext:4300:1
- no-export
- blackhole
- no-llgr
- 0:23
- 1.1.1.6
- For Expression:

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- 2:2 AND 200:300
- 2:2 OR 200:300
- 2:2 AND NOT 200:300
- 2:2 OR NOT 200:300
- 2:2 OR null
- (8:8 AND NOT 9:9) AND NOT (8:8)
- NOT 2:2 AND 200:200
- target:5:100 AND NOT target:9:9
- ((target:9:0 OR target:8:8) AND (target:7:4 OR target:4:5))

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2.20 Name of the Intent: policy-options-damping_msros_23-10-1_23-11

2.20.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS and 7250 IXR MD SR OS node version 23.10

2.20.2 Description

policy-options-damping_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Policy-Options Damping.

The following configurations are supported per deployment:

```
- Creation of Policy-Options Damping with Name
- Configuration of the following:
  - Half Life [1-45]
  - Max Suppress [1-720]
  - Reuse [1-20000]
  - Suppress [1-20000]
```

2.20.3 Advisory

N/A

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2.21 Name of the Intent: policy-options-prefix-list_msros_23-10-1_24-4

2.21.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950/7250 MD SR OS node version 23.10

2.21.2 Description

policy-options-prefix-list_msros_23-10-1_24-4 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of Policy Options Prefix List Policy.

The following configurations are supported per deployment:

```
- Prefix List Member : Prefix Type (IPv4 & IPv6), Type (Exact, Longer,
Through, Range, To, Address Mask)
```

2.21.3 Advisory

- Prefix List deployment should consist of at least one prefix.
- Deployment misalignment occurs on audit of IPv6 address with combination of IPv4 and IPv6. For example: <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.

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2.22 Name of the Intent: policy-options-statement_mixr_23-10-1_24-4

2.22.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR node version 23.10 MD SROS.

2.22.2 Description

policy-options-statement_mixr_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of PolicyStatement Policy with PolicyStatement Policy name as target identifier. The following configurations are supported per deployment:

```
- Configuration of Description, Entry Type, Default Action
- Configuration of the following attributes on Default Action:
    - Action Type, Add Paths Send Limit, Admin Tag Policy, Advertise
Label, BGP High Priority, Create MPLS Tunnel, Damping, Dest Class, Flex
Algo, Install Backup Path, Local Preference, Next Hop, Origin, Origin
Validation State, Preference, Resolve Static, Tag, Route Table Install,
Type, Sticky ECMP, SR Maintenance Policy, SRv6 Locator, SRv6 Micro
Segment Locator,AIGP Metric, AS Path, AS Path Prepend, BGP Tunnel Metric,
Metric, SR Label Index, BGP MED, Community

- Policy Statement Entry ID/Name, Description, Action, Enable From
Criteria, Enable To Criteria. Conditional Expression Route exists string.


- Policy Statement Entry From Criteria
    - Level, Prefix List, Area, Aggregate Contributor, EVPN Type,
External, Family, Group Address, Host IP,Interface, MVPN Type, Origin,
Origin Validation State, OSPF Type, Path Type, Policy, Route
Distinguisher List, State, Tag, Color, Distinguisher, Endpoint, SRv6 SID
Prefix, SRv6 TLV, Neighbor, Interface Subnets, AS Path, Length, Cluster
ID, Community, Count,Local Preference, Metric, Next Hop, Protocol, Source
Address,Prefix List Override, Policy Variables

- Policy Statement Entry To Criteria
    - Level, Prefix List, Neighbor, Protocol

- Policy Statement Entry Action
    - Action Type, Add Paths Send Limit, Admin Tag Policy, Advertise
Label, BGP High Priority, Create MPLS Tunnel, Damping, Flex Algo, Install
Backup Path, Local Preference, Next Hop, Origin, Origin Validation State,
Preference, Resolve Static, Sticky ECMP, Tag, Type, Route Table Install,
SR Maintenance Policy, SRv6 Locator, SRv6 Micro Segment Locator, AIGP
Metric, AS Path, AS Path Prepend, BGP Tunnel Metric, Metric, SR Label
Index, BGP MED, Community
```

CLI context: policy-options > policy-statement

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Prerequisite:

- Configure Prefix List, Community, Global variables policy, AS Path, AS Path Group, Admin Tag, Damping Profile, SR Maintenance Policy
- Configure SRv6 Locator Policy, Route Distinguisher policy
- Configure oper-group, VRPN Service
- Enable synch-e true on MDA

2.22.3 Advisory

- In Default Action and Policy entry Action, SRv6-locator and SRv6-micro-segment-locator are mutually exclusive.
- In Default Action and Policy entry Action, Action-type is required to configure the Action attributes
- Entry Action Action-type attribute is required to modify the Named/Numbred Entry attributes
- In Policy entry From Criteria, interface and interface-subnets are mutually exclusive.
- In Policy entry From Criteria, prefix-list and prefix-list-override are mutually exclusive.
- From ICM, modification of entry-type from named to numeric removes all the named entries automatically. Delete the corresponding entries while modifying entry-type from RestAPI.
- From ICM, modification of entry-type from numeric to named will remove all the numeric entries automatically. Delete the corresponding entries while modifying entry-type from RestAPI.
- Maximum number of elements allowed for prefix-list-override is 1.
- Maximum number of Cluster ID IP addresses is 5.
- In Policy entry From Criteria, AS-Path and AS-Path-Group are mutually exclusive.
- From critera Community Count accepts numeric value from 0-1024, or a String that starts and ends with an at-sign(@).
- From critera Local Preference accepts numeric value from 0-255, or a String that starts and ends with an at-sign(@).
- In Policy entry From Criteria, "Protocol instance set all" is allowed only on protocol ospf, ospf3, or isis.
- All the Policy/Global variable strings must start and end with an at-sign(@).
- Flex Algo Policy/Global Variable accepts numeric value from 128-255 or a String that starts and ends with an at-sign(@).
- From criteria prefix-address host bits must be 0 for Policy-variable of type Prefix.
- Policy Entry From Criteria attributes can be configured only when Enable From Criteria is set to true.
- Policy Entry To Criteria attributes can be configured only when Enable To Criteria is set to true.
- In PolicyStatment policy cannot have both numbered and Named entry.
- Segment-routing-v6 is not supported on 7250 IXR-s,IXR-R4,IXR-e,IXR-ec,IXR-R6,IXR-10,IXR-6 nodes.
- Community Expression is not supported in Policystatement.
- Entry From Policy-variable prefix is not supported on 7250 IXR-s, IXR-R4, IXR-e, IXR-R6, IXR-10, IXR-6 and creates an unknown element error in ICM.

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2.23 Name of the Intent: policy-options-statement_msros_23-10-1_24-4

2.23.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS for MD SROS node version 23.10

2.23.2 Description

policy-options-statement_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of PolicyStatment Policy with PolicyStatement name as targetIdentifier.

The following configurations are supported per deployment:

```
- Configuration of Description, Entry Type, Default Action
- Configuration of the following attributes on Default Action:
    - Action Type, Add Paths Send Limit, Admin Tag Policy, Advertise
Label, BGP High Priority, BGP Leak, Create MPLS Tunnel, Create UDP
Tunnel, Damping, Dest Class, Egress Statistics, Ingress Statistics, Flex
Algo, Install Backup Path, Local Preference, NAT Policy, Next Hop,
Origin, Origin Validation State, Preference, Resolve Static, Source
Class, Sticky ECMP, Tag, Route Table Install, Type, Sticky ECMP, SR
Maintenance Policy, SR Return Path BFD Label, SRv6 Locator, SRv6 Micro
Segment Locator, SRv6 Return Path BFD SID, AIGP Metric, AS Path, AS Path
Prepend, BGP Tunnel Metric, Metric, Multicast Redirection, SR Label
Index, BGP MED, Community

- Policy Statement Entry ID/Name, Description, Action, Enable From
Criteria, Enable To Criteria. Conditional Expression Route exists string.


- Policy Statement Entry From Criteria
    - Level, Prefix List, Area, Aggregate Contributor, EVPN Type,
External, Family, Group Address, Host IP,Interface, MVPN Type, Origin,
Origin Validation State, OSPF Type, Path Type, Policy, Route
Distinguisher List, State, Tag, Color, Distinguisher, Endpoint, SRv6 SID
Prefix, SRv6 TLV, Neighbor, Interface Subnets, AS Path, Length, Cluster
ID, Community, Count, Flowspec, Local Preference, Metric, Next Hop,
Protocol, Source Address,Prefix List Override, Policy Variables

- Policy Statement Entry To Criteria
    - Level, Prefix List, Neighbor, Protocol

- Policy Statement Entry Action
    - Action Type, Add Paths Send Limit, Admin Tag Policy, Advertise
Label, BGP High Priority, BGP Leak, Create MPLS Tunnel, Create UDP
Tunnel, Damping, Dest Class, Egress Statistics, Ingress Statistics, Flex
Algo, Install Backup Path, Local Preference, NAT Policy, Next Hop,
```

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`Origin, Origin Validation State, Preference, Resolve Static, Source Class, Sticky ECMP, Tag, Type, Route Table Install, SR Maintenance Policy, SR Return Path BFD Label, SRv6 Locator, SRv6 Micro Segment Locator, SRv6 Return Path BFD SID, AIGP Metric, AS Path, AS Path Prepend, BGP Tunnel Metric, Metric, Multicast Redirection, SR Label Index, BGP MED, Forwarding Class, Community`

Prerequisite: Configure the following:

- Prefix List
- Community
- Global variables policy
- AS Path
- AS Path Group
- Admin Tag
- Damping Profile
- SR Maintenance Policy
- SRv6 Locator Policy
- Route Distinguisher policy

CLI context: policy-options > policy-statement

2.23.3 Advisory

- In Default Action and Policy entry Action, SRv6-locator and SRv6-micro-segment-locator are mutually exclusive.
- In Policy entry From Criteria, interface and interface-subnets are mutually exclusive.
- In Policy entry From Criteria, prefix-list and prefix-list-override are mutually exclusive.
- From ICM, modification of entry-type from named to numeric will remove all the named entries automatically. Ensure to delete the corresponding entries while modifying entry-type from RestAPI.
- From ICM, modification of entry-type from numeric to named will remove all the numeric entries automatically. Ensure to delete the corresponding entries while modifying entry-type from RestAPI.
- Maximum elements allowed for prefix-list-override is 1.
- Maximum 5 entries of Cluster ID IP address is supported.
- In Policy entry From Criteria, AS-Path and AS-Path-Group are mutual exclusive.
- From critera Community Count accepts numeric value from 0-1024 or String must start and end with an at-sign(@).
- From critera Local Preference accepts numeric value from 0-255 or String must start and end with an at-sign(@).
- In Policy entry From Criteria, Protocol instance set all allowed only on protocol ospf, ospf3 or isis.
- All the Policy/Global variable string must start and end with an at-sign(@).

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- Flex Algo Policy/Global Variable accepts numeric value from 128-255 or String must start and end with an at-sign(@).
- SR Return Path BFD Label Policy/Global Variable accepts numeric value from 32..1048512 or string must start and end with an at-sign(@).
- From criteria prefix-address host bits must be 0 for Policy-variable of type Prefix.
- Policy Entry From Criteria attributes can be configured only when Enable From Criteria is set to true.
- Policy Entry To Criteria attributes can be configured only when Enable To Criteria is set to true.
- Default Action: Action Type attribute configuration is mandatory to configure the Default Action attributes.

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2.24 Name of the Intent: port-eth_mixr_23-10-1_24-4

2.24.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR node version 23.10 MD SROS.

2.24.2 Description

The port-eth_mixr_23-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of Port MD IXR with Port ID as target identifier. The following configurations are supported per deployment:

```
- Confiugre the port with port number
- Configuration of Admin State,Description,DDM Events,Monitor Oper Group
- Configure the Scheduler attributes as below
    a.Mode
    b.Low Latency
    c.VLAN QOS Policy
- Configure the Ethernet Attributes as below
    a.Accounting Policy,Collect Stats
    b.Autonegotiate,Duplex,Dot1q Etype,LACP Tunnel,MAC Address,Min Frame
Length,Mode,Encap Type,MTU,MTU Profile,PTP Asymmetry,
      QinQ Etype,Rs FEC Mode,Speed,Util Stats Interval,XGIG,Discard Rx
Pause Frames,Down on Internal Error
    c.Hold Time
    d.Loopback
    e.Report Alarm
    f.CRC Monitor
        Signal Degrade,Signal Failure
    g.Dampening
    h.Down When Looped
    I.Eth CFM
        MEP
    J.Ingress
    K.SSM
    L.Storm Control
    M.Access
    N.EFM OAM
        Discovery,Link Monitoring,Peer Rdi Rx
    O.Dot1x
        Re Authentication,Per Host Authentication
    P.Egress
        Port Scheduler Policy,Rate,Eth Bn Rate Changes,Monitor Port
Scheduler
    Q.LLDP
        Dest Mac
    R.Network
- Configuration the OTU
```

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```
    a.Report Alarm
    b.Fine Granularity BER
      Signal Degrade,Signal Failure
    c.Path Monitoring
      Trail Trace Identifier
    d.Section Monitoring
      Trail Trace Identifier
- Configuration the SONET-SDH
    a.Hold Time
    b.Report Alarm
    C.Section Trace
      Section Trace,Group
```

CLI context: port >>id

Prerequisite:

- Enable synch-e true on MDA
- Configure Oper-Group,Accounting Policy
- Configure macsec connectivity-association
- Configure system eth-cfm
- configure eth-cfm Domain
- Configure port-qos-policy,vlan-qos-policy

2.24.3 Advisory

- Please consult NSP PLM if this intent is required for NSP 24.8
- vlan-qos-policy policy does not support on IXR-6 & IXR-10
- Deletion[NSP and Network] of port intent fail when ethernet mtu-profile true
- ddm-events does not support connector breakout port type
- ethernet Report Alarm Block Not Locked/Alignment Marker Not Locked/Duplicate Lane does not support IXR-R6d
- ethernet rs-fec-mode not supported on connector breakout port type
- ethernet discard-rx-pause-frames & down-on-internal-error -not supported
- Speed,AutoNegotiate and duplex attributes can be configure on the applicable ports
- ethernet dampening reuse-threshold - reuse-threshold must be less than suppress-threshold
- Distribution of Port intent will fail when Eth-CFM configuration due to Unique values required
- Encap Type null is not supported for Hybrid Ports.
- ethernet dampening max-suppress-time - max-suppress-time must be greater than or equal to half-life
- down-when-looped admin-state must be disabled to have loopback direction
- digital-coherent-optics and optical-line-system cannot both be set
- TDM & ethernet GNSS does not support breakout port
- Port-control must be force-authorized to enable tunneling of all encaps

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- eth-cfm MEP is not allowed with down-when-looped configured on the port
- Sub-second MEPs do not support ccm-padding
- Per-host-authentication cannot be enabled - fp option must be enabled
- efm-oam admin-state must be disabled to have loopback direction internal
- efm-oam accept-remote-loopback must be false to have loopback direction
- ethernet efm-oam admin-state - Cannot enable both tunneling and efm-oam
- connector breakout Port does not support qinq encap in network mode
- ethernet qinq-etype - Not supported for network mode or null/dot1q encap-type
- ethernet dot1q-etype - Not supported for network mode or null encap-type
- loopback direction internal is not supported on network ports
- swap-src-dst-mac must be false when direction is none
- OTU sf-threshold must be less than sd-threshold
- signal-failure raise must be greater than clear
- Sonet-SDH configure only when Ethernet Xgig WAN
- Sonet-SDH supports only Ethernet Port
- clock-source node-timed not allowed with loopback line
- Sonet-sdh hold-time up/Down - Not supported when ethernet xgig is wan
- Xgig support only Ethernet port not XIOM/Connector Breakout port
- OTU support only Ethernet port not XIOM/Connector Breakout port
- Expected/Transmit "Auto Generated" in OTU Section Monitoring/Path Monitoring reset after close and reopen GUI
- AIS level must be higher than its own level - configure eth-cfm domain md-admin-name
- EFM-OAM "Grace Tx" & EFM-OAM "Grace Tx" does not support
- Port does not support qinq encap in network mode
- Accounting policy 1.Ethernet Accounting Policy 2.Access Accounting Policy support record access-egress-octets/record combined-access-egress, port mode access 3.ethernet network Accounting Policy support record record network-egress-packets, port mode network
- port-connector_gsros_23-10-1_23-11 intent should be used to configure connector port and port-eth_msros_23-10-1_24-4 intent can be used to configure breakout ports.

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2.25 Name of the Intent: port-eth_mixr_24-10-1_24-4

2.25.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR node version 24.10 MD SROS.

2.25.2 Description

The port-eth_mixr_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of Port MD IXR with Port ID as target identifier. The following configurations are supported per deployment:

```
- Confiugre the port with port number
- Configuration of Admin State,Description,DDM Events,Monitor Oper Group
- Configure the Scheduler attributes as below
    a.Mode
    b.Low Latency
    c.VLAN QOS Policy
        Policy Name, Egress Queue Agg Rate, Burst Limit
- Configure the Ethernet Attributes as below
    a.Accounting Policy,Collect Stats
    b.Autonegotiate,Duplex,Dot1q Etype,LACP Tunnel,MAC Address,Min Frame
Length,Mode,Encap Type,MTU,MTU Profile,PTP Asymmetry,
      QinQ Etype,Rs FEC Mode,Speed,Util Stats Interval,XGIG,Discard Rx
Pause Frames,Down on Internal Error
    c.Hold Time
    d.Loopback
    e.Report Alarm
    f.CRC Monitor
        Signal Degrade,Signal Failure
    g.Dampening
    h.Down When Looped
    I.Eth CFM
        MEP
    J.Ingress
    K.SSM
    L.Storm Control
    M.Access
    N.EFM OAM
        Discovery,Link Monitoring,Peer Rdi Rx
    O.Dot1x
        Re Authentication,Per Host Authentication
    P.Egress
        Port Scheduler Policy,Rate,Eth Bn Rate Changes,Monitor Port
Scheduler
    Q.LLDP
        Dest Mac
    R.Network
```

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```
- Configuration the OTU
    a.Report Alarm
    b.Fine Granularity BER
      Signal Degrade,Signal Failure
    c.Path Monitoring
      Trail Trace Identifier
    d.Section Monitoring
      Trail Trace Identifier
- Configuration the SONET-SDH
    a.Hold Time
    b.Report Alarm
    C.Section Trace
      Section Trace,Group
```

CLI context: port

Prerequisite:

- Enable synch-e true on MDA
- Configure Oper-Group,Accounting Policy
- Configure macsec connectivity-association
- Configure system eth-cfm
- configure eth-cfm Domain
- Configure port-qos-policy,vlan-qos-policy

2.25.3 Advisory

- Please consult NSP PLM if this intent is required for NSP 24.8
- vlan-qos-policy policy does not support on IXR-6 & IXR-10
- The burst-limit attribute under vlan-qos-policy configuration is not supported by IXR-R6d, IXR-e2, IXR-R6dl, IXR-e2c, IXR-xs, IXR-x3, IXR-x1 and IXR-x.
- Deletion[NSP and Network] of port intent fail when ethernet mtu-profile true
- ddm-events does not support connector breakout port type
- ethernet Report Alarm Block Not Locked/Alignment Marker Not Locked/Duplicate Lane does not support IXR-R6d
- ethernet rs-fec-mode not supported on connector breakout port type
- ethernet discard-rx-pause-frames & down-on-internal-error -not supported
- Speed,AutoNegotiate and duplex attributes can be configure on the applicable ports
- ethernet dampening reuse-threshold - reuse-threshold must be less than suppress-threshold
- Distribution of Port intent will fail when Eth-CFM configuration due to Unique values required
- Encap Type null is not supported for Hybrid Ports.
- ethernet dampening max-suppress-time - max-suppress-time must be greater than or equal to half-life
- down-when-looped admin-state must be disabled to have loopback direction
- digital-coherent-optics and optical-line-system cannot both be set

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- TDM & ethernet GNSS does not support breakout port
- Port-control must be force-authorized to enable tunneling of all encaps
- eth-cfm MEP is not allowed with down-when-looped configured on the port
- Sub-second MEPs do not support ccm-padding
- Per-host-authentication cannot be enabled - fp option must be enabled
- efm-oam admin-state must be disabled to have loopback direction internal
- efm-oam accept-remote-loopback must be false to have loopback direction
- ethernet efm-oam admin-state - Cannot enable both tunneling and efm-oam
- connector breakout Port does not support qinq encap in network mode
- ethernet qinq-etype - Not supported for network mode or null/dot1q encap-type
- ethernet dot1q-etype - Not supported for network mode or null encap-type
- loopback direction internal is not supported on network ports
- swap-src-dst-mac must be false when direction is none
- OTU sf-threshold must be less than sd-threshold
- signal-failure raise must be greater than clear
- Sonet-SDH configure only when Ethernet Xgig WAN
- Sonet-SDH supports only Ethernet Port
- clock-source node-timed not allowed with loopback line
- Sonet-sdh hold-time up/Down - Not supported when ethernet xgig is wan
- Xgig support only Ethernet port not XIOM/Connector Breakout port
- OTU support only Ethernet port not XIOM/Connector Breakout port
- Expected/Transmit "Auto Generated" in OTU Section Monitoring/Path Monitoring reset after close and reopen GUI
- AIS level must be higher than its own level - configure eth-cfm domain md-admin-name
- EFM-OAM "Grace Tx" & EFM-OAM "Grace Tx" does not support
- Port does not support qinq encap in network mode
- Accounting policy 1.Ethernet Accounting Policy 2.Access Accounting Policy support record access-egress-octets/record combined-access-egress, port mode access 3.ethernet network Accounting Policy support record record network-egress-packets, port mode network
- port-connector_gsros_23-10-1_23-11 intent should be used to configure connector port and port-eth_msros_23-10-1_24-4 intent can be used to configure breakout ports.

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2.26 Name of the Intent: port-eth_msros_23-10-1_24-4

2.26.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS MD SR OS node version 23.10

2.26.2 Description

The port-eth_msros_23-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of Port MD.

The following configurations are supported per deployment:

```
- Port and port number
- Configuration of the following attributes:
    - Admin State
    - Description
    - DDM Events
    - Monitor Agg Egress Queue Stats
    - Oper Group
    - Monitor Oper Group
    - Dist CPU Protection
    - Hybrid Buffer Allocation
    - Egress Weight,Ingress Weight
    - Modify Buffer Allocation
    - Percentage of Rate
    - Access
    - Egress,Ingress
- Configuration of the following Ethernet attributes:
    - Down on Internal Error
    - Hold Time
    - Loopback
    - Report Alarm
    - CRC Monitor
    - Signal Degrade
    - Signal Failure
    - Dampening
    - Down When Looped
    - Eth CFM
    - MEP
    - Ingress
    - ELMI
    - SSM
    - Symbol Monitor
    - Signal Failure
    - Signal Degrade
    - Access
    - Egress,Access
```

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```
        - EFM OAM
        - Discovery
        - Link Monitoring
        - Peer Rdi Rx
        - Dot1x
        - Re Authentication
        - Per Host Authentication
        - P.Egress
        - Port Scheduler Policy
        - Rate
        - Eth Bn Rate Changes,
        - Monitor Port Scheduler
        - Q.LLDP
        - Dest Mac
        - R.Network
        - Egress
    - Configuration of the following Network attributes:
        - Egress
    - Configuration the following OTU attributes:
        - Report Alarm
        - Fine Granularity BER
        - Signal Degrade,Signal Failure
        - Path Monitoring
        - Trail Trace Identifier
        - Section Monitoring
        - Trail Trace Identifier
    - Configuration of the following SONET-SDH attributes
        - Hold Time
        - Report Alarm
        - Section Trace
        - Section Trace
        - Group
```

Prerequisites:

- Configure Oper-Group, Monitor Oper Group
- Configure Dist CPU Protection, Slop Policy, Accounting Policy
- Configure system eth-cfm, Domain
- Configure queue-group-templates-Ingress/Egress, network-queue,slop policy
- Configure scheduler-policy, policer-control-policy, port-scheduler-policy
- Configure hw-agg-shaper-scheduler-policy

2.26.3 Advisory

- Host match entries only supported on port access egress queue groups with system default instance 1
- ethernet access egress queue-group Cannot have both scheduler-policy and agg-rate params configured

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- Accounting Policy ID association is successful if Accounting Policy already created on the NE with proper record type.
- Speed,AutoNegotiate and duplex attributes can be configured on the applicable ports
- ethernet access ingress queue-group monitor-queue-depth fast-polling cannot set
- single-fiber only supported in network mode
- single-fiber is not supported with autonegotiate
- single-fiber - single-fiber port is only supported with null or dot1q encap-type
- ethernet report-alarm block-not-locked/alignment-marker-not-locked/duplicate-lane true does not support on Breakout/Ethernet port
- ethernet ingress rate - Cannot exceed port bandwidth
- Distribution of Port intent will fail when Eth-CFM configuration due to Unique values required
- Encap Type null is not supported for Hybrid Ports.
- ethernet access egress virtual-port can only have one of: port-scheduler-policy, scheduler-policy or agg-rate params configured
- Hybrid Buffer Allocation supported only hybrid mode of port
- digital-coherent-optics and optical-line-system cannot both be set
- monitor-oper-group and oper-group cannot both be set
- access attribute amber-alarm-threshold cannot be greater than red-alarm-threshold
- eth-cfm MEP is not allowed with down-when-looped configured on the port
- Sub-second MEPs do not support ccm-padding
- Cannot enable MEP grace tx and mcc-ed tx at the same time
- vlan - MEP does not support ETH-GNM BN Rx
- csf only supported on primary-vlan port
- signal-degrade must be less than or equal to signal-failure
- Cannot enable both tunneling and efm-oam
- Signal fail threshold must be equal to or greater than signal degrade threshold
- efm-oam frame monitoring and port crc-monitor cannot both be enabled
- efm-oam symbol monitoring and port symbol monitoring cannot both be enabled
- Port-control must be force-authorized to enable tunneling of all encaps
- Ethernet network egress queue-group Cannot have both scheduler-policy and agg-rate params configured
- OTU sf-threshold must be less than sd-threshold
- signal-failure raise must be greater than clear
- Adaptive CBS sizing step cannot be specified when reserved cbs is default
- ethernet ingress rate - Cannot exceed port bandwidth
- Eth-cfm "vlan" attribute support only when QinQ Encap port
- Sonet-SDH configure only when Ethernet Xgig WAN
- Sonet-SDH supports only Ethernet Port

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- clock-source node-timed not allowed with loopback line
- Xgig support only Ethernet port not XIOM/Connector Breakout port
- OTU support only Ethernet port not XIOM/Connector Breakout port
- Expected/Transmit "Auto Generated" in OTU Section Monitoring/Path Monitoring reset after close and reopen GUI
- Some port attributes such as TDM/GNSS/scheduler does not support on the SR nodes
- Some of port Ethernet attributes does not support on the SR nodes as below
  - Loopback
  - Storm Control
  - Network Egress Queue Group -->HS Turbo
  - Egress:HS Scheduler Policy
  - Port QOS Policy
  - HS Port Pool Policy
  - HW Agg Shaper Scheduler
- Ethernet satellite uplink port configuration not supported
- AIS level must be higher than its own level - configure eth-cfm domain md-admin-name
- hw-agg-shaper-scheduler policy does not support 7950 XRS NEs
- the port-connector_gsros_23-10-1_23-11 intent should be used to configure connector ports and the port-eth_msros_23-10-1_24-4 intent can be used to configure breakout ports.

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2.27 Name of the Intent: port-eth_msros_24-10-1_24-4

2.27.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS MD SR OS node version 24.10

2.27.2 Description

The port-eth_msros_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a Port MD.

The following configurations are supported per deployment:

```
- Port and port number
- Configuration of the following attributes:
    - Admin State
    - Description
    - DDM Events
    - Monitor Agg Egress Queue Stats
    - Oper Group
    - Monitor Oper Group
    - Dist CPU Protection
    - Hybrid Buffer Allocation
    - Egress Weight,Ingress Weight
    - Modify Buffer Allocation
    - Percentage of Rate
    - Access
        - Egress
                Pool
                    Monitor Pool Depth: Admin State
                                Alarm Thresholds: Total Pool In Use,
Shared Pool In Use, Reserved Pool In Use

        - Ingress
                Pool
                    Monitor Pool Depth: Admin State
                                Alarm Thresholds: Total Pool In Use,
Shared Pool In Use, Reserved Pool In Use
    - Network
        - Egress
                Pool
                    Monitor Pool Depth: Admin State
                                Alarm Thresholds: Total Pool In Use,
Shared Pool In Use, Reserved Pool In Use

- Configuration of the following Ethernet attributes:
    - Down on Internal Error
    - Hold Time
```

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```
        - Loopback
        - Report Alarm
        - CRC Monitor
        - Signal Degrade
        - Signal Failure
        - Dampening
        - Down When Looped
        - Eth CFM
        - MEP
        - Ingress
        - ELMI
        - SSM
        - Symbol Monitor
        - Signal Failure
        - Signal Degrade
        - Access
                Egress
                    Queue Group
                         Aggregate Rate
                                      Adaptation Rule,burst-limit
        - Network
                Egress
                    Queue Group
                         Aggregate Rate
                                      Adaptation Rule,burst-limit
        - Ptp Timestamping: Admin State, Ipv4 Address, Ipv6 Address
        - Tx Pause Frames
        - Egress,Access
        - EFM OAM
        - Discovery
        - Link Monitoring
        - Peer Rdi Rx
        - Dot1x
        - Re Authentication
        - Per Host Authentication
        - P.Egress
        - Port Scheduler Policy
        - Rate
        - Eth Bn Rate Changes,
        - Monitor Port Scheduler
        - Q.LLDP
        - Dest Mac
        - R.Network
        - Egress
    - Configuration of the following Network attributes:
        - Egress
    - Configuration the following OTU attributes:
        - Report Alarm
        - Fine Granularity BER
```

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```
    - Signal Degrade,Signal Failure
    - Path Monitoring
    - Trail Trace Identifier
    - Section Monitoring
    - Trail Trace Identifier
- Configuration of the following SONET-SDH attributes
    - Hold Time
    - Report Alarm
    - Section Trace
    - Section Trace
    - Group
```

Prerequisites:

- Configure Oper-Group, Monitor Oper Group
- Configure Dist CPU Protection, Slop Policy, Accounting Policy
- Configure system eth-cfm, Domain
- Configure queue-group-templates-Ingress/Egress, network-queue,slop policy
- Configure scheduler-policy, policer-control-policy, port-scheduler-policy
- Configure hw-agg-shaper-scheduler-policy

2.27.3 Advisory

- Host match entries only supported on port access egress queue groups with system default instance 1
- ethernet access egress queue-group Cannot have both scheduler-policy and agg-rate params configured
- Accounting Policy ID association is successful if Accounting Policy already created on the NE with proper record type.
- Speed,AutoNegotiate and duplex attributes can be configured on the applicable ports
- ethernet access ingress queue-group monitor-queue-depth fast-polling cannot set
- single-fiber only supported in network mode
- single-fiber is not supported with autonegotiate
- single-fiber - single-fiber port is only supported with null or dot1q encap-type
- ethernet report-alarm block-not-locked/alignment-marker-not-locked/duplicate-lane true does not support on Breakout/Ethernet port
- ethernet ingress rate - Cannot exceed port bandwidth
- Distribution of Port intent will fail when Eth-CFM configuration due to Unique values required
- Encap Type null is not supported for Hybrid Ports.
- ethernet access egress virtual-port can only have one of: port-scheduler-policy, scheduler-policy or agg-rate params configured
- Hybrid Buffer Allocation supported only hybrid mode of port
- digital-coherent-optics and optical-line-system cannot both be set
- monitor-oper-group and oper-group cannot both be set

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- access attribute amber-alarm-threshold cannot be greater than red-alarm-threshold
- eth-cfm MEP is not allowed with down-when-looped configured on the port
- Sub-second MEPs do not support ccm-padding
- Cannot enable MEP grace tx and mcc-ed tx at the same time
- vlan - MEP does not support ETH-GNM BN Rx
- csf only supported on primary-vlan port
- signal-degrade must be less than or equal to signal-failure
- Cannot enable both tunneling and efm-oam
- Signal fail threshold must be equal to or greater than signal degrade threshold
- efm-oam frame monitoring and port crc-monitor cannot both be enabled
- efm-oam symbol monitoring and port symbol monitoring cannot both be enabled
- Port-control must be force-authorized to enable tunneling of all encaps
- Ethernet network egress queue-group Cannot have both scheduler-policy and agg-rate params configured
- OTU sf-threshold must be less than sd-threshold
- signal-failure raise must be greater than clear
- Adaptive CBS sizing step cannot be specified when reserved cbs is default
- ethernet ingress rate - Cannot exceed port bandwidth
- Eth-cfm "vlan" attribute support only when QinQ Encap port
- Sonet-SDH configure only when Ethernet Xgig WAN
- Sonet-SDH supports only Ethernet Port
- clock-source node-timed not allowed with loopback line
- Xgig support only Ethernet port not XIOM/Connector Breakout port
- OTU support only Ethernet port not XIOM/Connector Breakout port
- Expected/Transmit "Auto Generated" in OTU Section Monitoring/Path Monitoring reset after close and reopen GUI
- Some port attributes such as TDM/GNSS/scheduler does not support on the SR nodes
- Some of port Ethernet attributes does not support on the SR nodes as below
  - Loopback
  - Storm Control
  - Network Egress Queue Group -->HS Turbo
  - Egress:HS Scheduler Policy
  - Port QOS Policy
  - HS Port Pool Policy
  - HW Agg Shaper Scheduler
- Ethernet satellite uplink port configuration not supported
- AIS level must be higher than its own level - configure eth-cfm domain md-admin-name
- hw-agg-shaper-scheduler policy does not support 7950 XRS

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- port-connector_gsros_23-10-1_23-11 intent should be used to configure connector port and port-eth_msros_23-10-1_24-4 intent can be used to configure breakout ports.

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2.28 Name of the Intent: port-xc_msros_23-10-1_24-4

2.28.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950 node version 23.10 MD SROS.

2.28.2 Description:

port-xc_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of PXC port.

The following configurations are supported per deployment:

```
- Port ID
- Description
- Admin State
```

2.28.3 Advisory:

- User should ensure that, port-id has Mode = hybrid, Encap-type = dot1q and Tunneling = true.
- User should also make sure that, port-id should not support "otu".

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2.29 Name of the Intent: qos-egress-remark-policy_mixr_23-10-1_23-11

2.29.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR MD node version 23.10

2.29.2 Description

qos-egress-remark-policy_mixr_23-10-1_23-11 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of QoS Egress Remark Policy.

The following configurations are supported per deployment:

```
- Description, Scope
- FC Dot1p Mapping, FC DSCP Mapping, FC LSP-EXP Mapping
- Forwarding Class : In Profile Dot1p, Out Profile Dot1p, Exceed Profile
Dot1p, In Plus Profile Dot1p, In Profile DSCP, Out Profile DSCP, Exceed
DSCP, In Plus DSCP, LSP-EXP In Profile, LSP-EXP Out Profile, LSP-EXP In
Plus Profile, LSP-EXP Exceed Profile, Mark DE bit, Force DE Value.
```

CLI context: config > qos > egress-remark-policy

Prerequisite: FC Dot1p Mapping, FC DSCP Mapping, FC LSP-EXP Mapping policies should be pre configured.

2.29.3 Advisory

- FC Dot1p Mapping, FC DSCP Mapping and FC LSP-EXP Mapping are supported only on IXR-R6d/R6dl/x/x3/e2/e2c chassis models.
- Forwarding Classes are not supported on IXR-R6d/R6dl/x/x3/e2/e2c chassis.
- To configure Exceed/In Plus Profile attributes, ensure the fc-4 profile admin-state is enabled (config > system > fp > options > qos > fc-4-profile).

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2.30 Name of the Intent: qos-ingress-classification_mixr_23-10-1_23-11

2.30.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR node version 23.10 MD SROS.

2.30.2 Description

qos-ingress-classification_mixr_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of ingress-classification policy-name name as target identifier. The following configurations are supported per deployment:

```
- Configuration of Description,Scope,Dot1p FC Map,DSCP FC Map,LSP Exp FC
Map,Remark,Allow Attachment
- Configure the Default Action attributes as below
    a.FC
    b.Profile
- Configure the Dot1p atributes as below
    a.Dot1p Value
    b.FC
    c.Profile
- Configure the DSCP attributes as below
    a.DSCP name
    b.FC
    c.Profile
- Configure the LSP Exp attributes as below
    a.LSP EXP value
    b.FC
    c.Profile
```

CLI context: qos > ingress-classification-policy

Prerequisite:

- Configure dot1p-fc-map/lsp-exp-fc-map/dscp-fc-map in qos
- Enable the fc-4-profile [configure system fp options qos fc-4-profile]

2.30.3 Advisory

- Allow Attachment is supported only on 7250 IXR-x/x3/R6dl/R6d
- remark and default-action is not supported on IXR-x/x3/R6dl/R6d
- Dot1p/DSCP/LSP-EXP ius not supported on IXR-x/x3/R6dl/R6d
- Dot1p FC Map/DSCP FC Map/LSP-EXP FC Map is supported only on IXR-x/x3/R6dl/R6d
- Profile exceed or inplus requires that system fp options qos fc-4-profile admin-state are configured

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2.31 Name of the Intent: qos-ingress-classification_mixr_24-10-1_24-4

2.31.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR node version 24.10 MD SROS.

2.31.2 Description

The qos-ingress-classification_mixr_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of ingress-classification policy-name name as target identifier. The following configurations are supported per deployment:

```
- Configuration of Description,Scope,Dot1p FC Map,DSCP FC Map,LSP Exp FC
Map,Remark,Allow Attachment
- Configure the Dot1p atributes as below
    a.Dot1p Value
    b.FC
    c.Profile
- Configure the DSCP attributes as below
    a.DSCP name
    b.FC
    c.Profile
- Configure the LSP Exp attributes as below
    a.LSP Exp value
    b.FC
    c.Profile
- Configure the Prec attributes as below
    a.Prec value
    b.FC
    c.Profile
```

CLI context: qos > ingress-classification-policy

Prerequisite:

- Configure dot1p-fc-map/lsp-exp-fc-map/dscp-fc-map in qo
- Enable admin state of fc-4-profile [configure system fp options qos fc-4-profile admin-sate]

2.31.3 Advisory

- Allow Attachment is supported only on 7250 IXR-x/x3/R6dl/R6d/IXR-e2c/IXR-e2
- Allow Attachment is not supported on IXR-ec/IXR-R4/IXR-s/IXR-e/IXR-R6
- remark and default-action is not supported on IXR-x/x3/R6dl/R6d/IXR-e2c/IXR-e2
- remark and default-action supported on IXR-ec/IXR-R4/IXR-s/IXR-e/IXR-R6
- Dot1p/DSCP/LSP-EX/Prec is not supported on IXR-x/x3/R6dl/R6d
- Dot1p FC Map/DSCP FC Map/LSP-EXP FC Map is supported only on IXR-x/x3/R6dl/R6d/IXR-e2c/IXR-e2
- Dot1p/DSCP/LSP-EX/Prec is supported on IXR-ec/IXR-R4/IXR-s/IXR-e/IXR-R6

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- Profile exceed and inplus requires the system fp options qos fc-4-profile admin-state configured

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2.32 Name of the Intent: qos-ml-ip-prefix-list_msros_23-10-1_24-4

2.32.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7950/7750/7450 MD SR OS node version 23.10

2.32.2 Description

qos-ml-ip-prefix-list_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of prefix_list_md Policy.

The following configurations are supported per deployment:

```
- Description
- Prefix List
```

CLI context: qos > match-list > ip-prefix-list

2.32.3 Advisory

N/A

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2.33 Name of the Intent: qos-ml-ipv6-prefix-list_msros_23-10-1_24-4

2.33.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7950/7450/7750 MD SR OS node version 23.10

2.33.2 Description

qos-ml-ipv6-prefix-list_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of qos IPv6 Prefix-list.

The following configurations are supported per deployment:

```
- Prefix-List-Name
- Description
- Prefix
```

CLI context: qos > match-list > ipv6-prefix-list

2.33.3 Advisory

N/A

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2.34 Name of the Intent: qos-ml-port-list_msros_23-10-1_24-4

2.34.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS MD SR OS node version 23.10

2.34.2 Description

qos-ml-port-list_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of QoS match-list port-list Policy.

The following configurations are supported per deployment:

```
- Description
- Port Value
- Port Range (Start to End)
```

CLI context: qos > match-list > port-list

2.34.3 Advisory

N/A

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2.35 Name of the Intent: qos-network-queue_msros_23-10-1_24-4

2.35.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750-SR, 7950-XRS, 7450-ESS MD SROS of node version 23.10

2.35.2 Description:

qos-network-queue_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of QoS Network Queue.

The following configurations are supported per deployment:

```
- Description
- Queue
    - Queue ID, Multipoint, Queue Type, CBS, MBS, Avg Frame
Overhead
    - Rate: PIR, CIR, FIR
    - Port Parent: Level, CIR Level, Weight, CIR Weight
    - Adaptation Rule: PIR, CIR, FIR
    - Drop Tail: Percent Reduction From MBS
- FC: FC Name, Queue, Multicast Queue
```

CLI context: configure > qos > network-queue

2.35.3 Advisory:

- FC queue mapping to a multipoint queue will fail.
- FC multicast-queue mapping to a non-multipoint queue will fail.
- FC association to Queue has to be removed for Queue deletion.

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2.36 Name of the Intent: qos-policer-control-policy_msros_23-10-1_24-4

2.36.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7950/7450/7750 MD SR OS node version 23.10

2.36.2 Description

qos-policer-control-policy_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of policer-control policy.

The following configurations are supported per deployment:

```
- Name of the Policer-Control Policy
- Configuration of Description
- Root configuration:
  - Max Rate Or Max Percent Rate, Profile Preferred
  - Priority-MBS-Thresholds
    - Priority Level, MBS Contribution, Fixed MBS
- Tier configuration:
  - Tier ID
  - Arbiter
    - Rate Or Percent Rate, Reference Rate, Arbiter Name, Description
    - Arbiter Parent
      - Arbiter Name, Level, Weight
```

2.36.3 Advisory

- Policer-Control Policy supports Tier ID 1 and Tier TD 2.
- Arbiter Parent cannot be configured on Tier ID 1.

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2.37 Name of the Intent: qos-sap-egress_msros_23-10-1_23-11

2.37.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7450/7950 MD SR OS node version 23.10

2.37.2 Description

qos-sap-egress_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of QoS SAP egress.

The following configurations are supported per deployment:

```
- Creation of SAP Egress with Policy ID and Name
- Configuration of Description, Default FC, Scope [Exclusive, Template],
Ethernet Ctag, Policers HQoS Manageable, Parent Location [Auto, SLA],
Post Policer Mapping, HS Attachment Policy, Use Policer Result Marking
Dot1p Inner
- Configuration of Dot1p Value with the following supported attributes:
    a. Dot1p Value, FC: Valid FC Name, Profile [In, Out, DE, Exceed,
Inplus]
- Configuration of DSCP with the following supported attributes:
    a. DSCP Name: Drop Down List [BE, CP1, CP2, etc.], FC: Valid FC Name,
Profile [In, Out, DE, Exceed, Inplus]
- Configuration of Prec Value with the following supported attributes:
    a. Prec Value, FC: Valid FC Name, Profile [In, Out, DE, Exceed,
Inplus]
- Configuration of Queue with the following suppported attributes:
    a. Queue ID, Avg Frame Overhead, HS WRR Weight, Maximum Data
Transmission, Queue Type [Expedited, Auto Expedited, Best Effort] Adv.
Config Policy, AGG Shaper Weight, Burst Limit, Packet Byte Offset, CBS
(kilobytes), MBS (Bytes), Rate or Percent Rate, Sched Class, FIR Burst
Limit, HS Class Weight
    b. Parent Mapping:
        - Scheduler Parent with the following attributes: Scheduler Name,
Level, Weight, CIR Level, CIR weight
        - Port Parent with the following attributes: Level, Weight, CIR
Level, CIR Weight
    c. Adaptation Rule with PIR and CIR configurable options [Max, Min,
Closest]
    d. WRED Queue: Policy, Mode [Native, Pool Per Queue], Usage [Auto,
Exceed Low]
    d. Drop Tail:
        - High Plus: Percent Reduction from MBS or auto.
        - High: Percent Reduction from MBS or auto.
        - Low: Percent Reduction from MBS or auto.
        - Exceed: Percent Reduction from MBS or auto.
    e. HS WRED Queue: Policy
```

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```
    f. Boolean attributes: HS Alt Port Class Pool
- Configuration of Policer with the following supported attributes:
    a. Policer ID, Description, Stat-Mode, Adv. Config Policy, CBS
(bytes), MBS (bytes), Rate or Percent Rate
        - Stat-Mode can be configured with Drop-Down list values [No
stats, Minimal, Offered Profile No CIR, etc.]
    b. Adaptation Rule with PIR and CIR configurable options [Max, Min,
Closest]
    c. Parent Mapping:
        - Scheduler Parent with the following attributes: Scheduler Name,
Level, Weight, CIR Level, CIR weight
        - Port Parent with the following attributes: Level, Weight, CIR
Level, CIR Weight
    d. Boolean attributes: DSCP PREC Remarking, Profile Out Preserve,
Profile Capped
- Configuration of FC with the following supported attributes:
    a. FC Name: Drop Down List with the following values [L2, BE, AF, L1,
H2, EF, H1, NC], Policer
    b. De Mark: Force
    c. De Mark Inner: Force
    d. De Mark Outer: Force
    e. Queue Redirection Mapping [Port Redirect Group Queue, Queue, Queue
Group Queue]:
        - Port Redirect Group Queue: Queue
        - Queue Group Queue: Queue Group Name, Queue, Instance
    f. Dot1p, Dot1p Inner, and Dot1p Outer consist of the following:
        - Exceed Profile, In Profile, Out Profile
- Configuration of HS WRR Group with the following supported attributes:
    a. Group ID, HS Class Weight
    b. Adaptation Rule: PIR [Max, Min, closest], Rate or Percent rate
- Configuraton of IP Criteria with entry having the following
configurable attributes:
    a. Entry ID, Description
    b. Match Protocol Choice [Protocol or Protocol List], DSCP, Fragment,
SRC IP Address Mask/Prefix-List, DST IP Address Mask/Prefix-List
        - Protocol [TCP, UDP, TCP-UDP, IGMP, IP, EGP, IGP, RDP, IPv6,
IPv6-Route, etc.]
        - DSCP List of all supported values in Drop Down List
        - Fragment [True/False]
    c. SRC Port DST Port
        - The user needs to configure Source and Destination Port with
options, such as EQ, LT, GT, Range
    d. Action Type [accept, Ignore Match], FC, Profile [In, Out, Exceed,
Inplus], Policer, Queue
        - Use FC Mapped Queue Boolean
- Configuration of IPv6 Criteria with Entry having the following
configurable attributes:
    a. Entry ID, Description
    b. Match Next Header Choice [Protocol or Protocol List], DSCP, SRC
```

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```
IPv6 Address Mask/Prefix-List, DST IPv6 Address Mask/Prefix-List
        - Protocol [TCP, UDP, TCP-UDP, IGMP, IP, EGP, IGP, RDP, IPv6,
IPv6-Route, etc.]
        - DSCP list of all supported values in Drop Down List
    c. SRC Port DST Port
        - The user needs to configure Source and Destination Port with
options, such as EQ, LT, GT, Range
    d. Action Type [accept, Ignore Match], FC, Profile [In, Out, Exceed,
Inplus], Policer, Queue
        - Boolean [Use FC Mapped Queue, Port Redirect Group Queue]
- Configuration of HW AGG Shaper Queues: Queue Set Size
- Configuration of Scheduler Class Elevation: Class ID, Weight
- Configuration of Subscriber Mgmt:
    a. PCC Rule Entry Range Checkbox: Start, End
    b. Dynamic Policer, Policer ID Range Check Box Start, End, Stat Mode
[No stats, Minimal, Offered Profile No CIR, etc.], Packet Byte Offset,
MBS, CBS
    c. Arbiter Parent with the following attributes: Arbiter Name, Level,
Weight
    d. Dynamic Queue Parent Mapping:
        - Scheduler Parent with the following attributes: Scheduler Name,
Level, Weight, CIR Level, CIR Weight
        - Port Parent with the following attributes: Level, Weight, CIR
Level, CIR Weight
    e. Queue ID Range: Start, End
```

2.37.3 Advisory

- If "Parent-Location" is set to "SLA", then Policers-hQoS-Manageable cannot be configured.
- Policers-hQoS-Manageable cannot be set when a policer in the policy has "Exceed-PIR" attribute enabled.
- The "Arbiter Parent" attribute under Policer and Policers-hQoS-Manageable is mutually exclusive.
- Cannot map Queue under FC when Policers-hQoS-Manageable is configured.
- If Policer-hQoS-Manageable is set to False, then only Port Redirect Group Queue or FC Mapped Queue can be configured.
- Port-Redirect-Group-Queue or FC Mapped Queue can be configured.
- For SR-14s and XRS-20e nodes, the Parent Mapping [Scheduler Parent or Port Parent] attribute is not supported.
- Cannot set Policers-hQoS-Manageable when a Dynamic Policer range is configured.
- The Dynamic Policer range cannot overlap with the provided policers. For example, suppose the provided Policer ID is 12, then the Dynamic Policer range should be between 13 and 63.
- The Dynamic Queue range cannot overlap with the provided Queue.

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2.38 Name of the Intent: qos-sap-egress_msros_24-10-1_24-4

2.38.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950 MD SR OS node version 24.10

2.38.2 Description

The qos-sap-egress_msros_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of QoS SAP egress.

The following configurations are supported per deployment:

```
- Creation of SAP Egress with Policy ID and Name
- Configuration of Description, Default FC, Scope [Exclusive, Template],
Ethernet Ctag, Policers HQoS Manageable, Parent Location [Auto, SLA],
Post Policer Mapping, HS Attachment Policy, Use Policer Result Marking
Dot1p Inner
- Configuration of Dot1p Value with the following supported attributes:
    a. Dot1p Value, FC: Valid FC Name, Profile [In, Out, DE, Exceed,
Inplus]
- Configuration of DSCP with the following supported attributes:
    a. DSCP Name: Drop Down List [BE, CP1, CP2, etc.], FC: Valid FC Name,
Profile [In, Out, DE, Exceed, Inplus]
- Configuration of Prec Value with the following supported attributes:
    a. Prec Value, FC: Valid FC Name, Profile [In, Out, DE, Exceed,
Inplus]
- Configuration of Queue with the following suppported attributes:
    a. Queue ID, Avg Frame Overhead, HS WRR Weight, Maximum Data
Transmission, Queue Type [Expedited, Auto Expedited, Best Effort] Adv.
Config Policy, AGG Shaper Weight, Packet Byte Offset, Rate or Percent
Rate, Sched Class, FIR Burst Limit, HS Class Weight, Burst Limit Or Delay
Time, CBS Or Delay Time Or Delay Percent, MBS Or Delay Time Or Delay
Percent
    b. Parent Mapping:
        - Scheduler Parent with the following attributes: Scheduler Name,
Level, Weight, CIR Level, CIR weight
        - Port Parent with the following attributes: Level, Weight, CIR
Level, CIR Weight
    c. Adaptation Rule with PIR and CIR configurable options [Max, Min,
Closest]
    d. WRED Queue: Policy, Mode [Native, Pool Per Queue], Usage [Auto,
Exceed Low]
    d. Drop Tail:
        - High Plus: Percent Reduction from MBS or auto.
        - High: Percent Reduction from MBS or auto.
        - Low: Percent Reduction from MBS or auto.
        - Exceed: Percent Reduction from MBS or auto.
```

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```
    e. HS WRED Queue: Policy
    f. Boolean attributes: HS Alt Port Class Pool
- Configuration of Policer with the following supported attributes:
    a. Policer ID, Description, Stat-Mode, Adv. Config Policy, CBS
(bytes), MBS (bytes), Rate or Percent Rate
        - Stat-Mode can be configured with Drop-Down list values [No
stats, Minimal, Offered Profile No CIR, etc.]
    b. Adaptation Rule with PIR and CIR configurable options [Max, Min,
Closest]
    c. Parent Mapping:
        - Scheduler Parent with the following attributes: Scheduler Name,
Level, Weight, CIR Level, CIR weight
        - Port Parent with the following attributes: Level, Weight, CIR
Level, CIR Weight
    d. Boolean attributes: DSCP PREC Remarking, Profile Out Preserve,
Profile Capped
- Configuration of FC with the following supported attributes:
    a. FC Name: Drop Down List with the following values [L2, BE, AF, L1,
H2, EF, H1, NC], Policer
    b. De Mark: Force
    c. De Mark Inner: Force
    d. De Mark Outer: Force
    e. Queue Redirection Mapping [Port Redirect Group Queue, Queue, Queue
Group Queue]:
        - Port Redirect Group Queue: Queue
        - Queue Group Queue: Queue Group Name, Queue, Instance
    f. Dot1p, Dot1p Inner, and Dot1p Outer consist of the following:
        - Exceed Profile, In Profile, Out Profile
- Configuration of HS WRR Group with the following supported attributes:
    a. Group ID, HS Class Weight
    b. Adaptation Rule: PIR [Max, Min, closest], Rate or Percent rate
- Configuraton of IP Criteria with entry having the following
configurable attributes:
    a. Entry ID, Description
    b. Match Protocol Choice [Protocol or Protocol List], DSCP, Fragment,
SRC IP Address Mask/Prefix-List, DST IP Address Mask/Prefix-List
        - Protocol [TCP, UDP, TCP-UDP, IGMP, IP, EGP, IGP, RDP, IPv6,
IPv6-Route, etc.]
        - DSCP List of all supported values in Drop Down List
        - Fragment [True/False]
    c. SRC Port DST Port
        - The user needs to configure Source and Destination Port with
options, such as EQ, LT, GT, Range
    d. Action Type [accept, Ignore Match], FC, Profile [In, Out, Exceed,
Inplus], Policer, Queue
        - Use FC Mapped Queue Boolean
- Configuration of IPv6 Criteria with Entry having the following
configurable attributes:
    a. Entry ID, Description
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```
    b. Match Next Header Choice [Protocol or Protocol List], DSCP, SRC
IPv6 Address Mask/Prefix-List, DST IPv6 Address Mask/Prefix-List
        - Protocol [TCP, UDP, TCP-UDP, IGMP, IP, EGP, IGP, RDP, IPv6,
IPv6-Route, etc.]
        - DSCP list of all supported values in Drop Down List
    c. SRC Port DST Port
        - The user needs to configure Source and Destination Port with
options, such as EQ, LT, GT, Range
    d. Action Type [accept, Ignore Match], FC, Profile [In, Out, Exceed,
Inplus], Policer, Queue
        - Boolean [Use FC Mapped Queue, Port Redirect Group Queue]
- Configuration of HW AGG Shaper Queues: Queue Set Size
- Configuration of Scheduler Class Elevation: Class ID, Weight
- Configuration of Subscriber Mgmt:
    a. PCC Rule Entry Range Checkbox: Start, End
    b. Dynamic Policer, Policer ID Range Check Box Start, End, Stat Mode
[No stats, Minimal, Offered Profile No CIR, etc.], Packet Byte Offset,
MBS, CBS
    c. Arbiter Parent with the following attributes: Arbiter Name, Level,
Weight
    d. Dynamic Queue Parent Mapping:
        - Scheduler Parent with the following attributes: Scheduler Name,
Level, Weight, CIR Level, CIR Weight
        - Port Parent with the following attributes: Level, Weight, CIR
Level, CIR Weight
    e. Queue ID Range: Start, End
```

CLI context: qos > sap-egress

2.38.3 Advisory

- If "Parent-Location" is set to "SLA", then Policers-hQoS-Manageable cannot be configured.
- Policers-hQoS-Manageable cannot be set when a policer in the policy has "Exceed-PIR" attribute enabled.
- The "Arbiter Parent" attribute under Policer and Policers-hQoS-Manageable is mutually exclusive.
- Cannot map Queue under FC when Policers-hQoS-Manageable is configured.
- If Policer-hQoS-Manageable is set to False, then only Port Redirect Group Queue or FC Mapped Queue can be configured.
- Port-Redirect-Group-Queue or FC Mapped Queue can be configured.
- For SR-14s and XRS-20e nodes, the Parent Mapping (Scheduler Parent or Port Parent) attribute is not supported.
- Cannot set Policers-hQoS-Manageable when a Dynamic Policer range is configured.
- The Dynamic Policer range cannot overlap with the provided policers. For example, suppose the provided Policer ID is 12, then the Dynamic Policer range should be between 13 and 63.
- The Dynamic Queue range cannot overlap with the provided Queue.

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2.39 Name of the Intent: qos-sap-ingress_msros_23-10-1_23-11

2.39.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7450/7950 MD SR OS node version 23.10

2.39.2 Description

qos-sap-ingress_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of QoS SAP Ingress.

The following configurations are supported per deployment:

```
- Creation of SAP Ingress with Policy ID and Name
- Configuration of Description, Default FC, Scope [Exclusive, Template],
Default Priority [Low, High], Policers HQoS Manageable
- MAC Criteria or IP Criteria can be configured
- Configuration of MAC Criteria Type Normal with Entry having the
following configurable attributes:
        a. Entry ID, Description
        b. Match Frame Type [802DOT3, 802DOT2-LLC, 802DOT2-SNAP,
Ethernet-II]
            Frame Type 802DOT3 with the following configuration options:
                - SRC MAC Address, DST MAC address, Dot1p [Priority
[0-7], Mask [0-7]]
            Frame Type 802DOT2-LLC with the following configuration
options:
                - SRC MAC Address, DST MAC address, Dot1p [Priority
[0-7], Mask [0-7]], LLC DSAP [DSAP [0-255], MASK [0-255]], LLC SSAP [SSAP
[0-255], MASK [0-255]]
            Frame Type 802DOT2-SNAP with the following configuration
options:
                - SRC MAC Address, DST MAC address, Dot1p [Priority
[0-7], Mask [0-7]], SNAP OUI [Zero, Non-Zero], SNAP PID [0-65535]
            Frame Type Ethernet-II with the following configuration
options:
                - SRC MAC Address, DST MAC address, Etype, Dot1p
[Priority [0-7], Mask [0-7]]
        c. Action Type [Accept, Ignore Match], FC, Priority [Low, High],
Policer
                - The policer can be configured only with Action Type
Accept.
- Configuration of MAC Criteria Type VID with Entry having the following
configurable attributes:
        a. Entry ID, Description
        b. Match Frame Type Ethernet-II, No Etype is allowed
        c. Inner Tag
            - VLAN [0-4095]
```

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```
            - MASK [1-4095]
        d. Outer Tag
            - VLAN [0-4095]
            - MASK [1-4095]
        e. Action Type [Accept, Ignore Match], FC, Priority [Low, High],
Policer
            - The policer can be configured only with Action Type Accept.

- Configuraton of IP Criteria Type Normal with entry having the following
configurable attributes:
        a. Entry ID, Description
        b. Match Protocol Choice [Protocol or Protocol List], DSCP,
Fragment, SRC IP Address Mask/Prefix-List, DST IP Address
Mask/Prefix-List
            - Protocol [TCP, UDP, TCP-UDP, IGMP, IP, EGP, IGP, RDP, IPv6,
IPv6-Route, etc.]
            - DSCP List of all supported values in Drop Down List
            - Fragment [True/False]
        c. SRC Port DST Port
            - The user needs to configure Source and Destination Port
with an option, such as EQ, LT, GT, Range.
        d. Action Type [Accept, Ignore Match], FC, Priority [Low, High],
Policer
            - The policer can be configured only with Action Type
Accept.
- Configuration of IPv6 Criteria Type Normal with Entry having the
following configurable attributes:
        a. Entry ID, Description
        b. Match Next Header Choice [Protocol or Protocol List], DSCP,
Fragment, SRC IPv6 Address Mask/Prefix-List, DST IPv6 Address
Mask/Prefix-List
            - Protocol [TCP, UDP, TCP-UDP, IGMP, IP, EGP, IGP, RDP, IPv6,
IPv6-Route, etc.]
            - DSCP List of all supported values in Drop Down List
            - Fragment [True/False, First only, Non-First only]
        c. SRC Port DST Port
            - The user needs to configure Source and Destination Port
with an option, such as EQ, LT, GT, Range.
        d. Action Type [Accept, Ignore Match], FC, Priority [Low, High],
Policer
            - The policer can be configured only with Action type Accept.

- Configuration of IP/IPv6 Criteria Type as VXLAN-VNI only with
Protocol/Next Header as UDP (18)
        - VXLAN-VNI Params can be configured with EQ and Range
[1-16777215] attribute
- IP/IPv6 Criteria Type as Tagged have a Tag attribute with value range
[0-255].
- Configuration of Policer with the following supported attributes:
```

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```
        a. Policer ID, Description, Stat-Mode, Adv. Config Policy,
Profile Capped, CBS (bytes), MBS (bytes), Rate or Percent Rate
            - Stat-Mode can be configured with drop-down list values [No
stats, Minimal, Offered Profile No CIR, etc.]
        b. Adaptation Rule with PIR and CIR configurable options [Max,
Min, Closest]
        c. Scheduler Parent with the following attributes: Scheduler
Name, Level [1-8], Weight [0-100], CIR Level [0-8], CIR Weight [0-100]
        d. Arbiter Parent with the following attributes: Arbiter Name,
Level [1-8], Weight [1-100]
- Configuration of FC with the following supported attributes:
        a. FC Name: Drop Down List with the following values [L2, BE, AF,
L1, H2, EF, H1, NC]
        b. Egress FC: Drop Down List with the following values [L2, BE,
AF, L1, H2, EF, H1, NC]
        c. Unknown Queue Policer Mapping: Drop Down List options [Unknown
Policer, Unknown Queue, Unknown Queue Group Queue]
        d. Broadcast Queue Policer Mapping: Drop Down List options
[Broadcast Policer, Broadcast Queue, Broadcast Queue Group Queue]
        e. Multicast Queue Policer Mapping: Drop Down List options
[Multicast Policer, Multicast Queue, Multicast Queue Group Queue]
        f. Queue Policer Mapping: Drop Down List options [QPolicer,
Queue, Queue Group Queue]
        g. Profile: [In, Out]
        h. Boolean attributes: FP Redirect Group Unknown Policer, FP
Redirect Group Multicast Policer, FP Redirect Group Broadcast Policer, DE
In/Out Profile
        i. In/Out Profile: [DSCP, Prec]
- Configuration of Queue with the following suppported attributes:
        a. Queue ID, Queue Type, Queue Mode, Burst Limit, Adv. Config
Policy, Packet Byte Offset, CBS (kilobytes), MBS (Bytes), Rate or Percent
Rate
        b. Adaptation Rule with PIR and CIR configurable options [Max,
Min, Closest]
        c. Scheduler Parent with the following attributes: Scheduler
Name, Level [1-8], Weight [0-100], CIR Level [0-8], CIR Weight [0-100]
        d. Drop Tail: Low Percent reduction from MBS [0-100] or auto
        e. Boolean attributes: CIR Non-Profiling, Multipoint
- Configuration of DSCP with the following supported attributes:
        a. DSCP Name: Drop Down List [BE, CP1, CP2, etc.], FC: Valid FC
Name, Priority [Low, High, Auto]
- Configuration of Dot1p Value with the following supported attributes:
        a. Dot1p Value [0-7], FC: Valid FC Name, Priority [Low, High,
Auto]
- Configuration of LSP EXP Value with the following supported attributes:

        a. LSP EXP Value [0-7], FC: Valid FC Name, Priority [Low, High,
Auto]
- Configuration of Prec Value with the following supported attributes
```

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```
        a. Prec Value [0-7], FC: Valid FC Name, Priority [Low, High,
Auto]
- Configuration of Subscriber Mgmt:
        a. PCC Rule Entry range checkbox: Start [ 1-65535], end [1-65535]

        b. Dynamic Policer, Policer ID range checkbox Start [1-63], End
[1-63], Stat Mode [No Stats , Minimal, Offered Profile No CIR, etc.],
Packet Byte Offset [-32-31], MBS, CBS
        c. Arbiter Parent attributes: Arbiter Name, Level [1-8], Weight
[1-100]
```

2.39.3 Advisory

- MAC Criteria or IP Criteria can be configured per deployment.
- If MAC Criteria "Type" is "VID", then only "Frame-Type" is valid: "Ethernet-II" and no Etype are allowed.
- VXLAN-VNI Match Params can only be configured when the protocol is UDP (17).
- The user cannot configure SRC/DST Port along with VXLAN-VNI classification.
- The user cannot configure DST Port under Match when the Criteria Type is Tagged-Entries.
- If Policers-hQoS-Manageable attribute is true, then only Scheduler-Parent can be configured. Otherwise, Arbiter Parent can be configured.
- SAP Ingress Policy can only reference one unique Queue Group.
- XRS-20e node does not support the following CLIS: Policers-hQoS-Manageable <Boolean> Policer <Number> Scheduler-Parent Subscriber-Mgmt
- SR-14s node does not support the following CLIS: Policers-hQoS-Manageable <Boolean> Policer <Number> Scheduler-Parent
- SR-a4/a8 nodes do not support the following CLIS: FC <String> Broadcast-Policer <Reference> FC <String> Multicast-Policer <Reference> FC <String> FP-Redirect-Group-Broadcast-Policer FC <String> FP-Redirect-Group-Multicast-Policer FC <String> Policer <Reference> FC <String> Unknown-Policer <Reference> FC <String> FP-Redirect-Group-Unknown-Policer IP-Criteria Entry <Number> Action Policer <Reference> IPv6-Criteria Entry <Number> Action Policer <Reference> MAC-Criteria Entry <Number> Action Policer <Reference> Policer <Number> Policers-hQoS-Manageable <Boolean> Queue <Number> CIR-Non-Profiling <Boolean>
- SR3e node does not support the following CLIS: Queue <Number> CIR-Non-Profiling <Boolean>
- The Dynamic Policer range cannot overlap with the provided policers, for example, the PCC Rule Entry range cannot overlap with Dynamic Policer.

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2.40 Name of the Intent: qos-shared-policer_mixr_23-10-1_23-11

2.40.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7250 IXR MD SR OS nodes version 23.10

2.40.2 Description

qos-shared-policer_mixr_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of QoS Shared Policer.

The following configurations are supported per deployment:

```
- Create Shared Policer with Shared Policer Name:
  - Stat Mode
  - MBS (bytes)
  - CBS (bytes)
  - Rate: PIR (kilobps)
  - Rate: CIR (kilobps)
```

CLI context: configure > qos > shared-policers > policer

2.40.3 Advisory

- Configuration is only permitted when administrative state of the fc-4-profile option is Disabled (system > fp > options > qos > fc-4-profile > admin-state).
- Stat Mode attribute is not supported on IXR-R6d, IXR-R6dl, IXR-x1, IXR-x3, IXR-xs, IXR-e2, IXR-e2c nodes.

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2.41 Name of the Intent: qos-shared-queue_msros_23-10-1_23-11

2.41.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7450/7950 MD SR OS node version 23.10

2.41.2 Description

qos-shared-queue_msros_23-10-1_23-11 intent supports modification/discovery/audit/alignment.

The following configurations are supported per deployment:

```
- Description
- Queue: Queue ID, Multipoint, Queue Type, CBS, MBS, Rate, Drop Tail
```

2.41.3 Advisory

- Before configuring any policy, the user should ensure that "Shared Queue Policy" is present on a particular node and the user has to configure the policy.
- Policy Name: Default, Egress-PBR-Ingress-Queues, IOM-Assisted-IP-FWD-Queues, Policer-Output-Queues
- The user cannot configure the policy without these four policies.
- If the user wants to select "Multipoint", then "Queue ID" should be greater than 8.
- If the user wants to configure "Default" policy, then "Queue ID" should be in the range from 1 to 32.
- If the user wants to configure "Egress-PBR-Ingress-Queues" policy, then "Queue ID" should be in the range from 1 to 8.
- If the user wants to configure "IOM-Assisted-IP-FWD-Queues" policy, then "Queue ID" should be 1.
- If the user wants to configure "Policer-Output-Queues" policy, then "Queue ID" should be in the range from 1 to 16.

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2.42 Name of the Intent: qos-slope-policy_msros_23-10-1_23-11

2.42.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS MD SR OS node version 23.10

2.42.2 Description

qos-slope-policy_msros_23-10-1_23-11 creation/modification/deletion/discovery/audit/alignment of QoS Slope_Policy.

The following configurations are supported per deployment:

```
- Create Slope Policy with Slope Policy Name:
  - Description
  - Highplus Slope: Admin State [Enable/Disable], Start Avg, Max Avg, Max
Prob
  - High Slope: Admin State [Enable/Disable], Start Avg, Max Avg, Max
Prob
  - Low Slope: Admin State [Enable/Disable], Start Avg, Max Avg, Max Prob

  - Exceed Slope: Admin State [Enable/Disable], Start Avg, Max Avg, Max
Prob
```

CLI context: qos > slope-policy

2.42.3 Advisory

- Start Avg should not be more than Max Avg.

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2.43 Name of the Intent: qos-vlan-policy_mixr_23-10-1_23-11

2.43.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR node version 23.10 MD SROS.

2.43.2 Description

qos-vlan-policy_mixr_23-10-1_24-8 intent supports creation/modification/deletion/discovery/audit/alignment of VLAN QOS POLICY with VLAN QOS POLICY name as target identifier. The following configurations are supported per deployment:

```
- Description, Scope, Stat Mode
- Queue: Queue ID, Queue Mgmt, Queue Forwarding Type, PIR Weight,
Scheduling-priority, Low Latency, Adaptation Rule, Percent Rate
- Packet Byte Offset
```

CLI context: qos >>vlan-qos-policy

2.43.3 Advisory

- User need to pre configure Queue mgmt (qos>> queue-mgmt-policy).
- In order to configure packet byte offset ,user need to admin-enable packet-byte-offset.(qos>> packet-byte-offset >>admin-state enable)
- Vlan-qos-policy is not supported on IXR-6 and IXR-10 node.
- Scheduling-priority attribute only supported on IXR-R6d,IXR-R6dl nodes.
- Low latency attribute is only visible when queue ID 7 or 8 is selected by User.

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2.44 Name of the Intent: router-bgp-group_msros_24-10-1_24-11

2.44.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS and 7250 IXR for MD SROS node version 24.10

2.44.2 Description

router-bgp-group_msros_24-10-1_24-11 intent supports creation/modification/deletion/discovery/audit/alignment of BGP Group with Group Name as target identifier.

The following configurations are supported per deployment:

```
- Description, Connect Retry, Admin State, Static Group, Keep Alive,
Damping, Local Preference, Loop Detect, Loop Detect Threshold ,     TCP
MSS, Selective Label IPv4 Install, Min Route Advertisement, Aggregator ID
Zero, Preference, Block Prefix Sid, Multihop, Med Out ,Authentication
Key, Client Reflect,  VPN Apply Export, VPN Apply Import, AS Override,
ASN 4 Byte, Path MTU Discovery, Next Hop Self, Passive, Type, Peer AS,
Capability Negotiation, Enforce First AS, Initial Send Delay Zero, Local
Address, TTL Security, Default Route Target, AIGP, Advertise Inactive,
Authentication Keychain, BFD Liveness, Fast External Failover, Peer IP
Tracking, Dynamic Neighbor Limit, Split Horizon, Third Party Nexthop,
Label Preference, Def Recv EVPN Encap, Multipath Eligible, Egress Peer
Engineering Label Unicast, Hold Time, Family, Send Default, Default
Preference, Default Label Preference, Remove Private, Cluster, EBGP
Default Reject Policy, Local AS, Next Hop Unchanged, Origin Validation,
Link Bandwidth, Send Communities, Error Handling, Damp Peer Oscillations,
BFD Strict Mode.
- Add Paths
    - IPv4, IPv6, Label IPv4, Label IPv6, VPN IPv4, VPN IPv6, MVPN IPv4,
MVPN IPv6, Mcast VPN IPv4, Mcast VPN IPv6, EVPN
- Import, Export, Extended NH Encoding, Advertise IPv6 Next Hops.
- Graceful Restart
    - Long Lived,Family
- Outbound Route Filtering
    - Extended Community, Send ORF, Route Target, Prefix Limit
- Dynamic Neighbor
    - Match, Prefix, Interface
- Monitor
    - Egress Engineering
- Segment Routing V6
```

CLI context: router > bgp > group

Prerequisites:

- Enable BGP routing Protocol [configure router]
- Configure Network interface [configure router]

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- Configure Import/Export policy [configure policy-options policy-statement]
- Configure BMP station name [configure bmp station]
- Configure Security keychain [configure system security keychains keychain]

2.44.3 Advisory

- IXR: Graceful-Restart & Prefix-limt family "flow-ipv4, flow-ipv6, flow-vpn-ipv4, flow-vpn-ipv6" are not supported on 7250-IXR-10/6/R4/R6/e-BigGNSS/e-Big/e-small/e-smallGNSS/e/S/ec.
- Monitor 'all-stations' cannot be set when a station exists in the list.
- Prefix-Limit, log-only, idle-timeout and hold-excess attributes are mutually exclusive.
- Client Reflect/ASN 4 Byte/Fast External Failover/send-communities attributes are not supported.
- The following attributes are not allowed to be set to false on a BGP Group:
  - path-mtu-discovery
  - enforce-first-as
  - selective-label-ipv4-install
  - vpn-apply-export
  - aggregator-id-zero
  - vpn-apply-import
  - damping
  - static-group
  - split-horizon
  - bfd-liveness
  - peer-ip-tracking
  - third-party-nexthop
  - advertise-inactive.
- The following attributes are not supported on certain chassis in node Release 23.10:
  - family flow-ipv4
  - flow-ipv6
  - flow-vpn-ipv
  - flow-vpn-ipv6.

  The deployment should proceed successfully if these attributes are not included in the payload. In case the deployment fails due to the configuration of an unsupported boolean-type attribute, it should be removed from the API.

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2.45 Name of the Intent: router-bgp-peer_msros_24-10-1_24-11

2.45.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS and 7250 IXR MD SROS node version 24.10

2.45.2 Description

The router-bgp-peer_msros_24-10-1_24-11 intent supports the creation/modification/deletion/discovery/audit/alignment of bgp peers, with neighbor id as the target identifier. The following configurations are supported per deployment:

```
- Neighbor
    - Advertise LDP Prefix, Default Route Target, Advertise Inactive, BFD
Liveness, Peer IP Tracking, Third Party Nexthop, TTL Security
      Aigp, Authentication Keychain, Fast External Failover, Split
Horizon, Label Preference, Def Recv EVPN Encap, Multipath Eligible
      Group, Peer Creation Type, Connect, Retry, Description, Admin
State, Keepalive, Damping, Local Preference, Loop Detect, Loop Detect
Threshold
      TCP Mss, Selective Label IPv4 Install, Min Route Advertisement,
Aggreg, ator ID Zero, Preference, Block Prefix Sid, Multihop, MED Out,
Authentication Key
      Client Reflect, VPN Apply Export, VPN Apply Import, AS Override,
Asn 4 Byte, Path MTU Discovery, Local Address, L2VPN Cisco Interop, Next
Hop Self
      Passive, Type, Peer AS, Capability Negotiation, Enforce First AS,
Initial Send Delay Zero, Egress Peer Engineering Label Unicast
    - Next Hop Unchanged: Label IPv4, Label IPv6, VPN IPv4, VPN IPv6,
EVPN
    - Origin Validation: Label IPv4, Label IPv6, IPv4, IPv6
    - Link Bandwidth
        - Accept From Ebgp: IPv4, IPv6 Label IPv4, Label IPv6, VPN IPv4,
VPN IPv6
        - Add To Received Ebgp: IPv4, IPv6 Label IPv4, Label IPv6, VPN
IPv4, VPN IPv6
        - Aggregate Used Paths: IPv4, IPv6 Label IPv4, Label IPv6, VPN
IPv4, VPN IPv6
        - Send To Ebgp: IPv4, IPv6 Label IPv4, Label IPv6, VPN IPv4, VPN
IPv6
    - Send Communities: Standard, Extended, Large
    - Error Handling: Update Fault Tolerance
    - Damp Peer Oscillations: Error Interval
        - Idle Hold Time: Initial Wait, Second Wait, Max Wait
    - BFD Strict Mode: Next Hop Reachability
        - Advertise:Holdtime (seconds)
    - Hold Time: Seconds, Minimum Hold Time
```

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```
    - Family: IPv4, VPN IPv4, IPv6, Mcast IPv4, VPN IPv6 ,L2 VPN ,Mvpn
IPv4 ,Mdt Safi, Ms Pw, Flow IPv4, Route Target, Mcast VPN IPv4, Mvpn IPv6
Flow IPv6, EVPN Mcast IPv6 Label, IPv4 Label, IPv6 BGP, Ls Mcast, VPN
IPv6, SR Policy IPv4, SR Policy IPv6, Flow VPN, IPv4 Flow, VPN IPv6
    - Send Default: IPv4, IPv6, Export Policy
    - Default Preference: Ebgp, Ibgp
    - Default Label Preference: Ebgp, Ibgp
    - Remove Private: Limited, Skip Peer AS, Replace
    - Cluster: Cluster ID, Orr Location, Allow Local Fallback
    - Ebgp Default Reject Policy: Import, Export
    - Local AS: AS Number, Private, Prepend Global AS
    - Add Paths
        - IPv4: Send, Receive
        - IPv6: Send, Receive
        - Label IPv4: Send, Receive
        - Label IPv6: Send, Receive
        - VPN IPv4: Send, Receive
        - VPN IPv6: Send, Receive
        - Mvpn IPv4: Send, Receive
        - Mvpn IPv6: Send, Receive
        - Mcast VPN IPv4: Send, Receive
        - Mcast VPN IPv6: Send, Receive
        - EVPN: Send, Receive
    - Import: Policy
    - Export: Policy
    - Extended NH Encoding: VPN IPv4, Label IPv4 ,IPv4
    - Advertise IPv6 Next Hops: VPN IPv6, Label IPv6, EVPN, VPN IPv4,
Label IPv4, IPv4
    - Graceful Restart: Restart Time, Stale Routes Time, GR Notification
        - Long Lived: Advertised Stale Time, Helper Override Stale Time,
Helper Override Restart Time, Forwarding Bits Set
                      Advertise Stale To All Neighbors, Without No Export

            - Family: Family Type, Advertised Stale Time, Helper Override
Stale Time
    - Outbound Route Filtering:
        - Extended Community: Accept Orf
            - Send Orf
                -Route Target: Community Name
    - Monitor: All Stations, Admin State
        - Route Monitoring: Pre Policy, Post Policy
            - Station: Station Name
    - Prefix Limit: Family, Maximum, Log Only, Threshold, Idle Timeout,
Post Import, Hold Excess
    - Egress Engineering: Admin State
    - Segment Routing V6
        - Route Advertisement
            - Drop Routes With Srv6 Tlvs
                - Family: Family Type, Strip Srv6 Tlvs
```

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CLI context: router > bgp > neighbor

Prerequisite:

- Configure bgp group [configure router bgp group testing_bgp]
- Configure Import/Export policy [configure policy-options policy-statement]
- Configure BMP station name [configure bmp station]
- Configure Security keychain [configure system security keychains keychain]

2.45.3 Advisory

- BGP group must be configured before using the intent.
- Client Reflect, ASN 4 Byte, Fast External Failover and send-communities attributes cannot be true.
- log-only, idle-timeout, and hold-excess are mutually exclusive.
- Below attributes are not supported on certain chassis in node Release 23.10:
  - flow-ipv4
  - flow-ipv6
  - flow-vpn-ipv4
  - flow-vpn-ipv6
  - segment-routing-v6

  The deployment should proceed successfully if these attributes are not included in the payload. In case the deployment fails due to the configuration of an unsupported boolean-type attribute, it should be removed from the API.

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2.46 Name of the Intent: router-bgp_msros_24-10-1_24-11

2.46.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS and 7250 IXR for MD SROS node version 24.10

2.46.2 Description

The router-bgp_msros_24-10-1_24-11 intent supports the creation/modification/deletion/discovery/audit/alignment of BGP, with Router Name "Base" as target identifier.

The following configurations are supported per deployment:

```
- Description, Connect Retry, Admin State, Keepalive, Damping, Local
Preference, Loop Detect, Loop Detect Threshold, TCP MSS, Selective Label
IPv4 Install, Selective Label IP Prioritization, Min Route Advertisement,
Aggregator ID Zero, Preference, Block Prefix SID, Multihop, Med Out,
Authentication Key, Client Reflect, Vpn Apply Export, Vpn Apply Import,
ASN 4 Byte, Path MTU Discovery, Enforce First As, Initial Send Delay
Zero, Router ID, Inter As Vpn, Purge Timer, Route Table Install, IBGP
Multipath,Rr Vpn Forwarding, Subconfed Vpn Forwarding, BFD Liveness, Fast
External Failover, Advertise Inactive, Authentication Keychain, Mvpn VRF
Import Subtype New, Rapid Withdrawal, MP BGP Keep, Peer IP Tracking,
Split Horizon, Override Tunnel ELC, Link State Route Export, Link State
Route Import, Third Party Nexthop, Dynamic Neighbor Limit, Label
Preference, Peer Tracking Policy, Def Recv EVPN Encap, Sr Policy Import,
BGP Tunnel Preference, Selective Label IP.
- Hold Time, Family, Label Allocation, Send Default, Default Preference,
Default Label Preference,
- Remove Private, Cluster, EBGP Default Reject Policy, BFD Strict Mode,
Local As,
- Best Path Selection
    - Always Compare Med, As Path Ignore, Ignore Router ID, EBGP IBGP
Equal
- Send Communities, Error Handling, Damp Peer Oscillations, Rapid Update,
Advertise External, Backup Path, Flowspec, Neighbor Trust Convergence,
Add Paths, Import, Export, Extended NH Encoding, Advertise IPv6 Next
Hops, Graceful Restart, Optimal Route Reflection,
- Next Hop Resolution
    - Shortcut Tunnel, Labeled Routes, Use Leaked Routes
- Multipath
- Outbound Route Filtering
- Monitor
- Rib Management
- Segment Routing
- Egress Peer Engineering
```

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```
- BGP Tunnel Metric
- Segment Routing V6
```

CLI context: router > bgp

Prerequisites:

- Enable BGP routing Protocol [configure router]
- Configure Network interface [configure router]
- Configure Import/Export policy [configure policy-options policy-statement]
- Configure BMP station name [configure bmp station]
- Configure Security keychain [configure system security keychains keychain]
- Configure locator and micro-segment-locator [configure router "Base" segment-routing segment-routing-v6]
- Configure entropy-label [configure system fp options mpls entropy-label admin-state]
- Configure sr-labels [configure router mpls-labels sr-labels]

2.46.3 Advisory

- SR: segment-routing-srv6 is not supported on SR-a4, SR-a8, SR-1e, SR-2e and SR-3e.
- segment-routing-v6 requires network interface ports to be on FP4 or higher generation line cards.
- Both locator-name and micro-segment-locator-name cannot be configured at the same time.
- route-table-install and rr-use-route-table attribute cannot both be set to true.
- Deleting a BGP intent deletes BGP Groups and Neighbors.
- The following attributes are not supported on certain chassis using Release 23.10 node software:
  - legacy-mode
  - family ecmp
  - enforce-untagged-route
  - flow-ipv4
  - flow-ipv6
  - flow-vpn-ipv4
  - flow-vpn-ipv6
  - flowspec
  - flowspec validate-dest-prefix
  - flowspec validate-redirect-ip
  - segment-routing-v6 family resolution
  - segment-routing-v6 family upa-trigger
  - segment-routing-v6 family upa-trigger next-hop.

  The deployment should proceed successfully if these attributes are not included in the payload. In case the deployment fails due to the configuration of an unsupported boolean-type attribute, it should be removed from the API.

IXR:

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- Family and Graceful-Restart family "flow-ipv4, flow-ipv6, flow-vpn-ipv4, flow-vpn-ipv6" are not supported on the 7250-IXR-10/6/R4/R6/e-BigGNSS/e-Big/e-small/e-smallGNSS/e/S/ec.
- segment-routing-srv6 is supported on IXR-x, IXR-x3, IXR-Xs, IXR-R6d, IXR-R6dl, IXR-e2 and IXR-e2c.
- "override-tunnel-elc" attribute requires entropy-label configured.
- The following attributes are not supported on the 7250 IXR:
  - leak-import
  - neighbor-trust
  - allow-unresolved-leaking
  - resolution-filter mpls-fwd-policy/rib-api

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2.47 Name of the Intent: router-interface_mixr_23-10-1_24-4

2.47.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR MD SROS node version 23.10

2.47.2 Description

router-interface_mixr_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of routing interface with interface name as target-indentifier.

The following configurations are supported per deployment:

```
- Description, Administrative State, Flavor, IP-MTU, MAC, Port,
Accounting Policy

  Boolean attributes: Loopback, Collect Stats, IP MTU Profile
- Egress
    - Filter: IP, IPv6
    - QOS: Egress Agg Rate: Agg rate or Percent Agg rate
           VLAN QoS Policy: Policy Name, Port Redirect
           Egress Remark Policy: Policy Name
- Ingress
    - IP Filter Pair
    - Filter: IP, IPv6
    - QoS: Network Ingress
- If Attribute:
    - Admin-Group
    - Delay: Delay Selection, Static
    - Dynamic: Measurement Template
    - Twamp Light:
            - IPv4 Admin state, Destination, Source
            - IPv6 Admin state, Destination, Source
- LDP Sync Timer: Seconds, End of LIB
- PTP HW Assist: Admin State
- Cflowd Parameters Sampling: Sampling Type, Sample Profile
- Hold Time:
    - IPv4: Up, Down, Init Only
    - IPv6: Up, Down, Init Only
- ETH CFM: MEP ID, MD Admin Name, MA Admin Name, Admin State,
Description, MAC Address, Facility Fault, CCM LTM Priority
          Low Priority Defect, LBM SVC ACT Responder, One Way Delay
Threshold(seconds)
        - Alarm Notification: FNG Alarm Time, FNG Reset Time
        - ETH Test: Bit Error Threshold(bit errors)
                 : Test Pattern: Pattern, CRC TLV
- IPv4: Local DHCP Server, TCP MSS, Unnumbered Source
    - ICMP: Mask Reply,  URPF Check, Ignore Default, Mode
```

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```
        - Redirects: Admin State, Number, Seconds
        - TTL Expired: Admin State, Number, Seconds
        - Unreachables: Admin State, Number, Seconds
        - Param Problem: Admin State, Number, Seconds
    -DHCP: Admin State, Description, Server, Trusted, GI Address, SRC IP
Address, GI Address, Relay Plain BOOTP
            - Option 82: Action, Circuit ID, Remote ID
                - Vendor Specific Option: System ID, Client MAC Address,
Pool Name, Port ID, Service ID, String
    - BFD: Admin State, Transmit Interval(milliseconds), Receive
(milliseconds), MultIPlier, Echo Receive(milliseconds), Type
    - Primary: Address, Prefix Length, Broadcast
    - Secondary: Address, Prefix Length, IGP Inhibit, Broadcast
    - Neighbor Discovery: Timeout(seconds), Retry Timer(deciseconds),
Learn Unsolicited, Proactive Refresh, Local Proxy ARP, Remote Proxy ARP,
Proxy ARP Policy
        - Limit: Max Entries, Log Only, Threshold(percent)
        - Static Neighbor Unnumbered: MAC Address
        - Static Neighbor: IPv4 Address, MAC Address
    - VRRP: Virtual Router ID, Admin State, Authentication Key, Backup,
Message Interval(deciseconds), Owner, Passive, Init Delay(seconds),
MAC,Priority, Master Int Inherit, Ping Reply, Policy
            Preempt, TELNET Reply, Traceroute Reply, Standby Forwarding,
SSH Reply, NTP Reply, Oper Group, Monitor Oper Group
          - BFD Liveness: DEST IP, Service Name, Interface Name
- IPv6: TCP MSS, Duplicate Address Detection, Local DHCP Server
    - ICMP6:
        Packet Too Big: Admin State, Number, Seconds
        Param Problem: Admin State, Number, Seconds
        Redirects: Admin State, Number, Seconds
        Time Exceeded: Admin State, Number, Seconds
        Unreachables: Admin State, Number, Seconds
    - Link Local Address: Address, Duplicate Address Detection
    - BFD: Admin State, Transmit Interval(milliseconds), Receive
(milliseconds), MultIPlier, Echo Receive(milliseconds), Type
    - Address: IPv6 Address, Prefix Length, EUI 64, Duplicate Address
Detection, Primary Preference
    - Neighbor Discovery: Reachable Time,Stale Time,Learn Unsolicited,
Proactive Refresh,Local Proxy ND,Proxy ND Policy
        - Secure ND: Admin State, Allow Unsecured Msgs, Public Key Min
Bits, Security Parameter
        - Limit: Max Entries, Log Only, Threshold(percent)
        - Static Neighbor: IPv6 Address, MAC Address
        - VRRP: Virtual Router ID, Admin State, Backup, Message
Interval(deciseconds), Owner, Passive, Init Delay(seconds), Priority,
Master Int Inherit, Ping Reply, Policy,
                Preempt, TELNET Reply, Traceroute Reply, Standby
Forwarding, NTP Reply, Oper Group, Monitor Oper Group
          - BFD Liveness: DEST IP, Service Name, Interface Name
```

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```
- External Reference:
    - Openconfig: Subinterface
```

Prerequisite: The parameters below must be configured before using the intent:

```
- Port and LAG configurations
- Dist CPU Protection, Filter IP, Filter IPv6, Policy Accounting,
Measurement Template, IPv4 Local DHCP Server,
- ETH CFM: MD Admin Name
- If Attribute: Admin Group, SRLG Group
- VRRP: policy, oper-group, monitor-oper-group
- IPv6: IPv6 local-dhcp-server, proxy-nd-policy
- QoS: VLAN QoS Policy, Egress Remark Policy, Network Ingress Policy
```

CLI context: router > interface

2.47.3 Advisory

- This intent supports the creation of network and unnumbered interface.
- Only Ethernet ports with mode network and hybrid can be attached for this interface.
- Port configuration is not supported when Loopback is enabled.
- Combination of unnumbered ip-address and loopback is not supported
- Combination of vlan-qos policy-name and port-binding loopback is not supported.
- Configuration of ip-filter without port-binding is not supported.
- Combination of dist-cpu-protection and loopback is not supported.
- A static-neighbor-unnumbered MAC address configuration is mandatory on an unnumbered interface.
- Either IPv4 or IPv6 can be configured for Twamp Light.
- IPv6 VRRP owner cannot be configured with boolean attributes: priority, master-int-inherit, ping-reply, policy, telnet-reply, traceroute-reply, standby-forwarding, ntp-reply.
- IPv4 VRRP owner cannot be configured with boolean attributes: priority, ping-reply, policy, telnet-reply, traceroute-reply, standby-forwarding, ssh-reply, ntp-reply.
- VRRP authentication key attribute does not participate in audit operations.
- The following attributes will not be shown as enabled in the UI after being deployed:
  - Circuit ID: ASCII Tuple, If Name, If Index, None, Port ID, VLAN ASCII Tuple
  - Remote ID: MAC, None
- User is advised to be aware of unsupported attributes when changing the flavor of an interface i.e regular to unnumbered or loopback and vice versa.

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2.48 Name of the Intent: router-interface_mixr_24-10-1_24-4

2.48.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR MD SROS node version 24.10

2.48.2 Description

The router-interface_mixr_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a routing interface with the interface name as target-indentifier.

The following configurations are supported per deployment:

```
- Description, Administrative State, Flavor, IP-MTU, MAC, Port,
Accounting Policy, Dist CPU Protection

  Boolean attributes: Collect Stats, IP MTU Profile
- Egress
    - Filter: IP, IPv6
    - QOS: Egress Agg Rate: Agg rate or Percent Agg rate
           VLAN QoS Policy: Policy Name, Port Redirect
           Egress Remark Policy: Policy Name
           Burst Limit (2 or auto)
- Ingress
    - IP Filter Pair
    - Filter: IP, IPv6
    - QoS: Network Ingress
- If Attribute:
    - Admin-Group
    - Delay: Delay Selection, Static
    - Dynamic: Measurement Template
    - Twamp Light:
            - IPv4 Admin state, Destination, Source
            - IPv6 Admin state, Destination, Source
- LDP Sync Timer: Seconds, End of LIB
- PTP HW Assist: Admin State
- Cflowd Parameters Sampling: Sampling Type, Sample Profile
- Hold Time:
    - IPv4: Up, Down, Init Only
    - IPv6: Up, Down, Init Only
- ETH CFM: MEP ID, MD Admin Name, MA Admin Name, Admin State,
Description, MAC Address, Facility Fault, CCM LTM Priority, Low Priority
Defect, LBM SVC ACT Responder, One Way Delay Threshold(seconds)
        - Alarm Notification: FNG Alarm Time, FNG Reset Time
        - ETH Test: Bit Error Threshold(bit errors)
                 : Test Pattern: Pattern, CRC TLV
- IPv4: Local DHCP Server, TCP MSS, Unnumbered Source
    - ICMP: Mask Reply,
```

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```
        - Redirects: Admin State, Number, Seconds
        - TTL Expired: Admin State, Number, Seconds
        - Unreachables: Admin State, Number, Seconds
        - Param Problem: Admin State, Number, Seconds
    -DHCP: Admin State, Description, Server, Trusted, GI Address, SRC IP
Address, Relay Plain BOOTP
            - Option 82: Action, Circuit ID, Remote ID
                - Vendor Specific Option: System ID, Client MAC Address,
Port ID, Service ID, String
    - BFD: Admin State, Transmit Interval(milliseconds), Receive
(milliseconds), MultIPlier, Echo Receive(milliseconds), Type
    - Primary: Address, Prefix Length, Broadcast
    - Secondary: Address, Prefix Length, IGP Inhibit, Broadcast
    - Neighbor Discovery: Timeout(seconds), Retry Timer(deciseconds),
Learn Unsolicited, Proactive Refresh, Local Proxy ARP, Remote Proxy ARP,
Proxy ARP Policy
        - Limit: Max Entries, Log Only, Threshold(percent)
        - Host route:
            - Populate: Route Type, Route Tag
        - Static Neighbor Unnumbered: MAC Address
        - Static Neighbor: IPv4 Address, MAC Address
    - VRRP: Virtual Router ID, Admin State, Authentication Key, Backup,
Message Interval(deciseconds), Owner, Passive, Init Delay(seconds),
MAC,Priority, Master Int Inherit, Ping Reply, Policy
            Preempt, TELNET Reply, Traceroute Reply, Standby Forwarding,
SSH Reply, NTP Reply, Oper Group, Monitor Oper Group
          - BFD Liveness: DEST IP, Service Name, Interface Name
- IPv6: TCP MSS, Duplicate Address Detection, Local DHCP Server
    - ICMP6:
        Packet Too Big: Admin State, Number, Seconds
        Param Problem: Admin State, Number, Seconds
        Redirects: Admin State, Number, Seconds
        Time Exceeded: Admin State, Number, Seconds
        Unreachables: Admin State, Number, Seconds
    - Link Local Address: Address, Duplicate Address Detection
    - BFD: Admin State, Transmit Interval(milliseconds), Receive
(milliseconds), MultIPlier, Echo Receive(milliseconds), Type
    - Address: IPv6 Address, Prefix Length, EUI 64, Duplicate Address
Detection, Primary Preference
    - Neighbor Discovery: Reachable Time,Stale Time,Learn Unsolicited,
Proactive Refresh,Local Proxy ND,Proxy ND Policy
        - Secure ND: Admin State, Allow Unsecured Msgs, Public Key Min
Bits, Security Parameter
        - Limit: Max Entries, Log Only, Threshold(percent)
        - Host route:
            - Populate: Route Type, Route Tag
        - Static Neighbor: IPv6 Address, MAC Address
        - VRRP: Virtual Router ID, Admin State, Backup, Message
Interval(deciseconds), Owner, Passive, Init Delay(seconds), Priority,
```

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```
Master Int Inherit, Ping Reply, Policy,
                Preempt, TELNET Reply, Traceroute Reply, Standby
Forwarding, NTP Reply, Oper Group, Monitor Oper Group
          - BFD Liveness: DEST IP, Service Name, Interface Name
- External Reference:
    - Openconfig: Subinterface
- Port Binding:
    - Loopback
    - Port Encap
```

Prerequisite:

The parameters below must be configured before using the intent:

- Port and LAG configurations
- Dist CPU Protection, Filter IP, Filter IPv6, Policy Accounting, Measurement Template, IPv4 Local DHCP Server,
- ETH CFM: MD Admin Name
- If Attribute: Admin Group, SRLG Group
- VRRP: policy, oper-group, monitor-oper-group
- IPv6: IPv6 local-dhcp-server, proxy-nd-policy
- QoS: VLAN QoS Policy, Egress Remark Policy, Network Ingress Policy

CLI context: router > interface

2.48.3 Advisory

- This intent supports the creation of network and unnumbered interfaces.
- Only Ethernet ports with mode network and hybrid can be attached for this interface.
- Port configuration is not supported when Loopback is enabled.
- Combination of unnumbered ip-address and loopback is not supported
- Combination of vlan-qos policy-name and port-binding loopback is not supported.
- Burst limit is not supported for 7250 IXR-e2c/IXR-x/IXR-R6dl/IXR-X3/IXR-e2/IXR-R6d/IXR-X1/IXR-Xs
- Configuration of ip-filter without port-binding is not supported.
- Combination of dist-cpu-protection and loopback is not supported.
- A static-neighbor-unnumbered MAC address configuration is mandatory on an unnumbered interface.
- Either IPv4 or IPv6 can be configured for Twamp Light.
- IPv6 VRRP owner cannot be configured with boolean attributes: priority, master-int-inherit, ping-reply, policy, telnet-reply, traceroute-reply, standby-forwarding, ntp-reply.
- IPv4 VRRP owner cannot be configured with boolean attributes: priority, ping-reply, policy, telnet-reply, traceroute-reply, standby-forwarding, ssh-reply, ntp-reply.
- VRRP authentication key attribute does not participate in audit operations.

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- The following attributes will not be shown as enabled in the UI after being deployed:
  - Circuit ID: ASCII Tuple, If Name, If Index, None, Port ID, VLAN ASCII Tuple
  - Remote ID: MAC, None
- Be aware of unsupported attributes when changing the flavor of an interface, i.e regular to unnumbered or loopback and vice versa.

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2.49 Name of the Intent: router-interface_msros_23-10-1_24-4

2.49.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR,7950 XRS,7450 ESS MD SROS node version 23.10

2.49.2 Description

router-interface_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of routing interface with interface name as target-indentifier.

The following configurations are supported per deployment:

```
- Description, Administrative State, Flavor, CPU Protection, IP-MTU, MAC,
ToS Marking State, Port
  Boolean attributes: GRE Termination, Loopback, Ingress Stats, StrIP
Label, MAC Accounting, URPF Selected VPRNs
- Egress
    - Filter: IP, IPv6
- Ingress
    - Policy Accounting
    - Destination Class Lookup
    - Filter: IP, IPv6
- If Attribute:
    - Admin-Group, SRLG Group
    - Delay: Delay Selection, Static
    - Dynamic: Measurement Template
    - Twamp Light:
            - IPv4 Admin state, Destination, Source
            - IPv6 Admin state, Destination, Source
- Untrusted: Default Forwarding
- LDP Sync Timer: Seconds, End of LIB
- Load Balancing: LSR Load Balancing, IP Load Balancing, SPI Load
Balancing, TEID Load Balancing, Flow Label Load Balancing
- Lag: Link Map Profile
       Per Link Hash: Class, Weight
- PTP HW Assist: Admin State
- Cflowd Parameters Sampling: Sampling Type, Type, Direction, Sample
Profile
- Hold Time:
    - IPv4: Up, Down, Init Only
    - IPv6: Up, Down, Init Only
- ETH CFM: MEP ID, MD Admin Name, MA Admin Name, Admin State,
Description, MAC Address, Facility Fault, CCM LTM Priority, CCM Padding
Size,
           CCM TLV Ignore, Low Priority Defect, Collect LMM Stats, LBM
SVC ACT Responder, One Way Delay Threshold(seconds)
        - Alarm Notification: FNG Alarm Time, FNG Reset Time
```

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```
        - ETH Test: Bit Error Threshold(bit errors)
                : Test Pattern: Pattern, CRC TLV
        - GRACE ETH ED: Max RX Defect Window, Priority, RX ETH ED, TX ETH
ED
        - ETH VSM Grace: RX ETH VSM Grace, TX ETH VSM Grace
- IPv4: Allow Directed Broadcasts, IP Helper Address, Local DHCP Server,
TCP MSS, QoS Route Lookup
    - ICMP: Mask Reply, Unnumbered Source, URPF Check, Ignore Default,
Mode
        - Redirects: Admin State, Number, Seconds
        - TTL Expired: Admin State, Number, Seconds
        - Unreachables: Admin State, Number, Seconds
        - Param Problem: Admin State, Number, Seconds
        - DHCP: Admin State, Description, Python Policy, Server, Trusted,
GI Address, SRC IP Address, Release Include GI Address, Relay Plain BOOTP

            - Option 82: Action, Circuit ID, Remote ID
                - Vendor Specific Option: System ID, Client MAC Address,
Pool Name, Port ID, Service ID, String
        - BFD: Admin State, Transmit Interval(milliseconds), Receive
(milliseconds), MultIPlier, Echo Receive(milliseconds), Type
    - Primary: Address, Prefix Length, Broadcast, Track SRRP
    - Secondary: Address, Prefix Length, IGP Inhibit, Broadcast, Track
SRRP
    - Neighbor Discovery: Timeout(seconds), Retry Timer(deciseconds),
Learn Unsolicited, Proactive Refresh, Local Proxy ARP, Remote Proxy ARP,
Proxy ARP Policy
        - Limit: Max Entries, Log Only, Threshold(percent)
        - Static Neighbor Unnumbered: MAC Address
            - Static Neighbor: IPv4 Address, MAC Address
    - VRRP: Virtual Router ID, Admin State, Authentication Key, Backup,
Message Interval(deciseconds), Owner, Passive, Init Delay(seconds),
MAC,Priority, Master Int Inherit, Ping Reply, Policy,
            Preempt, TELNET Reply, Traceroute Reply, Standby Forwarding,
SSH Reply, NTP Reply, Oper Group, Monitor Oper Group
          - BFD Liveness: DEST IP, Service Name, Interface Name
- IPv6: Qos Route Lookup, Forward IPv4 Packets, TCP MSS, Duplicate
Address Detection, Local DHCP Server
    - ICMP6:
        Packet Too Big: Admin State, Number, Seconds
        Param Problem: Admin State, Number, Seconds
        Redirects: Admin State, Number, Seconds
        Time Exceeded: Admin State, Number, Seconds
        Unreachables: Admin State, Number, Seconds
    - Link Local Address: Address, Duplicate Address Detection
    - URPF Check: Ignore Default, Mode
    - BFD: Admin State, Transmit Interval(milliseconds), Receive
(milliseconds), MultIPlier, Echo Receive(milliseconds), Type
    - Neighbor Discovery:Reachable Time,Stale Time,Learn Unsolicited,
```

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```
Proactive Refresh,Local Proxy ND,Proxy ND Policy
        - Secure ND: Admin State, Allow Unsecured Msgs, Public Key Min
Bits, Security Parameter
        - Limit: Max Entries, Log Only, Threshold(percent)
        - Static Neighbor: IPv6 Address, MAC Address
        - VRRP: Virtual Router ID, Admin State, Backup, Message
Interval(deciseconds), Owner,Passive, Init Delay(seconds), MAC,Priority,
Master Int Inherit, Ping Reply, Policy,
                Preempt, TELNET Reply, Traceroute Reply, Standby
Forwarding, NTP Reply, Oper Group, Monitor Oper Group
          - BFD Liveness: DEST IP, Service Name, Interface Name
- QoS: Network Policy, Egress Port Redirect Group, Egress Instance,
Ingress FP Redirect Group, Ingress Instance
- Network Domains: Domain Name
- External Reference:
    - Openconfig: Subinterface
```

Prerequisites:

Configure the following parameters:

- Port and LAG configuration
- Dist CPU Protection
- CPU protection
- Filter IP
- Filter IPv6
- Policy Accounting
- Measurement Template
- IPv4 Local DHCP Server
- Python Policy
- ETH CFM:
  - MD Admin Name
- If Attribute:
  - Admin Group
  - SRLG Group
- VRRP:
  - policy
  - oper-group
  - monitor-oper-group
- IPv6:
  - IPv6 local-dhcp-server
  - proxy-nd-policy
- QoS: network-policy
  - qos egress-port-redirect-group
  - qos ingress-fp-redirect-group

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- Network Domains:
  - network-domain

CLI context: router>>interface

2.49.3 Advisory

- This intent supports creation of network and unnumbered mpls tp interface.
- Ethernet port with mode network and hybrid can only be attached for this interface.
- Outer encapsulation and Inner encapsulation values are supported only on port with encap type as QinQ.
- Port configuration is not supported when Loopback is enabled.
- With Loopback ETH-CFM config, IPv4 DHCP config is not supported.
- Strip-label can only be set if configured port is single-fiber.
- Combination of vas-if-type to-from-network and loopback is not supported.
- Combination of dist-cpu-protection and loopback is not supported.
- Egress Instance & Egress Port Redirect Group both should be configured.
- Ingress Instance & Ingress FP Redirect Group both should be configured.
- A static-neighbor-unnumbered MAC address configuration is mandatory on an unnumbered-mpls-tp interface.
- Either IPv4 or IPv6 can be configured for Twamp Light.
- Below attributes do not support flavor unnumbered-mpls-tp. external-reference openconfig subinterface non-default stale-time{} Relay plain BOOTP track-srrp gre-termination urpf-selected-vprns{} destination-class-lookup lag per-link-hash if-attribute untrusted IPv6 local-dhcp-server IPv6 address
- IPv6 VRRP owner cannot be configured with boolean attributes: priority, master-int-inherit, ping-reply, policy, telnet-reply, traceroute-reply, standby-forwarding, ntp-reply.
- IPv4 VRRP owner cannot be configured with boolean attributes: priority, ping-reply, policy, telnet-reply, traceroute-reply, standby-forwarding, ssh-reply, ntp-reply.
- User cannot configure MEP grace tx[TX ETH ED] and mcc-ed tx[TX ETH VSM Grace] at the same time.
- VRRP authentication key attribute will not participate in audit operation.
- The following attributes will not be shown as enabled in the UI after being deployed:
  - Circuit ID: ASCII Tuple, If Name, If Index, None, Port ID, VLAN ASCII Tuple
  - Remote ID: MAC, None
- User is advised to take care of unsupported attributes while changing the flavor.
- Deployment fails if flavor is changed and corresponding mutual exclusive elements are not removed.

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2.50 Name of the Intent: router-interface_msros_24-10-1_24-4

2.50.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR,7950 XRS,7450 ESS MD SROS node version 24.10

2.50.2 Description

The router-interface_msros_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a routing interface with interface name as the target-indentifier.

The following configurations are supported per deployment:

```
- Description, Administrative State, Flavor, CPU Protection, IP-MTU, MAC,
ToS Marking State, Port
  Boolean attributes: GRE Termination, Loopback, Ingress Stats, StrIP
Label, MAC Accounting, URPF Selected VPRNs
- Egress
    - Filter: IP, IPv6
- Ingress
    - Policy Accounting
    - Destination Class Lookup
    - Filter: IP, IPv6
- If Attribute:
    - Admin-Group, SRLG Group
    - Delay: Delay Selection, Static
    - Dynamic: Measurement Template
    - Lag IP Measurement: Lag IP Measurement Template
    - Twamp Light:
            - IPv4 Admin state, Destination, Source
            - IPv6 Admin state, Destination, Source
- Untrusted: Default Forwarding
- LDP Sync Timer: Seconds, End of LIB
- Load Balancing: LSR Load Balancing, IP Load Balancing, SPI Load
Balancing, TEID Load Balancing, Flow Label Load Balancing
- Lag: Link Map Profile
       Per Link Hash: Class, Weight
- PTP HW Assist: Admin State
- Cflowd Parameters Sampling: Sampling Type, Type, Direction, Sample
Profile
- Hold Time:
    - IPv4: Up, Down, Init Only
    - IPv6: Up, Down, Init Only
- ETH CFM: MEP ID, MD Admin Name, MA Admin Name, Admin State,
Description, MAC Address, Facility Fault, CCM LTM Priority, CCM Padding
Size,
           CCM TLV Ignore, Low Priority Defect, Collect LMM Stats, LBM
SVC ACT Responder, One Way Delay Threshold(seconds)
```

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```
        - Alarm Notification: FNG Alarm Time, FNG Reset Time
        - ETH Test: Bit Error Threshold(bit errors)
                : Test Pattern: Pattern, CRC TLV
        - GRACE ETH ED: Max RX Defect Window, Priority, RX ETH ED, TX ETH
ED
        - ETH VSM Grace: RX ETH VSM Grace, TX ETH VSM Grace
- IPv4: Allow Directed Broadcasts, IP Helper Address, Local DHCP Server,
TCP MSS, QoS Route Lookup
    - ICMP: Mask Reply, Unnumbered Source, URPF Check, Ignore Default,
Mode
        - Redirects: Admin State, Number, Seconds
        - TTL Expired: Admin State, Number, Seconds
        - Unreachables: Admin State, Number, Seconds
        - Param Problem: Admin State, Number, Seconds
        - DHCP: Admin State, Description, Python Policy, Server, Trusted,
GI Address, SRC IP Address, Release Include GI Address, Relay Plain BOOTP

            - Option 82: Action, Circuit ID, Remote ID
                - Vendor Specific Option: System ID, Client MAC Address,
Pool Name, Port ID, Service ID, String
        - BFD: Admin State, Transmit Interval(milliseconds), Receive
(milliseconds), MultIPlier, Echo Receive(milliseconds), Type
    - Primary: Address, Prefix Length, Broadcast, Track SRRP
    - Secondary: Address, Prefix Length, IGP Inhibit, Broadcast, Track
SRRP
    - Neighbor Discovery: Timeout(seconds), Retry Timer(deciseconds),
Learn Unsolicited, Proactive Refresh, Local Proxy ARP, Remote Proxy ARP,
Proxy ARP Policy
        - Limit: Max Entries, Log Only, Threshold(percent)
        - Static Neighbor Unnumbered: MAC Address
            - Static Neighbor: IPv4 Address, MAC Address
    - VRRP: Virtual Router ID, Admin State, Authentication Key, Backup,
Message Interval(deciseconds), Owner, Passive, Init Delay(seconds),
MAC,Priority, Master Int Inherit, Ping Reply, Policy,
            Preempt, TELNET Reply, Traceroute Reply, Standby Forwarding,
SSH Reply, NTP Reply, Oper Group, Monitor Oper Group
          - BFD Liveness: DEST IP, Service Name, Interface Name
- IPv6: Qos Route Lookup, Forward IPv4 Packets, TCP MSS, Duplicate
Address Detection, Local DHCP Server
    - ICMP6:
        Packet Too Big: Admin State, Number, Seconds
        Param Problem: Admin State, Number, Seconds
        Redirects: Admin State, Number, Seconds
        Time Exceeded: Admin State, Number, Seconds
        Unreachables: Admin State, Number, Seconds
    - Link Local Address: Address, Duplicate Address Detection
    - URPF Check: Ignore Default, Mode
    - BFD: Admin State, Transmit Interval(milliseconds), Receive
(milliseconds), MultIPlier, Echo Receive(milliseconds), Type
```

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```
    - Neighbor Discovery:Reachable Time,Stale Time,Learn Unsolicited,
Proactive Refresh,Local Proxy ND,Proxy ND Policy
        - Secure ND: Admin State, Allow Unsecured Msgs, Public Key Min
Bits, Security Parameter
        - Limit: Max Entries, Log Only, Threshold(percent)
        - Static Neighbor: IPv6 Address, MAC Address
        - VRRP: Virtual Router ID, Admin State, Backup, Message
Interval(deciseconds), Owner,Passive, Init Delay(seconds), MAC,Priority,
Master Int Inherit, Ping Reply, Policy,
                Preempt, TELNET Reply, Traceroute Reply, Standby
Forwarding, NTP Reply, Oper Group, Monitor Oper Group
          - BFD Liveness: DEST IP, Service Name, Interface Name
- QoS: Network Policy, Egress Port Redirect Group, Egress Instance,
Ingress FP Redirect Group, Ingress Instance
- Network Domains: Domain Name
- External Reference:
    - Openconfig: Subinterface
```

Prerequisites:

Configure the following parameters:

- Port and LAG configuration
- Dist CPU Protection
- CPU protection
- Filter IP
- Filter IPv6
- Policy Accounting
- Measurement Template
- Lag IP Measurement template (configure test-oam lag-ip-measurement lag-ip-measurement-template)
- IPv4 Local DHCP Server
- Python Policy
- ETH CFM:
  - MD Admin Name
- If Attribute:
  - Admin Group
  - SRLG Group
- VRRP:
  - policy
  - oper-group
  - monitor-oper-group
- IPv6:
  - IPv6 local-dhcp-server
  - proxy-nd-policy

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- QoS: network-policy
  - qos egress-port-redirect-group
  - qos ingress-fp-redirect-group
- Network Domains:
  - network-domain

CLI context: router > interface

2.50.3 Advisory

- This intent supports creation of network and unnumbered mpls tp interfaces.
- Ethernet port with mode network and hybrid can only be attached for this interface.
- Outer encapsulation and Inner encapsulation values are supported only on port with encap type as QinQ.
- Port configuration is not supported when Loopback is enabled.
- With Loopback ETH-CFM config, IPv4 DHCP config is not supported.
- Strip-label can only be set if configured port is single-fiber.
- Combination of vas-if-type to-from-network and loopback is not supported.
- Combination of dist-cpu-protection and loopback is not supported.
- Egress Instance & Egress Port Redirect Group both should be configured.
- Ingress Instance & Ingress FP Redirect Group both should be configured.
- A static-neighbor-unnumbered MAC address configuration is mandatory on an unnumbered-mpls-tp interface.
- Either IPv4 or IPv6 can be configured for Twamp Light.
- Below attributes do not support flavor unnumbered-mpls-tp. external-reference openconfig subinterface non-default stale-time{} Relay plain BOOTP track-srrp gre-termination urpf-selected-vprns{} destination-class-lookup lag per-link-hash if-attribute untrusted IPv6 local-dhcp-server IPv6 address
- IPv6 VRRP owner cannot be configured with boolean attributes: priority, master-int-inherit, ping-reply, policy, telnet-reply, traceroute-reply, standby-forwarding, ntp-reply.
- IPv4 VRRP owner cannot be configured with boolean attributes: priority, ping-reply, policy, telnet-reply, traceroute-reply, standby-forwarding, ssh-reply, ntp-reply.
- You cannot configure MEP grace tx[TX ETH ED] and mcc-ed tx[TX ETH VSM Grace] at the same time
- VRRP authentication key attribute will not participate in audit operation.
- The following attributes will not be shown as enabled in the UI after being deployed:
  - Circuit ID: ASCII Tuple, If Name, If Index, None, Port ID, VLAN ASCII Tuple
  - Remote ID: MAC, None
- User is advised to take care of unsupported attributes while changing the flavor.
- Deployment fails if flavor is changed and corresponding mutual exclusive elements are not removed.

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2.51 Name of the Intent: router-isis-interface_msros_23-10-1_24-4

2.51.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS MD SROS node version 23.10

2.51.2 Description

router-isis-interface_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of ISIS Interface with ISIS Instance and Interface name as target-indentifiers.

The following configurations are supported per deployment:

```
- Admin-state,Hello Padding,Passive,Interface Type,CSNP Interval
(seconds),Default Instance,Hello Authentication,IPv4 Multicast,IPv6
Multicast,Level Capability,IPv6 Unicast,Load Balancing Weight,LSP Pacing
Interval (milliseconds), Retransmit Interval,
  Tag
- Hello Interval (seconds), Dead Interval (seconds),Retransmit Interval
(seconds),Transit Delay (seconds),SID Protection
- Authentication Type, Authentication Key , Authentication Keychain
,Message Digest Key
- BFD Liveness:
    - IPv4 : Include BFD TLV
    - IPv6 : Include BFD TLV
- Loopfree Alternate
    - Exclude
    - Policy Map and Route NH Template
- IPV4/IPV6 Node SID
    - Type (Index or Label)
    - Clear N Flag
- Mesh Group
- Adjacency Set (ID)
- Flex Algo ID (128..255)
    - Node SID Type (Index or Label)
```

Prerequisites:

- ISIS Instance should be created on the node
- Interface should be enabled
- LFA route-nh-template (configure routing-options route-next-hop-policy template)
- Authentication Keychain (configure system security keychains keychain)
- Adjacency-sid label (configure router "Base" mpls-labels reserved-label-block)
- Adjacency set (configure router "Base" segment-routing)

CLI context: router>>isis>>interface

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2.51.3 Advisory

- Default-instance can't be enabled on ISIS instance 0
- Adjacency-sid label is not supported for broadcast interface type.
- flex-algo can only be defined on loopback or system interface
- Audit report will not be shown for authentication-key attributes
- Node sid can only be defined on loopback or system interface
- To configure adjacency-sid, segment routing in the ISIS instance should be enabled and srlb should be created
- Deployment will be failed if the authentication-key(hashed value) is modified from the intent.
- Distribute operation will be successful only if the preconfiguration also exists on the selected target-nodes

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2.52 Name of the Intent: router-isis-interface_msros_24-10-1_24-4

2.52.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS MD SROS node version 24.10

2.52.2 Description

The router-isis-interface_msros_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of ISIS Interface with ISIS Instance and Interface name as target-identifiers.

The following configurations are supported per deployment:

```
- Admin-state,Hello Padding,Passive,Interface Type,CSNP Interval
(seconds),Default Instance,Hello Authentication,IPv4 Multicast,IPv6
Multicast,Level Capability,IPv6 Unicast,Load Balancing Weight,LSP Pacing
Interval (milliseconds), Retransmit Interval, Tag.
- Hello Interval (seconds), Dead Interval (seconds),Retransmit Interval
(seconds),Transit Delay (seconds),SID Protection.
- Authentication Type, Authentication Key , Authentication Keychain
,Message Digest Key.
- BFD Liveness:
    - IPv4 : Include BFD TLV
    - IPv6 : Include BFD TLV
- Loopfree Alternate
    - Exclude
    - Policy Map and Route NH Template
- IPV4/IPV6 Node SID
    - Type (Index or Label)
    - Clear N Flag
- Mesh Group
- Adjacency Set (ID)
- Flex Algo ID (128..255)
    - Node SID Type (Index or Label)
- Delay Normalization
    - Delay Tolerance Interval (microseconds)
    - Minimum Delay (microseconds)
```

Prerequisites:

- ISIS Instance should be created on the node.
- Interface should be enabled.
- LFA route-nh-template (configure routing-options route-next-hop-policy template).
- Authentication Keychain (configure system security keychains keychain).
- Adjacency-sid label (configure router "Base" mpls-labels reserved-label-block).
- Adjacency set (configure router "Base" segment-routing).

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CLI context: router > isis > interface

2.52.3 Advisory

- Default-instance can't be enabled on ISIS instance 0.
- Adjacency-sid label is not supported for broadcast interface type.
- flex-algo can only be defined on loopback or system interface.
- Audit report will not be shown for authentication-key attributes.
- Node sid can only be defined on loopback or system interface.
- To configure adjacency-sid, segment routing in the ISIS instance should be enabled and srlb should be created.
- Deployment will be failed if the authentication-key(hashed value) is modified from the intent.
- Distribute operation will be successful only if the preconfiguration also exists on the selected target-nodes.

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2.53 Name of the Intent: router-mpls-interface_msros_23-10-1_24-4

2.53.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS MD SR OS node version 23.10

2.53.2 Description:

The router-mpls-interface_msros_23-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of MD for Router MPLS Interface.

The following configurations are supported per deployment:

```
- Admin State
- Te Metric
- Admin Group
- SRLG Group: Name
- Label Map: Label Map Choice, Ingress Label, Admin State, Pop, Swap,
Egress Label, Next Hop
```

Prerequisites:

- You can use an existing interface name to deploy the intent, to use a new interface you must first create it in Router Base.
- "Admin Group" and "SRLG Group" should be configured in routing-options first.

2.53.3 Advisory:

- "SRLG Group" and "Label Map" are not supported for system interface.

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2.54 Name of the Intent: router-mpls-labels_msros_23-10-1_24-11

2.54.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS and 7250 IXR MD SROS node versions 23.10

2.54.2 Description

The router-mpls-labels_msros_23-10-1_24-11 intent supports the creation/modification/deletion/discovery/audit/alignment of mpls labels, with name "Base" as the target identifier. The following configurations are supported per deployment:

```
- MPLS Labels: BGP Labels Hold Timer, Static Label Range
- SR Labels: Start, End
- Reserved Label Block: Start Label, End Label, Label Block Name
```

CLI context: router > mpls-label >

Prerequisite:

- N/A

2.54.3 Advisory

- The sr-label/start range must start with static-label-range plus 32.

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2.55 Name of the Intent: router-mpls-lsp_msros_24-10-1_25-4

2.55.1 Applicability

Supported From: NSP 25.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SR OS node version 24.10

2.55.2 Description:

The router-mpls-lsp_msros_24-10-1_25-4 intent supports the creation/modification/discovery/audit/alignment of mpls-lsp, using lsp name as the target identifier.

The following configurations are supported per deployment:

```
- LSP type, From, To, Admin State, Exclude Node, Hop Limit, P2MP ID, PCE
Control, PCE Report, Revert Timer, RVP Resv Style, VPRN Auto Bind,
  LDP Over RSVP, Metric, Entropy Label, Adaptive, Adspec, Binding SID,
Class type, Least Fill, Load Balancing Weight, Main CT Relay Limit,
  Propagate Admin Group, Retry Limit, Retry Timer, Path Computation
Method, Metric Type, Local SR Protection, Label Stack Reduction,
  LSP Self Ping, Prefer Transport FRR, Soft Preemption, Include Admin
Group, Exclude Admin Group.
- Class Forwarding:
    - CBF Choice (Default LSP and FC, Forwarding Set).
- IGP Shortcut: Admin State, Allow SR Over SRTE, IGP Choice.
- Fast Reroute: Hop Limit, FRR Method, Node Protect.
- Max SR Labels: Label Stack Size, Additional FRR Labels.
    - Admin Tag: Admin Tag Name.
- Auto Bandwidth: Max Bandwidth, Min Bandwidth, Monitor Bandwidth, FC
Name, Sampling Weight.
    - Adjust Down: Percent, Bw.
    - Adjust Up: Percent, Bw.
    - Multipliers: Sample Multiplier, Adjust Multiplier.
    - Overflow Limit: Threshold, Bw, Number.
    - Underflow Limit:  Threshold, Bw, Number.
    - Use Last Adj Bw: Secondary Retry Limit.
- Egress Statistics: Collect Stats, Accounting Policy, Admin State, Stat
Mode.
- BFD: BFD Liveness, BFD Template, LSP Ping Interval, Failure Action,
Wait For Up Timer, Source Address.
- Path Profile: Profile ID, Path Group.
- PCE Associations: Diversity, Policy.
- Primary: Path Name, Admin State, Adaptive, Bandwidth, Hop Limit,
Record, Record Label, Class Type, Delay Metric, Backup Class Type.
    - Priority: Setup Priority, Hold Priority.
    - Include Admin Group: Group Name.
    - Exclude Admin Group: Group Name.
    - BFD: BFD Liveness, BFD Template, LSP Ping Interval, Wait For Up
```

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```
Timer, Source Address.
- Secondary: Path Name, Admin State, Adaptive, Bandwidth, Hop Limit,
Record, Record Label, Class Type, Delay Metric, Backup Class Type.
    - Priority: Setup Priority, Hold Priority.
    - Include Admin Group: Group Name.
    - Exclude Admin Group: Group Name.
    - BFD: BFD Liveness, BFD Template, LSP Ping Interval, Wait For Up
Timer, Source Address.
- Primary P2MP Instance: Instance Name, Admin State, Adaptive, Bandwidth,
Hop Limit, Record, Record Label.
    - Priority: Setup Priority, Hold Priority.
    - Include Admin Group: Group Name.
    - Exclude Admin Group: Group Name.
    - S2l Path: Path Name, To, Admin State.
```

CLI context: router > mpls > lsp

Prerequisites: The Below profile/list must be pre-configured on the node:

- Loopback Interface must be configured.
- IGP configuration.
- Enable MPLS on interfaces, and enable IGP support for MPLS/TE.
- sr-labels < *configure router mpls-labels sr-labels* >.
- reserved-label-block < *configure router mpls-labels reserved-label-block* >.
- lsp-bsid-block < *configure router mpls lsp-bsid-block* >.
- seamless-bfd reflector < *configure bfd seamless-bfd reflector* >.
- seamless-bfd peer < *configure router bfd seamless-bfd peer* >.
- bfd-template < *configure bfd bfd-template* >.
- admin-group < *configure routing-options if-attribute admin-group* >.
- admin-tags < *configure routing-options admin-tags admin-tag* >.
- class-forwarding-policy < *configure router mpls class-forwarding-policy* >.
- path < *configure router mpls path* >.
- enable pcep pcc < *configure router pcep pcc admin-state enable* >.
- accounting-policy < *configure log accounting-policy* <policyName> record combined-mpls-lsp-egress >*.
- pce-associations diversity < *configure router pcep pcc pce-associations diversity* >.
- pce-associations policy < *configure router pcep pcc pce-associations policy* >.

2.55.3 Advisory:

- Before you can create an MPLS path from the config form, the template of **`router-mpls-path_msros_23-10-1_23-11`** must exist in ICM.
- The 'to' address type IPv4 is supported only for p2p-rsvp and p2p-rsvp-bypass LSPs, whereas both IPv4 and IPv6 address types are supported for p2p-sr-te LSPs.

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- Attributes that are not supported for:
  - **p2p-sr-te:** exclude-node, p2mp-id, rsvp-resv-style, ldp-over-rsvp, adaptive, adspec, class-type, least-fill, main-ct-retry-limit, propagate-admin-group, lsp-self-ping, soft-preemption, fast-reroute, auto-bandwidth, bfd: lsp-ping-interval, primary: adaptive, record, record-label, class-type, backup-class-type, bfd: lsp-ping-interval, secondary: adaptive, record, record-label, class-type, backup-class-type, bfd: lsp-ping-interval, primary-p2mp-instance.
  - **p2p-rsvp:** return-path-label, primary-p2mp-instance, delay-metric-limit.
  - **p2p-rsvp-bypass:** primary-p2mp-instance , binding-sid , bandwidth, class-type, backup-class-type, delay-metric-limit, bfd-template, bfd.
  - **p2mp-rsvp:** fast-reroute (method: one-to-one), entropy-label, override-tunnel-elc, ff-style, path-computation-method (pce), metric-delay, node-protect (true).
- The 'fallback-path-computation-method' is supported with the 'path-computation-method' pce only.
- The per-fc stat-mode for egress statistics is not supported for p2p-sr-te LSP.
- The bfd failure-action - 'failure-action failover' is not allowed for p2p-sr-te LSP.
- The propagate-admin-group is not supported for one-to-one fast-reroute method for p2p-rsvp LSP.
- The 'lsp-self-ping true' and 'bfd-template' configurations are mutually exclusive for p2p-rsvp LSP.
- You cannot configure preference for secondary path for a p2p-rsvp LSP.
- The record must be true if fast-reroute is configured for p2p-rsvp LSP.
- BFD can be configured for LSP or for primary/secondary path, but not both.
- The 'path-computation-method' must be specified if metric-type is 'te'.
- The 'return-path-label' is valid only if 'bfd-template' is configured.
- The 'source-address' is valid only if 'bfd-template' is configured.
- The 'bfd-liveness' - BFD can be enabled only if 'bfd-template' is configured.
- Cannot have 'path-profile' and 'pce-associations- diversity or policy' configurations together.
- The value of the 'binding-sid' must fall within the range of the associated label block.
- The 'entropy-label' and 'override-tunnel-elc' are not supported for IXR.
- The 'bfd source-address', primary (bfd source-address, delay-metric-limit), secondary (bfd source-address, delay-metric-limit) are not supported in node Release 23.10.
- To manually enter the destination ID on the NSP Release 24.11 or earlier, contact your Nokia support representative.

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2.56 Name of the Intent: router-mpls-path_msros_23-10-1_23-11

2.56.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS and 7250 IXR MD SR OS node version 23.10

2.56.2 Description:

The router-mpls-path_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of MD intent for Router MPLS Path, with path-name as target identifier.

The following configurations are supported per deployment:

```
- Admin State
- Hop:
    Hop Index, Hop Info (Hop IP Address, Hop SID), IP Address, Type, SID
Label
```

Prerequisites:

- N/A

CLI context: router > mpls > path

2.56.3 Advisory:

- A path cannot have both SID and IP Address configured across its hops.
- The hops configuration must not include a mix of IPv4 and IPv6 hop addresses.

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2.57 Name of the Intent: router-mpls_msros_24-10-1_24-11

2.57.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS and 7250 IXR MD SROS node version 24.10

2.57.2 Description

The router-mpls_msros_24-10-1_24-11 intent supports the creation/modification/deletion/discovery/audit/alignment of mpls configurations, with Base as target identifier. The following configurations are supported per deployment:

```
- MPLS
    - Admin Group FRR, Admin State, Bypass Resignal Timer (minutes), CSPF
on loose hop, Dynamic Bypass, Exponential Backoff Retry, FRR Object, Hold
Timer (seconds), Least Fill Min Thd (percent), Least Fill Reoptim Thd
(percent), Logger Event Bundling, LSP BSID block, LSP Init Retry Timeout
(seconds), Max Bypass Associations, Max Bypass Plr Associations, MBB
Prefer Current Hops, P2MP Resignal Timer, P2MP S2L Fast Retry (seconds),
P2P Active Path Fast Retry (seconds), Resignal Timer (minutes), Retry On
Igp Overload, Secondary Fast Retry Timer (seconds), Shortcut Local Ttl
Propagate, Shortcut Transit Ttl Propagate, SRLG FRR, Static LSP Fast
Retry (seconds), User Srlg Db, Resignal On Igp Overload, Strict Ero Nhop
Direct Resolution, Resignal On Igp Event, P22MP TTL Propagate
    - LSP Self Ping: Interval (seconds), Rsvp Te, Timeout (seconds),
Timeout Action
    - Auto Bandwidth Multipliers: Sample Multiplier, Adjust Multiplier
    - Entropy Label: Rsvp Te, Sr Te
    - PCE Report: Rsvp Te, Sr Te
    - Srlg Database: Router Address, Admin State
        -Interface: Interface Address, Srlg Group
    - Tunnel Table Pref: Rsvp Te, Sr Te
    - Class Forwarding Policy: Policy Name, Default Set
        -Fc: Fc Name, Forwarding Set
    - Forwarding Policies: Admin State, Reserved Label Block
        -Forwarding Policy: Policy Name, Admin State, Binding Label,
Endpoint, Revert Timer (seconds), Preference
            -Next Hop Group: Index, Admin State, Resolution Type, Load
Balancing Weight
                -Primary Next Hop: Next Hop
                    -Pushed Labels: Index, Label
                -Backup Next Hop: Next Hop
                    -Pushed Labels: Index, Label
            -Ingress Statistics: Admin State
            -Egress Statistics: Admin State
    - PCE INIT LSP
        -Sr Te: Admin State
```

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```
    - Ingress Statistics
        -LSP: Sender, Lsp Name, Collect Stats, Accounting Policy, Admin
State, Stat Mode
        -P2MP Template LSP: Sender, Rsvp Session Name, Collect Stats,
Accounting Policy, Admin State, Stat Mode, Max Stats
        -P2P Template LSP: Sender, Rsvp Session Name, Collect Stats,
Accounting Policy, Admin State, Stat Mode, Max Stats
    - Aux Stats: sr
    - Sr Te Resignal: Resignal Timer (minutes), Resignal On Igp Event,
Resignal On Igp Overload
    - LSP History: Admin State
```

CLI context: router > mpls

Prerequisite:

- Configure srlg group [configure routing-options if-attribute srlg-group <name>]
- Configure LSP BSID block [configure router mpls-labels reserved-label-block <label-block-name>]
- Configure Accounting Policy [configure log accounting policy]

2.57.3 Advisory

- To enable pce-init-lsp sr-te admin-state, you must configure 'p2p-sr-te-pce-init' lsp-template with the default template-id.
- retry-on-igp-overload and resignal-on-igp-overload are mutually exclusive attributes.
- You cannot configure both binding-label and endpoint attribute.
- Metric can only be set with an endpoint policy
- Tunnel-table-pref can only be set with an endpoint policy.
- Pushed labels must be configured in sequential order.
- Binding-label must be configured if ingress-statistics is enabled.
- Before you can delete the deployment from NSP and network, you must remove all the dependendent objects
- Below attributes are not supported on certain chassis using node Release 23.10:
  - resignal-on-igp-event

  The deployment should proceed successfully if these attributes are not included in the payload. In case the deployment fails due to the configuration of an unsupported boolean-type attribute, it should be removed from the API.

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2.58 Name of the Intent: router-ospf-area-virtual-link_msros_23-10-1_24-4

2.58.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750-SR, 7950-XRS, 7450-ESS MD SROS of node version 23.10

2.58.2 Description:

router-ospf-area-virtual-link_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of OSPFv2 VIrtual Link

The following configurations are supported per deployment:

```
- Administrative State
- Authentication Type
- password
- Authentication Keychain
- Hello Interval
- Router Dead Interval
- Retransmission Interval
- Transit Delay
- OSPF MD5 Key Index and Key.
```

Prerequisites: keychain Policy should be preconfigured.

CLI context: configure >router>ospf> area 0.0.0.0> virtual-link

2.58.3 Advisory:

- Audit will not show diff for Authentication Key and OSPF MD5 Key.
- For BrownfieldField Discovery user need to select the Virtual Neighbour Router and Transit Area combination configured on the node.
- Area ID targetIdentifier should be set to 0.0.0.0 only.

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2.59 Name of the Intent: router-ospf-area_msros_23-10-1_24-4

2.59.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950/7250 IXR node version 23.10 MD SROS.

2.59.2 Description:

router-ospf-area_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of MD intent for Router OSPFv2 Area.

The following configurations are supported per deployment:

```
- Import Policy
- Export Policy
- Advertise Router Capability, Blackhole Aggregate, Database Export
Exclude, Loopfree Alternate Exclude
- Area Range: IP Prefix Mask, Advertise
- BIER: Template, Admin State
- Stub: Default Metric, Summaries
- NSSA: Redistribute External, Summaries, Area Range, Originate Default
Route
```

Prerequisites:

- Import and Export policies should be configured in "Policy options >> Policy statement"
- BIER template should be configured in "Router Base >> BIER template"

2.59.3 Advisory:

- Audit will not show misalignment for additional attributes configured on the node.

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2.60 Name of the Intent: router-ospf-interface_msros_23-10-1_24-4

2.60.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS MD SROS node version 23.10

2.60.2 Description

router-ospf-interface_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of OSPF Interface with OSPF Instance, Area and interface name as target-indentifiers.

The following configurations are supported per deployment:

```
- Admin-state,Advertise Router Capability,Advertise Subnet,Interface
Type,Metric,LSA Filter Out,MTU,Passive,Priority,RIB Priority,Load
Balancing Weight
- Hello Interval (seconds), Dead Interval (seconds),Retransmit Interval
(seconds),Transit Delay (seconds),SID Protection
- Authentication Type, Authentication Key , Authentication Keychain
,Message Digest Key
- BFD Liveness:
    - Remain Down On Failure
    - Strict and Strict Mode Holddown (seconds)
- Loopfree Alternate
    - Exclude
    - Policy Map and Route NH Template
- Node SID
    - Type (Index or Label)
    - Clear N Flag
- Adjacency SID
    - Type (Label)
- Neighbor Address
- Adjacency Set (ID)
- Flex Algo ID (128..255)
    - Node SID Type (Index or Label)
```

Prerequisites:

- OSPF Instance and Area should be created
- Interface should be enabled
- LFA route-nh-template (configure routing-options route-next-hop-policy template)
- Authentication Keychain (configure system security keychains keychain)
- Adjacency-sid label (configure router "Base" mpls-labels reserved-label-block)

CLI context: router>>ospf>>area>>interface

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2.60.3 Advisory

- The dead-interval must be >= 2 * hello-interval
- Static adjacency sid is only allowed on point to point interfaces
- Node sid can only be defined on loopback or system interface
- The label option specifies that the value is assigned to an MPLS label
- Neighbors can be configured only if interface type is non-broadcast
- Cannot enable "strict" and "remain-down-on-failure" at same time in BFD-Liveness
- Cannot configure route-nh-template on secondary interface type
- Cannot configure loopfree-alternate exclude on secondary interface type
- To configure adjacency-sid, segment routing in the ospf instance should be enabled and srlb should be created
- Static adjacency sid is only allowed on point to point interfaces
- Audit report will not be shown for Authentication key attribute.

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2.61 Name of the Intent: router-ospf-interface_msros_24-10-1_24-4

2.61.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS MD SROS node version 24.10

2.61.2 Description

The router-ospf-interface_msros_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of OSPF Interface with OSPF Instance, Area and interface name as target-indentifiers.

The following configurations are supported per deployment:

```
- Admin-state
- Delay Normalization
    - Delay Tolerance Interval (microseconds)
    - Minimum Delay (microseconds)
- Advertise Router Capability,Advertise Subnet,Interface Type,Metric,LSA
Filter Out,MTU,Passive,Priority,RIB  Priority,Load Balancing Weight,
Hello Interval (seconds), Dead Interval (seconds),Retransmit Interval
(seconds),Transit Delay (seconds), SID Protection
- Authentication Type, Authentication Key, Authentication Keychain
- BFD Liveness:
    - Remain Down On Failure
    - Strict and Strict Mode Holddown (seconds)
- Loopfree Alternate
    - Exclude
    - Policy Map and Route NH Template
- Node SID
    - Type (Index or Label)
    - Clear N Flag
- Adjacency SID
    - Type (Label)
- Message Digest Key ID
- Neighbor Address
- Adjacency Set (ID)
- Flex Algo ID (128..255)
    - Node SID Type (Index or Label)
```

Prerequisites:

- OSPF Instance and Area should be created
- Interface should be enabled
- LFA route-nh-template (configure routing-options route-next-hop-policy template)
- Authentication Keychain (configure system security keychains keychain)
- Adjacency-sid label (configure router "Base" mpls-labels reserved-label-block)

CLI context: router > ospf > area id > interface

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2.61.3 Advisory

- The dead-interval must be >= 2 * hello-interval
- Static adjacency sid is only allowed on point to point interfaces
- Node sid can only be defined on loopback or system interface
- The label option specifies that the value is assigned to an MPLS label
- Neighbors can be configured only if interface type is non-broadcast
- Cannot enable "strict" and "remain-down-on-failure" at same time in BFD-Liveness
- Cannot configure route-nh-template on secondary interface type
- Cannot configure loopfree-alternate exclude on secondary interface type
- To configure adjacency-sid, segment routing in the ospf instance should be enabled and srlb should be created
- Static adjacency sid is only allowed on point to point interfaces
- Audit report will not be shown for Authentication key attribute.

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2.62 Name of the Intent: router-rsvp-interface_msros_23-10-1_24-4

2.62.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS MD SR OS node version 23.10

2.62.2 Description

router-rsvp-interface_msros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of RSVP Interface.

The following configurations are supported per deployment:

```
- Configure the RSVP interface with name
- Configuration of Admin State,Authentication Keychain,Authentication
Key,BFD Liveness,Graceful Restart Helper Mode,
      Graceful Shutdown,Hello Interval,Subscription,Implicit Null Label
- Configure the Class Type BW
      CT0 to CT7
- Configure the Refresh Reduction,TE Down Threshold,TE UP Threshold
```

Prerequisites:

- Configure Network interface and associated to MPLS/RSVP
- Configure diffserv-te on the RSVP
- Configure system Keychain

2.62.3 Advisory

- Unable to delete the RSVP interface alone from intent Before deletion of RSVP interface we need to remove interface from MPLS also.
- Minimum one value required TE Down Threshold & TE Down Threshold
- te-up-threshold values must be in ascending order
- te-down-threshold values must be in descending order

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2.63 Name of the Intent: routing-flex-algo-def_msros_24-10-1_24-4

2.63.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS and 7250 IXR for MD SROS node version 24.10

2.63.2 Description

The routing-flex-algo-def_msros_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of Flex Algo, with Flex Algo Name as the target identifier.

The following configurations are supported per deployment:

```
- Configuration of Flex Algo  attributes as below
  Admin State, Description, Flags TLV, Metric Type, Priority
  Exclude
    Admin Group
  Include Any
    Admin Group
  Include All
    Admin Group
```

CLI context: routing-options > flexible-algorithm-definitions > flex-algo

Prerequisites:

- Configure admin-group [configure routing-options if-attribute]

2.63.3 Advisory

- Cannot use the same admin group in both include and exclude

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2.64 Name of the Intent: sec-aaa-local-profile_msros_23-10-1_23-11

2.64.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR MD SROS node version 23.10

2.64.2 Description

The sec-aaa-local-profile_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a AAA Local Profile, using USER-PROFILE-NAME name as the target identifier.

The following configurations are supported per deployment:

```
- Profile:
   CLI Session Group, Combined Max Sessions, SSH Max Session, Telnet Max
Sessions, LI, Default Action
   Create Entry with Entry ID
    Configuration of the following attributes on Entry:
    Description, Match, Action
- GRPC RPC Authorization:
  GNMI Capabilities, GNMI Get, GNMI Set, GNMI Subscribe, GNOI Certificate
Management Rotate, GNOI Certificate Management Install, GNOI Certificate
Management Get Certificates, GNOI Certificate Management Revoke, GNOI
Certificate Management Can Generate, GNOI File Get, GNOI File Transfer To
Remote, GNOI File Put, GNOI File Stat, GNOI File Remove, GNOI System
Ping, GNOI System Tracer Route, GNOI System Time, GNOI System Set
Package, GNOI System Switch Control Processor, GNOI System Reboot, GNOI
System Reboot Status, GNOI System Cancel Reboot, MD CLI Session, RIB API
Get Version, RIB API Modify
- NetConf Base OP Authorization:
  Action, Cancel Commit, Close Session, Commit, Copy Config, Create
Subscription, Delete Config, Discard Changes, Edit Config, Get, Get
Config, Get Data, Get Schema, Kill Session, Lock, Validate
```

CLI context: system > security > aaa > local-profiles > profile

Prerequisite: - Configure CLI Session Group [system > security > aaa > cli-session-group]

2.64.3 Advisory

- LI, RIB API Get Version and RIB API Modify are not supported on 7250 IXR NEs.
- Match accepts CLI commands only.

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2.65 Name of the Intent: sec-client-tls-profile_msros_23-10-1_23-11

2.65.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SROS node version 23.10

2.65.2 Description

sec-client-tls-profile_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of client profile with CLIENT-PROFILE-NAME name as target identifier. The following configurations are supported per deployment:

```
- Client TLS Profile
  Configuration of Admin State,Cert Profile,Cipher List,Trust Anchor
Profile,Group List,Signature List,Protocol Version

- Status Verify
  Default Result
```

CLI context: system security tls > client-tls-profile

Prerequisite:

- Configure cert-profile,client-cipher-list,client-group-list,client-signature-list,trust-anchor-profile [configure system security tls]

2.65.3 Advisory

- Client-tls-profile requires client-cipher-list client-group-list and client-signature-list when admin-state enabled

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2.66 Name of the Intent: sec-maf_msros_23-10-1_23-11

2.66.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SROS node version 23.10

2.66.2 Description

The sec-maf_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a MAF Filter, using 'maf' name as the target identifier. The following configurations are supported per deployment:

```
- Configuration of IP-Filter, IPv6-Flter, MAC-Filter
- Configuration of IP-Filter:
   Admin-State, Default Action, Entry
   Create Entry with Entry ID
    Configuration of the following attributes on Entry:
    Description, Action, Log Events
    Match
      Router Instance, Protocol
      Src IP
        Address/IP Prefix List, Mask
      Mgmt Port
        CPM, Lag ID, Port ID
      Dst Port
        Port, Mask
      Src Port
        Port, Mask
- Configuration of IPv6-Filter:
  Admin-State, Default Action, Entry
  Create Entry with Entry ID
    Configuration of the following attributes on Entry:
    Description, Action, Log Events
    Match
      Router Instance, Protocol, Flow Label
      Src IP
        Address/IPv6 Prefix List, Mask
      Mgmt Port
        CPM, Lag ID, Port ID
      Dst Port
        Port, Mask
      Src Port
        Port, Mask
- Configuration of MAC Filter:
  Admin-State, Default Action, Entry
  Create Entry with Entry ID
    Configuration of the following attributes on Entry:
```

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```
Description, Action, Log Events
Match
  Service, Frame Type, Etype, SNAP PID, SNAP OUI
  Src MAC
    Address, Mask
  Dst MAC
    Address, Mask
  Dot1p
    Priority, Mask
  LLC SSAP
    SSAP, Mask
  LLC DSAP
    DSAP, Mask
  CFM Opcode
```

CLI context: system > security > management-access-filter

Prerequisite:

- Configure IP-prefix-list/IPv6-prefix-list [configure filter match-list]
- Configure LAG, Services VPRN/VPLS/IES/Epipe

2.66.3 Advisory

- Single MAF-Filter supports per node
- The Default Action attribute must be configured before an entry can be created
- MAC-Filter is not supported on the 7250 IXR
- In Match etype supports only frame-type ethernet-ii
- In Match llc-ssap, llc-dsap supports only frame-type 802dot2-llc
- In Match snap-pid, snap-oui supports only frame-type 802dot2-snap
- Changing the default action to deny can cause connectivity issues

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2.67 Name of the Intent: sec-pki_msros_24-10-1_24-4

2.67.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR MD (SROS version 24.10)

2.67.2 Description:

The sec-pki_msros_24-10-1_24-4 intent enables creation, modification, discovery, audit, and alignment of public key infrastructure (pki) configurations using the pki as the identifier.

The following configurations are supported per deployment:

```
- Certificate Display Format.
- Imported Format.
- Max Cert Chain Length.
- CA Profile:
    - CA Profile Name.
    - Admin State.
    - Certificate Name.
    - Cert File.
    - CRL File.
    - Revocation Check.
    - OSCP:
        - Responder URL.
        - Service Name.
        - Transmission Profile.
    - CMPv2:
        - Always Set Sender for IR.
        - Same Recipient Nonce For Poll Request.
        - Signing Cert Subject.
        - Recipient.
        - Response Signing.
        - Accept Unprotected Message:
            - Error Message.
            - PKI Conf Message.
        - HTTP:
            -Response Timeout.
            - Version.
        - URL:
            - URL String.
            - Transport.
        - Key List:
            - Reference Number.
            - Password.
    - Auto CRL Update:
        - Admin State.
```

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```
            - Periodic Update Interval.
            - Pre Update Time.
            - Retry Interval.
            - Schedule Type.
            - CRL URLs:
                - Entry ID.
                - Transmission Profile.
                - URL.
    - Certificate Auto Update:
        - Certificate File Name.
        - Key File Name.
        - Profile.
    - Certificate Expiration Warning:
        - Hours.
        - Repeat Hours.
    - Certificate Update Profile:
        - Name.
        - Hash Algorithm.
        - Retry Interval.
        - Key Generation.
        - Protocol.
        - Schedule.
    - CRL Expiration Warning:
        - Hours.
        - Repeat Hours.
    - EST Profile:
        - Name.
        - Transmission Profile.
        - Client TLS Profile.
        - Check ID KP CMCRA Only.
        - Server:
            - Port.
            - Address.
        - HTTP Authentication:
            - Username.
            - Password.
```

CLI context: configure > system > security > pki.

Pre-requisite: Below profile/list must be pre-configured on the node.

- transmission-profile < *configure system transmission-profile* >.
- client-tls-profile < *configure system security tls client-tls-profile* >.

2.67.3 Advisory:

- The signing-cert-subject setting is valid only if response-signing-use-extracert is set to true.
- If the user wants to associate the created ca-profile, est-profile, and profile in the same configuration, they must manually enter them under certificate-update-profile and certificate-auto-update.

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- The HTTP Authentication password & cmpv2 key-list password (as a password-protected attribute) is excluded from audit operation for security reasons.
- The crl-file must be provided in order to enable the admin-state of auto-crl-update.
- The crl-urls list cannot be deleted when the admin-state of auto-crl-update is set to enable.

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2.68 Name of the Intent: sec-server-tls-profile_msros_23-10-1_23-11

2.68.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SR OS node version 23.10

2.68.2 Description:

sec-server-tls-profile_msros_23-10-1_23-11 intent supports creation/modification/discovery/audit/alignment of server-tls-profile with server-profile-name as identifier.

The following configurations are supported per deployment:

```
- Admin State, Cert Profile, TLS-re-negotiate-timer, Cipher List, Group
List, Signature List, Protocol Version
- Status Verify : Default Result.
- Authenticate Client : Trust Anchor Profile, Common Name List.
```

CLI context: configure > system > security > tls > server-tls-profile.

Prerequisite: Below profile/list must be pre-configured on the node:

- authenticate-client common-name-list <configure system security pki common-name-list >
- authenticate-client trust-anchor-profile <configure system security tls trust-anchor-profile >
- cert-profile <configure system security tls cert-profile >
- cipher-list <configure system security tls server-cipher-list >
- group-list <configure system security tls server-group-list >
- signature-list <configure system security tls server-signature-list >

2.68.3 Advisory:

- For Protocol Version TLS 13 or all versions, the server-tls-profile requires cert-profile, server-cipher-list, server-group-list and server-signature-list when enabled.
- For Protocol Version TLS 12, the server-tls-profile requires cert-profile and server-cipher-list when enabled.

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2.69 Name of the Intent: sec-ssh_msros_23-10-1_23-11

2.69.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SROS node version 23.10

2.69.2 Description

The sec-ssh_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a system security ssh, using SSH name as the target identifier. The following configurations are supported per deployment:

```
- Preserve Key,Server Admin State,Permit Empty Passwords
- Server Cipher List V2 with Index and Name:
- Client Cipher List V2 with Index and Name:
- Server Mac List V2 with Index and Name:
- Client Mac List V2 with Index and Name:
- Key Re Exchange:
    Server: Admin State, Minutes, Mbytes
    Client: Admin State, Minutes, Mbytes
- Server Kex List V2 with Index and Name:
- Client Kex List V2 with Index and Name:
- Authentication Method
    Server: Public Key Only
```

CLI context: system > security > ssh

2.69.3 Advisory

- Configuring Authentication Method Server to Public Key Only can disrupt the SSH connection.

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2.70 Name of the Intent: sec-tls-cert-profile_msros_23-10-1_23-11

2.70.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SROS node version 23.10

2.70.2 Description

sec-tls-cert-profile_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Cert Profile with CERT-PROFILE-NAME name as target identifier. The following configurations are supported per deployment:

```
- Configuration of Admin State, Entry
- Configure Entry with ID
   Certificate File, Key File
- Send Chain
   CA Profile
```

CLI context: system > security > tls > cert-profile

Prerequisite:

- Configure CA Profile [configure system security pki]

2.70.3 Advisory

- Cert Profile supports upto 8 Entry ID
- Cert Profile admin-state enabled requires Entry ID with certificate-file and key-file attributes

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2.71 Name of the Intent: sec-tls-client-cipher-list_msros_23-10-1_23-11

2.71.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SROS node version 23.10

2.71.2 Description

The sec-tls-client-cipher-list_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a client cipher-list, using CLIENT-CIPHER-LIST-NAME name as the target identifier. The following configurations are supported per deployment:

`- Tls12 Cipher configurations with Id and Cipher name:`

`- Tls13 Cipher configurations with Id and Cipher name:`

CLI context: system > security tls > client-tls-cipher-list

2.71.3 Advisory

- N/A

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2.72 Name of the Intent: sec-tls-server-cipher-list_msros_23-10-1_23-11

2.72.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR MD SROS node version 23.10

2.72.2 Description:

The sec-tls-server-cipher-list_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a server-cipher-list, using name as the target identifier.

The following configurations are supported per deployment:

```
- Tls12 Cipher configurations with id and  Cipher name

- Tls13 Cipher configurations with id and Cipher name
```

CLI context: system > security > tls > server-cipher-list

Prerequisites: - N/A

Advisory: - N/A

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2.73 Name of the Intent: sec-tls-trust-anchor-profile_msros_23-10-1_23-11

2.73.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR MD SROS node version 23.10

2.73.2 Description:

The sec-tls-trust-anchor-profile_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a trust-anch-profile, using trust-anch-profile name as the target identifier.

The following configurations are supported per deployment:

`- ca-profile-name`

CLI context: system > security > tls > trust-anchor-profile

Prerequisites: - Configure ca-profile list instance [system security pki]

2.73.3 Advisory:

- N/A

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2.74 Name of the Intent: sec-user-params_msros_23-10-1_23-11

2.74.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SROS node version 23.10

2.74.2 Description

The sec-user-params_msros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of User Parameters, using 'user-params' name as the target identifier. The following configurations are supported per deployment:

```
- Configuration of Attempts attributes as below
  Count, Time, Lockout
- Configuration of Authentication Order
  Order, Exit On Reject
- Configuration of Local User
   Password
    Aging, Hashing, History Size, Minimum Age, Minimum Change

   Complexity Rules
      Credits/Required
        Lowercase, Uppercase, Numeric, Special Character, Allow User
Name, Minimum Classes, Minimum Length, Repeated Characters.

    User
     Access, Console, Public Keys[ECDSA,RSA], SNMP, SSH Authentication
Method
```

CLI context: system > security > user-params

Prerequisite: NA

2.74.3 Advisory

- Attributes should be modified after brownfield discovery
- The SSH Authentication Method client is not supported in NSP 23.11, NSP 25.4 or later is required.
- The "Authentication Order" attribute does not display in NSP 23.11, NSP 25.4 or later is required.
- Access LI is not supported for 7250 IXR nodes
- Do not modify the user password for the user node discovered by NSP, as connectivity issues or node shutdown can occur.
- Do not create or modify empty users in local-user.
- Deleting a user-params intent in NSP only deletes it in the NSP; access the node manually to delete the intent from the node
- SNMP authentication-key requirements for each authentication type are as follows: hmac-

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md5-96 authentication key must be 16 bytes long hmac-sha1-96 authentication key must be 20 bytes long hmac-sha2-224 authentication key must be 28 bytes long hmac-sha2-256 authentication key must be 32 bytes long hmac-sha2-384 authentication key must be 48 bytes long hmac-sha2-512 authentication key must be 64 bytes long

- SNMP privacy-key for each key type are as follows: cfb128-aes-256 privacy key must be 32 bytes long cfb128-aes-192 privacy key must be 24 bytes long cfb128-aes-128 privacy key must be 16 bytes long
- Authentication protocol hmac-sha1-96 is not allowed when using privacy protocol cfb128-aes-256

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2.75 Name of the Intent: segment-routing-mpls_msros_24-10-1_24-4

2.75.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS and 7250 IXR MD SROS node version 24.10

2.75.2 Description

segment-routing-mpls_msros_24-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of sr-mpls with SR-MPLS as target identifier. The following configurations are supported per deployment:

```
- Interface Name, Node SID
- IPv4 SID : Type [Index, Label],Index or Label
- IPv6 SID : Type [Index, Label],Index or Label
```

CLI context: router > segment-routing > sr-mpls

Prerequisite:

- configure router interface (configure router interface name)
- configure SR-MPLS labels (configure router mpls-labels sr-labels)

2.75.3 Advisory

- N/A

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2.76 Name of the Intent: segment-routing-policies_msros_24-10-1_24-4

2.76.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS , 7450 ESS and 7250 IXR for MD SROS node version 24.10

2.76.2 Description

The segment-routing-policies_msros_24-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of sr-policies with Name as the target identifier.

The following configurations are supported per deployment:

```
- Configuration of sr-policies attributes as below
   Admin State, Reserved Label Block, ECMP, Ingress Statistics, Egress
Statistics
- Configuration of the Static Policy
   Admin State, Color, Endpoint, Preference, Head End, Binding SID,
Distinguisher, Maintenance Policy, Prefer Transport FRR, Type
   -Segment Routing V6
     Return Path BFD SID, Binding SID
       Binding SID Type, Locator, Micro Segment Locator,IP Address
   -Segment List
       Weight, Admin State
       Segment Index
          MPLS Label, SrV6 SID, Behavior And Structure
```

CLI context: router > segment-routing > sr-policies

Prerequisites:

- Enable MPLS and RSVP protocol in a router [configure router "Base"]
- Configure reserved-label-block [configure router "Base" mpls-labels]
- Configure maintenance-policy [configure router "Base" segment-routing]
- Configure locator [configure router "Base" segment-routing segment-routing-v6]
- Configure system interface into RSVP [configure router "Base" rsvp interface "system"]
- Configure micro-segment [configure router "Base" segment-routing segment-routing-v6 micro-segment]
- Configure micro-segment-locator [configure router "Base" segment-routing segment-routing-v6 micro-segment-locator]
- Configure accounting-policy [configure log accounting-policy record combined-sr-policy-egress]
- Configure origination-fpe [configure router "Base" segment-routing segment-routing-v6]
- Configure fwd-path [configure fwd-path-ext fpe 1 application srv6]
- Configure function-value [configure router "Base" segment-routing segment-routing-v6 locator static-function max-entries]

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2.76.3 Advisory

- segment-routing-v6 requires network interface ports to be on FP4 or higher generation line cards
- egress-statistics and ingress-statistics attributes does not support on 7250-IXR-10/6/R4/R6/e-BigGNSS/e-Big/e-small/e-smallGNSS/e/S/ec
- IXR : segment-routing-srv6 is supported on IXR-x, IXR-x3, IXR-Xs, IXR-R6d, IXR-R6dl, IXR-e2 and IXR-e2c.
- SR : segment-routing-srv6 is not supported on SR-a4, SR-a8, SR-1e, SR-2e and SR-3e.
- ECMP and prefer-transport-frr attributes does not support on 7750/7950/7450 nodes.
- segment-routing-v6 Binding SID only one of locator and micro-segment-locator can be configured
- segment-routing-v6 can be enabled only for policy of type 'srv6'
- Endpoint address must be of type IPv6 when the policy is of type 'srv6'
- Cannot configure "Binding-SID" attribute both place in Static policy and segment-routing-v6
- To Enable static policy 'admin-state' configuration of color, endpoint, head-end and segment-routing-v6 for Type "srv6" is mandatory.
- To Enable static policy 'admin-state' configuration of color endpoint head-end and binding-sid for Type "sr-mpls" is mandatory
- Configuration of distinguisher is mandatory when head-end is remote and static policy 'admin-state' set to enable.
- Configure at least one segment before enabling the segment-list
- Configure at least one segment with srv6-sid before enabling the segment-list for type srv6
- mpls-label attribute allowed only with static-policy type 'sr-mpls'
- origination-fpe is not supported on IXR nodes.
- Segment Index value differ from chassis to chassis
- Admin-state should be set to disable before updating return path sid value in static-policy
- return-path-sbfd-sid valid for local policy only
- head-end must be remote ip-address when ip-address is configured
- head-end must be local when locator is configured
- To configure path and application under fwd-path-ext fpe, the application srv6 requires network interface ports on FP4 or higher generation line cards.
- Mutual exclusive "IP Address" and "Locator/Micro Segment Locator" in Binding SID Segment Routing V6

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2.77 Name of the Intent: segment-routing-srv6_msros_24-10-1_24-4

2.77.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SR OS node version 24.10

2.77.2 Description:

The segment-routing-srv6_msros_24-10-1_24-4 intent supports the creation/modification/discovery/audit/alignment of segment-routing-srv6.

The following configurations are supported per deployment:

```
- Source Address, Origination FPE
- Locator:
    - Locator Name, Block Length, Function Length, Termination FPE, Label
Block, Argument Length, Admin State, Algorithm.
        - Prefix: IP Prefix.
        - Static Function: Max Entries, Label Block.
- Micro Segment: Block Length, Global SID Entries, SID Length, Argument
Length.
    - Block: Block Name, Admin State, Termination FPE, Label Block.
        - Prefix: IP Prefix.
        - Static Function: Max Entries.
    - Micro Segment Locator: Locator Name, Admin State, Algorithm, Block.

        - UN: SRH Mode, Value.
- Base Routing Instance:
    - Locator: Locator Name.
        - Function:
            - End: Value, SRH Mode.
            - End X: Value, SRH Mode, Protection, Interface Name.
            - End X Auto Allocate: SRH Mode, Protection.
            - End DT4: Value.
            - End DT6: Value.
            - End DT46: Value.
    - Micro Segment Locator: Locator Name.
        - Function:
            - UA: Value, SRH Mode, Protection, Interface Name.
            - UA Auto Allocate: SRH Mode, Protection.
            - UDT4: Value.
            - UDT6: Value.
            - UDT46: Value.
```

CLI context: router > segment-routing > segment-routing-srv6

Prerequisites:

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`Below profile/list must be pre-configured on the node.`

- Enable MPLS and RSVP protocol in a router < *configure router "Base"* >.
- static-label < *configure router mpls-labels static-label-range* >.
- reserved-label-block < *configure router mpls-labels reserved-label-block* >.
- fpe < *configure fwd-path-ext fpe* >.
- interface < *configure router interface* >.

2.77.3 Advisory:

- IXR: segment-routing-srv6 is supported on IXR-x, IXR-x3, IXR-Xs, IXR-R6d, IXR-R6dl, IXR-e2 and IXR-e2c.
- SR: segment-routing-srv6 is not supported on SR-a4, SR-a8, SR-1e, SR-2e and SR-3e.
- label-block is not supported on locator, locator static-function and micro-segment block for IXR.
- origination-fpe is not supported for IXR.
- termination-fpe is not supported on locator and micro-segment block for IXR.
- Function length of 16 is supported on IXR.
- Only prefix lengths of 96 and 64 are supported on IXR.
- argument-length is not supported under the contexts of locator and micro-segment context for IXR.
- To configure path and application under fwd-path-ext fpe, the application srv6 requires network interface ports on FP4 or higher generation line cards.
- admin-state cannot be enabled when locator IP prefix is not set.
- Cannot delete static-function max entries when function configuration has been performed in the base-routing-instance.
- micro-segment global-sid-entries: must be multiple of 4.
- micro-segment block ip-prefix: prefix-length must be equal to block-length.
- micro-segment-locator: value must be configured when admin-state is 'enable'.
- Must configure micro-segment before configuring micro-segment-locator.
- Cannot configure dt4, dt6 or dt46 values when termination fpe is not configured.
- reserved-label-block (Label Block) cannot be used by other users if it is already used by segment-routing-v6.
- label-block must be configured to configure locator function.
- micro-segment-locator: block must be configured when admin-state is 'enable'.
- Cannot change label-block. If the locator has function allocated from label block range. Deconfigure all the function configuration before changing the label-block.
- function-length + prefix-length must be multiple of 8 when argument-length is not default.
- static-function 'label-block' is not supported with 16 bits function-length.
- static-function Max-entries must be less than reserved label block range.
- Cannot configure the label-block in locator and static-function context as it is mutual exclusive between locator and static-function context.

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- Before you configure any static SRv6 functions (like end-dt4), you must configure a label-block under the static-function context of the locator.
- Do not delete the locator, micro-segment block or locator if they have any reference in the base-routing-instance.

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2.78 Name of the Intent: service-system-bgp-evpn_msros_23-10-1_23-11

2.78.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS MD SR OS node version 23.10

2.78.2 Description:

service-system-bgp-evpn_msros_23-10-1_23-11 intent supports modification/discovery/audit/alignment of Service System BGP EVPN.

The following configurations are supported per deployment:

```
- Etree Leaf Label, Etree Leaf Label Value, Route Distinguisher, Ingress
Replication Bum Label Block
- Multicast Leave Sync Propagation
- Ethernet Segment:
        - Ethernet Segment Name, Admin State, Type, Auto ESI, ESI, Orig
IP, Route Next Hop, Multi Homing Mode, Oper Group, AC DF Capability
        - DF Election: ES Activation Timer, Service Carving Mode
                -Manual: EVI, ISID,
                    -Preference:  Mode, value
        - Association: Lag, Network Interconnect Vxlan, Port, Pw Port,
Sdp, VPRN Next Hop
                - Virtual Ranges:
                        -Dot1q  Q tag: Start, End
                        -Qinq: C Tag Start, S Tag, C Tag End
                        -Qinq S Tag: Start, end
        - PBB: Source BMAC LSB, ES BMAC Table Size
- AD Per ES Route: Route Target Type
- IP Prefix Routes: IFF Attribute Uniform Propagation, IFF Bgp Path
Selection, D Path Length Ignore
- Mpls: Vpls Control Word
- AD Per EVI Routes: D Path Ignore, Attribute Propagation, Bgp Path
Selection
```

CLI context: configure > service > system > bgp-evpn

2.78.3 Advisory:

- Before associating an ethernet segment, configure pw-port, lag, sdp, and port.
- RD address must be specified with evi-route-target-set only.
- esi or auto-esi must be configured before enabling ethernet-segment.
- multi-homing-mode must be configured before enabling ethernet-segment.
- ethernet-segment oper-group not supported with service-carving-mode auto.
- ethernet-segment oper-group not supported with virtual ethernet-segment.

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- When preference is set the configuration requires service carving manual mode.
- Port association not supported on virtual ethernet-segments.
- Port encapsulation type should be dot1q or qinq for virtual ethernet segments.
- virtual-ranges allowed only on virtual ethernet-segments.
- virtual-ranges qinq s-tag tag- - *start 0 or tag-* are not supported.
- auto-esi type-1 is not supported for virtual ethernet-segment.
- d-path-ignore cannot be true when bgp-path-selection is disabled.
- The attributes d-path-length-ignore, iff-attribute-uniform-propagation, and iff-bgp-path-selection are deprecated in node release 24.10. Configuring these attributes results in a deployment failure.

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2.79 Name of the Intent: subscriber-mgmt-sla-profile_msros_24-10-1_24-11

2.79.1 Applicability

Supported From: NSP 24.11

Verified on Node Type/Version: 7750 SR, 7450 ESS MD SROS node version 24.10

2.79.2 Description:

The subscriber-mgmt-sla-profile_msros_24-10-1_24-11 intent supports the creation/modification/discovery/audit/alignment of sla-profile, with name as the target identifier.

The following configurations are supported per deployment:

```
- Description, Credit Control Policy, Def instance Sharing, Alternate SLA
Profile.
- Control: Local, Cups.
- Idle Timeout: Category Map Name.
  - Category: Name, Timeout, Action.
- Egress: IP Filter, IPv6 Filter.
  - Bonding Selection:
    - Rate Thresholds: High, Low.
    - Weight: Initial, Change.
  - QoS:
    - Hs Agg Rate, QoS Marking From SAP, Use Ingress L2TP DSCP.
    - SAP Egress: Policy Name, Port Parent.
      - Overrides: Hs Queue Stat Mode.
        - Hs Wrr Group: ID, Hs Class Weight, Rate.
        - Policer: ID, CBS, MBS, Packet Byte Offset, Stat Mode.
                  - Rate: PIR, CIR.
        - Queue: ID, Avg Frame Overhead, CBS, MBS, High Prio Only, Hs
Class Weight, Hs Wrr Weight, Stat Mode.
                - Hs Wred Queue: Policy Name.
                - Rate: PIR, CIR.
    - Scheduler Policy: Name.
      - Overrides: Scheduler Name, Rate: PIR, CIR.
  - Report Rate.
- Session Limits: IPoE, Overall.
  - PPPoE: Local LAC, Overall.
  - L2TP: LNS, LTS, Overall.
- Host Limits:LAC Overall, Overall, Remove Oldest.
  - IPv4: ARP, DHCP, Overall, PPP.
  - IPv6: Overall, PD IPoE DHCP, PD Overall, PD PPP DHCP, WAN IPoE DHCP,
WAN IPoE SLAAC, WAN Overall, WAN PPP DHCP, WAN PPP SLAAC.
- Ingress: IP Filter, IPv6 Filter.
  - QoS:
    - SAP Ingress: Policy Name, Queuing Type.
      - Overrides:
```

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```
        - Policer: ID, CBS, MBS, Packet Byte Offset, Stat Mode.
                  - Rate: PIR, CIR.
        - Queue: ID, CBS, MBS, High Prio Only, Stat Mode.
                - Rate: PIR, CIR.
  - Report Rate.
- Aggregate QoS Statistics:
  - Egress:
    - Queues: Queue Selection.
    - Policers: Policer Selection.
    - Custom Statistics Group: Group Name.
                              - Egress: Policer Selection.
- One Time HTTP Redirection: IP Filter.
- PFCP Mappings:
  - Session QER: Uplink, Downlink.
```

CLI context: subscriber-mgmt > sla-profile

Prerequisites:

The Below profile/list must be pre-configured on the node.

- alternate-sla-profile < *configure subscriber-mgmt sla-profile* >.
- credit-control-policy < *configure subscriber-mgmt credit-control-policy* >.
- category-map < *configure subscriber-mgmt category-map* >.
- category < *configure subscriber-mgmt category-map* <policyName> *category* >.
- ip-filter < *configure filter ip-filter* >.
- ipv6-filter < *configure filter ipv6-filter* >.
- sap-egress policy < *configure qos sap-egress* >.
- sap-egress policer < *configure qos sap-egress* <policyName> *policer* >.
- sap-egress queue < *configure qos sap-egress* <policyName> *queue* >.
- sap-egress hs-wrr-group < *configure qos sap-egress* <policyName> *hs-wrr-group* >.
- scheduler-policy & scheduler < *configure qos scheduler-policy* <policyName> *tier scheduler*>.
- slope-policy < *configure qos slope-policy* >.
- sap-ingress policy < *configure qos sap-ingress* >.
- sap-ingress policer < *configure qos sap-ingress* <policyName> *policer* >.
- sap-ingress queue < *configure qos sap-ingress* <policyName> *queue* >.

2.79.3 Advisory:

- An sla-profile can have one alternate. The alternate itself cannot have its own alternate (chaining is not supported).
- hs-agg-rate, hs-queue-stat-mode, hs-wrr-group, hs-class-weight, hs-wrr-weight, hs-wred-queue are supported only on SR-12, SR-7 and SR-12e.
- pfcp-mapings, bonding-selection, overrides-policer (egress & ingress) are not supported on SR-a4 and SR-a8.
- custom-statistics-group can have a maximum of one entry.

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- pppoe-sessions-overall must not exceed session-overall.
- ipv-4 and ipv-6 overall must not exceed overall session limits.
- idle-timeout and credit-control-policy are mutually exclusive.
- To configure aggregate-qos-statistics or custom-statistics-group, CUPS must be enabled.
- Egress Queue Options: For aggregate-qos-statistics, only options all and none are allowed.
- Report Rate Options:
  - **egress:** policer, pppoe-actual-rate, rfc5515-actual-rate are not supported for CUPS profiles.
  - **ingress:** policer, agg-rate, pppoe-actual-rate, rfc5515-actual-rate are not supported for CUPS profiles.
- CUPS must not be combined with:
  - credit-control-policy.
  - egress bonding-selection.
  - host limits.
  - idle-timeout.
  - one-time-http-redirection.
- The properties **`aggregate-qos-statistics`**, **`alternate-sla-profile`**, and **`custom-statistics-group`** are not supported in SR OS Release 23.10.
  - The deployment should proceed successfully if these attributes are not included in the payload.

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2.80 Name of the Intent: subscriber-mgmt-sub-profile_msros_24-10-1_24-11

2.80.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750 SR, 7450 ESS SROS MD node version 24.10

2.80.2 Description:

The subscriber-mgmt-sub-profile_msros_24-10-1_24-11 intent supports the creation/modification/deletion/discovery/audit/alignment of Sub-profile-name as targetIdentifier.

The following configurations are supported per deployment:

```
-Description, Accounting Policy, Accu Stats Policy, Collect Stats,
Firewall Policy, Host Tracking Policy, HS SLA Mode, IGMP Policy, MLD
Policy, PIM Policy, Preference, Secondary Shaper Hashing, Sub Mcac
Policy, UPNP Policy, Volume Stats Type, Vport Hashing, Alternate Sub
Profile,
-Control:- local, Cups
-NAT:- Policy, Prefix List, Allow Bypass, Access Mode
-ANCP:- ANCP Policy
-Egress
   -Lag Per Link Hash:-Class, Weight
   -QoS
     -Hs Agg Rate (kilobps)
     -Hs Min Resv Bw (kilobps)
     -Hs Low Burst Max Class
     -Agg Rate:-Rate (kilobps), Min Resv Bw (kilobps), Adaptation Rule,
Burst Limit, Queue Frame Based Accounting
     -Encap Offset:-Type
     -Policer Control Policy
       -Policy Name
       -Overrides
         -Root
          -Max Rate (kilobps)
          -Priority MBS Thresholds
            -Min Thresh Separation
            -Priority:-Level, MBS Contribution
     -Scheduler Policy
       -Policy Name
       -Overrides
         -Scheduler:-Scheduler Name, Rate(PIR (kilobps), CIR
(kilobps))
-Ingress
  -QoS
     -Policer Control Policy
       -Policy Name
```

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```
        -Overrides
          -Root
           -Max Rate (kilobps)
           -Priority MBS Thresholds
              -Min Thresh Separation
              -Priority:-Level, MBS Contribution
     -Scheduler Policy
        -Policy Name
        -Overrides
          -Scheduler:-Scheduler Name, Rate(PIR (kilobps),CIR
(kilobps))
-Radius Accounting:-Policy, Session Optimized Stop
-SLA Profile Map
  -Use Direct Map AS Default
  -entry:-SLA Profile String, Sla Profile
-Session Limits
  -IPOE, Overall
  -PPPOE:-Local, LAC, Overall
  -L2TP:-LNS, LTS, Overall
-Host Limits
  -LAC Overall
  -Overall
  -IPv4:-ARP, DHCP, Overall, PPP
  -IPv6:-Overall, PD IPOE DHCP, PD Overall, PD PPP DHCP, WAN IPOE DHCP,
WAN IPOE SLAAC, WAN Overall, WAN PPP DHCP,WAN PPP SLAAC
```

CLI context: configure > subscriber-mgmt > sub-profile

Prerequisites:

- The following should be predefined:
  - Accounting Policy
  - Accu Stats Policy
  - Firewall Policy
  - Host Tracking Policy
  - IGMP Policy
  - MLD Policy
  - PIM Policy
  - Sub Mcac Policy
  - UPNP Policy
  - NAT(Policy, Prefix List)
  - ANCP Policy
  - Policer Control Policy(Policy Name)
  - Scheduler Policy(Policy Name)
  - Radius Accounting(Policy)
  - SLA Profile Map(Sla Profile)

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2.80.3 Advisory:

- CUPS control cannot be combined with:
  - NAT
  - radius-accounting
  - collect-stats
  - SLA-profile-map
  - ANCP Policy
  - Accu Stats Policy
  - Sub Mcac Policy
  - secondary-shaper-hashing
  - host-tracking-policy
  - Firewall policy
  - PIM policy
  - Accounting Policy
- Sessions-Overall limit should be larger than IPOE session
- PPPOE-Sessions-overall limit should be larger than PPPOE-LAC-Session
- L2TP-Sessions-overall limit should be larger than L2TP-INS-Session
- IPV4-Overall limit should be larger than IPV4-DHCP, ARP, PPP
- IPV6-Overall limit should be larger than IPV6-WAN-Overall, IPV6-PD-Vverall
- IPV6-PD-Overall limit should be larger than IPV6-PD-IPOE-DHCP
- IPV6-PD-Overall limit should be larger than IPV6-PD-PPP-DHCP
- IPV6-WAN-Overall limit should be larger than IPV6-WAN-PPP-DHCP
- IPV6-WAN-Overall limit should be larger than IPV6-WAN-PPP-SLAAC
- IPV6-WAN-Overall limit should be larger than IPV6-WAN-IPOE-SLAAC
- Overall limit should be larger than IPV4-Overall in Host limit
- PIM policy and nat/firewall policy are mutually exclusive
- Cannot have both egress scheduler-policy and egress agg-rate-limit configured
- NAT policy is required to use UPNP policy
- Control local must not be false when CUP is configured true
- Host Tracking Policy and IGMP Policy are mutually exclusive
- Host Tracking Policy and MLD Policy are mutually exclusive
- Host Tracking Policy and Sub Mcac Policy are mutually exclusive
- PIM policy and Sub Mcac Policy are mutually exclusive
- PIM policy and MLD policy are mutually exclusive
- PIM policy and IGMP policy are mutually exclusive
- PIM policy and Host tracking policy are mutual exclusive
- The following attributes are not supported on Release 23.10 nodes. The deployment proceeds successfully if the attributes are not in the payload.
  - Alternate Sub Profile

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2.81 Name of the Intent: system-lldp_msros_23-10-1_23-11

2.81.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS MD SR OS node version 23.10

2.81.2 Description

system-lldp_msros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of System-LLDP with "lldp" as target-identifier.

Below configurations are supported per deployment:

```
- Admin State
- Tx Credit Max
- Message Fast Tx
- Message Fast Tx Init
- Tx Interval
- Tx Hold Multiplier
- Reinit Delay
- Notification Interval
```

CLI context: configure > system > lldp

2.81.3 Advisory

N/A

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2.82 Name of the Intent: system-login-control_msros_23-10-1_23-11

2.82.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SR OS node version 23.10

2.82.2 Description:

The system-login-control_msros_23-10-1_23-11 intent supports the creation/modification/discovery/audit/alignment of system login control, using login-control as the identifier.

The following configurations are supported per deployment:

```
- Exponential Backoff, Idle Timeout, Login Banner.
- Login Scripts : Global Script, Per User Script [User Directory, File
Name].
- MOTD : Text or URL.
- Pre Login Message : Message, Name.
- FTP : Inbound Sessions.
- SSH : Graceful Shutdown, Inbound Max Sessions, Outbound Max Sessions,
TTL security.
- Telnet : Graceful Shutdown, Inbound Max Sessions, Outbound Max
Sessions, TTL security.
```

CLI context: configure > system > login-control

Pre-requisite: N/A

2.82.3 Advisory:

- The operator must enter login-control as the target identifier; otherwise, the deployment will fail.
- To configure SSH and Telnet TTL security, per-peer queuing must be enabled.
- IXR systems does not support per-peer queuing; hence, SSH and Telnet TTL security are not supported.
- To configure a per-user script, both the user directory and file name must be specified.

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3 SRL intent types

3.1 Name of the Intent: acl-ipv4-filter_srl_23-10-1_24-4

3.1.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.1.2 Description

The acl-ipv4-filter_srl_23-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a IPv4 filter.

The following configurations are supported per deployment:

```
- Create IPv4 Filter with Name
- Configuration of Description, Statistics Per Entry Admin State,
Subinterface Specific [Disabled, Input-Only, Output-Only,
Input-and-Output]
- Create Entry with Sequence ID
  - Description
  - Action [Accept/Drop]
      - Accept with Rate Limit or
      - Accept/Drop with Logging
    - Match
      - Fragment, First Fragment
      - Protocol, TCP Flags
      - Destination IP [Prefix, Address, Mask]
      - Source IP [Prefix, Address, Mask]
      - Destination Port
        - [Operator, Value, Range [Start, End]]
      - Source Port
        - [Operator, Value, Range [Start, End]]
```

Prerequisites:

- If the Policer is used, then the Policer Policy in ACL must be configured on the SRL node.

3.1.3 Advisory

- The 7220 IXR family SRL node does not support Accept Action with the Logging command from the node.
- The 7220 [IXR-H2/H3/H4] SRL node does not support the DSCP-Set command from the node.
- The 7220 [IXR-H2/H3/H4] and 7250 IXR family SRL node do not support the Rate-Limit command from the node.
- Host bits of the Source and Destination IP Prefix should be set to 0.

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- Sample Config for TCP Flags are "rst&ack|syn", "!!rst|ack&syn", "!!rst&ack", etc.
- The user has to ensure that the Destination Port Range start value/enumeration is smaller than the end value/enumeration.
- The dscp-set supported values are "range [0 to 63] | [AF11 | AF12 | AF13 | AF21 | AF22 | AF23 | AF31 | AF32 | AF33 | AF41 | AF42 | AF43 | CS0 | CS1 | CS2 | CS3 | CS4 | CS5 | CS6 | CS7 | EF | LE]".

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3.2 Name of the Intent: acl-ipv6-filter_srl_23-10-1_24-4

3.2.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.2.2 Description

acl-ipv6-filter_srl_23-10-1_24-4 intent supports creation/modification/deletion/audit/alignment/discovery of IPV6 Filter configuration.

The following configurations are supported per deployment:

```
- The DSCP Set range [0 to 63] [AF11 AF12 AF13 AF21 AF22 AF23 AF31 AF32
AF33 AF41 AF42 AF43 CS0 CS1 CS2 CS3 CS4 CS5 CS6 CS7 EF LE]
- Subinterface Specific [Disabled, Input-Only, Output-Only,
Input-and-Output]
- Statistics Per Entry
- Action [Accept Drop]
- Next Header
- Source-IP and Destination-IP
```

Prerequisites:

The ACL Policer must be configured on the node.

3.2.3 Advisory

- Deployment misalignment occurs on the audit of IPv6 Address with the combination of IPv4 and IPv6. For example, <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d> e.g. ::ffff:192.168.0.1 and ::ffab:c0a8:1.
- The DSCP Set and Rate Limit in Action Criteria are not supported for 7220 IXR-H nodes.
- The filter must be Subinterface-Specific "Input-and-Output" when Policer Scope Subinterface is used.
- The filter must be Subinterface-Specific "Disabled" when Policer Scope Global is used.
- It is always advisable to use the compressed format of IPv6 Address to avoid misaligned attributes in the audit report.

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3.3 Name of the Intent: acl-mac-filter_srl_23-10-1_24-4

3.3.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL) SRL node version 23.10

3.3.2 Description

acl-mac-filter_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of ACL-MAC filter policy.

The following configurations are supported per deployment:

```
- Description, Subinterface Specific, Statistics Per Entry
- Entry [Sequence ID, range 0 to 65535], Description, Action [Accept,
Drop {Log}], Match [Ethertype, Destination MAC {Address, Mask}, Source
MAC {Address, Mask}]
  - VLAN Outermost VLAN ID: Range with One Value - Value, Operator
                Range with Two Values - Range [Start, End]
                Special Value None To Match Untagged - None
```

3.3.3 Advisory

- Not supported on SRL H and D1 systems.
- An/A Address/Mask must be configured for use with the Mask/Address.
- Address/Mask must be configured in the format XX:XX:XX:XX:XX:XX, where X is (0-F).

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3.4 Name of the Intent: acl-system-filter_srl_23-10-1_24-4

3.4.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL) SRL node version 23.10

3.4.2 Description

acl-system-filter_srl_23-10-1_24-4 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of filter configuration under acl contest for SRL nodes with fixed targetIdentifier as "system-filter".

The following configurations are supported per deployment:

```
- IPv4 Filter Entry:
  - Sequence ID, Description
  - Action (Drop/Accept)
  - Match:
    - Fragment, First Fragment, Protocol, DSCP Set, ICMP Type, ICMP Code,
TCP Flags
    - Destination IP Prefix, Destination IP Address, Destination IP Mask
    - Source IP Prefix, Source IP Address, Source IP Mask
    - Destination Port Range, Start and End, Value, Operator
    - Source Port Range, Start and End, Value, Operator
- IPv6 Filter Entry:
  - Sequence ID, Description
  - Action (Drop/Accept)
  - Match:
    - Fragment, First Fragment, Next Header, DSCP Set, ICMPv6 Type,
ICMPv6 Code
    - Destination IP Prefix, Destination IP Address, Destination IP Mask
    - Source IP Prefix, Source IP Address, Source IP Mask
    - Destination Port Range, Start and End, Value, Operator
    - Source Port Range, Start and End, Value, Operator
```

3.4.3 Advisory

- Deployment misalignment occurs on the audit of IPv6 address with the combination of IPv4 and IPv6. For example, <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d> as well as ::ffff:192.168.0.1 and ::ffab:c0a8:1.
- The user has to ensure that the Destination Port Range start value/enumeration should be smaller than the end value/enumeration.
- The System-Filter is not supported on SRL 7220 H-Series and 7250 6/6e/10/10e.
- Host bits of the Source and Destination IPv4/IPv6 Address/Prefix should be set to 0.
- The user has to ensure not to exhaust the resources reserved for ACL.
- Sample configuration for TCP Flags are "rst&ack|syn", "!!rst|ack&syn", "!!rst&ack", etc.

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- It is advisable to use the compressed format of IPv6 address to avoid misaligned attributes in the audit report.

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3.5 Name of the Intent: interface_srl_23-10-1_24-4

3.5.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.5.2 Description

interface_srl_23-10-1_24-4 Intent type deployment can be created on ethernet interfaces. Single intent supporting both connector port and physical port.

The following configurations are supported per deployment:

```
- Connector-Port/Breakout-Port Configuration:
  - Description, Admin State, MTU, Loopback Mode, VLAN tagging, TPID
  - Transceiver: TX-Laser, DDM-Events, Forward Error Correction
  - Ethernet: Auto-Negotiate, Duplex Mode, DAC Link Training, LACP Port
Priority, Port Speed, PTP Asymmetry, Standby-Signaling, Reload-Delay
    - Flow-Control: Receive. Hold-Time: Up, Down. Storm Control: Units,
Broadcast Rate, Multicast Rate, Unknown Unicast Rate. Synce: SSM Admin
State.
    - L2CP Transparency: Tunnel-All-L2CP, LLDP, LACP, DOT1X, PTP, XSTP
- Sflow: Admin State
- Breakout Mode: Number of Breakout Ports, Breakout Port Speed
- QoS Output
  - Scheduler: Tier <level&gt;, Node Number, Strict Priority, Weight
  - Queue: Queue Name, Template, Scheduling Peak Rate Percent

- Connector-Port/Breakout-Port configuration with LAG association:
  - Description, Admin State
  - Transceiver: TX-Laser, DDM-Events, Forward Error Correction
  - Ethernet: Auto-Negotiate, Duplex Mode, Aggregate ID, DAC Link
Training, Forwarding-Viable, LACP Port priority, Port Speed, PTP
Asymmetry, Reload-Delay
    Hold-Time: Up, Down. Storm Control: Units, Broadcast Rate, Multicast
Rate, Unknown Unicast Rate. Synce: SSM Admin State.
    L2CP Transparency: Tunnel-All-L2CP, LLDP, LACP, DOT1X, PTP, XSTP
- Sflow: Admin state

- To enable Breakout-Mode:
  - Description, Admin State
  - Breakout Mode: Number of Breakout Ports, Breakout Port Speed
```

3.5.3 Advisory

- Audit information will only be displayed for modified attributes that are present in the intent.
- Deletion of a container or object present in the intent will not result in the deletion of the Config present in the node.

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- Audit information for newly added objects that are not present in the intent will not be shown.
- Breakout Mode is not supported on D1/D2/D2L/H2 systems.
- Breakout Port Speed and Number of Breakout Ports should be selected to match with the interface speed.
- Ethernet interface on SR Linux devices operate at different speeds and the operator has to configure the appropriate speed.
- Port-Speed is not configurable for a Breakout Port.
- QoS: Queue Template Name is not configurable on Multicast Queues for H2/H3/H4 systems.
- QoS: Only one Scheduler Level is supported.
- LACP-based Standby-Signaling is supported only on LACP enabled LAG interfaces.
- To configure Tunnel-All-L2CP and LLDP Tunnel, the interface must have LLDP Admin-State disabled at system context (system LLDP Admin-State disable).
- To configure LACP Tunnel, the interface must not be part of an LACP enabled LAG.
- PTP Asymmetry and Synce are only supported on D5.
- Forwarding-Viable is configured only when Aggregate ID is enabled.

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3.6 Name of the Intent: interface_srl_24-10-1_25-4

3.6.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220/7250/7215/7730 SRL node version 24.10.1

3.6.2 Description

The interface_srl_24-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of interface configuration for SRL nodes with interface name as the targetIdentifier.

The following configurations are supported per deployment:

```
- Interface/Breakout-Port Configuration:
  - Description, Admin State, MTU, Loopback Mode, VLAN tagging, TPID, Num
Physical Channels
  - Transceiver: TX-Laser, DDM-Events, Forward Error Correction
  - Ethernet: Auto-Negotiate, Duplex Mode, DAC Link Training, LACP Port
Priority, Port Speed, PTP Asymmetry, Standby-Signaling, Reload-Delay, MAC
Address, Link Loss Forwarding.
    - Flow-Control: Receive. Hold-Time: Up, Down. Storm Control: Units,
Broadcast Rate, Multicast Rate, Unknown Unicast Rate, Rising Threshold
Action. Synce: SSM Admin State.
    - L2CP Transparency: Tunnel-All-L2CP, LLDP, LACP, DOT1X, PTP, XSTP,
Esmc, Elmi, Efm Oam
  - Sflow: Admin State
  - P4rt: ID


- Interface/Breakout-Port configuration with LAG association:
  - Description, Admin State, Loopback Mode, Num Physical Channels
  - Transceiver: TX-Laser, DDM-Events, Forward Error Correction
  - Ethernet: Aggregate ID, Forwarding-Viable, Auto-Negotiate, DAC Link
Training,  LACP Port priority, Port Speed, PTP Asymmetry, Reload-Delay,
MAC Address.
    - Hold-Time: Up, Down. Storm Control: Units, Broadcast Rate,
Multicast Rate, Unknown Unicast Rate, Rising Threshold Action. Synce: SSM
Admin State.
    - L2CP Transparency: Tunnel-All-L2CP, LLDP, LACP, DOT1X, PTP, XSTP,
Esmc, Elmi, Efm Oam.
  - Sflow: Admin state
  - P4rt: ID

- To enable Breakout-Mode (for breakout connector):
  - Description, Admin State
  - Breakout Mode: Number of Breakout Ports, Breakout Port Speed
```

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3.6.3 Advisory

- This intent doesn't support the lifecycle managemtn of System/Management interfaces.
- Breakout Mode is not supported on D1/D2/D2L/H2/A1 systems.
- Breakout Port Speed and Number of Breakout Ports should be configured to match the interface speed.
- Ethernet interface on SR Linux devices operate at different speeds; configure the appropriate speed.
- Port-Speed is not configurable for a Breakout Port.
- LACP-based Standby-Signaling is supported only on LACP enabled LAG interfaces.
- To configure Tunnel-All-L2CP and LLDP Tunnel, the interface must have LLDP Admin-State disabled at system context (system LLDP Admin-State disable).
- To configure LACP Tunnel, the interface must not be part of an LACP enabled LAG.
- Forwarding-Viable is configured only when Aggregate ID is enabled.
- vlan-tagging true and link-loss-forwarding can not be configured together.
- vlan-tagging/mtu/tpid and aggregate-id can not be configured together.
- flow-control receive and aggregate-id can not be configured together.
- standby-signaling and aggregate-id cannot be configured together.
- link-loss-forwarding and aggregate-id cannot be configured together.
- duplex-mode and aggregate-id can not be configured together.
- For more information on Feature Considerations for breakout connectors and ports see the SRL Interfaces Guide
- Creation of break-ports is applicable on different platforms on different port numbers which cannot be handled at intent level. For more information see the SRL Interfaces Guide.
- Handling of Mutually Exclusive Attributes During Creation: If mutually exclusive attributes are configured in the creation form and/or subsequently modified, then the current create form should be discarded. During Update: If the user intends to change the values of mutually exclusive attributes, the existing intent or configuration must be deleted and recreated with the new values.

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3.7 Name of the Intent: lag-interface_srl_23-10-1_24-4

3.7.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.7.2 Description

lag-interface_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of interface LAG.

The following configurations are supported per deployment:

```
- Create LAG with Name, such as LAGN. For example, LAG10
- Configuration of Description, Admin State, MTU, VLAN Tagging, TPID
[TPID_0X88A8/TPID_0X8100/TPID_0X9100/TPID_0X9200/TPID_ANY]
- Ethernet MAC, Standby Signaling [LACP/Power-Off], Flow Control
- P4RT ID
- QoS
    - Scheduler
      - Tier Node [Node Number, Strict Priority, Weight]
    - Queue
      - Queue Name
-LAG
  - LAG Type, Mini Links, LACP Fallback Mode, LACP Fallback Time
Out

-LACP
  - Interval, LACP Mode, System ID MAC, System Priority
```

3.7.3 Advisory

- The 7250 SRL family and 7220 [IXR-H2 and IXR-H4] SRL node do not support "Standby Signaling" node CLI.
- The 7250 SRL family SRL node does not support "TPID" node CLI.
- The 7220 SRL family SRL node does not support Ethernet MAC, P4RT attributes node CLI.
- LAG Member speed on SRL devices operate at different speeds; therefore, the operator has to configure the appropriate speed.
- The 7250 and 7220 SRL family SRL node supports QoS, only one Scheduler Level is supported from node CLI.
- The 7220 and 7250 family QoS Scheduler "Node Number" and QoS Queue-Name "Multicast" Number will be diffrent based on the platform.
- The 7250 family QoS Queue "Multicast" is not supported.
- Limitation of maximum number of LAGs per platform.
- Audit information will only be displayed for modified attributes that are present in the intent.

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- Deletion of a container or object present in the intent will not result in the deletion of the configuration present in the node.
- Audit information for newly added objects that are not present in the intent will not be shown.

lag<N>, N=1..1000 [note1] [note1] The maximum number of LAGs per platform is as follows: D1: 32 D2-D3: 128 D4-D5: 64 H2-H3: 127 H4: 255 IXR: 128

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3.8 Name of the Intent: netw-inst-default-interface_srl_23-10-1_24-4

3.8.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR(D/DL/H) SRL node version 23.10

3.8.2 Description

netw-inst-default-interface_srl_23-10-1_24-4 intent supports creation/modification/deletion/audit/alignment/discovery of default network instance configuration.

`Interface and subinterface reference configurations are supported per deployment.`

Prerequisites:

Interface and subinterface configuration on node.

3.8.3 Advisory

- Interface and subinterface listing is not shown when multiple targets are selected.

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3.9 Name of the Intent: qos-classifier-dot1p-policy_srl_23-10-1_24-4

3.9.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.9.2 Description

The qos-classifier-dot1p-policy_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of Dot1p QoS Policy.

The following configurations are supported per deployment:

```
- Configuration of the Dot1p Policy with the following attributes:
  - Value, Forwarding Class, Drop Probability:
    - Value
    - FC [FC Type [Default Forwarding Class [FC0 to FC7], String]
    - Drop Probability [Low, Medium, High]
```

3.9.3 Advisory

- The 7250 SRL family and 7220 [IXR-H2/H3 and IXR-H4, IXR-D1] SRL nodes do not support Dot1p.

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3.10 Name of the Intent: qos-classifier-dscp-policy_srl_23-10-1_24-4

3.10.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.10.2 Description

qos-classifier-dscp-policy_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of DSCP QoS Policy.

The following configurations are supported per deployment:

```
- DSCP Value (0 to 63), Forwarding-Class [Named Forwarding Classes and
Default Forwarding Class (FC0 to FC7)], and Drop-Probability
(High-Low-Medium).
```

3.10.3 Advisory

- Named Forwarding Classes is not supported for 7220 IXR (D/DL/H) SRL, it is supported for 7250 IXR.
- The DSCP policy is not supported on the 7220 IXE D1.

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3.11 Name of the Intent: qos-classifiers-dot1p-policy_srl_24-10-1_25-4

3.11.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220/7730 SRL node version 24.10.1

3.11.2 Description

The qos-classifier-dot1p-policy_srl_24-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a Dot1p QoS Classifier Policy.

The following configurations are supported per deployment:

```
- Configuration of the Dot1p Policy with the following attributes:
    - Value
    - Forwarding Class
    - Drop Probability
    - Profile
    - De Out Profile
    - IP Rewrite Policy
```

CLI Context: qos classifiers

Prerequisites:

- To reference a forwarding-class in any QoS policy, the forwarding-class must first be explicitly mapped to an output queue. Example:

```
qos {
        forwarding-classes {
            forwarding-class fc_22 {
                forwarding-class-index 10
                output {
                    queue queue-7
                }
            }
        }
    }
```

3.11.3 Advisory

- The 7250 SRL family, 7215 IXS-A1 and 7220 [IXR-H2/H3/H4, IXR-D1] SRL does not support Dot1p QoS Classifier Policy.
- Since auto-suggestion is not supported for the 'Forwarding Class' attribute, you must enter a value manually.

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3.12 Name of the Intent: qos-classifiers-dscp-policy_srl_24-10-1_25-4

3.12.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220/7250/7730 SRL node version 24.10.1

3.12.2 Description

The qos-classifiers-dscp-policy_srl_24-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a Dscp QoS Classifier Policy.

The following configurations are supported per deployment:

```
    - Value:
      - DSCP Value
        - 0 to 63
      - DSCP Name
        - AF11, AF12, AF13, AF21, AF22, AF23, AF31, AF32, AF33, AF41,
AF42, AF43,CS0, CS1, CS2, CS3, CS4, CS5, CS6, CS7,EF, LE
    - Forwarding Class
    - Drop Probability:
        - low
        - medium
        - high
    - Profile:
      - in
      - out
      - exceed
      - in-plus
      - in-low
      - out-low
    - De out profile
    - Ip Rewrite Policy
```

CLI Context: qos classifiers dscp-policy

Prerequisites:

- Configure qos forwarding-classes forwarding-class
- Configure qos rewrite-rules ip-rewrite-policy
- To reference a forwarding-class in any QoS policy, the forwarding-class must first be explicitly mapped to an output queue. Example:

```
qos {
        forwarding-classes {
            forwarding-class fc_22 {
                forwarding-class-index 10
                output {
                    queue queue-7
```


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```
                    }
                }
            }
        }
```

3.12.3 Advisory

- DSCP policy is not supported for 7220 IXR-D1 and 7215 IXS-A1.
- Profile, IP Rewrite Policy, De Out Profile attributes are not supported for 7220 IXR-D3L, 7250 IXR-X1B, 7250 IXR-6e, 7250 IXR-X1B, 7250 IXR-X3B and 7220 IXR-D5.
- Drop Probability is not supported for 7730 SXR-1x-44S and 7730 SXR-1d-32D.
- Since auto-suggestion is not supported for the 'Forwarding Class' attribute, the user is required to enter a value manually.

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3.13 Name of the Intent: qos-multifield-classifier_srl_23-10-1_24-4

3.13.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL) SRL node version 23.10 except 7220 IXR-D1

3.13.2 Description

qos-multifield-classifier_srl_23-10-1_24-4 intent supports creation/modification/deletion/audit/alignment/discovery of Multifield Classifier configuration.

The following configurations are supported per deployment:

```
- Ipv4 or Ipv6 Classifier Type
- Match Criteria: DSCP Set, Fragment, Source/Destination IP,
Source/Destination Port, Protocols
- Action Criteria: Forwarding Class, Rewrite DSCP, and Drop Probability
```

3.13.3 Advisory

- ICMP Type/Code should be used for the Ipv4 Policy and ICMP6 Type/Code should be used for the Ipv6 Policy.
- The audit will show the misaligned attribute for the same protocol name and protocol number. For example, the audit will show a misaligned attribute if the intent has a protocol number of 6 and the node is configured with protocol TCP.
- The audit will show the misaligned attribute for the IPv4 Address mapped with the IPv6 Address. For example, 2001:db8::1234:5678:192.168.2.1 and 2001:db8::1234:5678:c0a8:201.
- It is always advisable to use the compressed format of the IPv6 Address to avoid any audit mismatch.

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3.14 Name of the Intent: qos-multifield-ipv4-policy_srl_23-10-1_24-4

3.14.1 Applicability

Supported from: NSP 24.4, 24.8 and 24.11. Intent is not supported from NSP 25.4 onwards.

Verified on Node Type/Version: 7220 IXR (D/DL) SRL node version 23.10 except 7220 IXR-D1

3.14.2 Description

The qos-multifield-ipv4-policy_srl_23-10-1_24-4 intent supports creation/modification/deletion/audit/alignment/discovery of Multifield Ipv4 Policy configuration.

The following configurations are supported per deployment:

```
- Match Criteria: Fragment and First Fragment
- Match Criteria: Protocol
- Match Criteria: DSCP Set Range [0 to 63] [AF11 AF12 AF13 AF21 AF22 AF23
AF31 AF32 AF33 AF41 AF42 AF43 CS0 CS1 CS2 CS3 CS4 CS5 CS6 CS7 EF LE]
- Match Criteria: Source-IP and Destination-IP
- Action [Forwarding Class and Drop Probability]
- Next Header
- Rewrite [Set DSCP]
```

3.14.3 Advisory

- The audit will show the misaligned attribute for the same protocol name and protocol number. For example, the audit will show a misaligned attribute if the intent has a protocol number of 6 and the node is configured with protocol TCP.

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3.15 Name of the Intent: qos-multifield-ipv6-policy_srl_23-10-1_24-4

3.15.1 Applicability

Supported from: NSP 24.4, 24.8 and 24.11. Intent is not supported from NSP 25.4 onwards.

Verified on Node Type/Version: 7220 IXR (D/DL) SRL node version 23.10

3.15.2 Description

The qos-multifield-ipv6-policy_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of QoS Multifield IPv6 Policy.

The following configurations are supported per deployment:

```
- Entry [Sequence ID, Range from 0 to 65535]
  - Match: Next Header, DSCP Set, Source IP [Prefix, Address, Mask],
Destination IP [Prefix, Address, Mask]
  - Action: Forwarding Class, Drop Probability, Rewrite [Set DSCP]
```

3.15.3 Advisory

- Not supported on SRL H and D1 systems.
- An/A Address/Mask must be configured to use with the Mask/Address.
- Prefix must be configured in xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx/d, where x is (0-9, a-f) and d is (1-128).
- Address/Mask must be configured in xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx, where x is (0-9, a-f).
- The user must ensure that the Source/Destination Port Range "start" value or enumeration is smaller than the "end" value or enumeration.
- Sample Config for TCP Flags are "rst&ack|syn", "!!rst|ack&syn", "!!rst&ack", etc.
- It is advisable to use the compressed format of the IPv6 Address to avoid misaligned attributes in the audit report.
- The audit will show the misaligned attribute for the same protocol name and protocol number. For example, the audit will show misaligned attributes if the intent has a protocol number of 6 and the node is configured with protocol TCP.
- DSCP-Set needs to be added manually. The DSCP Set range is from 0 to 63 [AF11 AF12 AF13 AF21 AF22 AF23 AF31 AF32 AF33 AF41 AF42 AF43 CS0 CS1 CS2 CS3 CS4 CS5 CS6 CS7 EF LE].

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3.16 Name of the Intent: qos-policer-template_srl_23-10-1_24-4

3.16.1 Applicability

Supported from: NSP 24.4, 24.8 and 24.11. Intent is not supported from NSP 25.4 onwards.

Verified on Node Type/Version: 7220 IXR (D/DL) SRL node version 23.10 except 7220 IXR-D1

3.16.2 Description

The qos-policer-template_srl_23-10-1_24-4 intent supports the creation/modification/deletion/audit/alignment/discovery of Policer Template configuration.

The following configurations are supported per deployment:

```
- Statistics Mode
- Peak Rate (Kbps), Committed Rate (Kbps), Maximum Burst Size (Bytes),
Committed Burst Size (Bytes)
- Forwarding Class and Forwarding Type
- Exceed Action Drop Probability [Low/Medium/High]
- Violate Action Drop Probability/Recolor [Green/Yellow/Red]
```

3.16.3 Advisory

- Forwarding Type "Multicast" is not supported for 7220 IXR-D4.
- The maximum number of supported QoS Policer instances (Sequence IDs) is 255.

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3.17 Name of the Intent: qos-policer-template_srl_24-10-1_25-4

3.17.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220 IXR-D2/D2L/D3/D3L/D4/D5 SRL node version 24.10.1

3.17.2 Description

The qos-policer-template_srl_24-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of QoS policer templates.

The following configurations are supported per deployment:

```
- Policer Templates:
    - statistics-mode
        - violating-focus
        - forwarding-focus
    - Policer:
        - peak-rate-kbps
        - committed-rate-kbps
        - maximum-burst-size
        - committed-burst-size
        - Forwarding Class
            - forwarding-type
                - unknown-unicast
                - broadcast
                - unknown-multicast
                - unicast
                - multicast
        - exceed-action
            - drop-probability
                - high
                - low
                - medium
        - violate-action
            - drop
            - drop-probability
                - high
                - low
                - medium
```

CLI Context: qos policer-templates policer-template

Prerequisites:

- Configure qos forwarding-classes forwarding-class
- To reference a forwarding-class in any QoS policy, the forwarding-class must first be explicitly mapped to an output queue. Example:

```
qos {
```

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```
        forwarding-classes {
            forwarding-class fc_22 {
                forwarding-class-index 10
                output {
                    queue queue-7
                }
            }
        }
    }
```

3.17.3 Advisory

- This intent-type is not supported for 7215 IXS-A1, 7730 SXR, 7250 IXR and 7220 IXR [H2/H3/H4/D1] nodes.
- Since auto-suggestion is not supported for the 'Forwarding Class' attribute, the user is required to enter a value manually.

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3.18 Name of the Intent: qos-queue-template_srl_23-10-1_24-4

3.18.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.18.2 Description

qos-queue-template_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/align of Queue-templates on SRL systems.

The following configurations are supported per deployment:

```
- Queue Depth: Maximum Burst Size, High Threshold Bytes
- Active Queue Management:
  - Weight Factor
  - WRED Slope: Drop Probability, Traffic Type, Min Threshold Percent,
Max Threshold Percent, Max Probability
  - ECN Slope: ECN Drop Probability, Min Threshold Percent, Max Threshold
Percent, Max Probability
```

3.18.3 Advisory

- The Queue Template policy is not supported on the D1 system.
- High Threshold Bytes is not supported on the D4/D5/H4 systems.
- Traffic Type "all" is supported only for the 7250 (6/6e/10/10e) systems.
- ECN Drop Probability "all" is supported only for the 7250 (6/6e/10/10e) systems.
- The maximum number of queue templates per system varies by platform.
  - On the 7250 IXR systems, the maximum is eight queue templates.
  - On the 7220 IXR systems, the maximum is 62 queue templates.

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3.19 Name of the Intent: qos-rewrite-rules-dot1p-policy_srl_23-10-1_24-4

3.19.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.19.2 Description

The qos-rewrite-rules-dot1p-policy_srl_23-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of rewrite Dot1p QoS policies.

The following configurations are supported per deployment:

```
- Create Rewrite Dot1p QoS Policy
- Configuration of the Rewrite Dot1p Policy with the following
attributes:
  - MAP:
    - Forwarding Class [FC0 to FC7], Dot1p
    - Drop Probability [Low, Medium, High], Dot1p
```

3.19.3 Advisory

- The 7220 IXR [H2/H3/H4/D1] and 7250 IXR family SRL node does not support Rewrite Dot1p.
- The Dot1p QoS Policy cannot be changed to be blank/empty from the NSP GUI.

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3.20 Name of the Intent: qos-rewrite-rules-dot1p-policy_srl_24-10-1_25-4

3.20.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220/7730 SRL node version 24.10.1

3.20.2 Description

The qos-rewrite-rules-dot1p-policy_srl_24-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a rewrite Dot1p QoS Policy.

The following configurations are supported per deployment:

```
- Forwarding Class
- Drop Probability: Drop Probability, Dot1p
- Profile: Profile, Inner Dot1p, Outer Dot1p, Inner De, Outer De
- Dot1p
- Inner Dot1p
- Outer Dot1p
- Inner De
- Outer De
```

CLI Context: qos rewrite-rules

Prerequisites:

- To reference a forwarding-class in any QoS policy, the forwarding-class must first be explicitly mapped to an output queue.
- Example:

```
qos {
        forwarding-classes {
            forwarding-class fc_22 {
                forwarding-class-index 10
                output {
                    queue queue-7
                }
            }
        }
    }
```

3.20.3 Advisory

- The 7250 SRL family, 7215 IXS-A1 and 7220 IXR [H2/H3/H4/D1] SRL does not support Rewrite Dot1p.
- Since auto-suggestion is not supported for the 'Forwarding Class' attribute, you must enter a value manually.

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- For SXR nodes, a policy named 'default' is present as read-only. While it is listed for 'Associate', it cannot be associated because the corresponding object is unavailable for configuration.

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3.21 Name of the Intent: qos-rewrite-rules-dscp-policy_srl_23-10-1_24-4

3.21.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.21.2 Description

The qos-rewrite-rules-dscp-policy_srl_23-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of QoS Rewrite DSCP policies.

The following configurations are supported per deployment:

```
- MAP:
  - Forwarding Class [Named Forwarding Class and Default Forwarding Class
(FC0 to FC7)] and DSCP [Value and Name]
    - Drop Probability [Low, Medium, High] and DSCP [Value and Name]
```

3.21.3 Advisory

- Named Forwarding Class is not supported for 7220 IXR (D/DL/H) SRL, it is supported for 7250 IXR.
- Rewrite Rules DSCP Policy is not supported on the D1 system.

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3.22 Name of the Intent: qos-rewrite-rules-dscp-policy_srl_24-10-1_25-4

3.22.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220/7250/7730 SRL node version 24.10.1

3.22.2 Description

The qos-rewrite-rules-dscp-policy_srl_24-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of the rewrite-rules of a QoS Policy.

The following configurations are supported per deployment:

```
    - DSCP:
      - DSCP Value
        - 0 to 63
      - DSCP Name
        - AF11, AF12, AF13, AF21, AF22, AF23, AF31, AF32, AF33, AF41,
AF42, AF43,CS0, CS1, CS2, CS3, CS4, CS5, CS6, CS7,EF, LE
    - Forwarding Class
    - Drop Probability:
        - low
        - medium
        - high
    - Profile:
      - in
      - out
      - exceed
      - in-plus
      - in-low
      - out-low
```

CLI Context: qos rewrite-rules dscp-policy

Prerequisites:

- Configure qos forwarding-classes forwarding-class
- To reference a forwarding-class in any QoS policy, the forwarding-class must first be explicitly mapped to an output queue.
- Example

```
qos {
        forwarding-classes {
            forwarding-class fc_22 {
                forwarding-class-index 10
                output {
                    queue queue-7
                }
```

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```
                    }
                }
            }
```

3.22.3 Advisory

- DSCP policy is not supported for 7220 IXR-D1 and 7215 IXS-A1.
- Profile attribute is not supported for 7220 IXR-D3L, 7250 IXR-X3B, 7250 IXR-X1B, 7250 IXR-6e, 7220 IXR-H2, 7220 IXR-D5.
- Drop Probability is not supported for 7730 SXR-1x-44S and 7730 SXR-1d-32D.
- Since auto-suggestion is not supported for the 'Forwarding Class' attribute, the user is required to enter a value manually.

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3.23 Name of the Intent: routing-as-path-set_srl_23-10-1_24-4

3.23.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.23.2 Description

routing-as-path-set_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of AS path set configuration.

The following configuration is supported per deployment:

```
- Expression (Length 1 to 65535)
- Expression (Regular Expression supports '|','*','?','+','( )','[
]','-','{}')
```

Sample Expressions:

1. 11* (Zero or more occurrences of AS number 11), for example, Null AS path/11/11 11/11 11 11 ....
2. 11+ 22+ 33+ (One or more occurrences of AS number 11, followed by one or more occurrences of AS number 22, followed by one or more occurrences of AS number 33)

For example, 11 22 33

```
11 11 22 33

11 11 22 22 33

11 11 22 22 33 33

11 ... 11 22 ... 22 33 ... 33
```

1. 11{1,4} 22 (One through four occurrences of AS number 11 followed by one occurrence of AS number 22)

For example, 11 22

```
11 11 22

11 11 11 22

11 11 11 11 22
```

3.23.3 Advisory

- Deployment deletion fails if the As-Path Set is attached to any routing policy, for example BGP.

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3.24 Name of the Intent: routing-community-set_srl_23-10-1_24-4

3.24.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.24.2 Description

routing-community-set_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of Community Set Policy.

The following configurations are supported per deployment:

```
- Addition of Community Members
  - A standard BGP Community Value, Regular Expression or Well-Known Name
or a large BGP Community Value or Regular Expression.
```

3.24.3 Advisory

- The template will have one default entry of Community Member, to support greenfield deployment. The user can delete this default entry during configuration. At least one entry should be present in the form for successful deployment.

Examples of Community Members Type:

1. BGP-STD-Community-Well-Known-Type: No-Advertise, No-Export, No-Export-Subconfed.
2. BGP-STD-Community-Type: Represents the full range of valid port numbers in the format of "XXXX:XXXX", where each X can be any digit from 0 to 9. Ensure that it ranges from 0 to 65535. For example: {65529:65530, 0:0}.
3. BGP-Large-Community-Type: Expression is expected to match three numbers separated by colons, matching numbers ranging from 0 to 4294967295. For example: {4294967295:2147483647:1000000000, 0:0:0}.
4. BGP-Ext-Community-Type: Represents Route Target, Route Origin and Color Tag. For example: {target:65535:429496729, target:192.168.0.1:65535, target:4294000000:1, origin:1:1, origin:172.16.254.3:4294967290, origin:429496729:655, color:11:4294967295}.
5. BGP-Ext-Community-Regexp-Type: Regular Expression patterns to define policies or configurations based on specific types of BGP Extended Communities. For example: {target:65000:100, origin:12345:231, bgp-tunnel-encap:VXLAN, blackhole:123:432}.
6. BGP-Large-Community-Regexp-Type.
7. BGP-STD-Community-Regexp-Type.

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3.25 Name of the Intent: routing-community-set_srl_24-10-1_25-4

3.25.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220/7250/7215/7730 SRL node version 24.10.1

3.25.2 Description

routing-community-set_srl_24-10-1_25-4 intent supports creation/modification/deletion/discovery/audit/alignment of community-set configuration under routing-policy context for SRL nodes with community-set name as the targetIdentifier.

The following configurations are supported per deployment:

```
- match-set-options (below is the list of supported enum values)
  - any
  - all
  - invert
- member (multiple member values are supported to be configured)
  - no-export
  - no-advertise
  - no-export-subconfed
  - string
```

CLI Context:

routing-policy

Prerequisites:

- N/A

3.25.3 Advisory

- The SRL NE CLI match-set-options context has default value of "all", however this default behavior is removed in this intent.
- The 'string' in the member can bean extended bgp community value or regular expression.
- See the SRL documentation for more information about supported community-set member formats.

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3.26 Name of the Intent: routing-policy_srl_23-10-1_24-4

3.26.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.26.2 Description

routing-policy_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of policy configuration under routing-policy contest for SRL nodes with policy name as the targetIdentifier.

The following configurations are supported per deployment:

```
- Default-Action:
  - BGP
    - As-Path Remove/Replace, Prepend (As-Number Repeat-N)
    - Communities Add/Remove/Replace
    - Local-Preference Set
    - MED Set
    - Origin Set
- Statement:
  - Name
  - Match
    - Prefix-Set, EVPN Route-Type, Family, Protocol
    - BGP
    - As-Path-Set (As-Path-Length, Value, Operator Unique):
      - Community-Set
    - ISIS (Level, Route-Type)
    - OSPF (Area-ID, OSPF-Area-Identifier, Route-Type, Instance-ID)
    - Action:
      - Policy-Result, BGP, As-Path (Remove/Replace, Prepend (As-Number,
Repeat-N)), Communities (Add/Remove/Replace), Local-Preference Set, MED
Set, Origin Set
```

3.26.3 Advisory

- The "Default-Action" for the "As-Path" can be configured to "Remove" or "Replace" configuration.
- The Statement Action As-Path can have "Remove" or "Replace" configured.
- The Family attribute only supports EVPN, IPv4-Unicast, and IPv6-Unicast on 7220 IXR.

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3.27 Name of the Intent: routing-prefix-set_srl_23-10-1_24-4

3.27.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.27.2 Description

routing-prefix-set_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of Prefix-set Policy.

The following configurations are supported per deployment:

```
- IP Prefix {IPv4 and IPv6} and Mask Length Range
```

3.27.3 Advisory

- IPv6 Prefix Set must be configured in xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx/d, where x is (0-9, a-f) and d is (1-128).
- Host bits of the prefix must be 0.

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3.28 Name of the Intent: routing-prefix-set_srl_24-10-1_25-4

3.28.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220/7250/7215/7730 SRL node version 24.10.1

3.28.2 Description

The routing-prefix-set_srl_24-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of Prefix Set configuration under routing-policy context for SRL nodes with the prefix-set name as the targetIdentifier.

The following configurations are supported per deployment:

`- IP Prefix {IPv4|IPv6} and Mask Length Range`

CLI Context:

routing-policy

Prerequisites:

- NA

3.28.3 Advisory

Ensure you follow RFC 5952 while providing IPv6 address format.

- example 1: 2001:db8:0:0:0:0:2:1 should be 2001:db8::2:1
- example 2: 2001:DB8::1 should be 2001:db8::1

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3.29 Name of the Intent: routing-tag-set_srl_24-3-1_24-4

3.29.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR-D3L SRL node version 24.3

3.29.2 Description

routing-tag-set_srl_24-3-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of Tag Set policy.

The following configurations are supported per deployment:

```
- Create Tag Set with Name
- Configuration of Tag Value
```

3.29.3 Advisory

- Tag Set is supported from 7220 IXR SRL node version 24.3.1 or later.
- Maximum of two Tag values are supported.
- Tag value supports the following format and the values will be [Number, range 1 to 4294967295]|[String, length 1 to 11].
- String: A or a to F or f
- Format: <XX>|<XX:XX>|<XX:XX:XX>|<XX:XX:XX:XX> XX: Integer Number 1 to 4294967295 XX: Character a to f or A to F XX: both

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3.30 Name of the Intent: subinterface-routed_srl_23-10-1_24-4

3.30.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7220 IXR (D/DL/H) SRL node version 23.10

3.30.2 Description

subinterface-routed_srl_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of subinterface of type routed on ethernet and lag interface.

The following configurations are supported per deployment:

```
- Type, Description, Admin State, IP-MTU
- IPv4:
  - Admin State, Allow Directed Broadcast, Address IP Prefix and Primary
  - DHCP Server, Admin State
  - ARP:
    - Duplicate Address Detection, Time Out, Learn Unsolicited, Proxy
ARP, Neighbor IPv4 Address and Link Layer Address
    - Host Route:
      - Populate Route Type, Datapath Programming
    - Unnumbered:
      - Admin State, Interface
- IPv6:
  - Address IP Prefix, Type, Primary
  - DHCPv6 Server, Admin State
  - Router Advertisement:
    - Router Role:
      - Admin State, Current Hop Limit, IP MTU, Managed Configuration
Flag, Other Configuration Flag, Max Advertisement Interval Timeout, Min
Advertisement Interval Timeout, Prefix, IPv6 Prefix, Autonomous Flag, On
Link Flag, Preferred Lifetime, Valid Lifetime
    - Neighbor Discovery:
      - Duplicate Address Detection, Reachable Time, Stale Time, Learn
Unsolicited, Proxy Nd
    - Host Route:
      - Populate Route Type, Datapath Programming
    - Limit:
      - Max Entries, Log Only, Warning Threshold Percentage, Neighbor
IPv6 Address and Link Layer Address
- VLAN:
  - Encap Single Tagged, VLAN ID
- ACL:
  - Input:
    - IPv4 Filter, IPv6 Filter
  - Output:
    - IPv4 Filter, IPv6 Filter, and MAC Filter
```

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```
- Ra Guard
- QoS:
  - Input:
    - Classifiers - DSCP Policy
    - Multifield - IPv4 Policy, IPv6 Policy
    - Policer Template
  - Output:
    - Rewrite Rules - DSCP Policy
```

3.30.3 Advisory

- Cannot enable both DHCP server and DHCP relay on the same subinterface.
- MAC ACLs and an IPv6 MFC Policy are mutually exclusive on the same subinterface.
- IP Address should not be configured when Unnumbered Admin State is set to enable.
- Only one IPv6 link local Address/Prefix is allowed per subinterface.
- VLAN Tagging True must be configured to create more than one subinterface on an interface.
- VLAN ID cannot be set in the subinterface if VLAN-Tagging is not enabled on the parent interface.
- QoS Input Classifiers Multifield IPv4-Policy/IPv6-Policy are not applicable on the IXR-H series.
- QoS Input and Output attributes are not applicable on IXR-D1.
- Unnumbered IPv4 is not applicable on the IXR-H series.
- IPv4 ARP Host Route -> Route Type 'EVPN' is not supported on the IXR-H series.
- Ra-Guard and MAC-Policy are not applicable on the IXR-H series.
- IPv4-Filter/IPv6-Filter/MAC-Policy/RA-Guard/Classifier DSCP Policy/Policer Template, and Rewrite Rules DSCP Policy should be preconfigured on the node.
- The MAC Output Filter cannot be configured when /ACL/Egress-MAC-Filtering is false.

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3.31 Name of the Intent: subinterface-routed_srl_24-10-1_25-4

3.31.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7220/7250/7215/7730 SRL node version 24.10.1

3.31.2 Description

The subinterface-routed_srl_24-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of routed-type subinterface configuration under Interface context of type Ethernet & Lag for SRL nodes with interface name and subinterface index as the targetIdentifier.

The following configurations are supported per deployment:

```
- Type, Description, Admin State, IP-MTU, MPLS-MTU,
Collect-detailed-stats, unidirectional-link-delay static-delay
- IPv4:
    - Admin State, Allow Directed Broadcast, Address IP Prefix and
Primary
    - Unnumbered: Admin state, ineterface
    - VRRP:
        - vrrp-group, admin-state, priority, preempt, preempt-delay,
init-delay, accept-mode, advertise-interval, master-inherit-interval
          version, virtual-address, authentication keychain,
interface-tracking track-interface priority-decrement
    - ARP:
        - duplicate-address-detection, timeout, learn-unsolicited,
proxy-arp, debug, neighbor <prefix> link-layer-address
        - host-route
            - populate static/dynamic/evpn
            - datapath-programming
            - internal-tags set-tag-set
    - DHCP Server: Admin state
    - DHCP Relay: Admin state, gi-address, use-gi-addr-as-src-ip-addr,
network-instance, option, server, trace-options
- IPv6:
    - Admin State, Address IP Prefix - type and primary
    - VRRP:
        - vrrp-group, admin-state, priority, preempt, preempt-delay,
init-delay, accept-mode, advertise-interval, master-inherit-interval,
          version, virtual-address, authentication keychain,
interface-tracking track-interface priority-decrement
    - Neighbor Discovery:
        - duplicate-address-detection, reachable time, stale-time,
learn-unsolicited, proxy-nd, debug, neighbor <prefix> link-layer-address
        - host-route
            - populate static/dynamic/evpn
            - datapath-programming
```

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```
            - internal-tags set-tag-set
        - limit
            - max-entries
            - log-only
            - warning-threshold-pct
    - Router Advertisement:
        - debug
        - router role
            - admin-state, current-hop-limit, ip-mtu,
managed-configuration-flag, other-configuration-flag,
max-advertisement-interval,
              min-advertisement-interval, reachable-time,
retransmit-time, router-lifetime
            - prefix
                - autonomous-flag
                - on-link-flag
                - preferred-lifetime
                - valid-lifetime
    - DHCP Server: Admin state
    - DHCP Relay: Admin state, source-address, network-instance, option,
server, trace-options
- VLAN Encap single-tagged vlan-id
- RA Guard Policy
```

CLI Context: interface

Prerequisites:

- Policies like ra-guard, authentication keychain, should be created before attachment to subinterface intent.
- vlan-tagging should be enabled in interface to support vlan-encap configuration in subinterface
  - Example:

```
interface ethernet-1/23 {
    vlan-tagging true
}
```

3.31.3 Advisory

- Ensure you follow RFC 5952 while providing IPv6 address format.
  - example 1: 2001:db8:0:0:0:0:2:1 should be 2001:db8::2:1
  - example 2: 2001:DB8::1 should be 2001:db8::1
- The configurations for the IRB subinterface are not compatible with this ICM intent; however, they are supported in the L3 EVPN IBSF intent for SRL.
- Cannot enable both DHCP server and DHCP relay on the same subinterface.
- IP Address should not be configured when Unnumbered Admin State is set to enable.
- DHCP relay config is supported only in an existing network-instance of type 'ip-vrf' or 'default'
- IPv4 ARP Host Route route-type 'EVPN' is not supported on the IXR-D1, IXS-A1 and IXR-H series.

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- IPv6 neighbor-discovery host-route route-type 'EVPN' is not supported on the IXR-D1, IXS-A1 and IXR-H series.
- Only one IPv6 link local address/prefix allowed per subinterface.

For Loopback Interface:

- Below attributes are not supported on loopback subinterfaces:
  - Subinterface type
  - vlan-tagging not configurable on loopback interface
  - mpls-mtu
  - collect-detailed-stats
  - unidirectional-link-delay.static-delay

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4 Classic mode intent types

4.1 Name of the Intent: bfd-sbfd-reflector_csros_23-10-1_23-11

4.1.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750-SR, 7950-XRS, 7450-ESS and 7250-IXR Classic SROS of node version 23.10

4.1.2 Description:

bfd-sbfd-reflector_csros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Seamless-BFD reflector.

The following configurations are supported per deployment:

```
- Description
- Admin State
- Local State
- Discriminator
```

CLI context: configure >bfd >seamless-bfd >reflector

4.1.3 Advisory:

- Unique discriminator should be assigned for every node.

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4.2 Name of the Intent: filter-ml-port-list_csros_23-10-1_23-11

4.2.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7450/7950 classic SR OS node version 23.10

4.2.2 Description

filter-ml-port-list_csros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Port List.

The following configurations are supported per deployment:

```
- Create Port List with Port List Name
- Description
- Create Port List Member with Key-ID:
  - Port [EQUAL/RANGE]
    - Port Lower Range
      - Port Higher Range
```

CLI context: filter > match-list > port-list

4.2.3 Advisory

- When Port is set to EQUAL, both Lower Port Range and Higher Port Range should have identical values.
- If Port is set to RANGE, ensure that the Higher Port Range value exceeds the Lower Port Range value.
- The user should ensure that the combination of Port, Port Lower Range, and Port Higher Range is unique. If multiple entries are created with the same combination in the Port List Member audit operation, it will cause misalignment.

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4.3 Name of the Intent: filter-ml-prefix-list_csros_23-10-1_23-11

4.3.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS classic SR OS node version 23.10

4.3.2 Description

filter-ml-prefix-list_csros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of IPv4 and IPv6 prefix list.

The following configurations are supported per deployment:

```
- Description
- Prefix List Member
- Prefix Type
- Prefix
- Mask

- Prefix List Apply Path Member
- Neighbor
- Group
- Index
- Source
- Router

- Prefix List Exclude Member
- Prefix Type
- Prefix
- Mask
```

Prerequisites: VPRN service for Apply-Path Member configuration

4.3.3 Advisory

- Prefix List Member entry, once created, cannot be modified.
- Apply-Path and Prefix-Exclude capabilities are mutually exclusive.
- Service Manager ID should be selected for Apply-Path Member configuration.
- Prefix List Apply Path Member is not applicable on 7250 IXR nodes.
- Prefix List Exclude Member is not applicable on 7250 IXR-R6d, IXR-R6dl, IXR-ec, IXR-x, IXR-x3 node variants

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4.4 Name of the Intent: filter-redirect-policy_csros_23-10-1_24-4

4.4.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950 classic SR OS node version 23.10

4.4.2 Description

filter-redirect-policy_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of filter redirect policy.

The following configurations are supported per deployment:

```
- Create Redirect Policy with Redirect Policy Name
- Description, Hold Time Up, Router Instance Type, Administrative State,
Destination Change Notify
- Create Destination List Member with Key-ID and Address:
  - Description
  - Administrative Priority
  - Administrative State
  - Enable Unicast Route Reachability Test
  - Enable Ping Test
    - Ping Test
      - Source Address
      - Drop Count
      - Interval
      - Hold Down
      - Time Out
    - URL Test
      - URL Test Name
      - HTTP Version
      - Drop Count
      - Interval
      - Hold Down
      - URL
      - TimeOut
    - SNMP Test
      - SNMP Test Name
      - OID
      - Community
      - TimeOut
      - Drop Count
      - Interval
      - Hold Down
      - SNMP Response
        - Response ID
        - Action
```

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```
        - Priority
        - Return Type
```

4.4.3 Advisory

- The user should ensure that each deployment must be configured to use a singular type of Destination Addresses exclusively, either IPv4 or IPv6, but not both.
- The user should ensure that both the Destination Address and Source Address are of the same type, either IPv4 or IPv6.
- The user should ensure that separate deployments must be performed for disabling "Enable Unicast Route Reachability" and enabling "Enable Ping Test" or disabling "Enable Ping Test" and enabling "Enable Unicast Route Reachability Test", requiring multiple deployments to manage this scenario effectively.
- The user should ensure that the address type selection must be determined exclusivley during the initial creation phase; thereafter, users are unable to change their chosen address type.

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4.5 Name of the Intent: lag_cixr_23-10-1_24-4

4.5.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 classic IXR node version 23.10

4.5.2 Description

lag_cixr_23-10-1_24-4 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of lag configuration on all variants of 7250 IXR with lag ID and lag Name as targetIdentifier.

The following configurations are supported per deployment:

```
- LAG Name, Description, Administrative State, MAC Address, Mode, Encap
Type, Monitor Oper Group Name, LAG BFD Disable Soft Reset, Port
Threshold, Port Threshold Action, Dynamic Costing, Access Bandwidth,
Access Booking Factor, Scheduler Mode, LAG Bundle Pool, Low Latency
- LAG LACP Configuration: LACP Administrative State, Standby Signaling,
LACP Transmit Interval, LACP Xmit Standby, Actor Admin Key, LAG System
Id, LAG System Priority, LACP Hold Time Down, LACP MUX Control, LACP Sel
Criteria, LACP Sel Criteria Slave To Partner, LAG LACP Hold Time Down
- LAG Port Termination: Member Name, Sub Group ID, Priority
- LAG BFD Configuration: Family Type, Remote IP, Multiplier, Max Setup
Time, Receive Interval, Admin State, Max Admin Down, Transmit Interval,
Local IP and BFD on Every Link
- Lag Shared Queue VLAN Policy: VLAN QoS Policy Name, Rate Type,
Aggregate Rate CIR, Aggregate Rate PIR, Percentage Rate PIR, Percentage
Rate CIR.
```

Prerequisites:

- Oper Group and Shared Queue VLAN Policy must exist before associating to LAG.
- Auto-Negotiation must be set to False/Limited on LAG Member Ports before associating to LAG.
- All the LAG Member Ports must work at the same rate.
- The Mode and Encap-Type of a LAG Member Port should be preconfigured to match the LAG Mode and Encap-Type.

4.5.3 Advisory

- LAG name should always start with "lag-" and can use either numeric or alphanumeric characters. However, if LAG name is lag-(only numeric), then the numeric value should match the LAG ID.
- LAG BFD entries should not be deleted from the intent configuration.
- LAG BFD IPv6 configuration is supported on IXR-R6d, IXR-R6dl, IXR-x and IXR-x3 variants. The entry needs to be added in the intent configuration for these nodes.
- Modification of Mode/Encap-Type fails when scheduler mode is set to "VLAN QoS".
- Deployment Deletion from Network fails when scheduler mode is set to "VLAN QoS".

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- Lag Shared Queue VLAN Policy is configurable only when scheduler mode is set to "VLAN QoS".
- Before changing the scheduler mode from "VLAN QoS" to "None", remove all Lag Shared Queue VLAN Policy associations except the default policy.
- Default Shared Queue VLAN Policy is associated to the LAG automatically when scheduler mode is set to "VLAN QoS".
- Aggregate Rate CIR and Aggregate Rate PIR are configurable when Rate Type is set to "Kbps". Percentage Rate PIR and Percentage Rate CIR are configurable when Rate Type is set to "Percent".
- LACP MUX control can be modified only when the Administrative State of LAG is Disable.
- LACP Administrative State cannot be enabled when Standby Signaling is Power Off.
- Standby Signaling Power Off is only supported with Selection Criteria Best_Port.
- The value of Port Threshold should be 0 when Signaling is set to Power Off.
- Configuring Actor Admin Key is supported when LACP Administrative State is enabled and LAG Mode is access.
- LACP Administrative State cannot be enabled when Selection Criteria is Best_Port.
- LAG Bundle Pool and Low Latency are not applicable on IXR-R4, IXR-R6, IXR-s, IXR-e, IXR-ec, IXR-6, IXR-10
- Scheduler Mode is not applicable on IXR-6, IXR-10
- LACP Admin Key must be unique per LAG.
- LAG Bundle Pool can be configured only during creation of LAG.

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4.6 Name of the Intent: lag_csros_23-10-1_24-4

4.6.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950 classic SR OS node version 23.10

4.6.2 Description

lag_csros_23-10-1_24-4 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of lag configuration on 7750, 7950 and 7450 nodes with lag ID as targetIdentifier.

The following configurations are supported per deployment:

```
- LAG Name, LAG Port Type, Administrative State, MAC Address, Mode, Encap
Type, Per Link Hashing, Per Link Hash Subscriber Hash, Per Link Hash SAP
Hash, Per Link Hash Subscriber Hash, LAG BFD Disable Soft Reset, Adaptive
Load Balancing, Tolerance, Adaptive LB Interval, Adaptive LB BW
Threshold, Monitor Oper Group Name, Port Threshold, Port Threshold
Action, Power Threshold Static Cost, Hash Weight Threshold, Hash Weight
Threshold Action, Hash Weight Threshold Static Cost, Dynamic Costing,
Enable Per Forwarding Path Ingress Queue, Enable Per Forwarding Path
Egress Queue, Enable SAP Per Flex Path, Adapt QoS, Enable LAG Access
Including Egress Hash Config, Access Bandwidth, Access Booking Factor
- LAG LACP Configuration: LACP Administrative State, Standby Signaling,
Actor Admin Key, LACP Transmit Interval, LACP Xmit Standby, LACP Hold
Time Down, LACP MUX Control, LACP Set Criteria, LACP Set Criteria Slave
To Partner, LAG LACP Hold Time Down
- LAG Port Termination: Member Name, Sub Group ID, Priority, Hash Weight
- LAG Link MAP Profile Port: Port ID, Type
- LAG BFD Configuration: Family Type, Remote IP, Multiplier, Max Setup
Time, Receive Interval, Admin State, Max Admin Down, Transmit Interval,
Local IP and BFD on Every Link
```

Prerequisites:

- Oper Group should pre-exist before associating to LAG.
- Auto-Negotiation should be set to False/Limited on LAG Member Ports before associating to LAG.
- All the LAG Member Ports must work at the same rate.
- The Mode and Encap-Type of a LAG Member Port should be preconfigured to match the LAG Mode and Encap-Type.

4.6.3 Advisory

- LAG name should always start with "LAG-" and it can be either numeric or alphanumeric. However, if LAG name is LAG-(only numeric), then the numeric value should match the LAG ID.
- LAG BFD entries should not be deleted from the intent configuration.
- LACP MUX control can be modified only when the Administrative State of LAG is Disable.

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- LACP Administrative State cannot be enabled when Standby Signaling is Power Off.
- Standby Signaling Power Off is only supported with Selection Criteria Best_Port.
- The value of Port Threshold should be 0 when Signaling is Power Off.
- Adaptive LB interval and Per Link Hashing are mutually exclusive.
- LAG Link MAP Profile configuration is not supported when Adopt QoS is set to Distribute.
- Enabling Per Link Hashing is not supported when Adopt QoS is set to Distribute.
- Adaptive Load Balancing and LAG Port Termination Hash Weight are mutually exclusive.
- Adaptive Load Balancing and Enable Per Forwarding Path Egress Queue are mutually exclusive.
- Configuring Actor Admin Key is supported when LACP Administrative State is enabled and LAG Mode is access.
- LACP Administrative State cannot be enabled when Selection Criteria is Best_Port.
- LACP Admin Key must be unique per LAG.
- Enable Per Forwarding Path Ingress Queue and Enable SAP Per Flex Path cannot be modified when LAG Members exist in LAG.
- LAG Mode cannot be modified when Enable Per Forwarding Path Egress Queue is configured.
- Align Operation will fail for audit differences with Enable Per Forwarding Path Ingress Queue, Enable Per Forwarding Path Egress Queue, and Enable SAP Per Flex Path when LAG Port Member configuration exists.

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4.7 Name of the Intent: log-syslog_csros_23-10-1_23-11

4.7.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7450/7950, 7250 IXR Classic 7210, 7210 SAS and 7705 SAR from node version 23.10

4.7.2 Description

The log-syslog_csros_23-10-1_23-11 intent supports creation/modification/deletion/brownfield discovery/audit/alignment operations, using Log Id and Log Name as target identifiers. The following configurations are supported per deployment:

`- Target Description, Target Address Type, Target Address, Target UDP Port, Facility, Severity, Message, HostName, Enter Value`

CLI context: configure > log > syslog

Prerequisites: - N/A

4.7.3 Advisory

- Valid range for Log Id is 1..10.
- Hostname and Value are not applicable for SAR and SAS nodes.
- Target Address Type can be set only during creation.
- For NFM-P 23.11 or earlier, creating a syslog policy fails if a global policy already exists for the log-id and the deployment target address type does not match the global policy target address type.
- For NFM-P 24.4 or later, the target address type and target address attributes are not supported. In ICM, the values must be set to default, (target address type = IPv4, and target address = 0.0.0.0)
- Association of tls-client-profile to syslog is not supported from Intent/NFMP.

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4.8 Name of the Intent: macsec-ca_csros_23-10-1_23-11

4.8.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7750/7450 and 7250 IXR classic SR OS node version 23.10

4.8.2 Description

macsec-ca_csros_23-10-1_23-11 intent type deployment creates Connectivity Association in NFM-P and supports distribution of PreShared-Keys to classic SR OS.

The following configurations are supported per deployment:

```
- AdminState, Description, Replay Window Size, Replay Protection, MACsec
Encrypt, Clear Tag Mode, Encryption Offset, Cipher Suite, Anysec
Reserved.
- Keying Parameters Skip Offline Nodes and Key Source
- Active PSK, MKA Hello Interval, MKA Key Server Priority,
Addition/deletion of Pre-Shared key
- Per-site configuration of AdminState, Delay Protection, Description,
Replay Window Size, Replay Protection, MKA Hello Interval, MKA Key Server
Priority.
```

4.8.3 Advisory

- SFTP credentials should be preconfigured to push the Pre-Shared Key configuration to classic NEs. SFTP failure will not be reflected in ICM.
- Before configuring the Key Source to HSM, ensure the HSM configuration exists in the NFM-P.
- Enabling Delay Protection requires MKA Hello Interval to be 500ms and Replay Protection to be enabled.
- Only the MACSec supported Site needs to be selected.
- One Pre-Shared Key per encryption type should be created.
- Active PSK can be configured only if Pre-Shared Key is configured.
- Only one Pre-Shared Key should be created per create/update operation when MACsec Key Source is set to "useHSMOnly". The second Pre-Shared Key should be created with some delay as Pre-Shared key creation will be In Progress for a while when MACsec Key Source is set to "useHSMOnly".
- Intent cannot be used in NSP releases earlier than 23.11, as certain enhancements to NSP have been completed to support this feature in NSP 23.11.

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4.9 Name of the Intent: policy-options-as-path-group_csros_23-10-1_23-11

4.9.1 Applicability

Supported from: NSP 23-11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7210 SAS, 7705 SAR, 7450 ESS Classic SROS node version 23.10

4.9.2 Description:

The policy-options-as-path-group_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of as-path-group, with name as-path-group as the target identifier.

The following configurations are supported per deployment:

```
- AS Path Group Entries
  - Entry ID
  - Regular Expression (max 255 char)
```

CLI context: configure > router > policy-options > as-path-group

Prerequisites:

```
- NA
```

Advisory:

```
- NA
```

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4.10 Name of the Intent: policy-options-as-path_csros_23-10-1_23-11

4.10.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7210 SAS, 7705 SAR, 7450 ESS Classic SROS node version 23.10

4.10.2 Description:

The policy-options-as-path_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of as-path, with name as the target identifier.

The following configurations are supported per deployment:

- Regular Expression(Max 255 char)

CLI context: configure > router > policy-options > as-path

Prerequisites:

- N/A

Advisory:

- N/A

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4.11 Name of the Intent: port-eth_cixr_23-10-1_24-4

4.11.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR Classic SR OS node version 23.10

4.11.2 Description

port-eth_cixr_23-10-1_24-4 intent supports discovery/modification/deletion/audit/alignment of Ethernet Ports.

The following configurations are supported per deployment:

```
- Admin State, Description, Configured MAC, Mode, Encap Type, Spped, MTU,
Load Balance Algorithm, DDM Event Suppression, Hold Time, Access
Bandwidth, Booking Factor,
  Operational Group, Accounting Policy, Port QoS Policy, Ethernet Access
Accounting Policy, Scheduler Mode.
- LLDP: Nearest Bridge, Nearest Customer, Nearest Non TPMR
- 802.1x port: Initialize, Reauthenticate Protocol.
- 802.1x Port Authenticator: Admin State, Controlled Port Control, Quiet
Period, Tx Period, Supplicant Timeout, Server Timeout, Max Req, Reauth
Enabled,
                            Radius Server Policy (Authentication &
Accounting), Tunnel Dot1q and Qinq enabled, Tunnelling, Per Host
Authentication.
- EFM-OAM: Admin State, Tunnelling, Mode, Transmit Interval, Multiplier,
Received Remote Loopback Requests, Hold Time, Ignore EFM State, Grace Tx
State Enable, Trigger Fault.
          Link Event Configuration.
- Ethernet: Duplex, Auto-Negotiate, Dot1q Ethertype, Qinq Ethertype, PBB
Ethertype, Report Alarms, Down on Internal MAC Tx Error, Down on Internal
MAC Tx Error Disable,
            PTP Asymmetry, Minium Frame Length, RS-FEC Mode, Egress Rate,
Ingress Rate, LACP Tunnel, Synchronous Ethernet, E-LMI, CRC Failure,
Symbol Monitor Failure, SFLOW, Utilization Stats, Exponential Port
Dampening, ETH-BN Egress Rate Changes, Discard Rx Pause Frames.
            Down When Looped: Keep-Alive Interval, Retry Timeout, Use
Broadcast Address, Type, Swap MAC address.
            MACsec Port: MAC Policy To Exclude MAC Addresses, Excluded
Protocols.
            MACsec Interface: ID, Dst EAPoL MAC Address, Encap Match, Max
Peers.
- Coherent Optics: Dispersion Control Mode, Dispersion, Dispersion Sweep
Start & End, Tx Target Power, Rx LOS Threshold, Rx LOS Reaction, CPR
Window Size, Compatibility Mode, Configured Alarms.
- DWDM Optics: Configured Frequency
```

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- VLAN QoS Policy: Rate Type, VLAN QoS Policy Name, Aggregate (CIR & PIR).

Prerequisite:

Below list of policies should be configured,

- Accounting Policy, Ethernet Access Accounting Policy, Port Scheduler Policy.
- VLAN QoS Policy, MAC Policy, macsec connectivity-association
- Service Operational Group

4.11.3 Advisory

- User should perform brownfield discovery of the port before configuring the port using the intent, as some properties/child-objects are dependent on MDA and other read-only properties on port.
- Distribution of the port intent will fail, as many objects are dependent on the port being distributed.
- Encap Type null is not supported for hybrid ports.
- Cannot change the port mode when accounting statistics is enabled.
- Cannot change mode/encap of port when the port is a LAG member.
- Port mode cannot be changed when scheduler mode vlan-qos is set.
- Speed attribute should not be configured for Fixed Speed Ports.
- MTU changes on the node whenever port Mode/ encap-type changes.
- DDM Event Suppression is not supported on Breakout/Ethernet port.
- Scheduler Mode: Default VLAN QoS entry should not be deleted from the config form.
- LLDP: Non TPMR Transmit Mgmt Address should not be deleted from the config form.
- LLDP: Cannot set tunnel mode when lldp-nearest-bridge admin state of Rx or Rx-Tx is enabled.
- Per-host-authentication cannot be enabled - fp option must be enabled
- DWDM frequency must be set first after enabling the DCO and to bring up the port.
- To bring up a Port, configuring DWDM frequency and enabling the port should be done in two separate operations. First configure the frequency and deploy. Later Enable the Port Admin state and deploy.
- DCO and OTU cannot be enabled at a time.
- Dispersion Sweep Start must be less than Dispersion Sweep End and the difference between them must be at least 1500.
- Auto Negotiate should be set to notApplicable for Fixed Speed Ports.
- Single fiber is not allowed with Auto Negotiate.
- Single fiber only supported in network mode with Null or Dot1q Encap type.
- Synchronous Ethernet(sync-e) must be enabled on the MDA, or XGig mode should be WAN, before the port can process sync status messages.
- Synchronous Ethernet(sync-e) should be enabled on the corresponding MDA to configure SSM AdminState, Code-Type ESMC Tunnel and Tx DUS attributes on Ports.

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- Synchronous status messages can only be enabled on ports that are synchronous ethernet capable.
- The CRC SD threshold value must be greater than or equal to CRC SF threshold.
- Modification of RS-FEC mode depends upon the ports configured.
- efm-oam loopback cannot be enabled when port is in loopback
- Dampening reuse-threshold must be less than suppress-threshold.
- Ethernet Report Alarms: No Block Lock/No AM Lock/Duplicate Lane are not supported on Breakout/Ethernet port.
- MACsec and Coherent Optics are not supported on Connector Breakout/Ethernet port.
- Port-control must be force-authorized to enable tunnelling of all encaps.

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4.12 Name of the Intent: port-eth_csros_23-10-1_24-4

4.12.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS Classic SR OS node version 23.10

4.12.2 Description

port-eth_csros_23-10-1_24-4 intent supports discovery/modification/deletion/audit/alignment of Ethernet Ports.

The following configurations are supported per deployment:

```
- Admin State, Description, Configured MAC, Mode, Encap Type, Spped, MTU,
Load Balance Algorithm, DDM Event Suppression, Hold Time, Port Percentage
Rates,
     Hybrid Ingress & Egress Buffer Allocation, Statistics, Access
Bandwidth, Operational Group, Network Queue, Accounting Policy, Ethernet
Accounting, Port Scheduler.
- LLDP: Nearest Bridge, Nearest Customer, Nearest Non TPMR
- 802.1x port: Initialize, Reauthenticate Protocol.
- 802.1x Port Authenticator: Admin State, Controlled Port Control, Quiet
Period, Tx Period, Supplicant Timeout, Server Timeout, Max Req, Reauth
Enabled,
                            Radius Server Policy (Authentication &
Accounting), Tunnel Dot1q and Qinq enabled, Tunnelling, Per Host
Authentication.
- EFM-OAM: Admin State, Tunnelling, Mode, Transmit Interval, Multiplier,
Received Remote Loopback Requests, Hold Time, Ignore EFM State, Critical
Event Notify, Critical Event Notify,
           Dying Gasp On Reset Enable, Grace Tx State Enable, Grace
Vendor OUI, Trigger Fault.
           Link Event Configuration (Errored Frame, Errored Frame Period,
Errored Frame Seconds Summary, Errored Frame Symbols).
- Ethernet: Duplex, Auto-Negotiate, Dot1q Ethertype, Qinq Ethertype, PBB
Ethertype, Report Alarms, Down on Internal MAC Tx Error, Down on Internal
MAC Tx Error Disable,
            PTP Asymmetry, Minium Frame Length, RS-FEC Mode, Egress Rate,
Ingress Rate, LACP Tunnel, Synchronous Ethernet, E-LMI, CRC Failure,
Symbol Monitor Failure, SFLOW, Utilization Stats, Exponential Port
Dampening, ETH-BN Egress Rate Changes, Discard Rx Pause Frames.
            Down When Looped: Keep-Alive Interval, Retry Timeout, Use
Broadcast Address.
            Access Ingress Queue Group: Overrides Ingress Queue.
            Access Egress Queue Group: Overrides Egress Queue, Host
Matching.
            Network Egress Queue Group: Overrides Network Egress Queue.
```

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```
            MACsec Port: MAC Policy To Exclude MAC Addresses, Excluded
Protocols.
            MACsec Interface: ID, Dst EAPoL MAC Address, Encap Match, Max
Peers.
            Egress Scheduling Virtual Port: Name, Port Scheduler Policy,
Scheduler Policy, Aggregate Rate Limit, Limit Unused Bandwidth, Status,
Host Matching.
            Egress HS Secondary Shapers: Name, HS Secondary shaper Rate,
HS Secondary shaper Lo Brst Max Class.
- Coherent Optics: Dispersion Control Mode, Dispersion, Dispersion Sweep
Start & End, Tx Target Power, Rx LOS Threshold, Rx LOS Reaction, CPR
Window Size, Compatibility Mode, Configured Alarms.
- SONET: Framing, Clock Source, Loopback, Single Fiber, Tx DUS/DNU, Hold
Time Up, Hold Time Down.
- SONET Overhead: SONET Section Trace Mode, J0 String.
- SONET Monitoring: Report Alarms, Suppress Low Order Alarms, BER Signal
Failure Threshold, BER Signal Degradation Threshold.
- DWDM Optics: Configured Frequency
```

Prerequisites:

Below list of policies should be configured,

- Network Queue Policy, Accounting Policy, Ethernet Accounting policy, Port Scheduler Policy.
- HS Port Pool Policy, Slope Policy, RADIUS Server Policy, MAC Policy To Exclude MAC Addresses.
- Connectivity Association, Queue Group Template Policy, Scheduler Policy, Queue Group Policy.
- Policer Control Policy, hw-agg-shaper-scheduler-policy.

4.12.3 Advisory

- User should perform Brownfield Discovery of the port before configuring the port using the intent, as few properties/child-objects are dependent of MDA & other read-only properties on port.
- Distribution of Port intent will fail, as many objects are dependent on the port being distributed.
- Encap Type null is not supported for Hybrid Ports.
- Cannot change Port Mode when accounting statistics is enabled.
- Modification of port mode or encap-type of Satellite port without active uplink will fail.
- Speed attribute should not be configured for Fixed Speed Ports.
- MTU changes on the node whenever port Mode/ encap-type changes.
- DDM Event Suppression is not supported on Breakout/Ethernet port.
- Hybrid Buffer Allocation supported only hybrid mode of port.
- Node can have only one Ethernet Accounting policy of record type complete Ethernet Port.
- LLDP: Non TPMR Transmit Mgmt Address should not be deleted from the config form.
- LLDP: Cannot set tunnel mode when lldp-nearest-bridge admin state of Rx or Rx-Tx is enabled.
- Port must be shut down before changing Digital Coherent Optics (DCO).
- DWDM frequency must be set first after enabling the DCO and to bring up the port.

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- To bring up a Port, configuring DWDM frequency and enabling the port should be done in two separate operations. First configure the frequency and deploy. Later Enable the Port Admin state and deploy.
- DCO and OTU cannot be enabled at a time.
- Dispersion Sweep Start must be less than Dispersion Sweep End and the difference between them must be at least 1500.
- QoS Pool reserved CBS should not be set to default in order to configure Amber Alarm Action Step & Amber Alarm Threshold.
- QoS Pool should not be created/deleted from Intent.
- QoS Pool: Amber alarm threshold cannot be greater than red alarm threshold.
- QoS Pool: Reserved CBS cannot be greater than Amber alarm action max.
- QoS Pool: Adaptive CBS sizing step cannot be specified when reserved cbs is default.
- Auto Negotiate should be set to notApplicable for Fixed Speed Ports.
- Auto-Negotiate is supported on Ethernet ports with speeds up to 1 Gbps.
- The Ethernet Ingress rate, cannot exceed port bandwidth.
- Single fiber is not allowed with Auto Negotiate.
- Single fiber only supported in network mode with Null or Dot1q Encap type.
- Synchronous Ethernet(sync-e) must be enabled on the MDA or XGig mode should be WAN before port can process sync status messages.
- Synchronous Ethernet(sync-e) should be enabled on the corresponding MDA to configure SSM AdminState, Code-Type ESMC Tunnel and Tx DUS attributes on Ports.
- Synchronous status messages can only be enabled on ports that are synchronous ethernet capable.
- The CRC SD threshold value must be greater than or equal to CRC SF threshold.
- The Symbol Monitor SD threshold value must be greater than or equal to SF threshold.
- The Symbol Monitor Failure is not supported on Breakout/Ethernet port.
- EFM-OAM frame monitoring and port crc-monitor cannot both be enabled.
- Modification of RS-FEC mode depends upon the ports configured.
- ELMI Mode can only be modified with port mode ACCESS and encap QTAG.
- ElMI T392 should be greater than T391.
- Dampening reuse-threshold must be less than suppress-threshold.
- Ethernet Report Alarms: No Block Lock/No AM Lock/Duplicate Lane are not supported on Breakout/Ethernet port.
- The EFM Link Monitoring Signal Failure (SF) Threshold must not exceed the corresponding window.
- Host match entries only supported on port access egress queue groups with system default instance 1.
- Access Egress Queue Group, cannot have both Scheduler Policy and agg-rate params configured.
- OTU support only Ethernet port not XIOM/Connector Breakout port

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- OTU SF Threshold must be less than SD Threshold.
- SONET, SONET Overhead & SONET Monitoring are configurable only on XGig mode WAN and not supported on XIOM/Connector Breakout port.
- MACsec and Coherent Optics are not supported on Connector Breakout/Ethernet port.
- Port-control must be force-authorized to enable tunnelling of all encaps.
- Cannot set mode to Network when port has access ingress queue group defined.
- Cannot set mode to Access when port has network egress queue group defined.
- Egress Scheduling Policy: Cannot have both scheduler-policy and egress-rate-modify configured.
- Egress Scheduling Virtual Port, cannot have both Scheduler Policy and agg-rate params configured.
- Egress Scheduling Virtual Port, cannot configure Port Scheduler Policy when Port scheduler policy is assigned at parent object.
- Access Ingress & Egress Queue Group: Overrides Scheduler is not supported from the intent.
- Values: -1 is referred as MAX and -2 is referred as No Override.

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4.13 Name of the Intent: port-xc_csros_23-10-1_24-4

4.13.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950 classic SR OS node version 23.10

4.13.2 Description

port-xc_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of Port XC.

The following configurations are supported per deployment:

```
- Administartive State
- Description
- PXC Port ID : Deafult value 0
```

CLI context: port-xc > pxc

4.13.3 Advisory

- User cannot update PXC port ID in single deployment. i.e Remove the PXC port ID and deploy, then update the Port ID.
- Before binding a port to a PXC port make sure port is in admin shutdown state.

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4.14 Name of the Intent: qos-egress-remark-policy_cixr_23-10-1_23-11

4.14.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR Classic SR OS node version 23.10

4.14.2 Description

The qos-egress-remark-policy_cixr_23-10-1_23-11 intent supports the creation/modification/deletion/brownfield discovery/audit/alignment of QoS Egress Remark Policy.

The following configurations are supported per deployment:

```
- Description, Scope
- FC Dot1p Mapping, FC DSCP Mapping, FC LSP-EXP Mapping
- Forwarding Class : In Profile, Out Profile, Exceed Profile, In Plus
Profile, In Profile DSCP, Out Profile DSCP, Exceed DSCP, In Plus DSCP,
LSP-EXP In Profile, LSP-EXP Out Profile,
LSP-EXP In Plus Profile, LSP-EXP Exceed Profile, Mark DE bit, Force DE
Value.
```

CLI context: config > qos > egr-remark-plcy

Pre-requisite: FC Dot1p Mapping, FC DSCP Mapping, FC LSP-EXP Mapping policies should be pre configured.

4.14.3 Advisory

- FC Dot1p Mapping, FC DSCP Mapping and FC LSP-EXP Mapping are supported only on R6d/R6dl/x/x3 chassis models.
- Forwarding Classes are not supported on R6d/R6dl/x/x3 chassis.
- To configure Exceed/In Plus Profile attributes, ensure the fc-4 profile is enabled (config > system > fp > options > qos > fc-4-profile).
- Forwarding Classes must not be deleted from the configuration form.
- When you perform a brownfield discovery of an 7250 IXR 10/6/R4/R6/s/e/ec, the deployment will misalign

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4.15 Name of the Intent: qos-ingress-classification_cixr_23-10-1_23-11

4.15.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR Classic SR OS node version 23.10

4.15.2 Description

The qos-ingress-classification_cixr_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of Ingress Classification Policy, with Displayed Name as the target Identifier.

The following configurations are supported per deployment:

```
- Description
- LSP-EXP FC Mapping
- Dot1p FC Mapping
- DSCP FC Mapping
- Allow Attachment
- Scope
- Forwarding Class
- Profile
- Remark
- Dot1p
- LSP-EXP
- DSCP
```

CLI context: configure > qos > ingress-classification-policy

Prerequisites:

- configure qos dot1p-fc-map
- configure qos lsp-exp-fc-map
- configure qos dscp-fc-map

4.15.3 Advisory

- Dot1p FC Mapping, DSCP FC Mapping, LSP-EXP FC Mapping and Allow Attachment are not supported on IXR-R4, IXR-6, IXR-10, IXR-R6, IXR-s, IXR-e and IXR-ec
- Dot1p, DSCP, LSP-EXP, Forwarding Class, Profile and Remark are not supported on IXR-x, IXR-x3, IXR-R6d, IXR-R6dl, IXR-e2 and IXR-e2c
- Profile cannot be set to exceed or inplus if fc-4-profile is not enabled (configure>system>fp>options>qos>fc-4-profile).
- The Allow Attachment attribute is applicable only during the creation of a new Ingress Classification Policy and should remain unchanged after the policy has been created. If a Global Policy already exists in the NFM-P, the value of Allow Attachment must match the value configured in the Global Policy.

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4.16 Name of the Intent: qos-ml-port-list_csros_23-10-1_24-4

4.16.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950 classic SR OS node version 23.10

4.16.2 Description

qos-ml-port-list_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of QoS match-list port-list.

The following configurations are supported per deployment:

```
- Create Port List with Port-List Name
- Description
- Create Port List Member with the following attributes:
  - Port [EQUAL/RANGE]
    - Port Lower Range
    - Port Higher Range
```

CLI context: qos > match-list > port-list

4.16.3 Advisory

- When Port is set to EQUAL, both Port Lower Range and Port Higher Range should have identical values.
- If Port is set to RANGE, then the Port Higher Range value must be greater than the Port Lower Range value.
- The user must configure a unique combination of Port, Port Lower Range, and Port Higher Range.

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4.17 Name of the Intent: qos-ml-prefix-list_csros_23-10-1_24-4

4.17.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS classic SR OS node version 23.10

4.17.2 Description

qos-ml-prefix-list_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of QoS Prefix List for IPv4 and IPv6 policies.

The following configurations are supported per deployment:

```
- Creation of QoS Prefix List with Name and Prefix List Type
- Configuration of QoS Prefix List:
  - Description
  - IP Prefix List Members
    - Prefix Type
    - Prefix
    - Mask
```

CLI context: qos > match-list

4.17.3 Advisory

- The entry created in IP Prefix List Members cannot be modified.
- The user should ensure that each deployment must be configured to use a singular type of member addresses exclusively, either IPv4 or IPv6 , but not both.

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4.18 Name of the Intent: qos-network-queue_csas_23-10-1_23-11

4.18.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7210 SAS [D/Dxp/M/S/Sx/T/Mxp/R6/K] classic/MD SR OS node version 23.10

4.18.2 Description

qos-network-queue_csas_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment on SR OS nodes.

The following configurations are supported per deployment:

```
- Description
- FC: FC Name and Queue
- Queue: ID [1 to 8], Name, Description, Adaptation rule [PIR and CIR],
Rate [PIR and CIR], Port Parent, Burst Size [cbs, mbs], Mode, Weight,
Priority, Queue Management Policy, Slope-Policy
```

4.18.3 Advisory

- SAS-K schema form should be used for SAS-K variant.
- Queue Management Policy and Slope-Policy should be preconfigured on the node.
- Default queue [1-8] should not be deleted from the template for other than SAS-K.
- Queue 1 should not be deleted from the template once it is created for SAS-K.
- For deployments on multiple nodes, the user must enter the Policy ID/Name in the suggest field.
- SAS-K: deletion of Queue associated to the FC should be done in a separate edit operation (delete FC - deploy and delete Queue - deploy).

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4.19 Name of the Intent: qos-network-queue_csros_23-10-1_23-11

4.19.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS and 7450 ESS for Classic SROS node version 23.10

4.19.2 Description

qos-network-queue_csros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment on SROS nodes.

The following configurations are supported per deployment:

```
- Description, HS Attachment Policy,WRR Policy
- FC: FC Name, Queue ID, Multipoint Queue ID and Egress HSMDA Queue ID
- Queue: ID [1 to 16], Name, Description, Adaptation rule [PIR and CIR],
Rate [PIR and CIR], Port Parent, Burst Size [cbs, mbs], HS Class Weight,
HS WRED Slope Policy,HS WRR Weight, Slope-Policy
```

Prerequisites:

- HS Attachment Policy
- WRR Policy
- Slope Policy
- HS WRED Slope Policy

4.19.3 Advisory

- An audit report is shown for Queue 1/9 if it is deleted from the intent.
- "-1" indicates the default values in NFMP.
- Queue type and multipoint configuration in queue configuration cannot be edited once created.
- Cannot assign multicast queue ID to the unicast queue property of a forwarding class.
- HS Config not supported on multicast queues or on unicast queue IDs above 8.
- Deleting a queue associated with the FC must be done using separate edit operations (first delete the FC, then deploy, then delete the queue, then deploy again). Additionally, the audit report will reflect the queue deletion only if the associated FC has not been deleted.

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4.20 Name of the Intent: qos-network_csas_23-10-1_23-11

4.20.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7210 SAS [SAS-T/SAS-Dxp/SAS-R6/SAS-M/SAS-MXP/SAS-Sx/SAS-K] classic node versions 9.0,11.0,22.9,23.3,23.10

4.20.2 Description

qos-network_csas_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Network QoS Policy.

The following configurations are supported per deployment:

```
- Configuration of the Network QoS Policy with Network Policy ID and
Network Manager ID
- Configuration of Description, Scope and Type [Port,IP-interface]
- Configuration of the following Ingress parameters:
    - Default Action [FC,Profile]
    - Dot1p [Dot1p Value,Fc,Profile]
    - Dscp [Dscp Name,Fc,Profile]
    - Lsp Exp [Lsp Exp Value,Fc]
    - If Type is Port then Classification Type [Dot1p/DSCP/MPLS LSP-EXP]
    - Fc [Fc Name,Fp Redirect Group[Queue,Multicast Queue,Use
DEI,Meter,Multicast meter]]
    - Meter [Meter ID, Adaptation Rule,Rate]
- Configuration of the following Egress parameters:
    - Enable Remarking, Remarking Type [use-Dot1p,use-dscp,all]
    - FC [FC name,Dot1p in,Dot1p out,DSCP in,DSCP out,Dot1p,DE Mark] if
network Type is port
    - FC [FC name,lsp-exp-in-profile,lsp-exp-out-profile] if network Type
is IP_interface
```

Prerequisites:

- Dot1p/DSCP/MPLS LSP-EXP Classification policy, Remarking policy, and MPLS LSP-EXP Profile must be configured in the node.

4.20.3 Advisory

- Use Default Template for 7210 SAS except SAS-K nodes.
- Use Default SasK Template for 7210 SAS-K nodes.
- Network Policy Type can be selected only at the time of policy creation, not later.
- SAS-K does not support Network Policy Type, Meter, Dot1p, DSCP, LSP EXP and mpls-lsp-exp-profile.
- Display is limited to supported attributes.
- Adaptation Rule PIR and Rate PIR are not applicaple if Meter mode is srtcm.

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- Creation of Network QoS deployment fails if Network QoS Network Manager ID and Network QoS Policy ID are not unique. That is, the Manager ID/Policy ID for the selected target should not be used already.
- Brownfield discovery will fail if both Network QoS Policy Manager ID and Network QoS Policy ID do not match.
- Remarking Policy is not supported on SAS-M/SAS-DXP NEs.
- Egress FC is supported only on SAS-M/SAS-DXP NEs.
- Remarking is supported on SAS-M/SAS-T/SAS-DXP NEs.
- User needs to choose proper Remarking Policy type such as dscp/Dot1p/lsp-exp.
- Deployment misalignment occurs for brownfield discovery of default Network Policy ID is 1.
- Before deleting a Queue/Meter/MultiCast Meter, its Association to FC should be removed and deployed to Target.
- For deployments on multiple nodes, the user must enter the Policy ID/Name in the suggest field.
- Deployment misalignment occurs on audit of IPv6 address with combination of IPv4 and IPv6. For example: <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.

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4.21 Name of the Intent: qos-policer-control_csros_23-10-1_23-11

4.21.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7950/7450 classic SR OS node version 23.10

4.21.2 Description

qos-policer-control_csros_23-10-1_23-11 Intent supports creation/modification/deletion/discovery/audit/alignment of QoS policer control.

The following configurations are supported per deployment:

```
- Description,Policy Mode, Maximum Frame Based Bandwidth Rate, Maximum
Frame Based Bandwidth Percent, Minimum Separation Buffer Space
- Policing Policer Control Level - Key Id, Level, Maximum Cumulative
Buffer Space, Fixed Cumulative Buffer Space
- Policing Policer Control Arbiter - Key Id, Arbiter Name, Description,
Rate Type, Tier, Priority Level, Parent Arbiter, Weight, Maximum Frame
Based Bandwidth Percent,Rate
```

4.21.3 Advisory

- If user wants to add Policing Policer Control Arbiter then user has to give Arbiter name along with key id then only deployment get success.
- If user wants to add multiple Policing Policer Control Level, then user has to select different level (eg.Level 1, Level 2)
- In Policing Policer Control Level, 8 levels are already auto populate so user has to edit only. User cannot add more than 8 levels in Policing Policer Control Level.

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4.22 Name of the Intent: qos-port-egress_csas_23-10-1_23-11

4.22.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7210 SAS [Dxp/M/Mxp/Sx/R6/T] classic SR OS node version 23.10

4.22.2 Description

qos-port-egress_csas_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment on classic SROS nodes.

The following configurations are supported per deployment:

```
- Description, Scope, Remarking, Remarking Type, Remarking Policy
- Queue: Id [1 to 8], Adaptation rule [PIR and CIR], Rate [PIR and CIR],
Queue Mode, Queue Management Policy, Weight
- FC: FC Name, Dot1p [Dot1p, In Profile, Out Profile], DSCP [In Profile,
Out Profile], DE [Mark DE Bit, Force DE]
```

4.22.3 Advisory

- FC configuration and Remarking Type are supported only for SAS (Dxp/M).
- Queue management and Remarking Policy should be preconfigured on the node.
- Default queue [1-8] should not be deleted from the template.
- For deployments on multiple nodes, the user must enter the Policy ID/Name in the suggest field.

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4.23 Name of the Intent: qos-port-ingress_csas_23-10-1_23-11

4.23.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: SAS [R6/R12 & Mxp] classic SR OS node version 23.10

4.23.2 Description

qos-port-ingress_csas_23-10-1_23-11 intent type deployment creates Port Ingress on SAS nodes.

The following configurations are supported per deployment:

```
- Description, Default-FC, Counter Mode, Num QoS Classifiers, DSCP
Classification, Table Classification Criteria, Dot1p Classification
- Meter supports Meter ID, Multipoint, Rate Mode, Color Mode, MBS Type,
MBS, CBS Type, CBS, Rate PIR, Rate CIR, Adaptation Rule PIR, and
Adaptation Rule CIR
- FC supports FC Name, Meter ID, and Multicast Meter ID
```

4.23.3 Advisory

- Meter ID creation and association of the same to FC in the same create/modify operation will fail. Meter ID needs to be created first, and association to FC should be done in the subsequent edit operation. (NFM-P limitation)
- Meter ID deletion should be done only after removing the meter ID association in FC. Meter association removal from FC and Meter ID deletion should be done in two separate edit operations. (NFM-P limitation)
- Meter IDs 1 and 9 are the default meters and should not be deleted.
- DSCP Classification and Dot1p Classification suggests the existing policies if single target is selected. When multiple network elements are selected as target, the user needs to manually enter the ID.

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4.24 Name of the Intent: qos-sap-egress_csas_23-10-1_23-11

4.24.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7210 SAS [Mxp/R6/K12/k5] classic SR OS node version 23.10

4.24.2 Description

qos-sap-egress_csas_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment on classic SR OS nodes.

The following configurations are supported per deployment:

```
- Description, Scope, Remarking, Remarking Policy, Dot1p Classification,
DSCP Classification, IP Prec Classification
- FC: FC Name and Queue
- Queue: ID [1 to 8], Adaptation Rule [PIR and CIR], Rate [PIR and CIR],
Burst size [cbs, mbs], Mode, Weight, Priority, Queue Management Policy,
Slope-Policy
```

4.24.3 Advisory

- SAS-K schema form should be used for SAS-K variant.
- Dot1p, DSCP, IP-Prec classification, Queue management, Remarking Policy, and Slope-policy should be preconfigured on the node.
- Default queue [1-8] should not be deleted from the template for SAS (Mxp/R).
- Queue 1 should not be deleted from the template for SAS-K.
- Dot1p, DSCP and IP Prec Classification are not applicable for the K5 variant.
- SAS-K, deletion of Queue associated to the FC should be done in separate edit operation (delete FC - deploy and delete Queue - deploy).
- For deployments on multiple nodes, the user must enter the Policy ID/Name in the suggest field.

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4.25 Name of the Intent: qos-sap-egress_csros_23-10-1_24-4

4.25.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750-SR, 7450-ESS, 7950-XRS, 7705-SAR classic SR OS node version 23.10

4.25.2 Description

The qos-sap-egress_csros_23-10-1_24-4 intent supports the creation, modification, deletion, brownfield discovery, audit, and alignment of SAP Egress configurations with Policy ID as the targetIdentifier.

The following configurations are supported per deployment:

```
- Displayed Name, Description, Parent Location, NE Policy Name, Post
Policer Mapping Policy, Policers HQoS Manageable, Use Policer Result
Marking Dot1p Inner, HW Aggregate Shaper Queues Queue Set Size, Ethernet
Ctag, HS Attachment Policy, Scope
- Scheduler Class Elevation: ID, Class Weight, Dot1p
- Queue: ID, Description, Displayed Name, Expedite, Port Average
Overhead, Shaper Weight, High Drop Tail, High Priority Reserved, Advanced
Configuration Policy, Use WRED Queue, High Plus Drop Tail, CIR, FIR Burst
Limit, Scheduler, Committed Burst Size, Rate Type, Burst Limit, HS Class
Weight, HS WRED Slope Policy, Scheduler Class, WRR Queue Mode, Packet
Byte Offset, PIR, WRED Queue Slope Usage, High Low Priority Only, Maximum
Burst Size, Maximum Data Transmission, Level, Weight, CIR Weight, WRED
Queue Slope Policy, HS WRR Weight, CIR Level, CIR Adaptation, HS
Alternate Port Class Pool, PIR Adaptation, Port Parent, Port Parent
Level, Port Parent CIR Level, Port Parent Weight, Port Parent CIR Weight,
Precedence, Forwarding Class, Profile, Precedence
- Policer: Description, ID, CIR, Packet Byte Offset, CIR Adaptation,
Priority Level, Committed Burst Size, High Priority Reserved, PIR,
Profile Capped, Parent Arbiter, PIR Adaptation, Maximum Burst Size,
Enable DSCP Precedence Remarking, Advanced Configuration Policy, Stats
Mode, Scheduler Parent, Rate Type, Profile Out Preserve, Enable Exceed
PIR, Port Parent
- IP Match Criteria: Displayed Name, Description, ID, Source IP,
Protocol, DSCP, Destination Net Mask, Destination Port, Source Net Mask,
SRC Mask, Port Redirect Queue Group, To Port DST, From Port DST, DST
Mask, Profile, To Port SRC, Policer ID, From Port SRC, Fragment,
Forwarding Class, Source Port, Queue ID, Destination IP, Destination IP
Prefix, Source IP Prefix
- HS WRR Group: ID, Rate Type, Class Weight, PIR, Adaptation
- DSCP: Forwarding Class, DSCP, Profile
- IPv6 Match Criteria: Displayed Name, Description, ID, Source IP,
Protocol, DSCP, Destination Net Mask, Destination Port, Source Net Mask,
SRC Mask, To Port DST, From Port DST, DST Mask, Profile, To Port SRC,
```

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Policer ID, From Port SRC, Forwarding Class, Source Port, Queue ID, Destination IP, Destination IP Prefix, Source IP Prefix
- Forwarding Class: Out Precedence, Queue ID, Exceed Precedence, Outer-Tag Force DE value, Policer ID, Outer-Tag Exceed Profile, Inner-Tag Exceed Profile, Outer-Tag Mark DE Bit, Port Redirect Queue Group, Queue Group, Template Policy, In profile, Out DSCP, In DSCP, Force DE Value, Exceed DSCP, Inner-Tag In Profile, Inner-Tag Force DE Value, Outer-Tag Out Profile, Inner-Tag Out Profile, Forwarding Class, Exceed Profile, Port Redirect Queue Group, Mark DE Bit, In Precedence, Inner-Tag Mark DE Bit, Out Profile, Outer-Tag In Profile, Dynamic Policer
- Dynamic Policer: Dynamic Policer Start Entry, Parent Arbiter, Shared PCC Rule Count, Shared PCC Rule Start Entry, Policy ID, Committed Burst Size, Dynamic Policer Count, Maximum Burst Size, Packet Byte Offset, Stats Mode

Prerequisites:

- HS Attachment Policy, Post Policer Mapping Policy, Parent Arbiter, Advanced Configuration Policy, Scheduler Parent, and IP-Prefix-List HS WRED Slope Policy configuration should pre-exist.

4.25.3 Advisory

- Use the **default schema form** for non-SAR systems, and **defaultSAR** for SAR systems.
- Default entries in Queue and Scheduler Class Elevation should not be deleted from the intent.
- NE Policy Name should not start with a number.
- Create, modify, align, and audit operations ignore configurations that are not applicable to the selected target. For example, HS WRR Group configuration is not applicable to XRS nods. When a SAP Egress Policy with HS WRR Group configuration is distributed or created on an XRS, the intent is created successfully while ignoring the HS WRR Group configuration.
- Enabling Policers HQoS Manageable with Parent Location is set to SLA is not supported.
- Policer Enable Exceed PIR cannot be enabled while the policy is configured when Policers HQoS Manageable is enabled.
- In an IP Match Entry, Queue ID and Policer ID should be configured to enable Port Redirect Queue Group.
- In an IPv6 Match Entry, Queue ID and Policer ID should be configured to enable Port Redirect Queue Group.
- In a Forwarding Class Entry, either DSCP or Precedence remarking can be enabled.
- In a Forwarding Class Entry, Outer-Tag Force DE Value cannot be configured when Outer-Tag Mark DE Bit is not enabled.
- In a Forwarding Class Entry, Inner-Tag Force DE Value cannot be configured when Inner-Tag Mark DE Bit is not enabled.
- In a Forwarding Class Entry, Force DE Value cannot be configured when Mark DE Bit is not enabled.
- In an IP Match Entry, Destination IP Prefix and Source IP Prefix removal and the addition of Destination IP and Source IP cannot be performed in a single edit operation.

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- In an IP Match Entry, Destination IP and Source IP removal and the addition of Destination IPv6 Prefix and Source IPv6 Prefix cannot be performed in a single edit operation.
- In an IPv6 Match Entry, Destination IPv6 Prefix and Source IPv6 Prefix removal and the addition of Destination IP and Source IP cannot be performed in a single edit operation.
- In an IPv6 Match Entry, Destination IP and Source IP removal and the addition of Destination IPv6 Prefix and Source IPv6 Prefix cannot be performed in a single edit operation.
- Creation of Policer with the ID greater than or equal to Dynamic Policer Start Entry will fail.
- Policer deletion should not be performed when it is associated with IP Match Criteria, Forwarding Class or IPv6 Match Criteria.
- Queue deletion should not be performed when it is associated with IP Match Criteria, Forwarding Class or IPv6 Match Criteria.
- Policer cannot have Port Parent enabled along with Scheduler Parent configuration as only one parent is allowed.
- Cannot set Enabling Policers HQoS Manageable when a dynamic policer range is configured.
- Creation of Policer with ID greater/ equal to Dynamic Policer Start Entry will fail.
- Dynamic policer range not allowed when Policers HQoS Manageable is enabled in the policy.
- **SAR** :
  - Deletion of a referenced queue should be performed in two operations:
  1. First, delete the **forwarding class** where the queue is associated and deploy.
  2. Then, delete the **queue** in the second operation and deploy.

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4.26 Name of the Intent: qos-sap-ingress_csas_23-10-1_23-11

4.26.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: SAS-R12/R6/Mxp/Sx/S/Dxp/K30/K12/K5 classic SR OS node version 23.10

4.26.2 Description

qos-sap-ingress_csas_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment on classic SR OS nodes.

The following configurations are supported per deployment:

```
- Description, Scope, Num QoS Classifiers, IP Match Criteria Type, Dot1p
Classification, DSCP Classification, MAC Match Criteria Type, Allow Any
IPv6 Match, Enable Inner Dot1p Match Criteria
- Forwarding Class, IP Criteria, IPv6 Criteria, MAC Criteria, Queue
- Queue: ID [1 to 8], Adaptation Rule [PIR and CIR], Rate [PIR and CIR],
Burst Size [cbs, mbs], Mode, Weight, Priority, Queue Management Policy,
Slope-Policy
- SAP Ingress configuration supports a single target identifier policy ID
in the range of 1-65535.
```

4.26.3 Advisory

- MAC Match Criteria Type should be selected as "any" to configure Src and Dst MAC address.
- Default Queue [1-8] should not be deleted from the template.
- Deletion of the Queue associated to the FC should be done in a separate edit operation (delete FC - deploy and delete Queue - deploy).
- For deployments on multiple nodes, the user must enter the policy ID/Name in the suggest field.
- Configuring on the correct supported node is the responsibility of the user. Deploying an unsupported configuration may result in deployment failure.
- Deployment misalignment occurs on audit of IPv6 address with combination of IPv4 and IPv6. For example: <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.

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4.27 Name of the Intent: qos-sap-ingress_csros_23-10-1_24-4

4.27.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7950/7750/7450/7705 SAR classic SR OS node version 23.10

4.27.2 Description

The qos-sap-ingress_csros_23-10-1_24-4 intent supports the creation/modification/deletion/discovery/audit/alignment of a SAP Ingress Policy.

The following configurations are supported:

```
- Description
- MAC Criteria Type
- IPv6 Criteria Type
- Ingress Packet Offset
- DSCP
- LSP Exp
- IP Match
- MAC Match
- IPv6 Match
- Dot1p
- Policer
- Precedence
- Forwarding Class
- Queue
- Dynamic Policer
```

Prerequisites:

- Ingress Classification Policy, Scheduler Parent, and Advanced Configuration Policy should pre-exist.
- Slop policy [configure qos slope-policy]

4.27.3 Advisory

- Use the default schema form for non-SAR systems and defaultSAR for SAR systems
- NE policy name should be unique on the node.
- Sub-Insert-Shared-PCC Rule, Dynamic Policer, and Policers HQoS Manageable attributes are not supported for the XRS node.
- Attribute value "-1" in the intent indicates a default value in NFM-P.
- Policer configuration is not supported for SRa4 nodes.
- Parent Arbiter cannot be assigned to an Ingress Policer with a Stat Mode of "No-Stats".
- Rate Type should be selected as "percentLocalLimit" for PIR and CIR Percent-Rate configuration.
- Rate Type should be selected as "kbps" for PIR and CIR values in kbps.

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- The SAP Ingress Policy Name should start with a character, not an integer.
- The sap-ingress policy name default and ID 1 intent cannot be created.

SAR

- Deletion of a referenced queue should be performed in two steps:
  1. Delete the forwarding class where the queue is associated and deploy.
  2. Delete the queue in the second operation and deploy.
- Un-supported:
  - Queue: The multipoint "true" attribute is not supported on the 7705 SAR.

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4.28 Name of the Intent: qos-schedulerpolicy_csas_23-10-1_23-11

4.28.1 Applicability

Supported from: NSP 23.4/23.8/23.11

Supported On: 7210 SAS-E/SAS-D/SAS-T/SAS-Dxp/SAS-M/SAS-Sx

Verified on Node Type/Version: 9.0, 11.0, 22.9, 23.3, 23.10

4.28.2 Description

qos-schedulerpolicy_csas_23-10-1_23-11 intent type deployment is supported for 7210 SAS nodes.

The following configurations are supported per deployment:

```
- Configuration of Description, Mode, and Queue Weights for eight queues
```

4.28.3 Advisory

- SAS-E Queue Weight can accept values from 1 to 5 only (only verified on 11.0.R11).
- Queue Weight configuration is applicable only when ModeObserved is set to "Weighted Deficit Round Robin"/"Weighted Round Robin".

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4.29 Name of the Intent: qos-shared-policer_cixr_23-10-1_23-11

4.29.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR Classic SR OS node version 23.10

4.29.2 Description

qos-shared-policer_cixr_23-10-1_23-11 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of QoS Shared Policer.

The following configurations are supported per deployment:

```
- Stat-mode
- CIR (kbps), PIR (kbps)
- Committed Burst Size (bytes), Maximum Burst Size (bytes).
```

CLI context: config > qos > shared-policers > policer

4.29.3 Advisory

- fc-4-profile should be disabled to configure the shared policers (system > fp > options > qos > fc-4-profile).
- Stat-mode is not supported on IXR R6d/R6dl/x/x3/e2 chassis.

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4.30 Name of the Intent: qos-vlan-policy_cixr_23-10-1_23-11

4.30.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR node version 23.10 all IXR variants

4.30.2 Description

qos-vlan-policy_cixr_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of VLAN QOS POLICY with VLAN QOS POLICY name as target identifier.

The following configurations are supported as per deployment

```
- Description, Scope, Stat mode, Packet Byte Offset
- Queue: Queue ID, Queue Management policy, Queue type, Scheduling
Priority, Low Latency, Adaptation Rule, CIR %, CIR Adaptation, PIR %, PIR
Adaptation, CIR weight profile, PIR weight
- CIR Weight profile
```

CLI context: qos > vlan-qos-policy

4.30.3 Advisory:

- User need to pre configure Queue mgmt (qos > queue-mgmt-policy).
- In order to configure packet byte offset, user need to admin-enable packet-byte-offset.(qos > packet-byte-offset > admin-state enable).
- Stat Mode cannot be set to 'Enqueued with Discard or Enqueued 4 profile with Discards' as forwarding path options fc-4-profile admin or oper state is enabled.
- Best effort should be selected in queue type in order to modification in queue.
- Low Latency is only supported on queues associated with FCs nc and h1.
- The 7250 IXR-R6d and 7250 IXR-R6dl do not support stat Mode, queue Type, CIR Weight Profile.
- The 7250 IXR-R6 do not support low Latency, PIR Weight, stat Mode, packet-byte-offset, SchedulingPriority.
- The 7250 IXR-e2 and 7250 IXR-e2c do not support queue Type, CIR Weight Profile.
- The 7250 IXR-R4, 7250 IXR-e, 7250 IXR-s and 7250 IXR-ec do not support packet-byte-offset, PIR Weight, lowLatency, Scheduling Priority.
- The 7250 IXR-x Series do not support CIR Weight Profile, Scheduling Priority.

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4.31 Name of the Intent: qos-wred-slope_csros_23-10-1_23-11

4.31.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950 XRS, 7750 SR, 7450 ESS classic SR OS node version 23.10

4.31.2 Description

qos-wred-slope_csros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of QoS wred slope Policy.

The following configurations are supported per deployment:

```
- Description, Time Average Factor
- High Slope: Admin State, Start Average (%), Start Probability (%), Max
Average (%)
- High Plus Slope: Admin State, Start Average (%), Start Probability (%),
Max Average (%)
- Low Slope: Admin State, Start Average (%), Start Probability (%), Max
Average (%)
- Exceed Slope: Admin State, Start Average (%), Start Probability (%),
Max Average (%)
```

Prerequisite: N/A

CLI context: qos > slope-policy

4.31.3 Advisory

- The Start Average should not be more than the Max Average.

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4.32 Name of the Intent: router-isis-interface_csros_23-10-1_24-4

4.32.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS and 7250 IXR Classic SROS node version 23.10

4.32.2 Description

router-isis-interface_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of ISIS Interface with ISIS Instance and Interface name as target-indentifiers.

The following configurations are supported per deployment:

```
- Description, Administrative State, Type, Level Capability, CSNP
Interval (seconds), LSP Pacing Interval, Retransmit Interval, Mesh Group
Status, Mesh Group, Passive, IPv6 Unicast Enabled
  Route Tag, IPv4 Multicast Enabled, IPv6 Multicast Enabled, Load
Balancing Weight, Hello Padding, Loopfree Alternate Exclude, Route
Next-Hop Policy,SID Protection
- IPV4/IPV6 SID Type
    -Type (Index or Label)
    -Clear N Flag
- BFD:
    - IPv4 : Include BFD TLV
    - IPv6 : Include BFD TLV
- Flexible Algorithm
    -Key ID
    -Flexible Algorithm ID
    -Ipv4 SID Type (Index or Label)
    -Ipv6 SID Type (Index or Label)
    -Ipv4 SID value
    -Ipv6 SID value
- Authentication:
    -Enable Authentication, Type, Key, Authentication Keychain
- Level 1:
    -Hello Interval, Hello Multiplier, Metric, IPv6 Unicast Metric, IPv4
Multicast Metric, IPv6 Multicast Metric, Passive, Priority, Signal
Degrade Threshold
     Signal Fail Threshold, Hello Padding
    -Authentication:
        -Type, Key, Authentication Keychain
- Level 2:
    -Hello Interval, Hello Multiplier, Metric, IPv6 Unicast Metric, IPv4
Multicast Metric, IPv6 Multicast Metric, Passive, Priority, Signal
Degrade Threshold
     Signal Fail Threshold, Hello Padding
    -Authentication:
```

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```
-Type, Key, Authentication Keychain
```

Prerequisite:

- ISIS Instance should be created on the node
- Interface should be created on the node.
- LFA route-nh-template (configure routing-options route-next-hop-policy template)
- Authentication Keychain (configure system security keychains keychain)

CLI context: router > isis >interface

4.32.3 Advisory

- Default-instance can't be enabled on ISIS instance 0
- Adjacency-sid label is not supported for broadcast interface type.
- flex-algo can only be defined on loopback or system interface
- Audit report will not be shown for authentication-key attributes
- Distribute operation will succeed only when the preconfiguration also exists on the selected target-nodes
- Load Balancing Weight is not supported on all variants of 7250 IXR.

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4.33 Name of the Intent: router-mpls-dynamic-lsp_csros_23-10-1_25-4

4.33.1 Applicability

Supported From: NSP 25.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR, 7705 SAR, 7210 SAS classic SR OS node version 23.10

4.33.2 Description:

The router-mpls-dynamic-lsp_csros_23-10-1_25-4 intent supports the creation/modification/discovery/audit/alignment of Dynamic LSP, with lsp name as the target identifier.

The following configurations are supported per deployment:

```
- Source IP address,Destination Node ID,Destination IP Address,
Description, Administrative State.
- Egress Statistics: Collect Accounting Statistics, Accounting Policy,
Administrative State, Statistics Mode.
- Tunnels
    - MPLS Path ID, Type (Primary/Secondary/Standby), Administrative
State.
        - Transport Properties: Setup Priority, Hold Priority.
        - TE Properties: TE Reserved Bandwidth, Hop Limit, Record Actual
Path, Record Label,   Diff-Serv Class Type, Diff-Serv Backup Class Type.
        - Make Before Break: Adaptive (Make Before Break), Resignal.
        - BFD: BFD Template, BFD Enabled, LSP Ping Interval, Wait For Up
Time.
        - Administrative Groups: Admin Groups Included, Admin Groups
Excluded.
- Properties:
    - Make Before Break, Path Computation Method, Metric Type, Fallback
Path Computation Method, Route Selection Metric, Least-Fill Path
Selection,
          LDP Over RSVP Include, Enable Auto Bind, LSP Self Ping, Load
Balancing Weight.
        - Traffic Engineering And Protection: TE & Protection Fast
Reroute, Hop Limit, IGP Shortcut Enabled, BGP Shortcut Enabled,
          BGP RSVP LSP Transport Tunnel, Diff-Serv Class Type, Main Class
Type Retry Limit.
        - Auto Bandwidth: Auto BW Adjust Up Threshold, Adjust Up
Bandwidth, Adjust Down Threshold, Adjust Down Bandwidth, Minimum
Bandwidth, Maximum Bandwidth,
          Monitor Bandwidth, Sample Multiplier, Adjust Multiplier,
Overflow Limit, Overflow Limit Threshold, Overflow Limit Bandwidth,
          FC (BE, L2, AF, L1, H2, EF, H1, NC) Sampling Weight, Underflow
Limit, Underflow Limit Threshold, Underflow Limit Bandwidth, Adjusted
Bandwidth, Secondary Limit.
        - IGP Shortcut: IGP Shortcut LFA Type, Relative Metric.
```

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```
        - Fast Reroute: FR Backup Type, Hop Limit, Reserved Bandwidth,
Node Protect, Propagate Administrative Groups.
        - Signalling: Signalling Retry Timer, Retry Limit, RSVP Reserve
Style, Propagate Admin Groups via Session Attribute, Revert Timer.
        - Administrative Groups: Admin Groups Included, Admin Groups
Excluded.
        - Class Forwarding: Policy Name, Set ID.
        - BFD: BFD Template, BFD Enabled, LSP Ping Interval, Failure
Action, Wait For Uptime/
        - PCE: PCE Control, PCE Report.
        - Entropy Label Capacity: Entropy Label.
- Path Profile: Path Profile ID, Path Group ID.
- PCE Associations: Association Type, Association Name.
- Admin Tags: Tag Name.
```

CLI context: router > mpls > lsp

Prerequisites:

The following profile/list must be pre-configured on the node.

- Loopback Interface must be configured.
- IGP configuration.
- Enable MPLS on interfaces.
- bfd-template < *configure router bfd bfd-template* >.
- admin-group < *configure router if-attribute admin-group* >.
- admin-tags < *configure router admin-tags admin-tag* >.
- class-forwarding-policy < *configure router mpls class-forwarding-policy* >.
- path < *configure router mpls path* >.
- enable pcep pcc < *configure router pcep pcc admin-state enable* >.
- accounting-policy < *configure log accounting-policy* <policyName> *record combined-mpls-lsp-egress* >.
- pce-associations diversity < *configure router pcep pcc pce-associations diversity* >.
- pce-associations policy < *configure router pcep pcc pce-associations policy* >.

Advisory:

- Before you can create an MPLS path from the config form, the template of **`router-mpls-path_csros_23-10-1_23-11`** must exist in ICM.
- Cannot configure 'from' and 'to' IPv6 addresses for **dynamic LSPs**.
- Tunnels: Only one **primary path** is allowed per LSP.
- Tunnels: Configure 'path preference' is supported only for **standby path type**.
- If 'metric-type' is set to 'TE', then 'path-computation-method' must be specified (it cannot be 'none').
- 'lsp-ping-interval' and 'BFD enabled' can only be configured if a 'bfd-template' is specified.
- Cannot disable 'path-computation-method pce' if 'fallback-path-computation-method' is enabled.

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- To enable 'least-fill', either 'CSPF' must be enabled or 'path-computation-method' must be set to a value other than 'none'.
- 'path-computation-method' cannot be disabled if 'least-fill' is enabled on the LSP.
- Cannot have 'path-profile' and 'pce-associations- diversity or policy' configurations together.
- 'backup-class-type' cannot be the same as the 'main class-type'.
- Cannot enable 'fast-reroute' when TE Properties Record label: 'no record' is already set on the primary path.
- The record must be true if fast-reroute is configured.
- BFD can be configured for LSP or for primary/secondary path but not both.
- Cannot enable 'lsp-self-ping' if a 'bfd-template' is already configured at the LSP path level.
- Cannot have 'backup type' set to 'Facility' when 'Record Label' set to 'No Record' & 'RSVP Reserve Style set' to 'Fixed-Filter'.
- 'setup-priority' cannot be higher than 'hold-priority'.
- LSP level setup-priority and hold-priority is not applicable on IXR and SAR.
- Deleting an LSP path from the configuration requires two steps:
  1. Set the administrative state of the LSP path to **`down`** and deploy the change.
  2. Delete the LSP path from the list and redeploy.
- To manually configure the destination ID using NSP Release 24.11 or earlier, please contact your Nokia support representative.

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4.34 Name of the Intent: router-mpls-interface_csros_23-10-1_24-4

4.34.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950/7705 SAR/7210 SAS and 7250 IXR classic SR OS node version 23.10

4.34.2 Description

The router-mpls-interface_csros_23-10-1_24-4 intent supports the creation/modification/deletion/brownfield discovery/audit/alignment of Router MPLS-Interface, with Interface Name as the target identifier.

The following configurations are supported per deployment:

```
- Administrative state
- Administrative group
- Shared Link Risk Group : SRLG Pointers
- TE Metric
- Static Label Maps : Label Action [Pop, Swap], Administrative State,
Ingress Label, Egress Label, Implicit Null Label, Next Hop Address
- TP MEP Interface: Validation, Alarm Indication Signal (AIS), TP Section
Layer ID
```

Prerequisites:

- Network Interface should be created in the routing instance.
- SRLG should be configured on the node [configure router if-attribute]
- Admin-group be configured on the node [configure router if-attribute]

4.34.3 Advisory

- MPLS and MPLS-TP must be enabled from the router.
- Static Label maps of type "unspecified" are not supported during the association.
- TE Metric & Label Maps are not supported on unnumbered MPLS-TP interfaces.
- TP MEP Interface configuration is not supported on 7250 IXR/SAR/SAS.
- Shared risk link groups are not allowed on the system interface.
- Label maps cannot be created on the system interface.
- Static Label Maps Swap: The "Enable Implicit Null Label" attribute is not applicable for SAR.

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4.35 Name of the Intent: router-mpls-path_csros_23-10-1_23-11

4.35.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR, 7705 SAR and 7210 SAS Classic SROS node version 23.10

4.35.2 Description

The router-mpls-path_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of mpls path, with path name as target identifier. The following configurations are supported per deployment:

```
- MPLS Tunnel: Administrative State, Description
- MPLS Provisioned Path
    MPLS Provisioned Hop: Hop ID, IP Address, SID-Label, Type
```

CLI context: router > mpls > path

Prerequisite:

- N/A

4.35.3 Advisory

- For multihop, if the NFM-P inserts or deletes a hop, an audit may report this as an issue
- A Path cannot have both SID and IP Address configured across its hops.
- IP address and SID-Label cannot be configured together.
- SID-Label and Type cannot be configured together.

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4.36 Name of the Intent: router-mpls-srte-lsp_csros_23-10-1_25-4

4.36.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR, 7705 SAR for Classic SROS node version 23.10

4.36.2 Description

The router-mpls-srte-lsp_csros_23-10-1_25-4 intent supports the creation/modification/deletion/discovery/audit/alignment of type SRTE lsp, with lsp name as the target identifier.

The following configurations are supported per deployment:

```
- Configuration of SRTE LSP as below
   Source IP Address, Destination Node ID, Destination IP Address,
Description, Administrative State
   -Statistics
    Accounting Policy, Administrative State, Collect Accounting
Statistics

- Configuration of the Properties

   Path Computation Method, Metric Type, Fallback Path Computation
Method, Route Selection Metric, Enable Auto-Bind, Load Balancing Weight,
Override Entropy Label Capability, Binding SID
   Hop Limit, IGP Shortcut Enabled, BGP RSVP LSP Transport Tunnel, BGP
Shortcut Enabled, Maximum Segment Routing Labels, Maximum FRR Overhead
Labels, Local SR Protection, Label Stack Reduction
   LFA Type, Relative Metric, Allow SR Over SRTE
   Retry Timer, Retry Limit, Revert Timer
   Admin Groups Included, Admin Groups Excluded
   Enable Class Forwarding, Policy Name, Set ID
   BFD Template, BFD Enabled, Wait For Uptime, Return Path Label
   PCE Control, PCE Report
   Entropy Label, PCE Computation, Name

- Configuration of Tunnel properties

   Type, Administrative State, Setup Priority, Hold Priority, Path
Preference
   Reserved Bandwidth, Hop Limit, Resignal
   BFD Enabled, BFD Template, Wait For Uptime, Return Path Label

- Path Profile: Profile ID, Path Group

- PCE Associations: Diversity, Policy.
```

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- Admin Tags: Tag Name

CLI context: router > mpls > lsp

Prerequisites:

- Enable MPLS protocol in a router [configure router]
- Configure systeme interface into MPLS [configure router mpls interface "system"]
- Configure sr-labels [configure router mpls-labels sr-labels]
- Configure reserved-label-block [configure router mpls-labels reserved-label-block]
- Configure bfd-template [configure router bfd bfd-template]
- Configure admin-group [configure router if-attribute admin-group]
- Configure admin-tags [configure router admin-tags]
- Configure class-forwarding-policy [configure router mpls class-forwarding-policy]
- Configure MPLS Path [configure router mpls path]
- Configure accounting-policy [configure log accounting-policy record combined-mpls-srte-egress]
- Configure pce-associations diversity and policy [configure router pcep pcc pce-associations]

4.36.3 Advisory

- Before you can create an MPLS path from the config form, the template of **router-mpls-path_csros_23-10-1_23-11** must exist in ICM.
- The 'fallback-path-computation-method' is supported with the 'path-computation-method' pce only.
- BFD can be enabled only if 'bfd-template' is configured.
- The value of the 'binding-sid' must fall within the range of the associated label block.
- Cannot have 'path-profile' and 'pce-associations- diversity or policy' configurations together.
- The 'path-computation-method' must be specified if metric-type is 'te'.
- The 'return-path-label' is valid only if 'bfd-template' is configured.
- SAR does not support Accounting Policy, Collect Accounting Statistics and Administrative State for SRTE LSP
- SAR does not support admin-tag and Class Forwarding for SRTE LSP
- SAR does not support bfd-template
- IXR nodes cannot have the following attributes configured: Class Forwarding, holdPriority, class-type, soft-preemption, failure-action.
- BFD can configured at the LSP level, cannot be configured at the path level.
- SR and SAR nodes cannot have the following attributes configured: Path MTU, Prefer Transport FRR, Path Setup Priority, Secondary Retry Limit, Hold Priority, Soft Preemption, Manual Switch LSP.
- To configure the destination ID using the NSP Release 24.11 or earlier, contact your Nokia support representative.
- Tunnels: only one primary path is allowed per LSP.

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- Tunnels: Configure 'path-preference' is supported only for standby path type.
- Cannot set 'metric-type' to 'TE' when 'path-computation-method' is set to 'none'.
- When 'metric-type' is set to 'TE' then 'path-computation-method' must be specified.
- 'Admin-group' is displayed at path-level only by enabling Overridden Properties.
- 'Setup priority' cannot be higher (numerically lower) than 'holding priority' at Path level.
- Cannot create more than one path for SR-TE LSP with 'pce-compute' enabled for SAR nodes
- The Path Preference and Enable SRLG attributes are supported on standby or secondary path.

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4.37 Name of the Intent: router-mpls_csros_23-10-1_24-11

4.37.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR, 7705 SAR and 7210 SAS Classic SROS node version 23.10

4.37.2 Description

The router-mpls_csros_23-10-1_24-11 intent supports the creation/modification/deletion/discovery/audit/alignment of mpls configurations, with "mpls" as the target identifier. The following configurations are supported per deployment:

```
- MPLS
    - Administrative State, Fast Reroute Signaling, Secondary Fast Retry
Timer (seconds), Dynamic Bypass, Static LSPs Fast Retry Timer (seconds),
Exponential Backoff Retry
    - Maximum Bypass Associations, Logger Event Bundling, Maximum Bypass
PLR Associations, Enable Administrative Group for FRR Path Computation
    - LSP Init Retry Timeout (seconds), Static Label Range, Make Before
Break Prefer Current LSP Hops, Start Segment Routing Label, End Segment
Routing Label
    - BGP Labels Hold Timer (seconds), Retry on IGP overload, PCE Report,
Entropy Label for RSVP-TE LSPs, Entropy Label for SR-TE LSPs, Strict ERO
Next-hop Direct Resolution
    - LSP Resignal Timer (min), Active Path Fast Retry Timer (seconds),
P2MP Resignal Timer (min), S2L Active Path Fast Retry Timer (seconds),
P2MP TTL Propagate, Bypass Resignal Timer (min)
    - SR-TE Resignal Timer (min), Resignal SR-TE LSPs On IGP Event,
Resignal SR-TE LSPs On IGP Overload, Hold Timer (seconds), Enable SRLG
for FRR, Strict, Use SRLG Static Configuration, Least Minimum Threshold,
Least Reoptimization Threshold
    - Shortcut Local TTL Propagate, Shortcut Transit TTL Propagate,
Sample Multiplier, Adjust Multiplier, CSPF On Loose Hop, Interval
(seconds), Enable RSVP-TE, Timeout (seconds)
    - Timeout Action, Resignal on IGP Overload, SR-TE, RSVP-TE, LSP BSID
Block Name, LSP History, Administrative State, Maximum Number of Static
SVC Labels, Auxiliary Statistics, Node SID, Maximum Number of Static LSP
Labels
    - Class Forwarding Policy: Policy Name, Default Set ID
        - Class Forwarding Policy FC: Forwarding Class, Set ID
    - MPLS LSP Ingress Stats: Sender Address, Lsp Name, Collect
Accounting Statistics, Ingress Accounting Policy, Admin State, Statistics
Mode
    - MPLS LSP Template Ingress Stats: LSP Template Type, Sender Address,
Lsp Name, Collect Accounting Statistics, Ingress Accounting Policy, Admin
State, Statistics Mode, Maximum Accounting Statistics
    - Forwarding Policy General: Admin State, Reserved Label Block,
```

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```
Tunnel Table Preference
        - Forwarding Policy: Policy Name, Admin State, Binding Label,
Endpoint IP Address, Revert Timer (seconds), Preference, Tunnel
Preference, Tunnel Metric
            - Next Hop Group: ID, Admin State, Resolution Type
                - Next Hop: Next Hop Type, Next Hop Address, Label 1 to
10
            - Forwarding Policy Statistics: Statistics Type,  Admin State

    - PCE-Initiated LSP
        - SR TE: Administrative State
```

CLI context: router > mpls

Prerequisite:

- Configure srlg group [configure routing-options if-attribute srlg-group <name>]
- Configure LSP BSID block [configure router mpls-labels reserved-label-block <label-block-name>]
- Configure Accounting Policy [configure log accounting policy]

4.37.3 Advisory

- To enable pce-init-lsp sr-te admin-state, you must configure 'p2p-sr-te-pce-init' lsp-template with default template-id.
- SR-TE LSP Resignal must be either 0 or between 30 and 10080.
- You cannot enable 'lsp-self-ping rsvp-te' if bfd-template is associated with an RSVP LSP.
- retry-on-igp-overload and resignal-on-igp-overload are mutually exclusive attributes.
- You cannot configure both binding-label and endpoint attribute.
- Metric can only be set with endpoint policy
- tunnel-table-pref can only be set with valid endpoint IP
- Pushed labels must be configured in sequential order.
- Binding-label must be configured if ingress-statistics is enabled.
- class-forwarding-policy related attributes are not supported on the 7250 IXR.
- In order to delete a deployment from the NSP and the network, you must remove all the dependendent objects
- The following attributes not supported on SAR/SAS nodes:
  - class-forwarding-policy
  - MPLS LSP Ingress Stats
  - MPLS LSP Template Ingress Stats
  - Forwarding Policy General
  - PCE-Initiated LSP
  - Secondary Fast Retry Timer (seconds)
  - Exponential Backoff Retry,
  - Enable Administrative Group for FRR Path Computatation

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- Make Before Break Prefer Current LSP Hops
- BGP Labels Hold Timer (seconds)
- Retry on IGP overload
- Strict ERO Next-hop Direct Resolution
- Active Path Fast Retry Timer (seconds)
- P2MP Resignal Timer (min)
- Node SID, Auxiliary Statistics
- LSP History
- LSP BSID Block Name
- Resignal on IGP Overload
- Enable RSVP-TE
- Use SRLG Static Configuration
- Resignal SR-TE LSPs On IGP Event
- Resignal SR-TE LSPs On IGP Overload

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4.38 Name of the Intent: router-network-interface_cixr_23-10-1_24-4

4.38.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 IXR Classic SROS node version 23.10

4.38.2 Description

router-network-interface_cixr_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of Router network Interface with interface name as target-indentifier.

The following configurations are supported per deployment:

```
- General
- ICMP and ICMP6 Configuration
- PTP Configuration
- BFD and BFD6 Configuration
- DHCP Relay Configuration
- ARP Configuration
- Delay Metric Configuration
- Router Advertisements
- DNS Options
```

Prerequisites:

- Port and LAG configurations
- Policy Accounting Template
- CPU protection and Dist-CPU Protection Policy
- Egress VLAN QoS Policy
- Egress Remark Policy
- Network Ingress Policy
- Accounting Policy
- Ingress/Egress IP ACL
- Cflowd Sampling >Sample profile
- VRRP/VRRP6 > Monitor Oper Group Name and Member Oper Group Name
- VRRP Policy
- Python policy
- Link Measurement Template Name

CLI context: configure > router > interface

4.38.3 Advisory

- This intent supports creation of network and unnumbered interface.
- Only Ethernet ports in Network or Hybrid mode can be attached to this interface
- 'IPv6 Allowed' should be selected for all IPV6 configuration.

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- Port configuration is not supported when Loopback is enabled.
- Suggest function for Proxy ARP configuration is not supported for remote routers.
- Suggest function is not shown for the attributes which are dependent on the configuration of other attributes.
- If arp-proactive-refresh is configured then the arp timeout should be greater than 45sec.
- Static ARP must exist prior to administratively turning up the Network Interface.
- Address id 1 is reserved for ipv4 address in DHCP config.
- IP address configuration should be deleted before modifying the interface to unnumbered.
- From NSP, GI IP address cannot support during interface creation. GI address can be added as part of update/modify operation. GI-address must be a local interface address.
- The first entry in DHCP Relay Configuration addresses table should be IPv4 address type and Address ID should be 1 and same IPv4 address should be configured with GI IP address.
- Address configuration is not supported on unnumbered mpls tp interface.
- DHCP Relay is not supported on Unnumbered MPLS-TP Network Interface.
- Message interval in VRRP Configuration must be in 100 ms increments.
- Master inherit interval must be turned off when message interval millisecond is non-zero in VRRP Configuration.
- 'Unnumbered Reference', 'Unnumbered Reference Type' is supported when the Interface Class is unnumbered.
- Ipv6 helper address should be configured when help address type is selected.
- Valid echo interval is 100-100000ms and 0 if disabled
- 'LDP End Of Lib' can configure only after 'LDP Synchronization Timer' is configured.
- Link local address should always starts with fe80:: eg: fe80::abcd:efff:fe12:3456
- The format to write IPv6z address is: < ipv6-link-local-address > % < egress-interface >
- Distribute operation will lead to deployment failure if the selected target does not support the attributes present in the template.
- Dist-cpu-protection configuration is not supported when loopback is enabled.
- Audit report will not be shown for VRRP authentication key attribute.
- While configuring DHCP Relay and DNS Option using REST API,IP address to server should be added in the sequential manner.
- In DNS Option, Include DNS should be false, and only RDNSS Lifetime can be configured.
- In DNS option, there are no separate checkboxes available for Infinite and Inherit from Site. To select Infinite and Inherit checkbox, configure the value in RDNSS Lifetime: "-1" for Infinite and "-2" for Inherit.
- Egress VLAN QoS Policy and Port Redirect is not supported on 7250 IXR-10 and IXR-6.
- Association of management Interface fails.

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4.39 Name of the Intent: router-network-interface_csros_23-10-1_24-4

4.39.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS Classic SROS node version 23.10

4.39.2 Description

router-network-interface_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of Router network Interface with interface name as target-indentifier.

The following configurations are supported per deployment:

```
- General
- ICMP and ICMP6 Configuration
- PTP Configuration
- BFD and BFD6 Configuration
- DHCP Relay Configuration
- ARP Configuration
- Delay Metric Configuration
- Router Advertisements
- DNS Options
```

Prerequisites:

- Port and LAG configurations
- Queue Group: Ingress FP Redirect Group and Ingress Queue Group Instance ID
- Queue Group: Egress Port Redirect Group and Egress Queue Group Instance ID
- Policy Accounting Template
- CPU protection and Dist-CPU Protection Policy
- Network Policy
- Ingress/Egress IP ACL
- Cflowd Sampling >Sample profile
- VRRP/VRRP6 > Monitor Oper Group Name and Member Oper Group Name
- VRRP Policy
- Python policy
- Link Measurement Template Name

CLI context: configure > router > interface

4.39.3 Advisory

- This intent supports creation of network and unnumbered mpls tp interface.
- Only Ethernet ports in Network or Hybrid mode can be attached to this interface.

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- Outer encapsulation and Inner Encapsulation value is supported only on ports with encap type as QinQ.
- 'IPv6 Allowed' should be selected for all IPV6 configuration.
- Port configuration is not supported when Loopback is enabled.
- Cannot enable 'PTP HW Assist' when interface is not bound to a port or interface does not have an IPv4 address.
- Suggest function for Proxy ARP configuration is not supported for remote routers.
- Suggest function is not shown for the attributes which are dependent on configuration of other attributes.
- If arp-proactive-refresh is configured then arp timeout should be greater than 45sec.
- Static ARP must exist prior to administratively turning up the Network Interface.
- Address id 1 is reserved for ipv4 address in DHCP config.
- IP address configuration should be deleted before modifying the interface to unnumbered.
- Lag link map profile is only supported on lag ports.
- From NSP, GI IP address cannot support during interface creation. GI address can be added as part of update/modify operation.GI-address must be a local interface address.
- The first entry in DHCP Relay Configuration addresses table should be IPv4 address type and Address ID should be 1 and same IPv4 address should be configured with GI IP address.
- Address configuration is not supported on unnumbered mpls tp interface.
- DHCP Relay is not supported on Unnumbered MPLS-TP Network Interface.
- Strip label configuration is supported only when the port is bound to the interface.
- Strip label is only supported on IOM3 IOM-XP and IMM cards.
- Message interval in VRRP Configuration must be in 100 ms increments.
- Master inherit interval must be turned off when message interval millisecond is non-zero in VRRP Configuration.
- 'Unnumbered Reference', 'Unnumbered Reference Type' is supported when Interface Class is unnumbered.
- Ipv6 helper address should be added if help address type is selected.
- Valid echo interval is 100-100000ms and 0 if disabled
- 'LDP End Of Lib' can configure only after 'LDP Synchronization Timer' is configured.
- 'Queue Group: Ingress FP Redirect Group' and 'Ingress Queue Group Instance ID' both should be configured at the same time.
- 'Queue Group: Egress Port Redirect Group' and 'Egress Queue Group Instance ID' both should be configured ate the same time.
- Link local address should always starts with fe80:: eg: fe80::abcd:efff:fe12:3456
- The format to write IPv6z address is: < ipv6-link-local-address>%<egress-interface >
- Distribute operation will lead to deployment failure if the selected target does not support the attributes present in the template.
- Value Added Service (eg :vas-if-type>to-from-network) is not supported when loopback is enabled.

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- Dist-cpu-protection configuration is not supported when loopback is enabled.
- Audit report is not shown for VRRP authentication key attribute.
- GRE Termination can be set on a single interface per router. Deployment will fail if it is set on any other interfaces on the same router.
- While configuring DHCP Relay and DNS Option using REST API, IP address to server should be added in the sequential manner.
- In DNS Option, Include DNS should be false, and then only RDNSS Lifetime can be configured.
- VRRP Policy should be added in the format < network:NEID:VRRP:policyname > eg: network:92.168.96.31:VRRP:2
- Sample Profile Pointer should be added in the format < network:NEID:cflowd:samProf-id> eg: network:92.168.96.31:cflowd:samProf-1
- In DNS option, there are no separate checkboxes available for Infinite and Inherit from Site. To select Infinite and Inherit checkbox, configure the value in RDNSS Lifetime: "-1" for Infinite and "-2" for Inherit.
- Association of management Interface fails.

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4.40 Name of the Intent: router-ospf-area-virtual-link_csros_23-10-1_24-4

4.40.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750-SR, 7950-XRS, 7450-ESS and 7250-IXR Classic SROS of node version 23.10

4.40.2 Description

router-ospf-area-virtual-link_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of OSPFv2 VIrtual Link.

The following configurations are supported per deployment:

```
- Administrative State
- Authentication Type
- Password
- Authentication Keychain
- Hello Interval
- Router Dead Interval
- Retransmission Interval
- Transit Delay
- OSPF MD5 Key Index and Key.
```

Prerequisite: keychain Policy should be preconfigured.

CLI context: configure > router > ospf > area 0.0.0.0 > virtual-link

4.40.3 Advisory

- Audit will not show diff for Authentication Key and OSPF MD5 Key.
- Authentication Key and OSPF MD5 Key needs to be configured manually while distributing a Brownfield discovered intent.
- Align will not set AuthenticationKey OSPF MD5 Key if it is blank in config instance.
- For BrownfieldField Discovery, select the Virtual Neighbour Router and Transit Area combination configured on the node.

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4.41 Name of the Intent: router-ospf-area_csros_23-10-1_24-4

4.41.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950 and 7250 IXR classic SR OS node version 23.10

4.41.2 Description

router-ospf-area_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of OSPFv2 area.

The following configurations are supported per deployment:

```
Area Type,BIER Template,BIER Adminstrative state,Default Cost,Originate
Default Route
Boolean attributes:
    - Loop-free Alternate Exclude,Blackhole Range,Enable Advertise Router
Capability,
    - Database Export Exclude,Adjacency Check,NSSA Redistribute External
Routes
Area Range can be configured with below supported attributes:
    - Key ID,Link State DB Type, Network, Prefix Length, Effect
OSPF Area Export Policy : Max 5 routing policies can be associated.
OSPF Area Import Policy : Max 5 routing policies can be associated.
```

Prerequisites:

- User need to pre-configure ospfv2 instance under which area will be created.
- BIER Template, Routing Policies should be pre configured for association in ospf area intent.

CLI context: router > ospf > instance > area

4.41.3 Advisory

- Do not modify the area type if the area type is backbone.
- Only for area type NSSA, Link State DB type can be configured as "NSSA".
- Policies must be imported and exported sequentially
- After the area range is configured, only the "Effect" attribute can be modified.
- BIER Template and BIER Adminstrative state are not applicable for 7250 IXR.

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4.42 Name of the Intent: router-ospf-interface_csros_23-10-1_24-4

4.42.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750/7450/7950 and 7250 IXR classic SR OS node version 23.10

4.42.2 Description

router-ospf-interface_csros_23-10-1_24-4 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of OSPF Interface configuration on 7750, 7950 and 7450 nodes with OSPFv2 Instance, OSPF AreaID and InterfaceName as targetIdentifier.

The following configurations are supported per deployment:

```
- Configured MTU (bytes)
- Interface Type
- Priority
- Advertise Subnet
- Passive
- BFD Enabled
- Remain Down on BFD Failure
- Metric, Loop-free Alternate Exclude
- LSA Filter Out
- Enable Advertise Router Capability
- RIB Priority
- Load Balancing Weight
- BFD Strict
- Strict Mode Hold Down
- Hello Interval (seconds)
- Retransmission Interval (seconds)
- Poll Interval (seconds)
- Router Dead Interval (seconds)
- Transit Delay (seconds)
- Route Next-Hop Policy
- Node SID Type
- Node SID Value
- SID Protection
- Adjacency SID Type
- Adjacency SID Value
- Authentication Type
- Authentication Keychain
- Password
- SPF MD5 Key
- OSPF Segment Routing Adjacency Set Binding- Adjacency Set ID
- OSPF Interface Flexible Algorithm -Flexible Algorithm ID, SID Type, SID
Value
```

Prerequisites: The following must be configured:

- Authentication Keychain

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- Route Next-Hop Policy
- Adjacency Set

4.42.3 Advisory:

- System interface and loopback interfaces must be passive enabled.
- BFD can only be enabled on OSPF Interface when BFD is enabled on the corresponding Network Interface.
- Audit will not show diff for Authentication Key and OSPF MD5 Key.
- AuthenticationKey and OSPF MD5 Key needs to be configured manually while distributing a Brownfield discovered intent.
- Align will not set AuthenticationKey OSPF MD5 Key if it is blank in config instance.
- SID Label/Index and OSPF Interface Flexible Algorithm can be configured only on system Interface or loopback interface.
- IPv4 Neighbor address should match the netmask of the Interface.
- Adjacency SID Type, Adjacency SID Value are not configurable on system and loopback Interface
- Adjacency SID Value should be within the range of the start and end label of Reserved label block assigned to the OSPF Instance.
- OSPF Interface Flexible Algorithm SID Value and Node SID Value should be within the range of Segment-routing Label range configured on the NE.
- Load Balancing Weight, Adjacency Set ID and Neighbor address are not supported on 7250 IXR.

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4.43 Name of the Intent: router-rsvp-interface_csros_23-10-1_24-4

4.43.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS classic SR OS node version 23.10

4.43.2 Description

router-rsvp-interface_csros_23-10-1_24-4 intent supports modification/discovery/audit/alignment of Router RSVP Interface.

The following configurations are supported per deployment:

```
- Description
- Administrative State
- Enable Graceful Shutdown
- Class Type BW
- Enable Implicit Null Label
- Inherit Implicit Null Label
- BFD Enabled
- Graceful Restart Helper Mode
- Enable Refresh Reduction
- Enable Reliable Delivery
- Hello Interval
- RSVP Authentication
- RSVP TE Thresholds
```

Prerequisites:

- User need to configure MPLS interface on NFMP or node. Then along with MPLS interface, RSVP interface will get create. Then we can do brownfield discovery of the same.
- Configure RSVP Keychain.

4.43.3 Advisory:

- We can delete RSVP interface from NSP, if user wants to delete RSVP interface from NFMP/node then user has to delete MPLS interface. So along with MPLS, RSVP interface will get deleted.
- Flow of execution: Create MPLS interface in NFMP>> RSVP interface gets created>>brownfield discovery>>modification>>audit>>align>>delete MPLS interface>>RSVP interface gets deleted
- User needs to add value in incremental order for Inherit TE Up Thresholds and decremental order for Inherit TE Down Thresholds otherwise will get misalignment during audit.
- User should make sure that the total sum of Class Type BW should not exceed 100.

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4.44 Name of the Intent: routing-community_csros_23-10-1_23-11

4.44.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750-SR, 7950-XRS, 7450-ESS, 7250-IXR, 7210 SAS and 7705 SAR SROS of node version 23.10.

4.44.2 Description:

The routing-community_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a routing community, with Community Name as the target identifier.

The following configurations are supported per deployment:

```
- Members
- Expression: Expression, Exact
```

CLI context: configure router > policy-options > community

4.44.3 Advisory:

- Members and expressions are mutually exclusive.
- Community operator/bracket/expressions must be separted by space characters.
- Adding community members or removing expressions causes a failed deployment, but retrying the deployment succeeds.
- The following are examples of members and expressions.
- Members:
  - ipv6-redirect:22:22:22:22:22:455:4542:876
  - ext:ffaf:fffffffffffe
  - target:2.2.2.2:4
  - origin:2.2.2.2:6
  - target:56:78
  - no-advertise
  - 4.4.4.4:1234
  - 65535:65535
  - llgr-stale
  - ext:4300:1
  - no-export
  - blackhole
  - no-llgr
  - 0:23
  - 1.1.1.6
- Expression:
  - 2:2 AND 200:300
  - 2:2 OR 200:300

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- 2:2 AND NOT 200:300
- 2:2 OR NOT 200:300
- 2:2 OR null
- (8:8 AND NOT 9:9) AND NOT (8:8)
- NOT 2:2 AND 200:200
- target:5:100 AND NOT target:9:9
- ((target:9:0 OR target:8:8) AND (target:7:4 OR target:4:5))

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4.45 Name of the Intent: routing-damping_csros_23-10-1_23-11

4.45.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7950/7450/7250 IXR classic SR OS node version 23.10

4.45.2 Description

routing-damping_csros_23-10-1_23-11 Intent supports creation/modification/deletion/discovery/audit/alignment.

The following configurations are supported per deployment:

```
- Reuse [0-20000]
- Half Life [0-45]
- Max Suppression [0-720]
- Suppress [0-20000]
```

4.45.3 Advisory

N/A

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4.46 Name of the Intent: routing-policy-statement_cixr_23-10-1_24-4

4.46.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7250 classic for IXR SROS node version 23.10

4.46.2 Description

routing-policy-statement_cixr_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of PolicyStatement with PolicyStatement name as targetIdentifier.

CLI context: router > policy-options > policy-statement

The following configurations are supported per deployment:

```
- Enable Default Action
    - Sticky ECMP, Install Backup Path, Advertise Label, Advertise Next
Hop Self, BGP Leaking, Create UDP Tunnels Enabled, Resolve Static,
Disable Route Table Install, BGP High Priority, Create MPLS Tunnel, OSPF
Route Type Value/Policy-Global variable, Route Origin Value/Policy-Global
variable, Route Preference Value/ Policy-Global variable, BGP Local
Preference Set/ Value/ Policy-Global Variable, Next Hop Type and Value,
Metric Action and Value, BGP Metric Source/ Expression/ Policy-Global
variable, OSPF/RIP/ISIS Tag Value and Policy-Global Variable, Accumulated
IGP Metric, Origin Validation State Value , Add Paths Send Limit, SR
Label Index Preference IGP, SR Label Index Policy-Global Variable, BGP
Tunnel Metric Preference AIGP, BGP Tunnel Metric Preference MED, BGP
Tunnel Metric Policy-Global Variable, Damping Profile Name, Flex Algo
Policy-Global variable, Source Class Index, Destination Class Index,
Admin Tag Policy Name, SR Maintenanace Policy,  BGP AS Path Name and
Action, BGP AS Prepend number, Prepend Count, BGP AS Prepend
Policy-Global Variable, Prepend Count Policy-Global Variable, Prepend
Most Recent AS, Interface, Add/Remove/Replace Community Name.
- Policy Statement Entry ID, Description, Action, Term Name, Enable From
Criteria, Enable To Criteria.
- Policy Statement Entry Action
    - Sticky ECMP, Install Backup Path, Advertise Label, Advertise Next
Hop Self, BGP Leaking, Create UDP Tunnels Enabled, Forwarding Class Name/
Priority, Resolve Static, Disable Route Table Install, BGP High Priority,
Create MPLS Tunnel, OSPF Route Type Value/Policy-Global variable, Route
Origin Value/Policy-Global variable, Route Preference Value/
Policy-Global variable, BGP Local Preference Set/ Value/ Policy-Global
Variable, Next Hop Type and Value, Metric Action and Value, BGP Metric
Source/ Expression/ Policy-Global variable, OSPF/RIP/ISIS Tag Value and
Policy-Global Variable, Accumulated IGP Metric, Origin Validation State
Value and Policy-Global Variable, Add Paths Send Limit, SR Label Index
Preference IGP, SR Label Index Policy-Global Variable, BGP Tunnel Metric
Preference AIGP, BGP Tunnel Metric Preference MED, BGP Tunnel Metric
Policy-Global Variable, Ingress Statistics, Egress Statistics, Damping
```

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```
Profile Name, Flex Algo Policy-Global variable, Source Class Index,
Destination Class Index, Admin Tag Policy Name, SR Maintenanace Policy,
BGP AS Path Name and Action, BGP AS Prepend number, Prepend Count, BGP AS
Prepend Policy-Global Variable, Prepend Count Policy-Global Variable,
Prepend Most Recent AS, Interface, Add/Remove/Replace Community Name,
Route Exists Strings.
-Policy Statement Entry From Criteria
    - AS Path Name, AS Path Group Name, Protocol Bits, Community List
Name, Origin, State, Interface Name, Static Route Tag, No Route Tag,
Family, Origin Validation State, Policy Statement Name, BGP Path Type, AS
Path Length, AS Path Length Unique, Community Count, Metric, Next Hop
Prefix-List, Next Hop Type/ Hop, Cluster Id None, Active MG VM-ID, EVPN
Type, Aggregate Contributor, Color, Distinguisher, End Point Type, End
Point, Route Distinguisher, ISIS Level, ISIS External Route, OSPF Route
Type, Area IP Addr Type,  OSPF Area, OSPF Area Set, Multicast Source IP
Address Type, Multicast Source IP Address, Multicast Source Prefix List
Name, Multicast Group Prefix List Name, IGMP Host Prefix List Name, BGP
MVPN Route Type, Neighbor IP Address, Neighbor IP Address Zone Index
Pointer, Source Prefix List Name, Destination Prefix List Name, Neighbor
Prefix List, 5 Cluster Address Addresses and Type, Interface Names,
Policy Variables list, Prefix List and Type Community Expression, Add
Prefix List Member Name, Route Exists Strings.
- Policy Statement Entry To Criteria
    - Protocol Bits, ISIS Level, Neighbor IP Address and Type, Neighbor
Prefix List, Prefix List Members,Route Exists Strings.
```

Prerequisites: : The following need to be configured:

- Prefix List
- Community
- Admin Tag
- Damping Profile
- SR Maintenance policy
- Route Distinguisher policy

4.46.3 Advisory

- Default Action attributes can be configured only when Default action is set to value other than None.
- Policy Entry Action attributes can be configured only when Action is set to value other than None.
- Policy Entry From Criteria attributes can be configured only when Enable From Criteria is set to true.
- Policy Entry To Criteria attributes can be configured only when Enable To Criteria is set to true.
- All the Policy/Global variable string must start and end with an at-sign(@).

In Default Action and Policy entry Action:

- OSPF Route Type value and OSPF Route Type policy variable are mutually exclusive.
- Route Origin value and Route Origin policy variable are mutually exclusive.

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- BGP Local Preference value and BGP Local Preference policy variable are mutually exclusive.
- Use Next Hop and Peer Address are mutually exclusive. Next Hop Policy/Global Variable and Use Next Hop are mutually exclusive.
- Metric Source can be configured only when Metric Action is set to value other than None. BGP policy/Global variable can be configured when BGP Metric source is set to MED value. BGP Metric Expression can be configured only when BGP Metric source is set to Expression.
- BGP Metric Expression must contain one valid variable igp or med or @parameter@ and one valid operator(+ or - or *) (max 64 chars).
- Origin validation Policy/Global variable can be configured when Origin validation State Value is set to Not Applicable.
- Origin validation State value and Origin Validation State policy variable are mutually exclusive.
- Flex Algo Policy/Global Variable accepts a numeric value from 128-255 or a string that must start and end with an at-sign(@).Flex Algo Policy-Global Variables accepts a Global Policy of type INT with a configured value in the range 128-255.
- Use Next Hop and Advertise Next Hop Self are mutually exclusive.
- Preference IGP cannot be set to True without SR Label Index value or Policy/Global variable configuration.
- When Prepend Most Recent AS is enabled either Prepend Count or Prepend count Policy variable must be configured.
- When Prepend Most Recent AS is enabled and Prepend Count is set to non-zero, BGP AS Prepend Number should be 0.
- Prepend Count value and Prepend Count Policy variable are mutually exclusive.
- Prepend AS value and Prepend AS Policy variable are mutually exclusive

In Policy entry From Criteria:

- Either AS-Path or AS-Path-Group should be selected
- AS path length are mutually exclusive to As path length indicator or AS path length unique
- Static route tag and No Route Tag are mutually exclusive.
- Community name cannot be set if community expression is configured.
- Community Count accepts numeric value from 0-1024 or String must start and end with an at-sign(@).
- Local Preference accepts numeric value from 0-255 or String must start and end with an at-sign(@)
- Metric accepts numeric value from 0-255 or String must start and end with an at-sign(@)
- OSPF Area can be configured when OSPF Area Set is set to true.
- Community Expression are defined expression that are using AND/OR/NOT operators with a maximum length of 900 characters. Community name should be enclosed within square bracket-[] and cannot be more than 32 characters. Ex: [aaa] AND [bbb] [aaa] OR NOT [bbb]

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4.47 Name of the Intent: routing-policy-statement_csros_23-10-1_24-4

4.47.1 Applicability

Supported from: NSP 24.4

Verified on Node Type/Version: 7950/7450/7750 classic SR OS node version 23.10

4.47.2 Description

routing-policy-statement_csros_23-10-1_24-4 intent supports creation/modification/deletion/discovery/audit/alignment of PolicyStatement with PolicyStatement name as targetIdentifier.

The following configurations are supported per deployment:

```
- Enable Default Action
    - Sticky ECMP, Install Backup Path, Advertise Label, Advertise Next
Hop Self, BGP Leaking, Create UDP Tunnels Enabled, Resolve Static,
Disable Route Table Install, BGP High Priority, Create MPLS Tunnel, OSPF
Route Type Value/Policy-Global variable, Route Origin Value/Policy-Global
variable, Route Preference Value/ Policy-Global variable, BGP Local
Preference Set/ Value/ Policy-Global Variable, Next Hop Type and Value,
Metric Action and Value, BGP Metric Source/ Expression/ Policy-Global
variable, OSPF/RIP/ISIS Tag Value and Policy-Global Variable, Accumulated
IGP Metric, Origin Validation State Value and Policy-Global Variable, Add
Paths Send Limit, SR Label Index Preference IGP, SR Label Index
Policy-Global Variable, BGP Tunnel Metric Preference AIGP, BGP Tunnel
Metric Preference MED, BGP Tunnel Metric Policy-Global Variable, Ingress
Statistics, Egress Statistics, Use BFD Sid, BFD Sid Address Type/ Value,
SRv6 Return Path BFD Policy/Global Variable, SR Return Path BFD Label
Policy/Global Variable, Damping Profile Name, Flex Algo Policy-Global
variable, Source Class Index, Destination Class Index, Admin Tag Policy
Name, SR Maintenanace Policy, SRv6 Locator, BGP AS Path Name and Action,
BGP AS Prepend number, Prepend Count, BGP AS Prepend Policy-Global
Variable, Prepend Count Policy-Global Variable, Prepend Most Recent AS,
Interface, Add/Remove/Replace Community Name.
- Policy Statement Entry ID, Description, Action, Term Name, Enable From
Criteria, Enable To Criteria. Conditional Expression Route exists string.

- Policy Statement Entry Action
    - Sticky ECMP, Install Backup Path, Advertise Label, Advertise Next
Hop Self, BGP Leaking, Create UDP Tunnels Enabled, Forwarding Class Name/
Priority, Resolve Static, Disable Route Table Install, BGP High Priority,
Create MPLS Tunnel, OSPF Route Type Value/Policy-Global variable, Route
Origin Value/Policy-Global variable, Route Preference Value/
Policy-Global variable, BGP Local Preference Set/ Value/ Policy-Global
Variable, Next Hop Type and Value, Metric Action and Value, BGP Metric
Source/ Expression/ Policy-Global variable, OSPF/RIP/ISIS Tag Value and
Policy-Global Variable, Accumulated IGP Metric, Origin Validation State
Value and Policy-Global Variable, Add Paths Send Limit, SR Label Index
```

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Preference IGP, SR Label Index Policy-Global Variable, BGP Tunnel Metric Preference AIGP, BGP Tunnel Metric Preference MED, BGP Tunnel Metric Policy-Global Variable, Ingress Statistics, Egress Statistics, Use BFD Sid, BFD Sid Address Type/ Value, SRv6 Return Path BFD Policy/Global Variable, SR Return Path BFD Label Policy/Global Variable, Damping Profile Name, Flex Algo Policy-Global variable, Source Class Index, Destination Class Index, Admin Tag Policy Name, SR Maintenanace Policy, SRv6 Locator, BGP AS Path Name and Action, BGP AS Prepend number, Prepend Count, BGP AS Prepend Policy-Global Variable, Prepend Count Policy-Global Variable, Prepend Most Recent AS, Interface, Add/Remove/Replace Community Name.

- Policy Statement Entry From Criteria

- AS Path Name, AS Path Group Name, Protocol Bits, Community List Name, Origin, State, Interface Name, Static Route Tag, No Route Tag, Family, Origin Validation State, Policy Statement Name, BGP Path Type, AS Path Length, AS Path Length Unique, Community Count, Community Count Indicator, Community Count Type, Local Preference Indicator, Metric, Metric Indicator, Next Hop Prefix-List, Next Hop Type/ Hop, Cluster Id None, Active MG VM-ID, EVPN Type, Aggregate Contributor, Color, Distinguisher, End Point Type, End Point, Route Distinguisher, SRv6 TLV, Instance ID, All Instances, ISIS Level, ISIS Externak Route, OSPF Route Type, Area IP Addr Type, OSPF Area, OSPF Area Set, Multicast Source IP Address Type, Multicast Source IP Address, Multicast Source Prefix List Name, Multicast Group Prefix List Name, IGMP Host Prefix List Name, BGP MVPN Route Type, Neighbor IP Address, Neighbor IP Address Zone Index Pointer, Source Prefix List Name, Destination Prefix List Name, Neighbor Prefix List, 5 Cluster Address Address and Type, Interface Names, Use SRv6 Sid Prefix, SRv6 Sid Prefix Type/ Address/ Prefix Length/ Policy-Global Variables, Policy Variables list, Community Expression, Add Prefix List Member Name.

- Policy Statement Entry To Criteria

- Protocol Bits, Instance ID, All Instances, ISIS Level, Neighbor IP Address and Type, Neighbor Prefix List, Prefix List Members, Custom Properties

Prerequisites:

The following need to be configured:

- Prefix List
- Community
- Global variables policy
- AS Path
- AS Path Group
- Admin Tag
- Damping Profile
- SR Maintenance policy
- SRv6 Locator policy

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- Route Distinguisher policy

4.47.3 Advisory

- Default Action attributes can be configured only when Default action is set to value other than None.
- Policy Entry Action attributes can be configured only when Action is set to value other than None.
- Policy Entry From Criteria attributes can be configured only when Enable From Criteria is set to true.
- Policy Entry To Criteria attributes can be configured only when Enable To Criteria is set to true.
- All the Policy/Global variable string must start and end with an at-sign(@).

In Default Action and Policy entry Action:

- OSPF RouteType value and OSPF Route Type policy variable are mutually exclusive.
- Route Origin value and Route Origin policy variable are mutually exclusive.
- BGP Local Preference value and BGP Local Preference policy variable are mutually exclusive.
- Use Next Hop and Peer Address are mutually exclusive. Next Hop Policy/Global Variable and Use Next Hop are mutually exclusive.
- Metric Source can be configured only when Metric Action is set to value other than None. BGP policy/Global variable can be configured when BGP Metric source is set to MED value. BGP Metric Expression can be configured only when BGP Metric source is set to Expression.
- BGP Metric Expression must contain one valid variable *igp* or *med* or @parameter@ and one valid operator(+ or - or *) (max 64 chars).
- OSPF, RIP or ISIS Tag value or Policy/Global variable can be configured only when Accumulated IGP Metric is set to add/replace
- Origin validation Policy/Global variable can be configured when Origin validation State Value is set to Not Applicable.
- Origin validation State value and Origin Validation State policy variable are mutually exclusive.
- SR Return Path BFD Label Policy/Global Variable accepts numeric value from 32..1048512 or string must start and end with an at-sign(@).
- Flex Algo Policy/Global Variable accepts a numeric value from 128-255 or a string that must start and end with an at-sign(@). Flex Algo Policy-Global Variables accepts a Global Policy of type INT with a configured value in the range 128-255.
- Use Next Hop and Advertise Next Hop Self are mutually exclusive.
- Preference IGP cannot be set to True without SR Label Index value or Policy/Global variable configuration.
- SR Maintenance Policy cannot be un-configured when SRv6 return path BFD is configured.
- When Prepend Most Recent AS is enabled either Prepend Count or Prepend count Policy variable must be configured.
- When Prepend Most Recent AS is enabled and Prepend Count is set to non-zero, BGP AS Prepend Number should be 0.
- Prepend Count value and Prepend Count Policy variable are mutually exclusive.
- Prepend AS value and Prepend AS Policy variable are mutually exclusive

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- SRv6 Return Path BFD SID Value can be configured only when Use BFD Sid is enabled.

In Policy entry From Criteria:

- AS-Path and AS-Path-Group are mutually exclusive.
- Static route tag and No Route Tag are mutually exclusive.
- Community name can not be set if community expression is configured.
- Either Neighbor IP Address or Neighbor Prefix List can be set but not both.
- Community Count accepts numeric value from 0-1024 or String must start and end with an at-sign(@).
- Local Preference accepts numeric value from 0-255 or String must start and end with an at-sign(@)
- OSPF Area can be configured when OSPF Area Set is set to true.
- Prefix-address host bits must be 0 for Policy-variable of type Prefix.
- SRv6 Sid Prefix Policy-Global Variables accepts Global Variable Policy of type Prefix with IPv6 Prefix address.
- Community Expression are defined expression that are using AND/OR/NOT operators with a maximum length of 900 characters. Community name should be enclosed within square bracket-[] and cannot be more than 32 characters. Ex: [aaa] AND [bbb] [aaa] OR NOT [bbb]

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4.48 Name of the Intent: routing-prefix-list_csros_23-10-1_23-11

4.48.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7705 SAR and 7210 SAS Classic SR OS node version 23.10

4.48.2 Description

The routing-prefix-list_csros_23-10-1_23-11 intent supports the creation/modification/deletion/brownfield discovery/audit/alignment of a Routing Prefix List Policy.

The following configurations are supported per deployment:

```
- Prefix List Member : Prefix Type (IPv4 & IPv6), Prefix, Mask, Type
(Exact, Longer, Through, Range, To, Address Mask)
```

4.48.3 Advisory

- Deployment misalignment occurs on audit of an IPv6 address with any combination of IPv4 and IPv6. For example: <::d.d.d.d > and <x:x:x:x:x:x:d.d.d.d > .
- Avoid configuring Prefix List with the Type To or Address Mask for SAS and SAR nodes, as these are not supported.

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4.49 Name of the Intent: sec-aaa-local-profile_csros_23-10-1_23-11

4.49.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR, 7705 SAR Classic SR OS node version 23.10

4.49.2 Description:

The sec-aaa-local-profile_csros_23-10-1_23-11 intent supports the creation/modification/discovery/audit/alignment of AAA Local Profile, using Profile Name as the target identifier.

The following configurations are supported per deployment:

```
- Description, Default Action, LI Profile
- Profile Entry : ID, Description, Action, Match String
```

CLI context: configure > system > security > profile

4.49.3 Advisory:

- Non-LI users cannot modify LI Profile.
- LI Profile is not supported on the 7250 IXR and 7705 SAR.
- Match String accepts CLI commands only.

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4.50 Name of the Intent: sec-client-tls-profile_csros_23-10-1_23-11

4.50.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR, 7705 SAR CSROS node version 23.10

4.50.2 Description:

sec-client-tls-profile_csros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of client-tls-profile wth name as client-tls-profile targetIdentifier.

The following configurations are supported per deployment:

```
- Client Signature List
- Admin state
- Cipher List
- Protocol Version
- Trust Anchor Profile
- Certificate Profile
- Group List
- Default Result
```

CLI context: system > security > tls > client-tls-profile

Prerequisites: - N/A

Advisory: - N/A

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4.51 Name of the Intent: sec-maf_csros_23-10-1_23-11

4.51.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR, 7705 SAR and 7210 SAS Classic SROS node version 23.10

4.51.2 Description

The sec-maf_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of security maf, using default-maf name as the target identifier. The following configurations are supported per deployment:

```
- Description, IPv4 Default Filter Action, IPv4 Administrative Status,
IPv6 Default Filter Action, IPv6 Administrative Status, MAC Default
Filter Action,
  MAC Administrative Status

- IPv4 Entries:
        - ID, Displayed Name, Description, Action, Source IP, Source IP
Mask, Source IP Prefix,  Source Port Type, Destination Port, Destination
Port Mask
          Source Port, Source Port Mask, Protocol,  Router, Fragment, L4
Source Port, L4 Source Port Mask
- Ipv6 Entries:
        - ID, Displayed Name, Description, Action, Source IP, Source IP
Mask, Source IP Prefix,  Source Port Type, Destination Port, Destination
Port Mask
          Source Port, Source Port Mask, Next Header, Log, Router,
Fragment, Flow Label, L4 Source Port, L4 Source Port Mask

- Mac Entries:
        - Entry ID, Displayed Name, Description, Action, Frame Type,
Source MAC, Src Mask, Destination MAC, Dst Mask,
          Dot1p, Dot1p Mask, Ether Type, CFM Opcode, CFM Val 1, CFM Val
2, Service Id, Log
```

CLI context: system > security > management-access-filter

Prerequisite:

- N/A

4.51.3 Advisory

- Before modifying a MAC Entry, the MAC Admin State muse be down and the default action must be permit.
- Changing the default action to Deny can cause connectivity issues.
- Only configure the Source Port ID parameter when the Source Port Type is set to Port

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4.52 Name of the Intent: sec-server-tls-profile_csros_23-10-1_23-11

4.52.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR Classic SR OS node version 23.10

4.52.2 Description:

sec-server-tls-profile_csros_23-10-1_23-11 intent supports creation/modification/discovery/audit/alignment of server-tls-profile with server-profile-name as identifier.

The following configurations are supported per deployment:

```
- Administrative State, Certificate Profile, Re-negotiate-timer, Cipher
List, Group List, Server Signature List, Protocol Version
- Status Verify : Default Result.
- Authenticate Client : Trust Anchor Profile, Common Name List.
```

CLI context: configure > system > security > tls > server-tls-profile.

Prerequisite: The following profiles or lists must be configured on the node:

- authenticate-client common-name-list <*configure system security pki common-name-list* >
- authenticate-client trust-anchor-profile <*configure system security tls trust-anchor-profile* >
- cert-profile <*configure system security tls cert-profile* >
- cipher-list <*configure system security tls server-cipher-list* >
- group-list <*configure system security tls server-group-list* >
- signature-list <*configure system security tls server-signature-list* >

4.52.3 Advisory:

- The Cipher List and Certificate Profile can only be added or removed when the TLS server profile is in the admin shutdown state.
- For Protocol Version TLS 1.3 or all versions, enabling the server-tls-profile requires a cert-profile, server-cipher-list, server-group-list, and server-signature-list.
- For Protocol Version TLS 1.2, enabling the server-tls-profile requires a cert-profile and server-cipher-list.

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4.53 Name of the Intent: sec-ssh_csros_23-10-1_23-11

4.53.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR, 7705 SAR, 7210 SAS-R12/DXP/MXP/Sx/SAS-K/SAS-T/SAS-M CSROS node version 23.10

4.53.2 Description:

The sec-ssh_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of ssh, using ssh as the target identifier. The following configurations are supported per deployment:

```
- Servers Configuration
- SSH Client Kex List
- SSH Server Kex List
- SSH Client MAC List
- SSH Server MAC List
- SSH Authentication Configuration
- SSH Client Cipher List
- SSH Server Cipher List
- SSH Client Key Re Exchange
- SSH Server Key Re Exchange
```

CLI context: system > security > ssh **CLI context:** system > login control

Prerequisites:

- N/A

4.53.3 Advisory:

- Enabling public key as the SSH authentication method may disrupt the SSH connection
- A strong and standard login password are recommended
- 7210 SAS nodes support SNMP v1 and V2, and 7250 IXR nodes support SNMP V2
- 7210 SAS nodes support only thw following password hasskey in server configuration: sha2-pbkdf2, bcrypt

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4.54 Name of the Intent: sec-tls-cert-profile_csros_23-10-1_23-11

4.54.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR, 7705 SAR CSROS node version 23.10

4.54.2 Description:

The sec-tls-cert-profile_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of cert-profile, using name as the cert-profile target identifier.

The following configurations are supported per deployment:

`- Tls Certificate profile entry configuration`

CLI context: system > security > tls > cert-profile

Prerequisites:

- N/A

Advisory:

- Configure the admin state to disable while adding a TLS cert profile certificate and key.
- Modifying a cert_profile while the administrative state is up causes deployment failures in NFM-P

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4.55 Name of the Intent: sec-tls-client-cipher-list_csros_23-10-1_23-11

4.55.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR, 7705 SAR, 7750 SR CSROS node version 23.10

4.55.2 Description:

The sec-tls-client-cipher-list_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a client-cipher-list, using name as the client-cipher-list target identifier.

The following configurations are supported per deployment:

- Tls 1.3 client cipher list configurations with id and Cipher name

- Tls cient Cipher list configurations with id and Cipher name

CLI context: system > security > tls > client-cipher-list

Prerequisites: - N/A

Advisory: - N/A

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4.56 Name of the Intent: sec-tls-server-cipher-list_csros_23-10-1_23-11

4.56.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR MD SROS node version 23.10

4.56.2 Description

The sec-tls-server-cipher-list_csros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of a client cipher-list, using CLIENT-CIPHER-LIST-NAME name as the target identifier. The following configurations are supported per deployment:

```
- Tls12 Cipher configurations with Id and Cipher name:

- Tls13 Cipher configurations with Id and Cipher name:
```

CLI context: system > security tls > server-tls-cipher-list

4.56.3 Advisory

- N/A

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4.57 Name of the Intent: sec-tls-trust-anchor-profile_csros_23-10-1_23-11

4.57.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7450 ESS, 7950 XRS, 7250 IXR classic nodes and 7705 SAR from node version 23.10

4.57.2 Description

The sec-tls-trust-anchor-profile_csros_23-10-1_23-11 intent supports the creation/modification/deletion/brownfield discovery/audit/alignment of a trust anchor profile, with Trust Anchor Profile Name as target identifier.

The following configurations are supported per deployment:

`- CA Profile Names`

CLI context: configure > system > security > tls > trust-anchor-profile

Prerequisites:

- CA Profiles should be pre-configured

4.57.3 Advisory

- N/A

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4.58 Name of the Intent: security-user_csros_23-10-1_23-11

4.58.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS, 7250 IXR ,7705 SAR and 7210 SAS Classic SROS node version 23.10

4.58.2 Description

The security-user_csros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of a security user, using User Name as the target identifier. The following configurations are supported per deployment:

```
- Public Key authentication method from Server to Client : System, False,
True
- Access: ne3s
            - NE3S Password, Confirm NE3S Password
          nwi3
            - LTE OMS Password, Confirm LTE OMS Password
          linux
            - Old Password, New Password, Confirm New Password
          console
            - Password, Confirm Password, Password Aging, Home Directory,
Restrict to Home, Save When Restricted, Console Login Exec File, User
Cannot Change Password
              Console New Password At Login, Console User Local Lockout,
Authorization Level, Status , Session Inactivity Timeout
              Profile [MAX 8]
          ftp
            - Password, Confirm Password, Password Aging, Home Directory,
Restrict to Home, Save When Restricted
          snmp
            - Authentication Protocol, Privacy Protocol, Status
          li
          grpc
          netconf
-
```

CLI context: system > security > user

4.58.3 Advisory

- Cannot create a new user with SNMPv3 compatiblity
- SNMPv3 password cannot be set if the Node is discovered using SNMPv2 mediation
- A non-LI user cannot modify the LI option of the access property.

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4.59 Name of the Intent: security-zone_csros_23-10-1_23-11

4.59.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7705 SAR classic SROS of node version 23.10

4.59.2 Description

The security-zone_csros_23-10-1_23-11 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of a security zone using targetIdentifier as the ID.

The following configurations are supported per deployment:

```
- Displayed Name, Description, Bypass Config, Zone Type
[Base/IES/VPRN/VPLS/MVPLS/EPIPE], Inbound TCP Session Limit, Inbound UDP
Session Limit, Inbound ICMP Session Limit, Inbound Other Session Limit,
Outbound TCP Session Limit, Outbound UDP Session Limit, Outbound ICMP
Session Limit, Outbound Other Session Limit, Security Log, IES Service,
VPRN Service, Auto-Bind, VPLS Service, MVPLS Service, EPIPE Service.
- NAT Pool:  Entry ID, Displayed Name, Description, Direction
[None/Inbound/Outbound].
- NAT Pool Entry: Entry ID, Source Network Interface, Source IES L3
Access Interface, Source VPRN L3 Access Interface, IP Operator
[NONE/EQUAL/RANGE], IP Address Start, IP Address End, Port Operator
[NONE/EQUAL/RANGE], Src Port Start, Src Port End.
- Security Policy: Security Policy.
```

CLI context:

configure > router > zone

configure > service > ies > zone

configure > service > vprn > zone

configure > service > vpls > zone

configure > service > epipe > zone

Prerequisites: The following must be configured before they can be associated with a security zone:

- Security Log
- Security Policy
- Network Interface
- IES Service and IES L3 Access Interface
- VPRN Service and VPRN L3 Access Interface
- VPLS Service, MVPLS Service, and EPIPE Service

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4.59.3 Advisory

- Security-zone is not supported on the following nodes: SAR-A T1/E1, SAR-A, SARM ASAP, SARM, SAR-Hm, SAR-Hmc.
- The source network interface must be configured in separate steps: first, deploy the zone policy with the zone type set to "base"; then, associate the zone with the network interface; and finally, add the source network interface.
- The Source IES L3 Access Interface must be configured in separate steps: first, deploy the zone policy with the zone type set to "IES"; then, associate the zone with the IES L3 Access Interface; and finally, add the Source IES L3 Access Interface.
- The Source VPRN L3 Access Interface must be configured in separate steps: first, deploy the zone policy with the zone type set to "VPRN"; then, associate the zone with the VPRN L3 Access Interface; and finally, add the Source VPRN L3 Access Interface.
- To delete an intent, first remove any associated NAT pool entries for the source network interface, source IES L3 access interface, or source VPRN L3 access interface. Then, disassociate the zone from the interface (network, IES L3 Access, or VPRN L3 Access) before deleting the intent.
- Modifying an existing NAT pool entry (from IP address to interface or vice-versa) causes a deployment failure on the NFM-P. To change a NAT pool entry, delete the existing entry and then create it with the new values.
- A security zone policy can have only one associated security policy at a time.

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4.60 Name of the Intent: service-bgp-evpn_csros_23-10-1_24-4

4.60.1 Applicability

Supported From: NSP 24.4

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7450 ESS classic SR OS node version 23.10

4.60.2 Description:

service-bgp-evpn_csros_23-10-1_24-4 intent supports modification/discovery/audit/alignment of Service System BGP EVPN.

The following configurations are supported per deployment:

```
- Enable Auto Route Distinguisher
- Multicast Leave Sync Propagation
- Route Distinguisher Type
- Route-Distinguisher IP Address
- Route-Distinguisher Community Value
- SRLB Reserved Label Block
- VPLS Control Word
- AD per-ES Routes Advertisement Type
- Allocate Leaf Label
- Allocate Static Leaf Label
- Attribute Uniform Propagation
- BGP Path Selection
- D-Path Length Ignore
- Attribute Propagation
- D-PATH Ignore
- BGP EVPN Ethernet Segment:
- Name
- Ethernet Segment Type
- Auto ESI
- ESI
- Route Originating Address
- Route Next Hop Address
- LAG
- Port
- SDP
- PW Port
- VPRN Next Hop Address
- Network Interconnect VXLAN
- Activation Timer
- Source BMAC LSB
- BMAC Table Size
- AC Influenced DF Capable
- Multi-homing
- Service Carving
```

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```
- Member Oper Group Name
- Administrative State
- EVI Range
- ISID Range
- L3 Next Hop EVI Range
- S-Tag Range
- S-Tag-C-Tag Range
- Q-Tag Range
- Service Range
- VCID Range
```

CLI context: configure > service > system > bgp-evpn

4.60.3 Advisory:

- Service System BGP EVPN cannot be deleted from the network.
- The align operation in NSP cannot remove an existing BGP EVPN Ethernet Segment from NFMP if it is associated with Port/PW Port/LAG/VPRN Next Hop Address/Network Interconnect VXLAN/SDP.
- User has to modify Auto Route Distinguisher IP Address and Start-End range seperately otherwise, it will cause deployment failure.
- Source BMAC LSB not supported for VPRN Next Hop Address and Network Interconnect VXLAN.
- L3 Next Hop EVI Range should be configured with VPRN Next Hop Address.
- VCID Range should be configured with SDP.
- Service Range should be configured with Network Interconnect VXLAN.
- Q-Tag Range should be configured with Port/LAG (Dot1Q Encapsulation type).
- S-Tag Range, S-Tag-C-Tag Range should be configured with Port/LAG (QinQ Encapsulation type).

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4.61 Name of the Intent: subscriber-mgmt-sla-profile_csros_23-10-1_24-11

4.61.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750 SROS, 7450 ESS CSROS node version 23.10

4.61.2 Description:

The subscriber-mgmt-sla-profile_csros_23-10-1_24-11 intent supports the creation/modification/deletion/discovery/audit/alignment of ike-policy, with name as the targetIdentifier.

The following configurations are supported per deployment:

```
- Application, Description,Credit Control Policy, Default Instance
Sharing, Access Ingress Policy, Use Shared Queue,Use Multipoint Shared
Queue, Report Rate Type,Report Rate Scheduler,Report Rate Policer ID,
Access Egress Policy, Scheduler Policy, Use Egress QoS Marking from SAP,
Scheduler Type, Report Rate Type, Report Rate Scheduler, Use Ingress L2TP
DSCP, HS Aggregate Rate Limit (kbps),HS Queue Stats Mode, Ingress IP
Filter, Egress IP Filter, Ingress IPv6 Filter, Egress IPv6 Filter,
One-Time HTTP Redirect Filter, VLAN per Subscriber, VLAN per Service,
VLAN for all Services, VLAN per ISP per Service, VLAN per Subscriber,
VLAN per Service, VLAN for all Services, VLAN per ISP per Service,
Initial (%), Change (%), High (%), Low (%), Overall Limit, Remove Oldest
Subscriber Host, Overall, PPP, DHCP, ARP, Overall, Overall, PPP SLAAC,
PPP DHCP, IPoE SLAAC, IPoE DHCP, Overall, PPP DHCP, IPoE DHCP, LAC
Overall, Category Map,hs-agg-rate-limit
- Access Ingress Queues
    - Access Policy Queue ID, PIR (kbps), CIR (kbps), Maximum Burst Size
(bytes), Committed Burst Size (KB), High Priority Reserved (%), Stats
Mode
- Access Egress Queues
    - Access Policy Queue ID, PIR (kbps), CIR (kbps), Maximum Burst Size
(bytes), Committed Burst Size (KB), High Priority Reserved (%), Port
Average Overhead (%), Stats Mode, WRR Weight, Class Weight, HS Slope
Policy
- Access Ingress Policer
    - Policer ID, PIR (kbps), CIR (kbps), Maximum Burst Size (bytes),
Committed Burst Size (bytes), Packet Offset, Stats Mode
- Egress Scheduler
    - Scheduler Name, Summed CIR, PIR (kbps)
- HS WRR Groups
    - HS WRR Group ID, PIR (kbps), Class Weight
```

CLI context: configure > subscriber-mgmt > sla-profile

Prerequisites:

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- Access Egress Policy, Access Ingress Policy, Scheduler Policy, Credit Control Policy, Policer ID, Scheduler Name, Category Map should be predefined
- Access Policy Queue ID, Access Ingress Policer, Access Egress Policer, HS Slope Policy, Egress Scheduler, HS WRR Group ID should be predefined

Advisory:

- Policer ID must be non-zero when the Report Rate Type is set to Policer ID and must be zero if the Report Rate Type is set to any other type.
- Qos Policy cannot be changed until Overrides have been deleted and the deletion has been Applied.
- When the Report Rate Type is set to Scheduler, the Scheduler field cannot be empty
- All the configured host limits must be less than or equal to Overall Limit
- The Category Map and Credit Control Policy are mutually exclusive on a SLA Profile
- IngressSharedQueuePolicyPointer cannot be set to value passed in
- Access egress queue configurable supported only on the SR-12, SR-7s chassis

The following attributes are supported only on 7750 SR-12, SR-7:

- Access Egress Queues:
  - WRR Weight
  - Class Weight
  - HS Slope Policy
- HS WRR Groups:
  - HS WRR Group ID
  - PIR (kbps)
  - Class Weight
- HS Aggregate Rate Limit
- HS Queue Stats Mode:
  - V4-V6

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4.62 Name of the Intent: subscriber-mgmt-sub-profile_csros_23-10-1_24-11

4.62.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: 7750, 7450 SROS ESS Classic node version 23.10

4.62.2 Description:

The subscriber-mgmt-sub-profile_csros_23-10-1_24-11 intent supports the creation/modification/deletion/discovery/audit/alignment of Sub-profile, using name as the targetIdentifier.

The following configurations are supported per deployment:

```
- Description, Preference, Virtual Port Hashing, Secondary Shaper
Hashing, Accounting Policy ID, Accounting Enabled, Volume Stats Type,
ANCP Policy, Host Tracking Policy, NAT Policy, IGMP Policy, MLD Policy,
Subscriber Multicast CAC Policy, PIM Policy, UPnP Policy Pointer, NAT
Prefix List Pointer, Firewall Policy, Frame-Based Accounting,
Encapsulation Offset, Adaptation Rule, Burst Limit, Class, Weight,
Ingress Scheduler, Egress Scheduler, HS Aggregate Rate Limit, HS Minimum
Reserved Bandwidth,Policy #1, Policy #2, Policy #3, Optimize for Session
Accounting, Aggregate Rate Limit, Minimum Reserved Bandwidth,Egress
Policer Control Policy,Ingress Policer Control Policy, Maximum Frame
Based Bandwidth, Minimum Separation Buffer Space, Maximum Frame Based
Bandwidth, Minimum Separation Buffer Space, Use Direct Map as Default, HS
SLA Profile Handling Mode, Encapsulation Offset Mode
- Egress Policer Level Override
  - Policer Control Name
  - Direction
  - Level
  - Policer Control Level Override Cum MBS
- Ingress Policer Level Override
  - Policer Control Name
  - Direction
  - Level
  - Policer Control Level Override Cum MBS
- Egress Scheduler Policy Entry Override
  - Scheduler Policy Entry Displayed Name
  - Summed CIR
  - PIR
  - CIR
- Ingress Scheduler Policy Entry Override
  - Scheduler Policy Entry Displayed Name
  - Summed CIR
  - PIR
  - CIR
- SLA Profile Entry
```

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```
- SLA Profile String
- SLA Profile
- Subscr Profile Name
```

CLI context: configure > subscriber-mgmt > sub-profile

Prerequisites:

- Configure the following:
  - Accounting Policy ID
  - ANCP Policy
  - Host Tracking Policy
  - IGMP Policy
  - NAT Policy
  - MLD Policy
  - Subscriber Multicast CAC Policy
  - PIM Policy
  - UPnP Policy Pointer
  - NAT Prefix List Pointer
  - Firewall Policy
  - Ingress Scheduler
  - Egress Scheduler
  - Policy #1, Policy #2, Policy #3
  - Egress Policer Control Policy
  - Ingress Policer Control Policy
  - SLA Profile
  - Policer Control Name
  - Scheduler Policy Entry Displayed Name.

Advisory:

- Cannot have both Egress Scheduler and Egress Aggregate Rate Limit/Frame Base Accounting configured
- Scheduler Policy cannot be changed until Overrides have been deleted and the deletion has been Applied
- Cannot enable Accounting On without an Accounting Policy
- Pim policy and nat/firewall policy are mutually exclusive
- Host tracking policy and IGMP policy are mutually exclusive
- Host tracking policy and MLD policy are mutually exclusive
- PIM policy and IGMP policy are mutually exclusive
- PIM policy and MLD policy are mutually exclusive
- PIM policy and Host tracking policy are mutually exclusive
- HS aggregate rate can not be smaller than HS minimum reserved bandwidth
- The Host Tracking Policy and Subscriber Multicast CAC Policy are mutually exclusive on a Subscriber Profile

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- Scheduler Policy encap offset mode only supports the Auto configuration
- AdaptationRule, burstLimit, firewall policy ARE not supported on the 7450-ESS7, 7750-SR1, SR-1s, SR-2s, SR-14s, SR-7s, SRc4, SRa4, SRa8
- nat policy, upnp policy are not supported on the 7750-SR1, SR-1s, SR-2s, SR-14s, SR-7s, SRc4, SRa4, SRa8
- Policer control Policy is not supported on the 7750-SRa4, SRa8, SR-1s, SR-2s, SR-7s, SR-14s
- HS Aggregate Rate Limit is not supported on the 7750-SR1, SR-1s, SR-2s, SR-14s, SR-7s, SRc4, SRa4, SRa8, SR-1-46s, SR-1x-48D, 7450-ESS7

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4.63 Name of the Intent: system-lldp_csros_23-10-1_23-11

4.63.1 Applicability

Supported From: NSP 23.11

Verified on Node Type/Version: 7750 SR, 7950 XRS, 7250 IXR, 7450 ESS and 7250 IXR classic SR OS node version 23.10

4.63.2 Description:

system-lldp_csros_23-10-1_23-11 intent supports creation/modification/discovery/audit/alignment of Classic System LLDP.

The following configurations are supported per deployment:

```
- Administrative Status
- Transmission Interval
- Transmission Multiplier
- Re-Init Delay
- Maximum Consecutive Transmissions
- Fast Transmission Interval
- PDUs in Fast Transmission
- Notification Interval
```

CLI context: configure > system > lldp

4.63.3 Advisory:

- For Classic System LLDP, this cannot be deleted from the network element.

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5 Unified intent types

5.1 Name of the Intent: icm-equipment-card_mda

5.1.1 Applicability

Supported from: NSP 23.4

Verified on Node Type/Version: 7950/7750/7250 classic/MD SR OS node version 21.10

5.1.2 Description

icm-equipment-card_mda intent supports creation/modification/deletion/discovery/audit/alignment of card, MDA/XIOM Card/MDA.

The following configurations are supported per deployment:

```
- Creation of Card Type with Admin State
- Choice of MDA or XIOM depends on the node type
- Configure the MDA with attributes such as MDA Slot, Admin State, MDA
Type, Sync E
- Configure the XIOM with attributes such as XIOM Slot, Admin State, XIOM
Type, MDA Slot, MDA Type, Sync E
```

5.1.3 Advisory

- Operator needs to choose Card Type based on the chassis.
- Operator needs to choose MDA Type based on the card.
- XIOM supports only 7750-SR-1s/2s/7s/14s chassis types.
- XIOM Slot starts from "X" letter such as X1/X2 for MD mode.
- MDA Slot and XIOM Slot may not co-exist in card due to node restriction. For example: Node does not allow MDA 1 and XIOM X1 in Card 6.
- ICM shows both [MDA & XIOM] options, so the operator needs to choose MDA or XIOM based on their chassis.
- Suggest Function will not work when multiple targets are selected.

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5.2 Name of the Intent: icm-equipment-port-access-c3794

5.2.1 Applicability

Supported from: NSP 22.6

Verified on Node Type/Version: SAR8/SAR18 21.10 (8x Voice protection and 8xC37.94 Voice Teleprotection port MDA)

5.2.2 Description

icm-equipment-port-access-c3794 intent type deployment can be created on TDM ports.

The following configurations are supported per deployment:

```
- Adminstatus and Description at Port level
- Adminstatus and Framing at DataChannel level
- Adminstatus and Timeslot selection at DS0Group level
```

5.2.3 Advisory

- Framing attribute value change should be done only after deleting the DS0Group and deployed to the network element. Deletion of DS0Group and modification of framing should not be done in a single deploy operation.
- On Voice Teleprotection Interface MDA only Port 1/1/1 and Port 1/1/2 support C37.94 configuration and support framing value as "Framed" only.
- On C37.94 Teleprotection Interface MDA odd ports support both "Unframed" and "Framed" framing configuration. Even ports support framing value as "Framed" only.
- Timeslot selection must start from the ts1 and must be consecutive. For Unframed framing, use the Select All option to select all the timeslots in DS0Grp.
- Audit will not show misalignment if deployment causes deployment failures on classic SR OS in NFM-P.

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5.3 Name of the Intent: icm-equipment-port-access-ce

5.3.1 Applicability

Supported from: NSP 22.6

Verified on Node Type/Version: SAR8/SAR18 node version 21.10

5.3.2 Description

icm-equipment-port-access-ce intent type deployment can be created on tdm ports.

The following configurations are supported per deployment:

```
- Deployment configuration of port Description, Port Type, and Admin
State
- Deployment creates DS1E1 Channel and supports configuration of Channel
Framing, CAS Signal Mode, and Admin State
- Deployment provides option to create/edit/delete DS0 Channel Group
- Configuration of Channel Group ID and enablement of EncapType: CEM,
Timeslots, and Admin State on DS0 Channel Group are supported
```

5.3.3 Advisory

- If DS0Grp configuration exists on the NE on a particular port, the timeslot selection might overlap with intent configuration and deployment will fail.
- If DS0Grp is configured from NE/NFM-P on port and if the intent-configured Channel Framing and Signal Mode do not match the configuration on the NE, deployment will fail.
- Timeslot changes on multiple DS0Grp in one Edit operation may fail. Initially: DS0Grp1.1 had time slots 2,3,4,5 and DS0Grp1.2 had time slots 6,7,8,9
- During edit, if the user clears the timeslot configuration and configures it as DS0Grp1.1 timeslots 6,7,8,9 and DS0Grp1.2 had time slots 2,3,4,5
- Channel framing should be selected as per the Port Type. If Port Type is DS1, then select "esf", "sf" and "DS1-unframed". If Port Type is E1, then select "G.704 no-CRC", "G.704 CRC", and "E1-unframed" as channel framing.
- For the Port Type DS1, Channel Group ID range to be selected between "1-24".

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5.4 Name of the Intent: icm-equipment-port-access-rs232

5.4.1 Applicability

Supported from: NSP 23.8

Verified on Node Type/Version: 12x Serial Data/ 12x Serial Data v2 / 12x Serial Data v3 MDAs on SAR8/SAR18 node version 21.10

5.4.2 Description

icm-equipment-port-access-rs232 intent supports creation/modification/deletion/discovery/audit/alignment on TDM ports.

The following configurations are supported per deployment:

```
- Description, Admin State of port
- Serial Channel: Speed, Device-mode, Admin State, and Serial Socket with
options to configure Admin State and Description
- Ds0 Channel Group: Admin state, Description, Encap Type
```

5.4.3 Advisory

- Operator has to consider the grouping of ports in trio manner for serial-type configuration. For example: Configuration of port type to rs232 on port 1/4/1 will fail when port 1/4/2 port type is not rs232. Grouping is different for different serial types.
- Modification of Serial Channel Speed requires Ds0 Channel Group Admin Status to be Down. This can be achieved by performing Admin Status Down using Edit operation followed by another Edit operation to modify the Speed.
- Device-mode synchronous is not supported on Serial Socket.
- Serial Channel Speed depends upon on the MDA (Serial Data v1, v2, and v3) type configured.

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5.5 Name of the Intent: icm-equipment-port-access-x21

5.5.1 Applicability

Supported from: NSP 22.6

Verified on Node Type/Version: 12x Serial Data/ 12x Serial Data v2 / 12x Serial Data v3 MDAs on SAR8/SAR18 node version 21.10

5.5.2 Description

icm-equipment-port-access-x21 intent type deployment can be created on TDM ports.

The following configurations are supported per deployment:

```
- Deployment configures Serial Type to X.21 and provides option to
configure Description and AdminStatus of port
- Deployment creates a Serial Channel and supports configuration of Speed
and AdminStatus
- Deployment provides option to create/edit/delete DS0 Channel Group.
Configuration of EncapType and AdminStatus on DS0 Channel Group is
supported.
```

5.5.3 Advisory

- Operator has to consider the grouping of ports for Serial Type configuration. For example, configuration of port type to X21 on port 1/4/1 will fail when port 1/4/2 port type is not X.21. Grouping is different for different MDA types.
- Modification of Encap Type of DS0 Channel Group is not supported. DS0 Channel Group should be deleted and recreated with the required Encap Type. Deletion and creation should be two separate Edit operations.
- Modification of Serial Channel Speed requires DS0 Channel Group Admin Status to be Down. This can be achieved by performing Admin Status Down using Edit operation followed by another Edit operation to modify the Speed.
- Audit will not show misalignment if deployment causes deployment failures on classic SR OS in NFM-P.

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5.6 Name of the Intent: icm-equipment-port-ethernet

5.6.1 Applicability

Supported from: NSP 23.4

Verified on Node Type/Version: 7750/7950/7450 node version 21.10, 7250 IXR classic/MD SR OS node version 21.5/21.7, 7210 SAS-M/SAS-T classic SR OS node version 20.0/21.0, 7705 SAR-18/SAR-8 classic SR OS node version 22.0

5.6.2 Description

icm-equipment-port-ethernet intent type deployment can be created on ethernet ports.

The following configurations are supported per deployment:

```
- Adminstatus, Description, Accounting Policy, Collect Stats,
AutoNegotiate, Dot1q EType Encap Type, MTU, PBB Etype, QinQ EType, Speed,
Mode, HoldTime Down, Hold Time Units and Hold Time Up, Down When Looped
AdminStatus, Down When Looped Keep Alive, Down When Looped Retry Timeout,
Use Broadcast Address, SSM Admin State, Code Type, ESMC Tunnel, and Tx
DUS
- LLDP MAC Type, Notification, Receive, Transmit Tunnel Nearest Bridge,
Port Id Subtype, Tx Mgmt Address Admin status and Tx Mgmt Address System
Type, Tx TLVs Port Desc, Tx TLVs Sys Cap, Tx TLVs Sys Desc and Tx TLVs
Sys Name
- Port QoS Policy and Access Accounting Policy for Default IXR
Configuration Form
- SSM AdminState, Code-Type, ESMC Tunnel and Tx DUS attributes of SyncE
- CRC Monitor attributes Window Size, Signal Degrade Threshold, Signal
Degrade Multiplier, Signal Failure Threshold, Signal Failure Multiplier
```

5.6.3 Advisory

- Accounting Policy ID association is successful if Accounting Policy is already created on the NE.
- Speed attribute should not be configured for Fixed Speed Ports on classic/MD SR OS.
- AutoNegotiate should be set to notApplicable for Fixed Speed Ports on classic/MD SR OS.
- Deleting the intent from the network will not remove the configurations made as part of deployment on classic SR OS.
- Discrepancy will not be shown in Audit if deployment causes deployment failures on classic SR OS.
- LLDP entry does not show delete option; workaround in the GUI is to zoom out 110%.
- Use default IXR Configuration Form in Template creation for IXR nodes.
- Use default SAR Configuration Form in Template creation for SAR nodes.
- Use default SAS Configuration Form in Template creation for SAS nodes.
- Dot1q Etype and QinQ Etype configurations have system limits of 4 Etypes per IXR.
- QinQ Etype value should not be entered when encap-type is Dot1q/null for IXR and SAS ports.

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- Dot1q Etype value should not be entered for network port on 7250 IXR nodes. Refer to the NE documentation for Dot1qEtype and QinQEtype restrictions on IXR NEs.
- Accounting Policy association should be removed and Collect Stats should be disabled before the modification of Port Mode (first disable statistics collection, remove the accounting policy, and deploy; then change the mode and deploy).
- Synchronous Ethernet (SyncE) should be enabled on the corresponding MDA to configure SSM AdminState, Code-Type ESMC Tunnel and Tx DUS attributes on ports.
- For Encap Type enumeration "QinQ" is not supported on network ports of IXR, SAR, and SAS.
- Encap Type null is not supported for Hybrid Ports.
- Down-when-looped should be disabled before the modification of Port Mode for SAR.
- "QinQ Etype" and "Dot1q Etype" value should not be entered for network ports on SAR and SAS.
- Cannot set Encap Type to any value other than QinQ if the l2Uplink is enabled.
- Dot1q Etype can be modified only when the port encapsulation type is Dot1q in access/hybrid mode for SAR ports.
- Signal Degrade Threshold value must be greater than or equal to Signal Failure threshold.
- Signal Degrade Threshold/Signal Failure Threshold will be disabled when no value is entered.
- Signal Degrade Multiplier should not be modified when Signal Degrade Threshold value is not entered.
- Synchronous status messages can only be enabled on ports that are synchronous-Ethernet-capable.
- Cannot set tunnel mode when lldp-nearest-bridge admin state of Rx or Rx-Tx is enabled.
- The Speed should not be entered for GigE ports, and Auto Negotiate is not available on GigE type of port.
- Dot1q Etype and QinQ Etype configuration has a system limit of 3 Etypes per SAS.
- For LLDP dest Mac entries, discrepancy is not shown for classic nodes when node has default entries for LLDP.

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5.7 Name of the Intent: icm-equipment-port-e_m

5.7.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 6 x EM MDA on SAR8/SAR18 node version 22.10

5.7.2 Description

icm-equipment-port-e_m intent type deployment can be created on E and M ports.

The following configurations are supported per deployment:

```
- Deployment can configure AdminStatus, Description, Tlp-Rx, Tlp-Tx of
port
- Deployment allows to create/edit/delete a Voice Channel and supports
configuration of AdminStatus with Signal Mode
- Deployment provides option to create/edit/delete DS0 Channel Group
- Modification of AdminStatus on DS0 Channel Group is supported and
Encap-Type is set to CEM
```

5.7.3 Advisory

- Operator can create/edit Voice Channel/DS0Grp only when Voice Channel is enabled.
- Signal Mode cannot be modified if there is a channel on the trio group of ports that uses a different transmission mode.
- Modification of Signal Mode cannot be done when a channel group exists under the Voice Channel.
- Audit will not show discrepancy if Voice Channel/DS0Grp exists on NFM-P but not on intent.

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5.8 Name of the Intent: icm-filter-ip_filter

5.8.1 Applicability

Supported from: NSP 23.4

Verified on Node Type/Version: 7750/7250/7950 classic/MD SR OS node version 22.10.R2

5.8.2 Description

icm-filter-ip_filter intent supports creation/modification/deletion/discovery/audit/alignment of ip-filter.

The following configurations are supported per deployment:

```
- Creation of IP Filter with ID and name
- Configuration of Description, Default Action
- Creation of Entry ID with Description
- Configuration of Match criteria and Action
- Configuration of the Action option, such as Accept/Drop, with
corresponding checkbox
- Choice of the following configuration options:
  a. Match Protocol choice [protocol or Protocol list]
  b. Protocol [Tcp, udp, tcp-udp, igmp, ip, egp, igp, rdp, ipv6,
ipv6-route etc.]
  c. IP Selector [IP, Src Dst IP]
     1. when "IP" is selected, user needs to configure single IP
     2. when "Src Dst IP" is selected, source and destination IP need to
be configured with Port address
  d. Port Selector[Port, Src Dst Port]
     1. when Port is selected, user needs to configure Port option such
as eq, lt, gt,range, port-list
     2. when Src Dst Port is selected, user needs to configure source and
destination port with option such as eq, lt, gt, range, port-list
```

5.8.3 Advisory

- 7250 IXR nodes do not support below attributes:
  - Match Protocol Choice: protocol-list
  - IP Selector: ip
  - Match Address Choice: ip-prefix-list
  - Port Selector: port
- Only filters created with Filter Name and Filter ID on node can be discovered in ICM.
- Port Selector supports only a few protocols, such as tcp, udp, tcp-udp, sctp and protocol-list.
- Match protocol action criteria is accept/drop for MD nodes, however for classic nodes it is forward/drop. Forward action is equivalent to accept.
- Action checkmark is missing after brownfield discovery of filter intent.

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5.9 Name of the Intent: icm-logical-lag-access

5.9.1 Applicability

Supported from: NSP 22.6

Verified on Node Type/Version: 7950/7450/7750 classic/MD SR OS node version 21.10

5.9.2 Description

icm-logical-lag-access intent supports creation/modification/deletion/discovery/audit/alignment of Access LAG.

The following configurations are supported per deployment:

```
- Configuration of Description, Admin State, Encap Type
[null/Dot1q/qinq], Mode [Access,Network,Hybrid], Port-Type [hs/Standard],
LACP Xmit Interval, LACP Xmit Stdby
- LACP configuration
    Enable LACP, Mode, Administrative Key, System Id, System Priority
- Port Threshold configuration
- Per Link Hash
- BFD
    IPv4/IPv6[Admin State,Local IP Address,Remote IP Address,Receive
Interval,Transmit Interval,Max Admin Down Time (seconds),Max Setup Time
(milliseconds),Multiplier]
- Associate port to the LAG
- Configure some LAG attributes on port such as priority, subgroup, hash
weight on
```

Preconfiguration:

- Port Mode and Encap should be configured in the node

5.9.3 Advisory

- To create LAG deployment from ICM, the operator must first enable the port and encap type from NFM-P/node CLI.
- Enable LACP is not applicable for MD nodes.
- The ports in a LAG must have the same characteristics, speed, duplex etc.
- In all, 8 ports can be bundled in a LAG, and up to 64 LAGs can be on a chassis.
- All involved ports must work at the same rate.
- All involved ports must have the same configurations regarding AutoNegotiation and Flow Control.
- All aggregated links must be point-to-point (multicast is not allowed).
- All involved interfaces must work in full-duplex modality.
- Port Type and Encap Type act as filters to list the available ports in the "Add Port" configuration.
- Hybrid Port Type does not support Null Encap Type.

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- If System Priority, Sub group, Hash group is not added during deployment creation, ICM will take default values and create the deployment for classical node.
- LAG ID should always be a numeric value.
- LAG name should always start with "lag-" and it can be either numeric or alphanumeric. However, if LAG name is lag-(only numeric) then the numeric value should match the LAG ID.
- "Add Port" section in deployment creation window will list all the ports; it is the user's responsibility to add relevant ports as LAG members.
- Both greenfield and brownfield deployments allow the same target identifier deployments in different templates.
- Pre-21.2 SR OS nodes support only LAG ID, and post-21.2 SR OS nodes support LAG Name for MD node.
- Deployment will be created in misaligned state if LACP Administrative Key is not unique.
- Max-port restriction is applicable only to SR MD mode and LAG ID.
- Max-port value for LAG ID: lag-ID 1 to 64 max-port value is 64 and lag-ID 65 to 800 max-port value is 32.
- Max-port restriction is applicable only to SR family SR OS MD nodes, not 7250 IXR nodes.
- Default value of max-port is 32 for SR and IXR nodes.

Updates for 2209:

- Use Default IXR Template for IXR nodes.
- Display only supported attributes.
- Static cost is not supported for IXR nodes.
- Port threshold value for IXR-EC range is 0-25.
- Deployment will be created in misaligned state if the LAG ID is not in the supported range for IXR nodes.

Updates for Port Type attribute from NSP 23.4:

- Port Type is not supported on 7950 XRS/7750 SR-1s/7s/14s/7250 IXR nodes.
- Port Type is not supported below 7750 21.10 R5 node version.
- Port Type is supported on 7750 SR-12/1e etc.
- Verified on Node Type/Version 7750 SR-12/1e 22.10R1.
- Port Type is silently ignored on unsupported node versions.
- Port Type is supported from NSP 23.4 onwards.

Update for LAG BFD support from NSP 23.11:

- BFD IPv6 family does not support on IXR nodes except IXR-X NE.
- LAG mode "Network" is supported.
- Deployment misalignment occurs on audit of IPv6 address with combination of IPv4 and IPv6. For example: <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.

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5.10 Name of the Intent: icm-nge-key_group

5.10.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: SAR node version 22.10

5.10.2 Description

icm-nge-key_group intent type deployment creates KeyGroup in NFM-P and supports encryption/decryption of SDPs.

The following configurations are supported per deployment:

```
- Configuration of KeyGroup Name, Description, Encryption Algorithm,
Authentication Algorithm, Re-Key
- Addition/deletion of SDP to KeyGroup
```

5.10.3 Advisory

- Group Encryption Label should pre-exist in NFM-P.
- SFTP credentials should be preconfigured to push the KeyGroup configuration to SAR nodes. SFTP failure will not be reflected in ICM.
- Disabling Encryption of SDP removes the SDP from KeyGroup in NFM-P. Delete the SDP instead of disabling encryption to avoid the intent going into Misaligned state.
- Deletion of KeyGroup fails from ICM if VPRN/PW-Template/Network Domain is associated to the KeyGroup in NFM-P.
- Distribute action of ICM cannot be used for keygroup deployments.
- Intent cannot be used in older NSP releases as certain enhancements to NSP have been made to support this feature in NSP 23.11.
- SDP ID will not be shown in misaligned attributes if SDP does not exist in NFM-P.

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5.11 Name of the Intent: icm-qos-sapegress-srqos

This intent has been withdrawn.

5.11.1 Withdrawal information

Reason for Withdrawal: The functions of this intent have been replaced with a device-specific configuration intent. Contact your Nokia support representative or information about device configuration intents.

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5.12 Name of the Intent: icm-qos-sapingress-srqos

This intent has been withdrawn.

5.12.1 Withdrawal information

Reason for Withdrawal: The functions of this intent have been replaced with a device-specific configuration intent. Contact your Nokia support representative or information about device configuration intents.

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5.13 Name of the Intent: icm-router-bgp_group

5.13.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7750/7450/7250/7705/7210 classic/MD SR OS node version 22.10

5.13.2 Description

icm-router-bgp_group intent supports creation/modification/deletion/discovery/audit/alignment of BGP Group & Neighbor.

The following configurations are supported per deployment:

```
- Creation of BGP Group with Name
- Creation of BGP Group and configuration of the following related
attributes:
    - Admin State, Connect Retry, Keepalive, Local Preference, Loop
Detect, Min Route Advertisement, Reference, Authentication Key, Multihop
- Enablement of applicable attributes such as Aggregator ID Zero,
Damping, VPN Apply Export, VPN Apply Import, Path MTU Discovery, ASN 4
Byte, Client Reflect
- Configuration of Hold Time, Family, Cluster ID, Local AS, Remove
Private, Import, Export, Graceful Restart, Outbound Route Filters
- Creation of BGP Neighbor and configuration of the following related
attributes:
    - IP Address [IPv4/IPv6], Admin State, Group, Peer Creation Type
[Dyamic, Dynamic-if-remote, Dynamic-if-local, Static], Connect Retry,
Keepalive, Local Preference, Loop Detect, Min Route Advertisement,
Reference, Authentication Key, Multihop, Peer AS
- Enablement of applicable attributes such as Aggregator ID Zero,
Damping, VPN Apply Export, VPN Apply Import, Path MTU Discovery, ASN 4
Byte, Client Reflect, L2VPN Cisco Interop, Next Hop Self, Capability
Negotiation, Passive
- Configuration of Hold Time, Family, Cluster ID, Local AS, Remove
Private, Import, Export, Graceful Restart, Outbound Route Filters
- Configuration of Prefix Limit
```

5.13.3 Advisory

- Use Default Template for 7750 SR/7450 ESS/7950 XRS nodes.
- Use Default IXR Template for IXR nodes.
- Use Default SAS Template for SAS nodes.
- Use Default SAR Template for SAR nodes.
- BGP Protocol should be enabled prior to the BGP Group creation for classic nodes.
- Local Address should be IPv4 Address for IPv4 BGP Peer.
- Local Address should be IPv6 Address for IPv6 BGP Peer.

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- Peer Creation Type attribute is ignored for classic SR OS.
- Peer Creation Type attribute for MD nodes can only be modified from dynamic to static.
- Import and Export policy suggests the preconfigured Policy-Statement on the node only when a single target is selected for deployment.
- BGP Peer configuration should be added after BGP Group creation is successful on classic nodes.
- The audit does not reveal which BGP Peer is missing and only shows that the BGP Peer Object is missing.
- Not all the family supported in the default SAS Template are supported on all the 7210 SAS nodes.
- Not all the PrefixList family supported in the default SAS Template are supported on all the 7210 SAS nodes.
- ICM Associate operation should be used to discover the BGP Groups existing on the node.
- Send-Orf attribute configuration is not supported for MD nodes.
- The audit will not show misalignment if additional configuration is performed on the node.
- BGP Neighbor Auth key will be displayed in the BGP Neighbor row table.
- Deployment misalignment occurs on the audit of IPv6 Address with the combination of IPv4 and IPv6. For example, <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.

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5.14 Name of the Intent: icm-router-communitypolicy-srrouter

This intent has been withdrawn.

5.14.1 Withdrawal information

Reason for Withdrawal: The functions of this intent have been replaced with a device-specific configuration intent. Contact your Nokia support representative or information about device configuration intents.

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5.15 Name of the Intent: icm-router-network-interface

5.15.1 Applicability

Supported from: NSP 23.4

Verified on Node Type/Version: 7750/7950/7450 MD SR OS 22.10 node version and 7250 IXR

5.15.2 Description

icm-router-network-interface intent type deployment can be used to create routing interface on Base routing instance and also to associate the interface to OSPFv2 and OSPFv3.

The following configurations are supported per deployment:

```
- Description, Admin State, CPU Protection (default Template), Port
Binding, Port, Outer Tag, Inner Tag, Loopback, LDP Sync Timer
- IPv4 Attributes: Primary Address and Prefix-Length, Secondary Address
and Prefix-Length, BFD attributes: Admin State, Transmit Interval,
Receive, Multiplier, Type
- IPv6 Attributes: Address and Prefix-Length, BFD attributes: Admin
State, Transmit Interval, Receive, Multiplier, Type
- Network Policy (applicable only for default schema)
- VLAN QoS Policy, Egress Remark Policy, Network Ingress (applicable only
for default IXR schema)
- OSPF instance and Area selection for both OSPF v2/v3
- OSPF v2 and v3 Interface attributes: Interface Name, Admin State,
Interface Type, Metric, MTU, Hello Interval, Dead Interval
- LDP Interface attributes: Admin State
- MPLS Interface attributes: Admin State
- RSVP Interface attributes: Interface Name, Admin State, Hello Interval,
BFD Liveness, Refresh Reduction, and Reliable Delivery
- ISIS Interface attributes: Interface Name, Admin State, Level
Capability, Interface Type, Hello Authentication Type, Hello
Authentication Key CSNP Interval, BFD Liveness IPv4, Include BFD TLV,
IPv6, Include BFD TLV, Metric
```

5.15.3 Advisory

- Use default Template for 7750/7450/7950 SR and use default IXR Template for 7250 IXR.
- Outer and Inner Tag has to be entered if the selected port is QinQ encapsulated-Type. Outer Tag has to be entered only if the selected port is Dot1q encapsulated-Type.
- Allow-QinQ-Network-Interface (configure>system>ip context) needs to be enabled to configure Inner Encap.
- Transmit Interval and Receive should not exceed 1000 when BFD Type is set to FP for IXR nodes.
- On MD SROS modification of IPv4, Secondary Address and IPv6 Address will create additional addresses instead of modifying the address. The address has to be deleted and a new address entry should be added.

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- VLAN QoS Policy is not applicable on IXR-10/IXR-6.
- IXR-10/IXR-6 does not support BFD Type FP.
- Creation of network interface and association of interface should be performed in different operations.
- SROS-oc-logical-inventory adaptor needs to be installed for the intent to list the configured OSPF v2/v3 Instance, Area ID, and ISIS instance configured on the node.
- Chassis types SR-2s/SR-7s/SR-14s do not support the CPU Protection policy other than Default policy.
- Select the OSPF/OSPF3 Instance and then select the Area suggested for associating Network Interface to OSPF/OSPF3 Area configured on the node.
- Intent does not support OSPF/OSPF3/LDP/MPLS/RSVP/ISIS protocol configuration. Only the association of network interface to these protocols is supported.
- Association of Network Interface to both MPLS and RSVP protocol should be performed in the same create/modify operation.
- For the association of Network Interface to OSPF v2/v3, LDP, MPLS, RSVP, and ISIS protocol, the Interface Name should be first specified as the target identifier, and the same has to be selected during the association to protocol.
- OSPF/ISIS Multi Instance (associating the interface to multiple instances of ISIS/OSPF) is not supported.
- Align operation fails if the intent configured OSPF v2/v3 Instance; Area is misaligned with the node.
- Deletion of interface association from MPLS and RSVP protocol should be performed in the same Modify operation.
- Network Policy ID should be entered for classic SR OS, and Network Policy name should be entered for MD SR OS.
- Policy Names of VLAN QoS Policy, Egress Remark Policy, and Network Ingress should be entered if default IXR schema form is used.
- Area ID suggestion will not work if the association of interface to OSPF v2 and OSPF v3 protocol is attempted in a single create/update operation.
- The audit will not show misalignment if deployment causes deployment failures on classic SR OS in NFM-P.
- ISIS BFD Liveness IPv4 and RSVP BFD Liveness should be disabled and then the Interface IPv4 BFD Admin State should be disabled in separate edit operations.
- ISIS BFD Liveness IPv6 should be disabled and then the Interface IPv6 BFD Admin State should be disabled in separate edit operations.
- Deployment misalignment occurs on the audit of IPv6 address with the combination of IPv4 and IPv6. For example, <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.

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5.16 Name of the Intent: icm-router-policystatement-srrouter

5.16.1 Applicability

Supported from: NSP 22.6

Verified on Node Type/Version: 7950/7450/7750 classic/MD SR OS node version 21.10

5.16.2 Description

icm-router-policystatement-srrouter intent supports creation/modification/deletion/discovery/audit/alignment of PolicyStatement Policy.

The following configurations are supported per deployment:

```
- Enable Default Action
- Create Entry ID with Description
- Configuration of From, To, and Action attributes under Entry ID:
    - From attributes such as Family, Attach Prefix Lists, Attach
Community Name/Expression, Protocol
    - To attributes such as Protocol
    - Action attributes such as Action Type, Local Preference,
Preference, Attach Multiple Community
```

Prerequisites:

- Prefix List and Community must be configured in the node if list/community is used.

5.16.3 Advisory

- Creation of PolicyStatement deployment fails if the PolicyStatement Policy Name is not unique. That is, Policy Name for the selected target should not be used already.

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5.17 Name of the Intent: icm-router-prefixlist-srrouter

This intent has been withdrawn.

5.17.1 Withdrawal information

Reason for Withdrawal: The functions of this intent have been replaced with a device-specific configuration intent. Contact your Nokia support representative or information about device configuration intents.

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5.18 Name of the Intent: icm-service-customer

5.18.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR/7705 SAR/7210 SAS/IXR classic/MD SR OS node version 9.0, 11.0, 22.9, 23.3

5.18.2 Description

The icm-service-customer intent supports the creation/modification/deletion/discovery/audit/alignment of customer intent.

The following configurations are supported per deployment:

```
- Configuration of Customer Name and ID
- Configuration of Description, Contact, and Phone
- Addition of sites
```

Prerequisites:

- The MDT intent artifact bundle must be installed.
- SR OS MD NEs are managed via GNMI protocol.
- IntentBasedServiceFulfillment Feature Pack should be enabled in the NSP.

5.18.3 Advisory

- Customer intents are not deleted from IBSF after being deleted from IM/ICM.
- SR OS MD nodes are not listed in the customer sites of service management. See the workaround below.
- Deployment misalignment occurs for name mismatch between the NSP and NE.

To work around the advisory above and include SR OS MD NEs in the customer sites of service management, resynch using the postman collection.

For example:

URL:https://{{server}}:443/restconf/operations/nsp-admin-resync:trigger-resync

Body: { "nsp-admin-resync:input": { "plugin-id": "mdm", "network-element": [{ "ne-id": "fded:fade:face:ffff::1c1", "sbi-classes": [{ "class-id": "nokia-conf:/configure/service/customer" }] }] } }

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5.19 Name of the Intent: icm-service-system_evpn

5.19.1 Applicability

Supported from: NSP 22.6

Verified on Node Type/Version: 7950/7750/7450/7250/7705 classic/MD SR OS node version 21.10

5.19.2 Description

icm-service-system_evpn intent supports creation/modification/deletion/discovery/audit/alignment of BGP EVPN Ethernet Segment.

The following configurations are supported per deployment:

```
- Creation of BGP EVPN Ethernet Segment with Name
- Provision to provide Ethernet Segement Identified (ESI) with validation

- Admin State option is available with Enable/Disable selection
- Choice of "Multi Homing Mode" options "Single-Active",
"single-active-no-esi-label", "All-Active", and "None"
- Provision to provide Ethernet Segement Activation Timer <0-100&gt; with
validation
- Choice of selection for Association type, either LAG or Port
- Addition of LAG/Port is available on the node based on the selection
- Brownfield discovery of BGP EVPN Ethernet segement that is already
created on NFM-P/NE
- Configuration of all CRUD operations on greenfield discoveries, and
edit and delete operations on brownfield discoveries
```

5.19.3 Advisory

- If there is a LAG/Port association in the configuration, deployment is supported on a single node at a time.
- The user must ensure that the same LAG/Port is available before the distribution of the configuration to other nodes.
- Distribution can be applied to one node at a time.

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5.20 Name of the Intent: icm-system-ptp

5.20.1 Applicability

Supported from: NSP 22.6

Verified on Node Type/Version: 7950/7750/7250 classic/MD SR OS node version 21.10

5.20.2 Description

icm-system-ptp intent supports creation/modification/deletion/discovery/audit/alignment of PTP.

The following configurations are supported per deployment:

```
- Creation of ptp with Clock Type options Master-Only, Slave-Only,
Boundary
- Choice of profile options ieee1588-2008, g8265dot1-2010, and
g8275dot1-2014
- Admin State option is available with Enable/Disable selection
- Provision to provide Announce Receipt Time Out, Log Announce Interval,
Network Type, Domain, Local Priority, Priority1, Priority2, and TX While
Sync Uncertain attributes
- Association of Port to the Port
- Configuration of the supported ptp attributes on the Port, such as
Admin State, Address, Local Priority, Log Delay Interval, Log Sync
Interval, Master only
```

Prerequisites:

- SyncE must be enabled on MDM level SyncE true.

5.20.3 Advisory

- Choose "ptp" as a unique identifier during the creation of ptp deployment.
- The ptp Admin State is enabled only when Port is associated with ptp.
- Support for some attribute/attribute ranges depends on the Clock Type/profile; for example, attributes such as Local Priority, Domain, Priority1, Log Sync Interval, and Master only.
- Cannot enable ptp on "Clock-Type Boundary" unless a timing reference under sync-if-timing configuration is enabled.
- PTP is supported from SR OS node version 21 or later, and TX While Sync Uncertain PTP attribute is supported from 22.2 or later on SR OS 7750.

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5.21 Name of the Intent: icm-system-security_cpm

5.21.1 Applicability

Supported from: NSP 23.4

Verified on Node Type/Version: 7950/7750/7450 classic/MD SR OS node version 22.10

5.21.2 Description

icm-system-security_cpm intent supports creation/modification/deletion/discovery/audit/align of CPM Filter.

The following configurations are supported per deployment:

```
- IP-Filter Attributes: Admin State, Protocol, IP Option, Source and
Destination IP addresses with (Prefix List and Address), port selections
based on the supported protocols
- IPv6-Filter Attributes, Admin State, Protocol, Source and Destination
IP addresses with (Prefix List and Address), port selections based on the
supported protocols
- CPM-Queue Attributes: CBS, MBS, PIR and CIR rates
```

5.21.3 Advisory

- If IPv4-Filter, IPv6-Filter or cpm-queue configuration exist on the node, then brownfield discovery of the intent should be performed.
- For IP and IPv6 Filter entries, Prefix List and Port List should be preconfigured on the node before use in deployment configuration.
- IP Address and Prefix for IP Filter and IPV6-Filter has to be provided in the IP Address Field as (x.x.x.x/prefix). Mask field will be ignored for classic nodes.
- During the distribution from MD to classic, if the MD node has entries associated with Mask, then make the necessary changes for the IP Address Field (x.x.x.x/prefix) as Mask will be ignored for classic nodes. The audit will show the discrepancy in Mask is not removed from Intent config for classic nodes.
- IP Administrative State should be up for both IPv4 and IPv6 if Default Action is selected as "Drop".
- CPM Queue: The value of CIR cannot be greater than the value of PIR.
- CPM Queue: The value of CBS should not be greater than MBS.
- Creation of CPM filter deployment fails if the target identifier is given other than "cpm" (all in lower case).
- Modification of IP-Filter entry protocol from tcp-up/udp/tcp to other protocols should not be performed.
- Align Operation will fail if there is a discrepancy for IP-Filter entry protocol between the ICM and node. The IP-Filter entry needs to be deleted and recreated.
- Modification of IPv6-Filter entry next header from tcp-up/udp/tcp to other supported next headers should not be performed.

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- Align operation will fail if there is a discrepancy for IP-Filter entry next header between the ICM and SR MD Nodes. The IPv6-Filter entry needs to be deleted and recreated.
- MD: The audit will not be shown as misaligned for new entries added from the node.
- Node will convert IPv4-Compatible IPv6 Address to IPv6 Address; the audit will be shown as misaligned. For example, ::192.0.2.1.

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5.22 Name of the Intent: icm-te-tunnel

5.22.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 Classic/MD SR OS node version 22.10

5.22.2 Description

The icm-te-tunnel intent supports the creation/modification/deletion/discovery/audit/alignment of LSPs and Paths based on the IETF-TE-TUNNEL Model.

The following configurations are supported per deployment:

```
- Creation of MPLS LSP with Name (only RSVP and SR-TE LSPs can be
created)
- Creation of MPLS Paths based on the primary and secondary path names
provided
- The supported configurations and mapping of IETF-TE-TUNNEL model to SR
node for LSPs is as follows:
    - Source, Destination, Admin State, Admin Tag (color), Protection
(fast reroute), Setup Priority, Hold Priority, Association Objects (Path
Profile and Path Group)
    - Supported configurations for Primary Path:
        - Path Computation Method, Path Metric Bound (Hop Limit), Setup
Priority, Hold Priority, Bandwidth, Path Affinity Names (Admin Groups to
include and exclude)
    - Supported configurations for Secondary Path:
        - Path Computation Method, Path Metric Bound (Hop Limit), Setup
Priority, Hold Priority, Bandwidth, Path Affinity Names (Admin Groups to
include and exclude), Restoration (Standby), Disjointness (Srlg)
- The Hops for MPLS Paths will be created based on Explicit Route Objects
configured for the primary and secondary paths
- Egress-Statistics attribute is deployed on the node by default with
admin up, which is currenltly not the configuration from the form
```

5.22.3 Advisory

- Only one template and one target can be selected for this intent.
- For brownfield deployment, the primary path created on the node should have a naming format <lspname>_<pathname>.
- For brownfield deployment, the secondary path created on the node should have a naming format <lspname>*secondary*<pathname>.
- The source address must be a valid IP address: either the system or loopback IP address.
- While provisioning the source address with the loopback address, ensure that the mdm-adp-openconfig-logical-inventory-sros-shared and mdm-adp-openconfig-interfaces adaptors are installed in the NSP.

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- For the color attribute, the Color-Mapping file must be updated according to the Admin-Tag configuration on the node.
- The LSP distribution feature is supported only for nodes using the same type of management, either classic to classic or MD to MD.
- The audit does show misalignment if additional configuration is performed on the node.

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6 Global intent types

6.1 Name of the Intent: bfd-bfd-template_gsros_23-10-1_23-11

6.1.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7450/7750/7250/7210 classic/MD SR OS node version 23.10

6.1.2 Description

bfd-bfd-template_gsros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of BFD Templates.

The following configurations are supported per deployment:

```
- Echo Receive (milliseconds)
- Multiplier (milliseconds)
- Receive Interval (milliseconds)
- Transmit Interval (milliseconds)
- Template Type
```

6.1.3 Advisory

- Only 7210 SAS-R6, 7210 SAS-R12, and 7210 SAS-T nodes support BFD templates.
- For 7250 IXR R4/R6/s/x/ec, the node versions should be 23.3R1 to deploy template type "fp".
- Multiplier values for 7210 SAS are 3-20.
- Once the template type is selected it cannot be reset to automatic for classic nodes.
- Template type should be selected based on the node support.
- For template type "cpm-np", the valid TX/RX interval is 10-100000 (milliseconds), else it is 100-10000 (milliseconds).

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6.2 Name of the Intent: file-accounting_gsros_23-10-1_23-11

6.2.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR/7450 ESS/7950 XRS/7250 IXR MD/Classic & 7210 SAS/7705 SAR Classic SR OS node version 23.10

6.2.2 Description

file-accounting_gsros_23-10-1_23-11 intent supports creation/modification/deletion/brownfield discovery/audit/alignment of File and Accounting Policy with Accounting Policy ID as target identifier.

The following configurations are supported per deployment:

```
- File Policy: File ID, File Name, Description, Rollover, Retention
- File Policy: Compact Flash Location (Primary Location and Backup
Location)
- Accounting Policy: Accounting Policy Name, Admin State, Description,
Collection Interval, Record, Log Memory, Default, Include System Info,
Align
- Custom Record: Significant Change, Reference AA Specific Counter,
Reference Queue (Egress-counters, Ingress-counters), Reference Policer
(Egress-counters, Ingress-counters
- Queue Counter : Queue ID, Egress-counters, Ingress-counters
- Policer Counter : Policer ID, Egress-counters, Ingress-counters
- AA Specific Counter: AA Sub Counters, AA Sub Attributes, From AA Sub
Attributes, To AA Sub Attributes
```

6.2.3 Advisory

- The accounting custom record can only have a threshold of 0 or 1 for the record Application Assurance type.
- The minimum collection interval is 5 minutes, except for policies using record type NE Schedulable Tests (saa) and Service TestHead.
- The custom record can only set the egress counters for queues 1 to 8.
- Collection interval must be in increments of 5 minutes for AA records.
- Compact Flash Location & Accounting Record Types: Vary across NEs, operator must select based on the NE configuration.
- Intent will not work for record types "combined-mpls-srte-egress" and "combined-sr-policy-egress" since these records cannot have a destination as a file.
- Classic: AA Counters, configurable only for "Custom Record AA Subscriber" and supported on ESS/SR NEs.
- Classic: XRS does not support AA Counters.
- Classic: IXR/SAR/SAS NEs, do not support AA Counters and Custom Record.
- Classic: Log Memory supported only on SAS NEs.

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- Classic: Include System Info and Align not supported on SAR and SAS NEs.
- Classic: Deletion of Queue/Policer entry from the list which is associated to the Reference-queue and Reference-policer has to be done in two separate operations. First remove the referenced Queue/Policer and deploy. Later remove the Queue/Policer from the list and deploy
- MD: File ID and Accounting Policy Name are blank after Brownfield discovery.

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6.3 Name of the Intent: filter-ipv6-filter_gsros_23-10-1_23-11

6.3.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7950/7250/7705/7210 classic/MD SR OS node version 23.10

6.3.2 Description

filter-ipv6-filter_gsros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of deployments.

The following configurations are supported on SR/XRS/ESS nodes:

```
Schema form: default.schemaForm
- Description, default-action, scope, type and filter entries
- Filter entry can have Description, "Pbr Down Action Override", Egress
PBR, Sticky Destination, Log, Filter Sample, Interface Sample, Sample
Profile configuration
- Match criteria supports all tcp flags, ip address filtering, next
header supports all IANA assigned protocols, extension header, flow
label, packet length, and hop limit
- Action supports FC, rate-limit, remark, pattern, drop-when and
accept-when criteria
- Filter configuration supports Flowspec, Openflow and embed filter
- Filter with type src-mac is not allowed when scope is system
```

The following configurations are supported on SAS/SAR/IXR nodes:

```
SAS Schema form: defaultSAS.schemaForm
SAR Schema form: defaultSAR.schemaForm
IXR Schema form: defaultIXR.schemaForm

- Description, default-action and scope (template and exclusive)
- Match criteria supports syn and ack tcp flags, ip address filtering,
next header supports all IANA assigned protocols
- Action supports drop or accept criteria
- Ignore-match action criteria cannot be configured with
pbr-down-action-override for IXR nodes
- Statistics collection is not allowed for action accept for IXR nodes
- Pbr Down Action Override and statistics collection are only supported
for IXR nodes
```

6.3.3 Advisory

- Destination class cannot be configured when filter type is src-mac.
- Packet-length match criteria is not allowed with filter type normal.
- Hop limit match criteria can be configured only with filter type src-mac.

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- Chain-to-system filter can be configured with filter type normal and packet-length.
- Shared Policer can only be configured when scope is template or exclusive.
- Egress PBR cannot be configured when interface sample is disabled.
- Log entries cannot be configured with egress PBR.
- Sample profile configuration is mutually exclusive with Egress PBR.
- Prefix List and Port List in IP address match criteria should be preconfigured on the node.
- Configuring on the correct supported node is the responsibility of the user, and deploying an unsupported configuration may result in deployment failure.
- For deployments on multiple nodes, the user must enter the policy ID/Name in the suggest field.
- MD: Audit will not be shown as misaligned for new entries added from the node.
- Openflow config for both greenfield and brownfield is not supported for classic nodes.
- Deployment misalignment occurs on audit of IPv6 address with combination of IPv4 and IPv6. For example: <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.

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6.4 Name of the Intent: filter-mac-filter_gsros_23-10-1_23-11

6.4.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7450/7750/7250/7705/7210 classic/MD SR OS node version 23.10

6.4.2 Description

The filter-mac-filter_gsros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of MAC Filter Policy.

The following configurations are supported per deployment:

```
- Creation of MAC Filter with ID and Name
- Configuration of Description, Default Action, Type[normal, vid, isid],
Scope
- Creation of Entry ID with Description, Pbr Down Action Override, Sticky
Dest, Log
- Configuration of Match with Frame Type[802dot3,802dot2-llc,
802dot2-snap,ethernet-ii]
- Configuration of the Action option with corresponding checkbox
- Choice of the following configuration options:
   Etype
   Snap Pid,Snap Oui [zero,non-zero]
   Src Mac [Address,Mask], Dst Mac [Address,Mask]
   Dot1p [Priority,Mask]
   Llc Dsap [Dsap,Mask], Llc Ssap [Ssap,Mask]
   Inner Tag [Tag,Mask], Outer Tag [Tag,Mask]
   Isid [Isid Range,Isid Value]
   Action [Accept, Drop, Forward, Http Redirect, Ignore Match]
   Rate limit
   Secondary Action Forward
```

Prerequisites:

- When using Log/Action Forward: ESI ethernet segment, VPLS service, SDP, and Log ID must be configured on the node

6.4.3 Advisory

- Use Default Template for 7750 SR/7450 ESS/7950 XRS nodes.
- Use Default IXR Template for 7250 IXR nodes.
- Use Default SAR Template for 7705 SAR nodes.
- Use Default SAS Template for 7210 SAS nodes.
- Brownfield discovery fails if the MAC Filter Name and MAC Filter ID do not match.
- Type and frame type support specific setup attributes or combination from node CLI.

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- Action should only be set to Forward SAP/Ignor Match when PBR Down Action Override/Sticky Dest attributes are configured.
- IXR nodes do not support certain attributes from node, such as: Type, Frame Type, Rate Limit, PBR Down Action Override, Sticky Dest, Log, and Secondary.
- Action Forward/Secondary VPLS ID and SAP ID need to be provided for MD node.
- Only IXR-X nodes support FC in the Action.
- The Frame Type can only be Ethernet II when the MAC Filter Type is VID.
- When the filter type is set to Normal, Action Foward SAP and Secondary Forward SAP can be configured only when the MAC Filter is already associated to the VPLS SAP.
- When the filter type is VID/ISID, Action Foward SAP and Secondary Forward SAP can be configured to any VPLS SAP.
- Before changing the Action from Forward to any other action, use an Edit operation to clear the Secondary Action, then perform another Edit operation to change the Action.
- MAC Filter name cannot be numeric for SAR nodes.
- For SAS/SAR nodes, do not edit the MAC Filter name. Modification of the MAC Filter fails with the error "Cannot read property \filterName\ from undefined" on SAR/SAS nodes.
- For classic SR OS, the Service Manager ID of VPLS should be entered when Action is set to ESI L2.
- Filter of type "isid" can only be applied on a b-vpls.
- Audit will not show misalignment if additional configuration is done on the MD node.
- Deployment misalignment occurs on audit of IPv6 address with combination of IPv4 and IPv6. For example: <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.
- Inner and outer dot1p attributes are supported on SAS-K nodes but are not currently supported by Intent.

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6.5 Name of the Intent: mc-peer-lag-endpoint_gsros_23-10-1_23-11

6.5.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750 SR/7450 ESS/7950 XRS/7250 IXR MD/Classic & 7210 SAS/7705 SAR Classic SR OS node version 23.10

6.5.2 Description

The mc-peer-lag-endpoint_gsros_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of multi-chassis redundancy between two peer nodes, using the End Point Site ID and Interface Address as target identifiers.

The following configurations are supported per deployment:

```
- MC LAG configure between nodeA/nodeB Site ID with interface address
- Configuration of the following attributes on End Point A Details/End
Point B Details:
    - Admin State
    - Description
    - Authentication Key
    - Peer Name
    - Warm Standby
- Configuration of the following attributes on MC LAG End Point A/Point B
Details :
    - Admin State
    - Keep Alive Interval
    - Hold On Neighbor Failure
- Associate LAG and Configuration of the following LAG attributes:
    - End Point A LAG ID/Name
    - LACP Key
    - System ID
    - System Priority
    - End Point B LAG ID/Name
    - Source Backbone MAC Least Significant Bits
- Enable MC-Endpoint and configuration of the End Point A/B Details
attributes as following:
    - Description
    - Admin State
    - System Priority
    - Keep Alive Interval
    - Hold On Neighbor Failure
    - Passive Mode
    - Boot Timer
    - BFD
- Enable MC IPsec and configuration of the End Point A/B Details
attributes as following:
        - BFD Liveness, Keep Alive Interval, Hold On Neighbor Failure
```

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```
        - Discovery Interval
            Interval Secs, Boot
            Domain
        - MC IPsec Group
           Synchronization Tag, First Tunnel Group ID, First Site
Administrative State, First Site Priority, Second Tunnel Group ID, Second
Site Administrative State, Second Site Priority
- Sync:
      - Admin-State, IGMP, IGMP Snooping, Subscriber IPoE Management,
Subscriber PPPoE Management, SRRP, MC Ring, MLD, MLD Snooping, PIM
Snooping,
        Subscriber Host Tracking, MC IPsec, MC Local DHCP Server, Python,
Diameter Proxy, L2TP, Sync Keychain.
      - MC Sync Group :
          - Sync Group : Synchronization Tag, Description, Syn Tag Config
Level.
      - End Point A/B Details:
           - NAT: Group ID, Sync Tag.
           - PIM Snooping: SAPs, Spoke SDPs
           - Subscriber Management: IPoE, PPPoE, WLAN GW
           - Transport Encryption Application: Application ID, Keychain
           - Tags
              - Port: ID, Sync Tag, Range
              - LAG: LAG Name, Sync Tag, Range
              - Pw Port: ID, Sync Tag, Range
              - SDP: ID, Sync Tag, Range
           - Tunnel Group: Tunnel Group ID, Sync Tag
           - Track SRRP
              - ID, L2TP Tunnel ID Range
           - Diameter Node: Host Name, Sync Tag
- MC Ring:
        - Ring Group: Ring Name,
          - End Point A/B Details:
            - Ring: Admin State, Type.
              - In Band Control Path: Admin State, Debounce, Max Debounce
Time, Destination IP Address, Service Name, Interface.
              - Path B: Wildcard SAPs, Range.
              - path Excl: Wildcard SAPs, Range.
              - SRRP Instance: ID.
              - Ring Node: Name, Admin State, Interval, SAP Service ID,
SAP Outer Encap Value, SAP Inner Encap Value, Service Name, Destination
IP Address, Source IP Address, Source MAC Address, VLAN.
```

CLI context: redundancy > multi-chassis > peer

Prerequisites:

- LAG with LACP Configuration
- Router Network Interfaces for both Site A and Site B
- Configure ipsec-domain [configure redundancy multi-chassis]

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- Configure Tunnel Group [configure isa]
- Configure IES or VPRN service [configure service]
- The IES or VPRN interface for the In-Band Control Path is configured.
- Configure nat-group [configure isa nat-group]
- Configure pw-port [configure pw-port]
- Configure diameter [configure aaa diameter node]
- Configure keychain [configure system security keychains]

6.5.3 Advisory

- Endpoint address is a network interface or system IP address.
- Deployment creation/association fails if the Site A interface address is higher than Site B.
- Distribute opeartion is not supported to other nodes.
- The source IP type must match the peer IP type. For example, MC-LAG/Ring creation or association fails if the Site A address type is IPv6, and the Site B address type is IPv4.
- Brownfield discovery fails if the Endpoint Interface Address and Site ID do not match.
- The sros-oc-logical-inventory adaptor suite is required for the suggest function to list the interface IP.
- Audit report is not shown for the Authentication key attribute.
- Greenfield deployment creation fails for more than one deployment for the same Endpoint Interface Address and Site ID using the same template.
- **MC-Peer:**
  - **Classic:**
  - "Warm Standby" attribute is supported on 7750/7950/7450/7250 SROS nodes in Classic mode only.
  - The administrative state of the peer must be down to enable or disable warm standby for a Classic SROS node.
- **MC-LAG:**
  - SROS nodes supports either MD mode or Classic mode, not a combination of Classic and MD mode.
  - MC-LAG can be configured between different chassis.
  - MC-LAG is not supported between pre-21.2 and post-21.2 SR OS version nodes.
  - "End Point B" LAG should be unique when more than one LAG is associated to the MC-LAG.
  - Deployed Misaligned BF Discovery of MC-LAG without discovery-interval, priority attributes.
  - **Classic:**
  - MC-LAG intent should be created with Sync Admin-state down for classic node [limitation of NFMP]
  - **MD:**
  - Pre-21.2 SROS nodes support only LAG ID and post-21.2 SROS nodes support LAG Name for MD node.
  - BF Discovery of MC-LAG without any configuration result in Deployed Misaligned state on MD node.

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- **MC-Endpoint:**
  - **Classic:**
  - Configuration is not supported for IXR Classic mode.
  - "Description" attribute in MC Endpoint configuration is supported on 7750/7950/7450/7250 SROS nodes in Classic mode only
- **MC-Ring:**
  - Configuration is not supported on IXR and SAS.
  - Configuration is supported on SAR-HM.
  - MC-Ring should be created between two nodes that support MC-Ring.
  - **Classic:**
  - MC Sync Group must exist between peers and sync protocol MC-Ring must be enabled for Classic nodes.
  - Description is not supported for classic nodes.
  - SRRP instance is not applicable for Classic nodes.
  - In-Band Control Path Service Name is not applicable for Classic nodes (read-only).
  - Ring Node- RNCV SAP Service Name & LAN value is not applicable for Classic nodes.
  - Once the In-Band Control Path Interface is configured, it should not be removed.
  - To bring up a MC-ring, user should has to check/configure,
  - The synchronization tag is configured on the BSA.
  - The VLAN-level synchronization tag is valid.
  - The IES or VPRN interface for the In-Band Control Path is configured.
  - BFD is enabled on the In-Band Control Path interface.
  - The IES or VPRN interface is operationally up.
  - The In-Band Control Path interface is not in use by another MC ring.
  - The In-Band Control Path interface is on the port used for MC synchronization.
  - The In-Band Control Path destination IP address is configured.
  - The In-Band Control Path destination IP address is not the same as the In-Band Control Path source IP address.
  - The In-Band Control Path destination IP address is not in the same subnet as the In-Band Control Path source IP address.
  - The MVPLS or VPLS site is not an endpoint for redundant spoke SDPs.
  - **MD:**
  - srrp-instance must be configured when the Type attribute is layer-3 ring.
  - path-b must be configured when the Type attribute is layer-2 ring.
  - Service Name & Interface accept any name on MD nodes.
  - Attributes, SAP Service ID/SAP Outer Encap Value/SAP Inner Encap Value are not supported.
  - Manually delete the path-b/path-excl entry before updating or modifying the type attribute from layer-2 to layer-3.
  - Manually delete the srrp-instance entry before updating or modifying the type attribute layer-3 to layer-2.
  - Configuration of attributes must be configured on both endpoints, not a single endpoint.
  - "In Band Control Path" admin-state is not supported on MD nodes.

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- **MC-Sync:**
  - Sync Admin State is applicable for Classic node only.
  - Configuration is supported on SAS [M, X, T Mxp, S/Sx, R].
  - **Classic:**
  - Diameter Proxy, Subscriber PPPoE Management, MLD, Subscriber Host Tracking, My Local DHCP Server, Python, L2TP are not supported for Classic 7950-XRS.
  - **MD:**
  - Sync End Point A and End Point B Admin State are applicable for MD node only.
  - transport-encryption application support only IPsec.
  - diameter-node/ipsec/nat/sub-host-trk/sub-mgmt/transport-encryption/tunnel-group attributes are not supported on the MD 7950 XRS node.
  - "sub-mgmt" wlan-gw is not supported on the 7450 ESS node.
  - Admin State, IGMP, IGMP Snooping, Subscriber IPoE Management, Subscriber PPPoE Management, SRRP, MC Ring, MLD, MLD Snooping, PIM Snooping, Subscriber Host Tracking, MC IPsec, MC Local DHCP Server, Python, Diameter Proxy, L2TP, Sync Keychain attributes are not supported for MD nodes.
- **MC-IPsec:**
  - MC-IPSec is not supported on 7705 SAR, 7250 IXR, 7950 XRS nodes.
  - MC-IPsec tunnel-group admin-state enable only when peer tunnel group is configured.
  - Tunnel groups admin-state should be down in order to modify IPsec properties.
  - The "source-bmac-lsb" attribute is not supported for 7250 IXR nodes.
  - **Classic:**
  - MC-IPsec tunnel-group "Synchronization Tag" applicable for classic node only.
  - MC-IPSec Tunnel Group cannot be created with IPv6 address managed classic node [limitation of NFMP]
  - peer sync must be configured when the peer contains enabled ipsec-domains.
  - **MD:**
  - MC-IPSec Tunnel Group "First Site Peer Group" and "Second Site Peer Group" attributes applicable only MD node.

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6.6 Name of the Intent: port-connector_gsros_23-10-1_23-11

6.6.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7950/7450/7250 node version 23.10

6.6.2 Description

port-connector_gsros_23-10-1_23-11 intent supports configuration of Breakout Type on connector port.

The following configurations are supported per deployment:

```
- Breakout
- Admin State
- RS FEC Mode
```

CLI context: configure > port > connector

6.6.3 Advisory

- Breakout list supports breakout types that are supported on all SR OS nodes. Assigning an unsupported breakout causes a deployment failure.
- Breakout type "None" is not applicable for MD nodes.
- Modification or deletion of deployment in ICM results in Deployment Error in NFM-P on classic SR OS if the breakout port is associated to any configuration such as Interface, LAG, SAP, etc.
- Audit operation will not show the discrepancy if modification of breakout results in deployment error.
- Deletion of deployment with "From NSP and Network" option will delete the breakout ports.
- Deletion of deployment fails if breakout port is associated to any configuration such as Interface, LAG, SAP etc.
- RS FEC mode is applicable for some breakout types and on some MDA types.

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6.7 Name of the Intent: qos-network-ingress_gixr_23-10-1_23-11

6.7.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR classic/MD SR OS node version 23.10

6.7.2 Description

qos-network-ingress_gixr_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of network ingress QoS policy on SR OS nodes.

The following configurations are supported per deployment:

```
- Description, Policer-Allocation, Ingress Classification Policy and
Scope
- FC: FC name and Multicast Policer
- Policer: Policer ID, CBS, MBS, Stat-Mode, Adaptation Rule [PIR and
CIR], Rate [PIR and CIR] in kbps
```

6.7.3 Advisory

- To allow multicast policer, confg>system>fp options>qos bum-policer must be administratively up. (To enable bum-policer, VPLS EVPN MPLS or FC-4 Admin state should be down).
- For deployments on multiple nodes, the user must enter the Policy ID/Name in the suggest field.
- MD: Audit will not be shown as misaligned for new entries added from the node.
- Ingress classification policy should be preconfigured on the node.
- Policer allocation is not supported for non - IXR-x/IXR-e2/IXR-R6d/IXR-r6dl chassis.
- FC is not supported for IXR-x/IXR-e2/IXR-R6d/IXR-r6dl chassis.
- Policer is supported with Policer Allocation type as other than [none] for IXR-x/IXR-e2/IXR-R6d/IXR-r6dl chassis.
- Policer Allocation type [none, per-fc] supports up to 8 policers.
- Audit report will be shown as misaligned for classic nodes if any of the default fc classes and policers have not been configured from the template during greenfield deployment.
- Default FC and policers should not be deleted from the intent for classic nodes.
- Creation of policer and association of the policer to FC should be done in separate edit operations for classic SR OS.
- Deletion of policer associated to the FC should be done in a separate edit operation for classic SR OS.

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6.8 Name of the Intent: qos-network_gsros_23-10-1_23-11

6.8.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7450/7750/7705 classic/MD SR OS node version 23.10

6.8.2 Description

qos-network_gsros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Network QoS Policy.

The following configurations are supported per deployment:

```
- Configuration of Description, Ingress, and Egress
- Configuration of the following Ingress attributes:
    - Enable "Ler Use Dscp"
    - Default Action [FC,Profile]
    - Dot1p [Dot1p Value,Fc,Profile]
    - Dscp [Dscp Name,Fc,Profile]
    - Lsp Exp [Lsp Exp Value,Fc,Profile]
    - Fc [Fc Name,Fp Redirect Group[Multicast Policer,Policer,Broadcast
Policer,Unknown Policer]]
    - Ip Criteria [Entry [Entry Id,Description], Match [Protocol,Dscp,
Fragment,Dst Ip,Src Ip,Src Port,Dst Port], Action [Type,Fc,Profile]]
    - Ipv6 Criteria [Entry [Entry Id,Description], Match [Next
Header,Dscp,Fragment,Dst Ip,Src Ip,Src Port,Dst Port], Action
[Type,Fc,Profile]]
- Configuration of the following Egress attributes:
    - Enable Remark Trusted, Force Egress Marking
    - Prec [Prec Value,Fc,Profile]
    - Dscp [Dscp Name,Fc,Profile]
    - Fc [Fc Name,Dot1p In Profile,Dot1p Out Profile,Dscp In Profile,Dscp
Out Profile,Lsp Exp In Profile,Lsp Exp Out Profile,Enable De Mark,Force,
Port Redirect Group [Queue,Policer]]
    - Ip Criteria [Entry [Entry Id,Description], Match[Protocol,Dscp,
Fragment,Dst Ip,Src Ip,Src Port,Dst Port], Action[Type,Fc,Profile]]
    - Ipv6 Criteria [Entry [Entry Id,Description], Match[Next
Header,Dscp,Fragment,Dst Ip,Src Ip,Src Port,Dst Port], Action[Type,Fc,
Profile]]
```

CLI context: configure > qos > network

Prerequisites:

- The ip-prefix-list/ipv6-prefix-list/port-list must be configured on the node if using a list.

6.8.3 Advisory

- Use Default Template for 7750 SR/7450 ESS/7950 XRS nodes.
- Use Default SAR Template for 7705 SAR nodes.

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- Once Ingress FC are created, they should not be deleted from the template for classic nodes.
- For classic nodes, user inputs for subnet mask should be of the format <IPAddress/prefix>; full-length mask should not be provided.
- During distribution from MD to classic, if MD node has entries associated with mask, then make necessary changes for IP address field (IPAddress/prefix).
- Do not change the IP-Filter entry protocol from tcp-up/udp/tcp to any other protocol.
- Do not change the IPV6-Filter entry protocol from tcp-up/udp/tcp to any other protocol.
- Display is limited to supported attributes.
- Network Egress IP/IPv6 Criteria Entries and Network Egress DSCP/Prec cannot be configured together.
- Creation of Network QoS deployment fails if Network QoS Policy Name and Network QoS Policy ID are not unique. That is, Policy Name/Policy ID for the selected target must not be used already.
- Brownfield discovery will fail if both Network QoS Policy Name and Network QoS Policy ID do not match.
- MD: Audit will not be shown as misaligned for new entries added from the node.
- Deployment misalignment occurs on audit of IPv6 address with combination of IPv4 and IPv6. For example: <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.
- Either one port or prefix list policy can be associated in match list of IPv4/v6 criteria entry in network.

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6.9 Name of the Intent: qos-port-policy_gixr_23-10-1_23-11

6.9.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR classic/MD SR OS node version 23.10

6.9.2 Description

The qos-port-policy_gixr_23-10-1_23-11 intent supports the creation/modification/deletion/discovery/audit/alignment of port QoS policy on SR OS nodes.

The following configurations are supported per deployment:

```
- Description, Scope
- FC: FC Name and Queue (will be defaulted in the template)
- Queue: ID [1 to 8], Queue Management, WRR-weights [value: 1],
Adaptation Rule [PIR and CIR], Scheduler mode-WFQ, PIR Weight[1 to 255],
Percent Rate: PIR[0.01 to 100.00] and CIR[0.00 to 100.00], Low Latency
- WRR-weights: ID [valid value: 1], Multicast Weight[1 to 255] & Unicast
Weight[1 to 255]
- Packet Byte Offset: Add [0 to 64, step:4]/Subtract [0 to 60, step:4]
```

6.9.3 Advisory

- A separate viewConfig form exists for MD nodes, which should be configured using the md-default Config Form.
- MD: Audit is not shown as misaligned for new entries added from the node.
- Default FC and Queue should not be deleted from the intent for classic nodes.
- Queue Management policy must be preconfigured on the node.
- Packet Byte Offset should be set as administratively up.
- Only one WRR-weights (ID should be 1) entry is supported per deployment.
- Deployment misalignment occurs during brownfield discovery of Port QoS policy intent for MD node if fc/queue/WRR weights are not configured.
- For deployments on multiple nodes, the user must enter the Policy ID/Name in the suggest field.
- Low Latency is only supported on queues associated with FCs "nc" and "h1".
- Low Latency is supported for IXR-x/IXR-e2/IXR-R6d/IXR-R6dl chassis.
- Packet Byte Offset's range is Add [0 to 8, step:4]/Subtract [0 to 8, step:4] for IXR-6/IXR-10/IXR-R6/IXR-s/IXR-R4/IXR-ec/IXR-e chassis.

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6.10 Name of the Intent: qos-sap-ingress_gixr_23-10-1_23-11

6.10.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7250 IXR classic/MD SROS node version 23.10

6.10.2 Description

qos-sap-ingress_gixr_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of SAP-Ingress configuration with Policy name and Policy ID as the target identifiers.

The following configurations are supported per deployment based on the Config Form:

`- Policer Allocation, Ingress Classification Policy, MAC Criteria, IP Criteria, IPv6 Criteria, Policer and FC configuration`

Prerequisites: Configure Ingress Classification Policy

6.10.3 Advisory

- Chassis supporting Policer allocation should be configured using "no-default-policer" Config Form, while those not supporting Policer allocation should use the "default" Config Form.
- The IXR-x, IXR-x3, IXR-R6d, IXR-e2 and IXR-R6dl do not support mac-criteria, ip-criteria and ipv6-criteria entries.
- In the configuration of a policer, the Committed Burst Size and Maximum Burst Size values are specified in bytes.
- In the configuration of a policer, within the rate settings, the Peak Information Rate (PIR) and Committed Information Rate (CIR) values are expressed in kilobits per second (kbps).
- Policer allocation config is not supported for IXR-s,IXR-R4,IXR-R6,IXR-10,IXR-e,IXR-ec,IXR-6 nodes.
- Deployment misalignment occurs on audit of IPv6 address with combination of IPv4 and IPv6. For example: <::d.d.d.d> and <x:x:x:x:x:x:d.d.d.d>.

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6.11 Name of the Intent: qos-schedulerpolicy_gsros_23-10-1_23-11

6.11.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7450/7750 classic/MD SR OS node version 23.10

6.11.2 Description

qos-schedulerpolicy_gsros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of QoS Scheduler Policy.

The following configurations are supported per deployment:

```
- Configuration of Description and Frame Based Accounting
(Enable/Disable)
- Choice of Tier ID and Parent Location
- Adding Schedulers under Tier ID and Parent Location
- Scheduler Name and Description
   - Enable/Disable "Limit Unused Bandwidth"
   - Choice of Parent Mapping options "Port Parent", "Scheduler Parent"
   - Provision to provide Port Parent Level, Weight, CIR Level, and CIR
Weight with range validation
   - Provision to provide Scheduler Parent Scheduler Name (allow to
choose from existing valid schedulers), Level, Weight, CIR Level, and CIR
Weight with range validation
   - Choice of Rate or Percent Rate options with input provision for PIR,
CIR, and Reference Rate (to choose "Local-Limit" or
"Reference-Port-Limit" only for Percent Rate)
```

6.11.3 Advisory

- For all the Policy QoS and router intents, the Admin user should have the OSS Management Role assigned in the NFM-P (NFM-P GUI Administration->Security->NFM-P User Security-> Scope of Command tab-> "Admin" user -> Roles).
- Scheduler Policy cannot have both Tier 1 Parent Location and Frame-based Accounting configured.
- Tier 1 Scheduler cannot have any Parent Scheduler when Tier 1 Parent is set to none.
- Creation of QoS Scheduler deployment fails if the Scheduler Policy Name is not unique. That is, the Policy Name for the selected target should not be used already.
- Brownfield deployment misalignment occurs for the Scheduler policy due to the percentage rate CIR attributes from the ICM app.
- MD: The audit will not be shown as misaligned for new entries added from the node.

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6.12 Name of the Intent: routing-admin-grp-and-srlg_gsros_23-10-1_23-11

6.12.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7250/7950/7450/7705 classic/MD SR OS node version 23.10

6.12.2 Description

routing-admin-grp-and-srlg_gsros_23-10-1_23-11 Intent supports creation/modification/deletion/discovery/audit/alignment on SROS nodes.

The following configurations are supported per deployment:

```
-  Creation of Administrative Groups: Group Name and Group Value
-  Creation of Shared Risk Link Groups (srlg): Group Name, Group Value,
and Penalty-Weight
```

6.12.3 Advisory

- Deployment will fail if the Target-Identifier is given other than "Admin-grp-srlg".
- MPLS has to be enabled on the router.
- SRLG: Penalty weight is not supported on SAR and SAS.
- Admin Group values supported up to 0..255 for SR/IXR/ESS/XRS and 0..31 for SAR/SAS.
- Group value for Admin-Group and srlg should be unique. No two entries can have the same group value.
- Group name/value should not be modified for admin-group/srlg.
- Admin-Group and srlg names should not be modified once created.
- The audit report will not be shown as misaligned if the user creates a new entry for Admin-Groups or srlg from the node.
- Since the intent is applicable per NE:
- It is recommended to the user to perform brownfield discovery of the intent so that all the created Admin-Groups and srlg will be available in the intent.
- It is advisable to the user to delete the Admin-Group/srlg manually from the intent.

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6.13 Name of the Intent: routing-admin-tags_gsros_23-10-1_23-11

6.13.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7750/7250/7950/7450 classic/MD SR OS node version 23.10

6.13.2 Description

routing-admin-tags_gsros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment on SROS nodes.

The following configurations are supported per deployment:

```
- Creation of Admin-Tags
- Creation of Admin-Tag-Policies
```

6.13.3 Advisory

- Deployment will fail if the Target-Identifier is given any value other than "Admin-Tag".
- The audit report will not show as misaligned if the user creates a new entry for Admin-Tags or Route-Admin-Policy from the node.
- It is recommended that the user perform brownfield discovery of the intent so that all the created Admin-Tags and policies will be available in the intent.
- Since the intent is applicable per NE, all the required Admin-Tag and Admin-Tag-Policies can be configured using a single intent. Therefore, it is advisable that the user delete the Admin-Tags or Admin-Tag-Policies manually from the intent.

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6.14 Name of the Intent: security-cpu-protection_gsros_23-10-1_23-11

6.14.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7750/7450 classic/MD SR OS node version 23.10

6.14.2 Description

security-cpu-protection_gsros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of CPU protection policy.

The following configurations are supported per deployment:

```
- Description, Alarm, Overall Rate, Per-source Rate, Output Profile Rate,
Log Events
- Per-source attributes: IP Src Monitoring - limit-dhcp-ci-addr-zero
- Eth-cfm attributes: Entry ID, PIR, Level, Opcode
```

6.14.3 Advisory

- Target Identifier will accept only numeric values in the range from 1 to 225.
- Security CPU Protection is not supported on 7250 IXR, 7750 SR-1-48D, 7750 SR-1x-48D, 7750 SR-1-24D, 7750 SR-a4, 7750 SR-2e, 7750 SR-1-46S, 7750 SR-a8, 7750 SR-2s, 7750 SR-1, 7750 SR-1x-92S, 7750 SR-1se, 7750 SR-2se, 7750 SR-3e, 7750 SR-1-92S, VSR, VSR NRC, 7750 SR-1e, 7750 SR-1s nodes.
- There can be no more than four DoS Protection Policies with CFM Rate Limiting Policies assigned on a specific node.
- MD: The audit will not be shown as misaligned for new entries added from the node.

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6.15 Name of the Intent: security-dist-cpu-protection_gsros_23-10-1_23-11

6.15.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7750/7250/7450 classic/MD SR OS node version 23.10

6.15.2 Description

security-dist-cpu-protection_gsros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of Distributed CPU Protection policy.

The following configurations are supported per deployment:

```
- Creation of Distributed CPU Protection Policy with Name
- Provision Description, choose type such as Access-Network or Port
- Creation of Static Policer/Local Monitoring Policer/Protocol
- Creation of Static Policer with Name, Description, Detection Time
- Choice of configuration for:
    - Log Events [True, False, Verbose]
    - Exceed Action [Action [Discard, Low-Priority, None], Hold Down]
    - Rate Type [Kbps, Packets]
- Creation of Local Monitoring Policer with Name, Description, Exceed
Action [Discard, Low-Priority, None]
- Choice of configuration for:
    - Log Events [True, False, Verbose]
    - Rate Type [Kbps, Packets]
- Creation of Protocol with Name
- Choice of configuration for:
    - Enforcement Type [Dynamic, Dynamic Local Mon Bypass, Static]
    - Dynamic attributes such as Detection Time, Log Events
    - Exceed Action [Action [Discard, Low-Priority, None], Hold Down]
    - Rate Type [Kbps, Packets]
```

6.15.3 Advisory

- Use Default IXR Template for IXR nodes.
- Once the policy is deployed, Policy Type cannot be changed from Access-Network to Port or vice-versa.
- Per deployment, only one Local Monitoring Policer entry is supported.
- Static Policer and Local Monitoring Policer are not supported for IXR nodes.
- Enforcement Type "Shared" is supported only for IXR and it should be available on the node.
- "Dynamic Local Mon Bypass" checkmark is missing after brownfield discovery of dist CPU Protection.
- MD: The audit will not be shown as misaligned for new entries added from the node.

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- For deployments on multiple nodes, the user has to enter the Policy ID/Name in the suggest field.

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6.16 Name of the Intent: sync-if-timing_gsros_23-10-1_23-11

6.16.1 Applicability

Supported from: NSP 23.11

Verified on Node Type/Version: 7950/7750/7250 classic/MD SR OS node version 23.10

- Verified Default3 template on 23.10 IXR-ec/IXR-X/IXR-e with ref-order (ref1 ref2 ptp)
- Verified Default4 template on 23.10 IXR-10/IXR-6/IXR-R4 with ref-order (bits ref1 ref2 ptp)
- Verified Default5 template on 23.10 ESS/SR-2s/SR12/XRS with ref-order (SyncE ref1 ref2 bits PTP) and IXR-R6 with ref-order (bits ref1 ref2 ptp gnss)
- Verified Default6 template on SR nodes with FP5 chassis with ref-order (bits SyncE ref1 ref2 PTP gnss)

6.16.2 Description

sync-if-timing_gsros_23-10-1_23-11 intent supports creation/modification/deletion/discovery/audit/alignment of SyncE.

The following configurations are supported per deployment:

```
- Creation of sync timing ref with "timing" naming
- Choice of the supporting Ql Minimum profile and Enable/Disable of Ql
Selection and Revert and Wait To Restore.
   - QL profile: prs, stu, st2, tnc, st3e, st3, prc, ssua, ssub, sec,
eec1, eec2
- Selection of the reference order based on the template chosen
- Configuration of the timing references based on the template and
chassis type
   a. Ref1: Admin-State, Ql Override: Choose the Ql profile, Source Port
   b. Ref2: Admin-State, Ql Override: Choose the Ql profile, Source Port
   c. PTP: Admin-State, Ql Override: Choose the Ql profile
   d. SyncE: Admin-State, Ql Override: Choose the Ql profile
   e. GNSS: Admin-State, Ql Override: Choose the Ql profile
   d. BITS: Interface Type, Ql Override
        1. Input: Admin-State
        2. Output: Admin-State, Ql Minimum, Source and Squelch
```

CLI context:

MD : configure > system > central-frequency-clock Classic : configure > system > sync-if-timing

Prerequisites:

- SyncE must be enabled on MDM level SyncE true.

6.16.3 Advisory

- Ql profile depends on the chassis type.
- SyncE supports four templates [Default/Default3/Default4/Default6]. Template selection should be based on the number of timing references supported by the node.

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- Use Default Template for five timing references such as ref1, ref2, ptp, bits, SyncE, gnss.
- Use Default3 Template for three timing references such as ref1, ref2, ptp.
- Use Default4 Template for four timing references such as ref1, ref2, ptp, bits.
- Use Default6 Template for six timing references such as ref1, ref2, ptp, bits, SyncE, gnss.
- Choose "timing" as unique identifier during the creation of SyncE deployment.
- The bits and SyncE cannot co-exist; the user cannot configure both at the same time due to SR node restriction.
- Admin State of ptp timing reference should not be enabled if ptp Clock Type is Ordinary Master.
- The ptp QL Override should be set to { prs, stu, st2, tnc, st3e, st3 } when ptp Network-Type is configured as SONET.
- The ptp QL Override should be set to {prc, ssub, ssua, sec} when ptp Network-Type is configured as SDH.
- The bits output line-length should be set to length-not-applicable when the Interface-Type is { e1-pcm30crc/ e1-pcm31crc }.
- The bits QL Override should be set to { prs, stu, st2, tnc, st3e, st3 } when the Interface-Type is configured as { ds1-esf / ds1-sf }.
- The bits QL Override should be set to {prc, ssub, ssua, sec} when the Interface-Type is configured as { e1-pcm30crc/ e1-pcm31crc /2048khz-g703 }.
- Ref1/Ref2 source port must be set to configure the QL Override and to set Admin State to enable.
- Admin State should be enabled on the port that is selected as the Ref1/Ref2 source port.
- SyncE must be enabled on the MDA of the selected Ref1/Ref2 source port.
- Reference sources must be on different cards for SR nodes on SR-12, and XRS and has to be be on different MDA for SR-2s.
- Reference source ports cannot be of the same modulo 4 if on the same MDA for IXR-6.
- If the NE is upgraded to a new version that supports additional timing reference, the existing intent in ICM should be deleted from the NSP and associated with the correct template.
- MD: Audit will not be shown as misaligned for new entries added from the node.

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7 Wavence intent types

7.1 Name of the Intent: icm-wavence-backup-restore-policy-assignment

7.1.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: MSS 4/8/1, MSS-E/HE/XE, UBT-NIM, MPRe, MSS1c, MSS-O and UBT SA nodes

7.1.2 Description

The icm-wavence-backup-restore-policy-assignment intent supports the deployment, creation, and association of Wavence backup restore policy assignment to the NEs.

The following configurations are supported per deployment with supported configuration template:

```
- User can configure the following parameters for deployment:
  - Select the template
  - Select the target nodes
  - Select Intent as icm-wavence-backup-restore-policy-assignment
- Click on VIEW/EDIT TEMPLATE CONFIG and enter the below parameter.
  - Select Policy ID

- User can configure the following parameters for associate to network:
-   Template is already pre-selected.
-   Select the target nodes
-   Select Intent as icm-wavence-backup-restore-policy-assignment
```

Prerequisite: Required intent artifacts are imported and installed

7.1.3 Advisory

- Import the new intent types for Wavence Backup and Restore Policy
- This intent is a pre-requisite for the Wavence Backup and Restore operations performed in Device Management
- Assign the Wavence backup restore policy to NEs using the configuration template in this intent
- Associate NEs with the Wavence backup restore policy already assigned in NFM-P using the configuration template in this intent.
- See the NSP Wavence Device Support Guide for information about assigning backup and restore policies to Wavence devices.

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7.2 Name of the Intent: icm-wavence-backup-restore-policy

7.2.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: MSS 4/8/1, MSS-E/HE/XE, UBT-NIM, MPRe, MSS1c, MSS-O, and UBT SA nodes

7.2.2 Description

The icm-wavence-backup-restore-policy intent supports the creation/discovery/audit/alignment of Wavence backup restore policies.

The following configurations are supported per deployment with supported configuration template:

```
- User can configure the following parameters for deployment:
-   Select the template
-   Enter new policy ID
-   Click on VIEW/EDIT TEMPLATE CONFIG and enter the below parameters.
         - Policy Name
         - Policy Type
         - Backup Purging by filling below details:
               - Auto Purge Scheme
               - Number Of Backups
               - Maximum Backup Ages(days)
         - Backup Transfer Settings by filling below details:
               - Transfer Protocol
               - User ID
               - Password
               - Server IP
               - Server Port
               - Hostkey Fingerprint for SFTP
               - Root Directory
-   User can configure the following parameters for associate to network:

-   Template is already pre-selected.
-   Enter the already existing policy ID.
```

Prerequisite: Required intent artifacts are imported and installed

7.2.3 Advisory

- Import the new intent types for Wavence Backup and Restore
- This intent is required for performance backup and restore operations for Wavence using Device Management
- Create a new Wavence Backup and Restore Policy from NSP using the configuration template on the Wavence Backup and Restore Policy intent
- Associate existing Wavence Backup and Restore policies in NFM-P using configuration template on Wavence Backup and Restore Policy intent

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- See the Wavence Device Support Guide for information about using backup and restore operations with Wavence devices.

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7.3 Name of the Intent: icm-wavence-epc-power-saving

7.3.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: W25 MSS, MSS-E/HE/XE, UBT-NIM, and UBT SA nodes

7.3.2 Description

The icm-wavence-epc-power-saving intent supports deploy/align/audit and configuration of all radio port parameters for Wavence nodes.

The following configurations are supported per deployment:

```
- Light Sleep: Enable, Disable
- Light Sleep Timeout (Minutes): 1 to 2880
- Deep Sleep: Enable, Disable
- Deep Sleep Timeout (Minutes): 1 to 2880
```

7.3.3 Advisory

- Light sleep is enabled prior to Deep Sleep.
- Deep Sleep is disabled prior to Light Sleep.

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7.4 Name of the Intent: icm-wavence-ethernet-port-configuration

7.4.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: MSS, MSS-E/HE/XE, UBT-NIM and UBT SA nodes

7.4.2 Description

The icm-wavence-ethernet-port-configuration intent supports deployment/alignment/audit to configure and update all the editable ethernet parameters for Wavence nodes.

The following configurations are supported per deployment with the three supported configuration templates:

```
For MSS-based nodes
- User can configure the following parameters:
-   States - Administrative State - Enable/Disable
-   LLDP - LLDP State - Enable/Disable
-   Ethernet - Auto Negotiate/Eth Hold Off Mode
-   Advertised Capability With Auto Negotiate Enabled - Advertised
Capability
-   Storm Control  - Ingress Storm Control
-   Ingress and Egress Mapping
-   Port Rate Limiter - Ingress Enable/Egress Enable
-   Provider Bridge Configurations - Port Type/S-VLAN TPID
-   EFM-OAM - Administrative State/Mode/Set Remote Loopback
-   CoS Rate Limiter
-   VLAN Rate Limiter
-   VLAN Port Configuration

For MSS-E/HE/XE and UBT-NIM nodes
- User can configure the following parameters:
-   States - Administrative State - Enable/Disable
-   LLDP - LLDP State - Enable/Disable
-   Ethernet - Auto Negotiate/Eth Hold Off Mode
    -   Advertised Capability With Auto Negotiate Enabled - Advertised
Capability
    -   Provider Bridge Configurations - Port Type/S-VLAN TPID

UBTSA-Based - For UBTSA-based nodes
- User can configure the following parameters:
-   States, Administrative State, Enable/Disable
-   Ethernet,  Auto Negotiate/Eth Hold Off Mode
    -   Advertised Capability With Auto Negotiate Enabled - Advertised
Capability
    -   Provider Bridge Configurations - Port Type/S-VLAN TPID
-   VLAN Port Configuration - Admit Frames
    - VLAN Configuration  Priority/S-VLAN ID
```

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7.4.3 Advisory

- Not all listed attributes are applicable to all ports (for example, due to different node types). Only configure attributes that are supported on the node being configured.
- When associating a configuration template, ensure it is the correct template. If there are extra parameters or if the values are not aligned with the actual network configuration, then the deployment is discovered in a misaligned state. Aligning the deployment will cause the value from the configuration template to be pushed to the network, even if that is not the desired result. If the wrong template was used, then delete the deployment instance from the NSP and associate a new template.
- For CPL(Coupling) ports on UBT-SA nodes only a few parameter updates are supported as per node behavior and they are listed after associate to network is performed on that target.
- For CPL(Coupling) ports on NIM nodes no parameter updates are supported.
- You must associate the configuration template with the ethernet port targets of the NEs
- Audits can be performed to verify that the configuration from the templates matches what is currently configured in the network. Misaligned attributes can be identified and aligned for the configuration to be deployed back on the network.
- Audits can be performed on individual configuration deployments, or they can be performed on configuration templates. If performed on configuration templates, then all deployments associated with the template will be audited.

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7.5 Name of the Intent: icm-wavence-ethernet-ring-element

7.5.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: W24 MSS, MSS-E/HE/XE, UBT-NIM

7.5.2 Description

The icm-wavence-ethernet-ring-element intent supports the deployment/alignment/audit/configuration/update of all editable radio port parameters for Wavence nodes.

The following configurations are supported per deployment:

- Radio Ring Topology - Topology Pointer - New ERP Label - Administrative State (Up, Down) - Wait-To-Restore Time(min) - RPL Type (None , Owner) - R-APS VLAN ID
- Service - R-APS MEG Level (0 - 7)
- OAM Switch Criteria - Switch Input (--Select All-- , CLE/ODNC , Loss of Continuity)
- Path Endpoints - Path ID (Path A , Path B) - Path Endpoint Type (Normal , Ring Protection Link End) - Path Pointer - R-APS MEG ID - CCM Interval

7.5.3 Advisory

- Path Endpoints cannot be deleted.
- Do not add more than two Path Endpoints

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7.6 Name of the Intent: icm-wavence-feature-capacity

7.6.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: All Wavence Node variants

7.6.2 Description

The icm-wavence-feature-capacity intent supports deployment/alignment/audit to configure and update for all Wavence nodes.

The following configurations are supported per deployment with the two supported configuration templates

```
- Enable Generation
- Generation Frequency
- Generation Time
```

7.6.3 Advisory

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7.7 Name of the Intent: icm-wavence-file-retrieval-policy-assignment

7.7.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: CorEvo MSS4/8 , MSS-E/HE/XE, UBT-NIM

7.7.2 Description

The icm-wavence-file-retrieval-policy-assignment intent supports the deployment/creation/associate/audit/alignment of a Wavence File Retrieval policy (Log/RSL/I and C) to the NEs. - As part of this intent, Nodes are assigned a Wavence File Retrieval policy.

The following configurations are supported per deployment:

```
- Policy-Type
- Targets
- Policy Id
```

Prerequisite: Required intents arftifacts are imported and installed

7.7.3 Advisory

- File Retrieval (Log) is supported only for CorEvo MSS4/8
- File Retrieval (RSL) is supported from W23A for CorEvo MSS4/8, MSS-E/HE/XE, UBT- NIM
- File Retrieval (I and C) is supported only for CorEvo MSS4/8

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7.8 Name of the Intent: icm-wavence-file-retrieval-policy

7.8.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: CorEvo MSS4/8, MSS-E/HE/XE, UBT-NIM

7.8.2 Description

The icm-wavence-file-retrieval-policy intent supports the creation/discovery/audit/alignment of File Retrieval policies (Log/RSL/I and C)

The following configurations are supported per deployment:

```
- Policy Id

- General:
   - Name
   - Policy Type

- SFTP Settings:
   - SFTP User Id
   - SFTP Password
   - SFTP Server Port
   - Hostkey Fingerprint
   - SFTP server IP
   - Root Directory
```

Prerequisite: Required intent artifacts are imported and installed

7.8.3 Advisory

- File Retrieval (Log) is supported only for CorEvo MSS4/8
- File Retrieval (RSL) is supported from W23A for CorEvo MSS4/8 , MSS-E/HE/XE NIM
- File Retrieval (I and C) is supported only for CorEvo MSS4/8

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7.9 Name of the Intent: icm-wavence-l3-config-network-interface

7.9.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: CoreEvo MSS-8/4

7.9.2 Description

The icm-wavence-l3-config-network-interface intent supports the deployment/alignment/audit/configuration/update of all editable network interface parameters for Wavence nodes.

The following configurations are supported per deployment with the supported configuration template

- L3 network interface - Port - IP Address - Administrative State (Up, Down)

7.9.3 Advisory

- L3VPN VLAN ID must be onfigured.
- System interface must be configured

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7.10 Name of the Intent: icm-wavence-l3-config-system-interface

7.10.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: CoreEvo MSS-8/4

7.10.2 Description

The icm-wavence-l3-config-system-interface intent supports the deployment/alignment/audit/configuration/update of system interface parameters for Wavence nodes.

The following configurations are supported per deployment:

- L3 System interface - IP Address - Administrative State (Up, Down)

7.10.3 Advisory

- L3VPN VLAN ID must be configured.

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7.11 Name of the Intent: icm-wavence-network-element-config

7.11.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: MSS 8, MSS 4, MSS E/HE/XE, UBT-NIM, and UBT SA nodes

7.11.2 Description

The icm-wavence-network-element-config supports deploy/align/audit to configure the network element level configurable parameters for Wavence nodes.

Two templates provided for this intent: - Multi-device support: for bulk configurations - Default: for single node configuration

Multi device support template: use this template to configure the generic parameters which can be done in bulk. Below are the bulk configuration parameters supported: - General details: Location/ Latitude and Longitude - Polling: Enable/Disable - NTP: Enable/Disable and Main/Spare server address - MPR System settings - IP Data Plane Settings: L3 VLAN ID - QoS: WRED Status (Enable/Disable) - OLC: Current OLC State (In Service, Maintenance)

Default template: use this template to configure specific parameters on a single node. - General details: Location/Latitude/Longitude and SiteName - Polling: Enable/Disable - NTP: Enable/Disable and Main/Spare server address - BNM configurations - MPR System settings - IP Data Plane Settings: Virtual Router IP address ,L3 VLAN ID - Radio Ring Component: Labe/East port and West - QoS: WRED Status (Enable/Disable , Color Classification, Dot1p upon selection of color classification as PCP) - OLC: Current OLC State (In Service, Maintenance)

The following configurations are supported per deployment:

```
- General details: Site name/Location/Latitude and Longitude
- Polling: Enable/Disable
- NTP: Enable/Disable and Main/Spare server address
- BNM Configuration
- MPR System settings
- IP Data Plane Settings
- Radio Ring Component: Labe/East port and West
- QoS:  WRED Status - Enable/Disable , Color Classification, Dot1p upon
selection of color classification as PCP
- OLC: Current OLC State - In Service, Maintenance
```

7.11.3 Advisory

- QOS and OLC configuration are supported in NSP Release 25.4 or later
- Configuring BNM using the NSP is a two step process: If the BNM is already enabled and you need to configure port/VLAN or any other parameters:
  - Disable the BNM and send the deployment to node
  - Configure/change your required parameters and deploy
  - Enable the BNM and deploy the configs on the node

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- Do not push the BNM config to the NE if the BNM is still enabled on the NE. Follow the process described above.
- If BNM is enabled on the NE after associating the template, AUDIT and ALIGN does not align the configs. Delete the deployment and associate a new template to NE. A new deployment is created with the BNM details.
- System settings: UBT SA nodes only supports DHCP server setting, Event and Alarm log and WebCT initial password reset.
- System settings: Ethernet LOS Criteria and Static LAG criteria are only supported for Core Evo protection
- System Settings: Single-shelf MSS-E/HE/XE/UBT-NIM NEs do not support the System priority settings
- System Settings: Only UBT SA nodes support the WebCT initial password reset.
- NTP setting: The Loopback IP and 0.0.0.0 are forbidden IPs
- BNM settings are not supported for single-shelf MSS-E/HE/XE/UBT-NIM NEs.
- IP Data Plane Settings is only supported for Core Evo nodes
- Do not delete BNM rows
- Under QoS, Only WRED Status - Enable/Disable can be updated for bulk template
- Under QoS, WRED Status - Enable/Disable, Color Classification - PCP/DEI, Dot1p upon selection of color classification as PCP
- An audit only examines attributes that were a part of the original deployment; if a change adds new attributes (for example, enabling QoS classification, which enables configuration of QoS-related attributes) these attributes are not compared when the deployment is audited and any changes are not detected.

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7.12 Name of the Intent: icm-wavence-node-password-policy-assignment

7.12.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: W24 CorEvo MSS4/8, MSS-E/HE/XE, UBT-NIM and UBT-SA

7.12.2 Description

The icm-wavence-node-password-policy-assignment intent supports the deployment/creation/associate/audit/alignment of Wavence Node Password policy assignment to the NEs. - As part of this intent Nodes are assigned a Wavence Node Password Management Policy

The following configurations are supported per deployment:

```
- Intent
- Targets
- Policy Id
```

Prerequisite: Required intent artifacts are imported and installed

7.12.3 Advisory

- N/A

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7.13 Name of the Intent: icm-wavence-node-password-policy-management

7.13.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: W24 MSS, MSS-E/HE/XE, UBT-NIM, UBT-SA

7.13.2 Description

The icm-wavence-node-password-policy-management intent supports the creation/discovery/audit/alignment of Wavence Node Password policies.

The following configurations are supported per deployment:

```
- Policy Id

- General:
   - Policy name
   - User name
   - Old Password
   - New Password
   - Confirm Password

- SFTP Settings:
   - SFTP User Id
   - SFTP Password
   - SFTP Server Port
   - Hostkey Fingerprint
   - SFTP server IP
   - SFTP Root Directory
```

Prerequisite: Required intent artifacts are imported and installed

7.13.3 Advisory

- N/A

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7.14 Name of the Intent: icm-wavence-port-timing-config

7.14.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: MSS 8, MSS 4, MSS E/HE/XE, UBT-NIM, and UBT SA nodes

7.14.2 Description

The icm-wavence-port-timing-config intent supports deploy, align, audit, and configure of all port Timing parameters for Wavence nodes.

The following configurations are supported per deployment:

```
- Sync-E
    - Operation Mode: Synchronous and Non-synchronous
    - Operation Role: Master or slave
- SSM: Enabled/Disabled
- Sync In-Out SFPs values
```

7.14.3 Advisory

- If Operation Mode is Non-synchronous, Operation Role and SSM can not be enabled.
- SSM mode and Operation mode cannot be modified together
- For Radio ports: Only SSM is supported
- For SFPs: Operation Mode config is not supported

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7.15 Name of the Intent: icm-wavence-radio-lag-specifics

7.15.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: W25 MSS and UBT SA nodes

7.15.2 Description

The icm-wavence-radio-lag-specifics intent supports deploy/align/audit and configuration of all radio LAG parameters for Wavence nodes.

The following configurations are supported per deployment:

```
- Radio PM
- Radio Queue
```

7.15.3 Advisory

- Only one LAG at a time can be used for association for each deployment using this intent.
- Radio Queue configuration is allowed only when WRED is enabled on the node and the radio LAG is Up.
- As there is no pull from network option for intents and WRED is part of Network Element configuration intent, if WRED is disabled on the node later, the radio queue attributes are not shown under misalignments when audit is triggered on the radio-lag-specifics intent deployment
- You must perform an associate to network on the radio port before deploying the configuration

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7.16 Name of the Intent: icm-wavence-radio-port-specifics

7.16.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: W25 MSS, MSS-E/HE/XE, UBT-NIM and UBT SA nodes

7.16.2 Description

The icm-wavence-radio-port-specifics intent supports deploy/align/audit and configuration of all radio port parameters for Wavence nodes.

The following configurations are supported per deployment:

```
- Profile: Design RSL, Install RSL
- TX Mute: On , Off , Timed
- Radio PM
- Loopback
- Radio Queue
```

7.16.3 Advisory

- Do not delete listed Loopbacks.
- Only configure Loopback if the radio port is online.
- Radio Queue configuration is allowed only when WRED is enabled on the node and radio port is equipped/online.
- As there is no pull from network option for intents and WRED is part of Network Element configuration intent, when WRED is disabled on the node later , the radio queue attributes are not shown under misalignments when audit is triggered on the radio-port-specifics intent deployment
- You must perform an associate to network on the radio port before deploying the configuration
- Radio Queue settings are availble in Release 25.4 or later

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7.17 Name of the Intent: icm-wavence-radio-qos

7.17.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: MSS-8/MSS-4/MSS-1/MSS-E/MSS-HE/MSS-XE/UBT-NIM and UBT SA nodes

7.17.2 Description

The icm-wavence-radio-qos intent supports deployment/alignment/audit to configure and update all editable radio QoS attributes for Wavence nodes. WRED and radio queue settings are not supported with this intent.

The following configurations are supported per deployment with the supported configuration template, the below list is updated dynamically based on the QoS classification chosen by the user:

```
Default - for all nodes

- User can configure below radio qos parameters:
-   General - QoS Classification - 802.1p/DiffServ/Disabled
-   Dot1p/DSCP - Configuration - Dot1p/DSCP and Forwarding Class
-   Global Queue Setting  - Configuration - Queue Number and Algorithm
-   FC Queue Mapping - Configuration - Forwarding Class and Queue Number
-   UBT Global Queue Setting - Configuration - Queue Number and Algorithm

-   UBT FC Queue Mapping - Configuration - Forwarding Class and Queue
Number
```

7.17.3 Advisory

- Radio QoS configuration attributes are updated dynamically on the deployment based on the QoS classification selected by the user. Not all the above attributes are shown when configuring a deployment; availability is based on QoS classification and applicability to the selected target.
- WRED and radio queue features are not supported as part of this intent.
- Aligning the deployment will cause the value from the configuration template to be pushed to the network.
- An audit only examines attributes that were a part of the original deployment; if a change adds new attributes (for example, enabling QoS classification, which enables configuration of QoS-related attributes) these attributes are not compared when the deployment is audited and any changes are not detected.
- You must associate the configuration template to the target Network Elements before modifying the configuration in the deployment
- Audits can be performed to verify that the configuration from the templates matches what is currently configured in the network. Misaligned attributes can be identified and aligned for the configuration to be deployed back on the network.
- Audits can be performed on individual configuration deployments, or they can be performed on

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configuration templates. If performed on configuration templates, then all deployments associated with the template will be audited.

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7.18 Name of the Intent: icm-wavence-radio-queue-map

7.18.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: MSS-8/MSS-4/MSS-1/MSS-E/MSS-HE/MSS-XE/UBT-NIM and UBT SA nodes

7.18.2 Description

The icm-wavence-radio-queue-map intent supports associate/alignment/audit to configure and update all editable radio queue map attributes for Wavence nodes.

The following configurations are supported per deployment with the supported configuration template:

```
Default for all nodes

- User can configure below Queue Size parameters for all queues from
Queue 1 to Queue 8
-
-   Queue 1 Size
-   Queue 2 Size
-   Queue 3 Size
-   Queue 4 Size
-   Queue 5 Size
-   Queue 6 Size
-   Queue 7 Size
-   Queue 8 Size
```

7.18.3 Advisory

- Only configure the Queue Size parameter when the radio is equipped and online.
- Aligning the deployment causes the value from the configuration template to be pushed to the network.
- You must associate the configuration template to the required radio ports (targets) of Network Elements
- Audits can be performed to verify that the configuration from the templates matches what is currently configured in the network. Misaligned attributes can be identified and aligned for the configuration to be deployed back on the network.
- Audits can be performed on individual configuration deployments, or they can be performed on configuration templates. If performed on configuration templates, then all deployments associated with the template will be audited.

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7.19 Name of the Intent: icm-wavence-radio-ring

7.19.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: W24 MSS, MSS-E/HE/XE NIM

7.19.2 Description

The icm-wavence-radio-ring intent supports the deployment/alignment/audit of configurations on all supported Wavence nodes.

The following configurations are supported per deployment with the supported configuration template:

- General - Name - Description
- Components - R-APS VLAN ID
- Ring Elements

7.19.3 Advisory

- n/a

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7.20 Name of the Intent: icm-wavence-software-upgrade-policy-assignment

7.20.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: MSS, MSS-E/HE/XE, UBT-NIM and UBT SA nodes

7.20.2 Description

icm-wavence-software-upgrade-policy-assignment intent supports deployment creation and association of wavence software upgrade policy assignment to the NEs.

The following configurations are supported per deployment with supported configuration template:

```
- User can configure the following parameters for deployment:
-   Select the template
-   Select the target nodes
-   Select Intent as icm-wavence-software-upgrade-policy-assignment
-   Select Policy ID

- User can configure the following parameters for associate to network:
-   Template is already pre-selected
-   Select the target nodes
-   Select Intent as icm-wavence-software-upgrade-policy-assignment
```

Prerequisite: Required intent artifacts are imported and installed

7.20.3 Advisory

- Import the new intent type for Wavence Software Upgrade policy assignment.
- This intent is pre-requisite for the wavenece software upgrade execution from operations manager.
- Assign NEs to a software upgrade policy using configuration template on Wavence Software Upgrade Policy Assignment intent.
- Associate NEs with software upgrade policy already assigned in NFM-P using configuration template on Wavence Software Upgrade Policy Assignment intent.

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7.21 Name of the Intent: icm-wavence-software-upgrade-policy

7.21.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: MSS, MSS-E/HE/XE, UBT-NIM and UBT SA nodes

7.21.2 Description

The icm-wavence-software-upgrade-policy intent supports the creation/discovery/audit/alignment of Wavence software upgrade policies.

The following configurations are supported per deployment with supported configuration template:

```
- User can configure the following parameters for deployment:
-   Select the template
-   Enter new policy ID
-   Click on View/Edit and enter the below parameters.
         - Policy Name
         - FTP/SFTP Transfer Settings by filling below details:
               - Transfer Protocol
               - User ID
               - Password
               - Server IP
               - Root Directory
               - Hostkey Fingerprint for SFTP
               - Forced Download if required

-   User can configure the following parameters for associate to network:

-   Template is already pre-selected.
-   Enter the already existing policy ID.
```

Prerequisite:

Required intent artifacts are imported and installed.

7.21.3 Advisory

- Import the intent type for Wavence Software Upgrade policy.
- This intent is a pre-requisite for the Wavence software upgrade execution from operations manager.
- Create new Wavence Software Upgrade policy from NSP using configuration template on Wavence Software Upgrade policy intent.
- Associate existing Wavence software upgrade policies in NFMP using configuration template on Wavence Software Upgrade policy intent.

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7.22 Name of the Intent: icm-wavence-system-preference

7.22.1 Applicability

Supported from: NSP 25.4

Verified on Node Type/Version: All Wavence Node variants

7.22.2 Description

The icm-wavence-system-preference intent supports deployment/alignment/audit to configure and update all the editable radio port parameters for Wavence nodes.

The following configurations are supported per deployment with the supported configuration template

```
- LAC Management
      - Enable LAC Management
      - Enable LAC Alarms


- RSL Monitoring Management
      - Enable RSL Monitoring And Deviation Alert
      - Maximum Allowed Number of Deviations
```

7.22.3 Advisory

- n/a

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7.23 Name of the Intent: icm-wavence-tdm-port-specifics

7.23.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: W24A CoreEvo MSS-8/4

7.23.2 Description

The icm-wavence-tdm-port-specifics intent supports deployment/alignment/audit to configure and update all the editable tdm port parameters for Wavence nodes.

The following configurations are supported per deployment:

- for T1 - Mode - Line Code - Signal Mode - Line Lenght (ft)
- for E1 - Mode - Signal Mode
- PDH PM - Incoming Current 15Min - Incoming Current 24Hrs - Outgoing Current 15Min - Outgoing Current 24Hrs

7.23.3 Advisory

- All nodes in target should be of same type. Either all are ANSI or ETSI.

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7.24 Name of the Intent: icm-wavence-time-synchronization

7.24.1 Applicability

Supported from: NSP 24.8

Verified on Node Type/Version: MSS 8, MSS 4, MSS E/HE/XE, UBT-NIM, and UBT SA nodes

7.24.2 Description:

The icm-wavence-time-synchronization intent supports deploy, align, audit, and configure for all Timing parameters on Wavence nodes.

The following configurations are supported per deployment:

```
- Role and  Reference Input mode
- WTR (minutes ) and WTR(seconds)
- Timing reference type: Primary and Secondary
- 1588 enable/disable
```

7.24.3 Advisory :

- UBT SA nodes only support 1588 TC enable/disable.
- If Role is slave:
  - Reference Input mode should be always Revertive
  - Secondary Reference Type should not be Radio LAG
  - Primary Reference Type cannot be E1/DS1
  - Primary Reference Type cannot be Free Run Local Oscillator
- If Role is Master:
  - Primary Reference Type Radio LAG/Radio Port cannot be configured.

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7.25 Name of the Intent: icm-wavence-1588-ptp

7.25.1 Applicability

Supported from: NSP 24.11

Verified on Node Type/Version: MSS 8, MSS 4, MSS-E/HE/XE, UBT-NIM, and UBT-M SA nodes

7.25.2 Description:

The icm-wavence-1588-ptp intent supports deploy/align/audit and configure all the IEEE 1588 PTP synchronization parameters for Wavence nodes.

The following configurations are supported per deployment:

```
- Clock type -- Boundary clock/ Ordinary Clock
- Administartive state -- enable/disable
- Local priority -valid values 1-255
- Clock domain -valid values 24-43
- Frequency source
- Network PDV
- Transport protocol -- Non-configurable
- Multicast MAC address
- Priority 2 - valid values 0-255
- IEEE PTP Port configurations
```

7.25.3 Advisory :

- When the Clock Admin status is Enabled, no configurations can be performed.
- Admin State can not be modified when the port is synchronous
- The same VLAN IDs cannot be used on the same PTP Port
- Admin State must be configured in a separate step from other parameters. For example, general parameters should be configured first and later the admin state should be set to UP.
- For UBT-m SA nodes:
  - 2 PTP ports can be configured.
  - Clock profile and priority 1 clock cannot be configured from NSP.
  - NEC should be enabled for clock and synchronization parameters to be configured.
  - NEC can not be disabled when PTP clock is enabled.
  - PTP clock can be enable donly when NEC is enabled.
  - PTP port can be configured only after the VLAN ID and interface is configured.
- Boundary has 16 ports and Master/Slave has 1 port. When switching from boundary to master and vice-versa, it may take some time to create or delete the ports. If port configuration fails, it is advised to retry the operation.
- UBT-m SA configurations are supported in NSP Release 25.4 or later.

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